



Ministry of Natural Resources of Suriname

Strategic Environmental and Social Assessment (SESA) for the Mining Sector of Suriname

Scoping Report - FINAL

Doc. No. P0032241-SR Rev. 2 – January, 2023

Rev.	Description	Prepared by	Controlled by	Approved by	Date
2	Second Issue	JV, NP, TS, JCC	JV, JH	Julie Evans	January 2023

All rights, including translation, reserved. No part of this document may be disclosed to any third party, for purposes other than the original, without written consent of RINA Consulting Inc.

TABLE OF CONTENTS

	Page
LIST OF TABLES	4
LIST OF FIGURES	4
ABBREVIATIONS AND ACRONYMS	6
1 INTRODUCTION	9
1.1 BACKGROUND	10
1.1.1 Background Documents	11
1.2 ASSUMPTIONS AND LIMITATIONS	11
1.3 STRUCTURE OF THE SCOPING REPORT	12
2 SCOPING OF THE SESA	13
3 APPLICABLE STANDARDS	15
3.1 WORLD BANK OPERATIONAL POLICIES	15
3.2 IFC EHS GUIDELINES – GENERAL AND MINING	18
3.2.1 Air Quality Guidelines	18
3.2.2 Noise and Vibration Guidelines	20
3.2.3 Treated Wastewater Discharge Quality Guidelines	23
3.3 FREE, PRIOR AND INFORMED CONSENT (FPIC)	25
4 LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK IN SURINAME'S MINING SECTOR	27
4.1 INTRODUCTION	27
4.2 LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK	27
4.2.1 National constitution and laws	27
4.2.2 Court rulings from cases that affect directly or indirectly related to the mining sector and related stakeholders	45
4.2.3 Sectoral and cross-sectoral policies	47
4.2.4 International agreements	50
4.2.5 National institutional framework for environmental and social management	59
5 ENVIRONMENTAL AND SOCIAL CHARACTERISTICS	66
5.1 INTRODUCTION	66
5.2 PHYSICAL ENVIRONMENT	66
5.2.1 Climatic Conditions	66
5.2.2 Climate change adaptation	70
5.2.3 Soil and Land Management	72
5.2.4 Water Resources	79
5.3 BIOLOGICAL ENVIRONMENT AND BIODIVERSITY	90
5.3.1 Ecosystems	90
5.3.2 Flora and Fauna	93
5.3.3 Forest Management	95
5.3.4 Deforestation	95
5.3.5 Biodiversity management	98
5.3.6 Protected Areas	98
5.3.7 Key Biodiversity Areas	101
5.4 SOCIO-ECONOMIC	104
5.4.1 Demographics and Population	104
5.4.2 Education	107
5.4.3 Health and Sanitation	107

5.4.4	Labor market	110
5.4.5	Economic Sectors	111
5.4.6	Poverty	113
5.4.7	Indigenous and Tribal Peoples	114
5.4.8	Archaeological and Cultural Heritage	114
5.4.9	Gender	121
6	OVERVIEW OF THE MINING SECTOR OF SURINAME	122
6.1	FUTURE MINING DEVELOPMENT SCENARIOS	124
6.1.1	Sector 1: Bauxite	124
6.1.2	Sector 2: Large-scale Gold Mining	125
6.1.3	Sector 3: Formalization of Small-scale and Artisanal Mining	126
6.2	MINING DEVELOPMENT SCENARIOS	126
6.2.1	Introduction	126
6.2.2	Definition and Assumptions	127
6.2.3	Preliminary Analysis	127
6.3	POTENTIAL CONFLICTS AROUND MINING	133
6.3.1	IAMGOLD and Ndyuka Maroon community of Nieuw Koffiekamp	133
6.3.2	Merian and the Paamaka traditional territory	135
7	IDENTIFICATION OF ENVIRONMENTAL AND SOCIAL RISKS AND VULNERABILITIES	136
8	STAKEHOLDER ENGAGEMENT	147
9	PROPOSED CONTENTS OF THE SESA REPORT	148
10	MECHANISM TO MONITOR IMPLEMENTATION OF SESA	149
	REFERENCES	150

APPENDIX A: Inception Report – SESA for the Mining Sector.

APPENDIX B: Macro-Economic Development Outlook.

APPENDIX C: General display map

APPENDIX D: Stakeholders' inputs for the Scoping Report

LIST OF TABLES

Table 3-1:	WBG Operational Policies	15
Table 3-2:	Comparison of WHO Ambient Air Quality Levels and Targets (2005 vs 2021).	19
Table 3-3:	WBG's Noise Level Guidelines	20
Table 3-4:	WBG's ESH Guidelines on Noise limits for Various Working Environments	21
Table 3-5:	ACGIH TLVs for Hand-Arm Vibration (HAV) Exposure	22
Table 3-6:	ACGIH TLVs for Whole Body Vibration (WBV) Exposure	22
Table 3-7:	EU Directive 2002/44/EC Limit and Action Levels for HAV and WBV	23
Table 3-8:	WBG's EHS Guidelines for Treated Sanitary Sewage Discharges	25
Table 4-1:	National Regulatory Framework	30
Table 4-2:	Court cases and rulings	45
Table 4-3:	Gold mining sector SWOT Analysis	48
Table 4-4:	International Agreements Relevant to Environmental and Social Issues	50
Table 5-1:	Hydrological characteristics of the main rivers in Suriname	81
Table 5-2:	Production stations of the SWM	85
Table 5-3:	SWM water distribution / supply branches	88
Table 5-4:	Water Consumption through SWM network in 2019	89
Table 5-5:	Water Consumption per type of connection	89
Table 5-6:	Taxonomic groups occurring in Suriname with extinction risk assessments published on The IUCN Red List (Aug 2022)	94
Table 5-7:	Post- Deforestation LULC information Deforestation Area (2000 – 2017)	97
Table 5-8:	Total number of Protected Area types designated at a national level and under international conventions and agreements	99
Table 5-9:	Suriname Protected Areas	101
Table 5-10:	Summary of IBAs located in Suriname	102
Table 5-11:	Distribution of cultures by districts and ressorts.	104
Table 5-12:	Environment-related health cases recorded between 2015 and 2019	107
Table 5-13:	Proportion of the population in Suriname that have access to improved sanitation facilities per district (2010-2018)	108
Table 5-14:	Estimated mercury releases into the environment	109
Table 5-15:	Merchandise Exports 2021: Ten Most Important Products	111
Table 6-1:	Base Case Scenario – Mining Sector Development	127
Table 6-2:	Accelerated (Optimistic) Case Scenario – Mining Sector Development	130
Table 7-1:	Preliminary Risks and Impacts and Strategic Objectives	137

LIST OF FIGURES

Figure 2-1:	Technical Approach and Methodology for SESA	13
Figure 4-1:	Development of strategic action item A6.14.7	49
Figure 4-2:	MNR organizational structure	60
Figure 5-1:	Orography of Suriname	67
Figure 5-2:	Monthly Climatology (Temperature and Precipitation) for Suriname between 1991 - 2020	68
Figure 5-3:	Annual and monthly rainfall distribution in Suriname (1961-1985)	69
Figure 5-4:	Modelled Sea Level Rise (+1 m) extent for Coastal Suriname	71
Figure 5-5:	Geological provinces of South America	74

Figure 5-6:	Geographic regions of Suriname	76
Figure 5-7:	Lithological map of Suriname	77
Figure 5-8:	Map of Suriname's Geology and Mineral Deposits.	78
Figure 5-9:	Surface waters of Suriname	80
Figure 5-10:	Groundwater Aquifers of Suriname	83
Figure 5-11:	Groundwater	84
Figure 5-12:	Potential Water Bodies Contamination from ASGM activities upstream	87
Figure 5-13:	Ecosystems of Suriname	91
Figure 5-14:	Mangrove Distribution along the Suriname coastline	93
Figure 5-15:	Distribution of red and black mangrove species dominated mangrove areas, MUMAs (red hatching) and Nature Reserves (purple hatching) along the Suriname coastline	93
Figure 5-16:	Sources of deforestation in Suriname 2000 – 2017	96
Figure 5-17:	Forest Cover in Suriname since 1990 to 2019	96
Figure 5-18:	Annual Deforestation Area (2000 – 2021)	97
Figure 5-19:	Deforestation extent (2000-2019)	98
Figure 5-20:	Protected areas of Suriname	100
Figure 5-21:	Overlap of Protected Areas with KBAs within Suriname	103
Figure 5-22:	Estimated (mid-year) Population, 2019 by sex and age groups	106
Figure 5-23:	Composition of Exports (US\$Mn)	112
Figure 5-24:	Total Government Revenue - by Sector	112
Figure 5-25:	Mining Sector Government Revenue - by Major Industry (%)	113
Figure 5-26:	Mining Sector Employees	113
Figure 5-27:	The largest mound in Suriname, Hertenrits (left) and Pre-Columbian raised fields visible in a flooded field	115
Figure 5-28:	Previously Recorded Archaeological Sites in Suriname	115
Figure 5-29:	Abandoned sugar mill (left) and train engine (right) - Marenburg Sugar Factory	116
Figure 5-30:	Example of Built Heritage Resources in Paramaribo - World Heritage Site	116
Figure 5-31:	Ruins at Joden Savanne and Cassipora Cemetery. Synagogue ruins (left) and cemetery (right)	117
Figure 5-32:	Examples of 19th and 20th century Built Heritage in Suriname.	118
Figure 5-33:	Indigenous Territories	119
Figure 5-34:	Traditional Wayana Tukusipan	120
Figure 5-35:	Maroon Communities in Suriname	120
Figure 6-1:	Greenstone belt, mining titles and current Industrial Mining sites operating	123

Disclaimer

This document has been prepared for the titled project or named part thereof and should not be relied upon or used for any other project without an independent check being carried out as to its suitability and prior written authority of RINA Consulting Inc (RINA) being obtained. RINA accepts no responsibility or liability for the consequence of this document being used for a purpose other than those for which it was commissioned. Any person using or relying on the document for such other purpose will by such use or reliance be taken to confirm her/his agreement to indemnify RINA for all loss or damage resulting therefrom. RINA Consulting accepts no responsibility or liability for this document to any party other than the person by whom it was commissioned.

As provided for in RINA's proposal, to the extent that this report is based on information supplied by other parties, RINA Consulting accepts no liability for any loss or damage suffered by the client, whether contractual or tortious, stemming from any conclusions based on data supplied by parties other than RINA and used by RINA in preparing this report.

ABBREVIATIONS AND ACRONYMS

ACGIH	American Conference of Governmental Industrial Hygienists
ACT	Amazon Conservation Team
AMD	Acid Mine Drainage
AQG	Air Quality Guidelines
ARD	Acid Rock Drainage
ASGM	Small-scale gold mining
ASM	Small-scale mining
AZE	Alliance for Zero Extinction Site
BIS	Bauxite Institute of Suriname
BOD	Biochemical oxygen demand
BOG	Bureau for Public Health
°C	Degree Celsius
CELOS	Center for Agricultural Research in Suriname
CEPF	Critical Ecosystem Partnership Fund
CITES	Convention on the International Trade of Endangered Species of Wild Fauna and Flora
CO	Carbon Monoxide
COD	Chemical oxygen demand
COP	Conference of the Parts
CPS	Country Partnership Strategy
CR	Critically Endangered
CS	Country Strategy
CSNR	Central Suriname Nature Reserve
Db	Decibel
DbA	Decibel A Scale
DD	Data Deficient
DIS	Mineral Institute of Suriname
DWV	Water Supply Service
EA	Environmental Assessment
EAS	Energy Authority Suriname
EIA	Environmental Impact Assessment
EC	European Council Directives
EHS	Environmental Health Safety
EIS	Environmental Impact Statement
EITI	Extractive Industries Transparency Initiative
EN	Endangered
ENSO	El Niño Southern Oscillation
ESMF	Environmental and Social Management Framework
EU	European Union
EX	Extinct
EW	Extinct in the Wild
FAO	Food and Agriculture Organization
FOB	Funding Development Interior
FPIC	Free, Prior, and Informed Consent
GBB	Ministry of Land Policy and Forest Management
GBS	General Statistics Office of Suriname
GDP	Gross Domestic Product
GHG	Greenhouse gas
GIS	Geographic Information System
GLiM	Global Lithological Map
GMD	Geological Mining Service
GOMIAN	Small Scale Gold Mining in the Amazon
GOS	Government of Suriname
ha	Hectare
HAV	Hand-Arm Vibration
HFLD	High Forest Cover, and Low Deforestation
HFO	Heavy Fuel Oil

Hz	Hertz
IBA	Important Bird and Biodiversity Area
IBAT	Integrated Biodiversity Assessment Tool
IBRD	International Bank for Reconstruction and Development
ICMM	International Mining Standards
IDB	Inter-American Development Bank
IFC	International Finance Corporation
IICA	Inter-American Institute for Cooperation on Agriculture
ILO	International Labour Organization
IP	Indigenous People
ITPPF	Indigenous and Tribal Peoples Planning Framework
IUCN	International Union for Conservation of Nature
KAMPOS	Collaboration of Tribal Peoples in Suriname
KBA	Key Biodiversity Area
km ²	Square kilometer
KOPI	Statistical Data for the National Forest Monitoring System
KSMH	Kaloti Suriname Mint House
LAeq	Equivalent Continuous Sound Pressure Level
LAm _{ax}	Maximum sound level at an instant in time
LBB	Suriname Forest Service
LC	Least Concern
LMO	Live Modified Organism
LR/CD	Lower Risk/Conservation Dependent
LSM	Large-scale Mining
LULC	Land Use Land Cover
LVV	Ministry of Agriculture, Animal Husbandry and Fisheries
m	Meter
m ³	Cubic Meter
MARPOL	International Convention for the Prevention of Pollution from Ships
mg/l	Milligram per liter
mg/m ³	Microgram per cubic meter
MI-GLIS	Management Institute for Land Registration and Land Information System
MinOWC	Ministry of Education, Science and Culture
MNR	Ministry of Natural Resources of the Government of Suriname
ML	Metals Leaching
MPN	Most Probable Number
MSG	Suriname Multi-Stakeholder Group
MUMA	Multi Use Management Areas
N. A	Not applicable
NAP	National Action Plan
NCCR	National Coordination Center for Disaster Relief
NCD	Nature Conservation Division
NDC	Nationally Determined Contribution
NDP	National Development Plan
NGO	Non-Governmental Organizations
NIMOS	National Institute for Environment and Development in Suriname
NMA	National Environment Authority
NMR	National Council for the Environment
NO ₂	Nitrogen dioxide
NT	Near Threatened
NZCS	National Zoological Collection Suriname
O ₃	Ozone
OGS	Suriname Gold Sector Regulation Commission
OHS	Occupational Health and Safety
OIS	Organization of Indigenous Communities of Suriname
OP	Operational Policies
OW	Ministry of Public Works
PBL	Policy-Based Loan
PCR	Physical Cultural Resources
pH	Potential of hydrogen

PM	Particulate Matter
POP	Persistent Organic Pollutants
REDD+	Reducing emissions from deforestation and forest degradation
RGB	Ministry of Spatial Planning, Land and Forestry Management
RIL	Reduced Impact Logging
RINA	RINA Consulting inc.
ROM	Ministry of Spatial Planning and the Environment
RPF	Resettlement Policy Framework
RSA	Rapid Social Assessment
SBB	Foundation for Forest Management and Production Control
SCSD	Suriname Competitiveness and Sector Diversification Project
SESA	Strategic Environmental and Social Impact Assessment
SLR	Sea Level Rise
SMEs	Small and medium-sized enterprises
SO ₂	Sulfur dioxide
SRD	Surinamese Dollars
SSM	Small-scale Mining
STINASU	Foundation for Nature Conservation in Suriname
SWM	Surinamese Water Company
TDS	Total Dissolved Solids
TLV	Threshold Limit Values
ToR	Terms of Reference
UN	United Nations
UNCLOS	United Nations Convention for the Law of the Sea
UNDP	United Nations Development Program
UNESCO	United Nations Educational, Scientific and Cultural Organization
UNFCC	United Nations Framework Convention on Climate Change
USAID	United States Agency for International Development
VU	Vulnerable
WASH	Access to safe drinking water, sanitation and hygiene
WBG	World Bank Group
WBV	Whole Body Vibration
WDPA	World Database on Protected Areas
WHO	World Health Organization
WSRIS	Suriname Water Resources Information System
WTO	World Trade Organization

1 INTRODUCTION

RINA Consulting, Inc. (RINA) has been retained by the Ministry of Natural Resources (MNR) of the Government of Suriname (GOS) to conduct a Strategic Environmental and Social Assessment (SESA) of the Mining Sector financed by the World Bank Group's (WBG) International Bank for Reconstruction and Development (IBRD). The development of this SESA is part of an overarching goal to strengthen the environmental and social standards, and impact management under the Country Partnership Strategy (CPS) between the WBG and the GOS.

As part of this consultancy, RINA will prepare the following:

Deliverable	Deliverable/Output	Main Objective
1	Inception Report and PPT (Methodology and Approach and Stakeholder list)	Include the workplan for consultations that will ensure that the stakeholders' concerns and advice are considered in the SESA.
2	Scoping Report	Frame the context and methodology for the SESA. This will include the Table of Contents for the SESA report.
3	Regulatory and Institutional Framework and Assessment of the GOS Capacity	Assess the adequacy of the framework, based on international experience, and the capacity of public agencies at national and local levels, and governmental organizations at national and community levels to address the potential impacts of developments in mining, especially for licensing and enforcement of legal ESHS requirements.
4	Identification of the general, strategic environmental and social consequences - and - Proposed Mitigation and Monitoring Measures report	Identify and evaluate the environmental and social consequences and risks, including climate change risks, associated with existing developments and plans for investment in mining, and safety and occupational health risks and issues relevant to mining, considering the forecasts and scenarios developed. This will include mapping to identify overlapping areas of natural resources / forests/ other areas of high biodiversity value, with potential and actual extraction sites, and identify potential mitigation measures and monitoring requirements for each potential strategic impact on mining. Mitigation measures for strategic impacts may include, among others: implementation of appropriate legislation, including legal environmental discharge standards for emissions into the air, water and soil; measures specifically targeting engagement with Indigenous and tribal communities, drawing on initial mitigation measures to enable effective benefit sharing with such communities identified in the Indigenous and Tribal Peoples Planning Framework (ITPPF) and Rapid Social Assessment; revised institutional organization, streamlining environmental responsibility, creation of new GOS agency/ies; strengthening of specific GOS Agencies; review or creation of environmental and social licensing agencies or mechanisms; enforcement of international best practices and operational standards guidelines for each sector; creation of

Deliverable	Deliverable/Output	Main Objective
		protected areas or exclusion zones (No Go zones, buffer zones); etc. Specific indicators may be selected to allow for follow-up of the effectiveness of proposed mitigation measures. Indicators may be individual/stand-alone or combination (complex index), such as established indices, e.g., representing poverty reduction, migratory bird species, water quality, etc.
5	Draft SESA Report and Action Plan	Based on the analysis conducted, and the recommendations developed, an Action Plan will be developed to facilitate the SESA's implementation and associated responsibilities. Priorities will be structured into immediate, short-term (1-2 years), and medium-term (3-5 years), including the Action Plan Cost Estimate.
6	Final SESA Report and SESA Action Plan, including Stakeholder Consultation Report	Final SESA report will be presented after validation with stakeholders. The SESA will be accompanied by the Stakeholders' Consultation Report.

1.1 BACKGROUND

The SCSD project consists of three components:

- ✓ Component 1: Strengthening the mining sector governance, transparency, accountability, and administration.
- ✓ Component 2: Investing in Small and medium-sized enterprises (SMEs) and value chains in targeted emerging industries.
- ✓ Component 3: Project management and evaluation.

The first component of the SCSD project (SCSD - C1) is focused on technical assistance and seeks to specifically strengthen the mining sector governance, transparency, accountability, and administration. This component focuses on supporting improvements to the legal, regulatory, and institutional framework governing mining in Suriname. The purpose is to improve social and environmental impact management and to align Suriname's framework to international best practice by providing its institutions with knowledge and tools to carry out their mandated functions. Specific activities planned in the SCSD, in addition to the development of the SESA, include the following:

1. Strengthening the legal, regulatory, and institutional frameworks governing the mining sector through support towards the revision of relevant legislation and regulations and the establishment of the Minerals Institute, which will integrate and strengthen mining cadaster, geological data management, and mining inspectorate functions. This includes defining a roadmap and implementing institutional reforms; supporting the development of required legal and regulatory frameworks; and supporting the establishment and operationalization of the Minerals Institute, including the collection of production and export statistics and setting up a Minerals Statistics Database.
2. Sector administration capacity building to assist the GOS, in particular staff from the Ministry of Natural Resources, the Minerals Institute, and other relevant ministries including the Ministry of Finance, to build the required technical skills and managerial capabilities to regulate and monitor the mining sector.
3. Strengthening revenue assessment, collection, and forecasting through support to the Extractive Industries Transparency Initiative (EITI) and setting up systems to improve revenue collection in a

transparent and accountable manner. This includes the development of documentation systems and standardized procedures for the submission and analysis of monitoring reports, in conformity with the Mining Decree and specific investment agreement commitments, as well as for the calculation of mining royalties and establishment of export controls.

4. Building independent oversight capacity and stakeholder consultation mechanisms, by providing support to non-state actors involved in the EITI process in Suriname (including Indigenous and tribal peoples' representatives, non-governmental organizations, community-based organizations, technical experts, and other civil society representatives), to strengthen their participation in increasing transparency and disclosure in the mining industry.
5. Enhancing environmental, health, and social performance of the mining sector, including:
 - a) Preparation of a SESA covering mining, which will entail a comprehensive sector-wide examination of potential impacts, both positive and negative, of future development in the mining sector, and the identification of gaps in regulations, institutional capacity, and public consultation mechanisms, and areas that can be strengthened.
 - b) Capacity support to enhance social and environmental impact management, including technical assistance for the implementation of select reforms recommended by the SESA. This support will be consistent with international standards such as the WBG's on Environmental, Health, Safety and REDD+ guidelines, and will promote improved mining practices that aim to reduce impacts on forest, reduce the sector's vulnerability to climate change risks, help mitigate climate change - through improved water efficiency and substitution of inputs/enhanced energy efficiency, to reduce GHG emissions in existing and future mining operations - and improve Indigenous and tribal community engagement.

This document is the Scoping Report for the SESA study for the Mining Sector of Suriname, described in item 5(a) above. It should be noted that according to the Terms of Reference (ToR) for this consultancy, the Inception Report or Work Plan was prepared simultaneously and should be read in combination with this Scoping Report. Both the Inception report and this Scoping Report will be presented before the stakeholders for consultation in the events to take place between October 24 and November 5, 2022, for review and comments.

1.1.1 Background Documents

Various instruments were prepared by the GOS as part of SCSD lending requirements. These prior studies have been reviewed and incorporated as references as part of the desktop review of this Scoping stage. The studies are listed below.

- ✓ Country Partnership Strategy (WBG, April 2015);
- ✓ SCSD Environmental and Social Management Framework – ESMF (WBG, June 2019a);
- ✓ SCSD Indigenous and Tribal Peoples Planning Framework – ITPPF (WBG, February 2019b);
- ✓ SCSD Rapid Social Assessment – RSA (WBG, June 2019c); and
- ✓ SCSD Resettlement Policy Framework – RPF (WBG, June 2019d).

In addition, the SCSD project has strong and compelling links with Suriname's National Development Plan (Nationaal Ontwikkelings Plan - NOP) 2022-2026 (NDP), which outlines the development trajectory for Suriname for upcoming years. For the development of this scoping report, RINA reviewed the NDP 2022-2026 to understand the objectives related to the development of the mining sector.

1.2 ASSUMPTIONS AND LIMITATIONS

The SESA Scoping process was subject to a number of assumptions and limitations, which should be considered when reading the information presented in this report. The Consultant is confident that these

assumptions and limitations do not compromise the integrity of the SESA Scoping report. The relevant assumptions and limitations are as follows:

- Limited Geographic Information Systems data to spatially analyze limits of possible impacts;
- The SESA Scoping report provides a general context about the potential impacts: additional specialized studies may be necessary following the identification of impacts that will need further detailed analysis during the SESA phase;
- The SESA Scoping process provides expert analysis and should not be categorized as a full SESA study.

1.3 STRUCTURE OF THE SCOPING REPORT

The structure of the scoping report was designed to aid in framing the content and methodology of the SESA. The scoping report consist of the following sections:

- Introduction: Introduces the scope of the consultancy and background documents used as references.
- Scoping of the SESA: Provides an overview of the SESA Scoping process.
- Applicable Standards: Provides an overview of the international standards that will guide the SESA.
- Legal and Institutional Framework: Provides an overview of the national legal and regulatory framework (decrees, acts), relevant institutions and latest updates to the regulatory framework and responsibilities of the relevant institutions.
- Environmental and Social Characteristics: Provides a baseline of the physical, biological and socio-economic conditions of the area most likely to be affected by future mining.
- Overview of the Mining Sector: Provides a description of the mining resources, history of exploitation, potential conflicts around mining, scenarios for potential development. The scenarios will assist to kick start the dialogue on the options available for the sector development.
- Identification of Environmental and Social Risks, Impacts and Vulnerabilities: Provides a general description of potential impacts centered in themes (water, air, land use, geology soils, biodiversity, cultural heritage, etc.) and preliminary mitigation measures. In conjunction with this section, a display map (Appendix C) was prepared to aid in consultations with stakeholders and especially with the Indigenous Populations and Tribal Communities.
- Stakeholder Engagement: Provides a snapshot of the agenda for the consultation activities and validation to occur with stakeholders. As noted earlier, the Inception Report contains the complete timeline and agenda for the consultation events.
- Proposed Terms of Reference for the SESA: Provides the proposed terms of reference for the development of the SESA to be validated.
- Mechanism to monitor implementation of the SESA: Provides a discussion of the mechanism to be drafted for the monitoring of the implementation for the SESA.
- Appendix A - Inception Report - SESA for the Mining Sector: At the time of this writing the Inception Report has been submitted to the client for comments. It is included here as reference.
- Appendix B – Macro-Economic Development Outlook: Provides an overview of current economic and business environment, imports and exports, linkages with donor and GOS development plans. This outlook will allow us to understand the potential cumulative impacts of the country's proposed developments. This analysis will be included in the draft SESA.

2 SCOPING OF THE SESA

The SESA will entail a comprehensive mining sector-wide examination of potential impacts, both positive and negative, of future potential investments in the mining sector, and identify gaps in regulations, institutional capacity, and public consultation mechanisms that can be strengthened. For this purpose, the SESA requires a strategic look at environmental and social aspects, and continuous stakeholder consultations. The SESA will provide guidance to the MNR and NIMOS on systematically integrating environmental, socio-economic, health and safety concerns in policy, regulation and planning, project development, operations, maintenance, and decommissioning of mining sector activities. Furthermore, the SESA will provide a benchmark for cumulative environmental and social effects with other ongoing sector programs and plans. The SESA will ensure that the concerns and expectations of different stakeholders are taken into account in the decision-making process for equity and sustainable development of Suriname's mining sector.

As a summary, the development of the SESA will follow 6 Stages. The general approach to be used is presented in Figure 2-1.

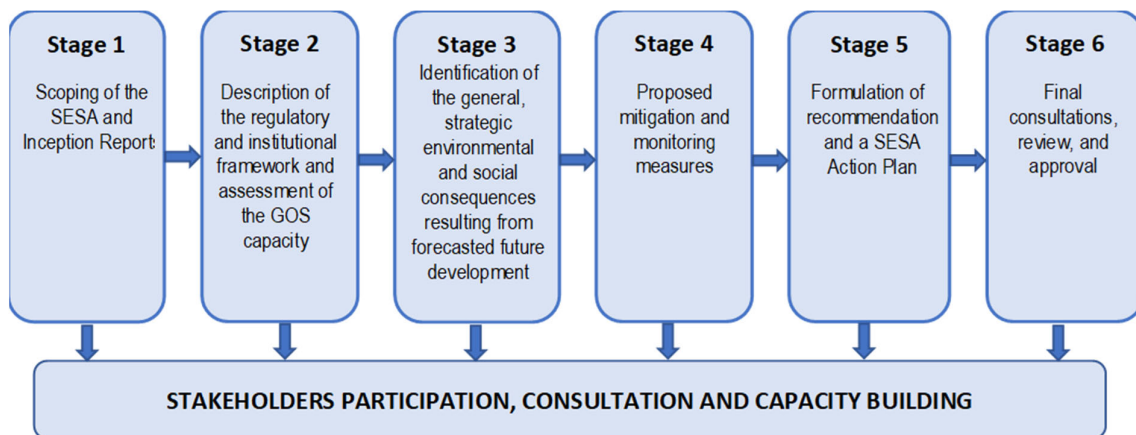


Figure 2-1: Technical Approach and Methodology for SESA

Source: RINA, 2022

The first stage of the SESA requires the development of a Scoping Report (this document) to frame the content and methodology with substantial stakeholder inputs. To frame the SESA study, scenarios of the mining sector's development have been proposed and will be presented in the relevant section of this report. The purpose of presenting these scenarios is to initiate dialogue with the stakeholders.

To this end, the following process is proposed in accordance with the ToR of this Consultancy:

- ✓ Identify the preliminary environmental and social characteristics of those areas most likely to be affected by future mining activities, identifying sensitive ecosystems, protected areas, areas of high biodiversity value, areas inhabited by Indigenous peoples, Maroons, fishermen, etc. This will include the identification of available and relevant Geographic Information Systems (GIS) data for the mining sector.
- ✓ Assemble preliminary desktop information relevant to the scope of the SESA, including but not limited to: current projects, plans and trends in the applicable mining sectors, including artisanal and small-scale mining (ASM), and to present the forecasts of future mining development scenarios to

be developed subsequently, and scenarios for sector development including constraints and opportunities.

- ✓ Identify the preliminary environmental and social risks and vulnerabilities to be considered and assessed (e.g., potential Indigenous and tribal community land use, water quality, air quality, geology and soils, biodiversity, marine resources, socio-economic, coastal activities, etc.).
- ✓ Ensure that the SESA be a participatory process, and that the Scoping Report be consulted/validated with stakeholders according to the stakeholder consultation plan presented in the Inception report. For this purpose:
 - Key stakeholders in the mining sector have been preliminary mapped. These stakeholders will be consulted, their interests, concerns and incentives will be analyzed. The specific plan for stakeholder consultations is presented in the Inception Report. The aim of the Inception report is to make sure that stakeholder's interests, concerns and advice are considered in the SESA.
 - The Scoping Report and Inception Report will be presented to the stakeholder in the first validation meeting(s). They will provide input to both reports prior to finalization.

3 APPLICABLE STANDARDS

The SESA is a Strategic Environmental Social Assessment process that uses analysis and public participation that engages various stakeholders in a dialogue on environmental and social issues they consider priorities and that are or may be affected by mining sector reforms. Stakeholders include government, public and private organizations, and vulnerable sectors such as local communities, Indigenous and tribal peoples, women and youth, and makes sure their voices are heard. Specific activities and general implementation of the SESA will be directed by the WBG Operational Policies, international standards, and best practices. These policies and standards are described below.

3.1 WORLD BANK OPERATIONAL POLICIES

The applicable Operational Policies of the WBG are described in the following table.

Table 3-1: WBG Operational Policies

Operational Policies (OPs)	Description	Applicability
OP 4.01 Environmental Assessment	OP 4.01 deals with environmental assessment and covers potential impacts on the natural environment (air, water, and land); human health and safety; social aspects (involuntary resettlement, Indigenous peoples, and physical cultural resources); and transboundary and global environmental aspects. Although this OP considers natural and social aspects in an integrated way, social aspects (involuntary resettlement, Indigenous peoples) as well as natural habitats, forests, and pest management are covered by separate policies with their own requirements and procedures, as presented in the lines below. Environmental management plan (EMP) and environmental and social management framework (ESMF) are presented on the OP as instruments for environmental assessments.	Considering that (i) one of the SESA's purposes is to improve social and environmental impact management and to align Suriname's framework with international best practices by providing its institutions with knowledge and tools to carry out their mandated functions, (ii) the nature of the mining activities (including associated facilities like rail, port, airstrips, etc.) as well as their potential social and environmental impacts OP 4.01 is triggered. To inform activities, an Environmental and Social Management Framework (ESMF) was prepared for the overall SCSD project, which will be considered during the SESA. In addition, for the SESA's purpose, IFC Performance Standard 1 (Assessment and Management of Environmental and Social Risks and Impacts) EHS Guidelines (both general and mining specific) will be considered.
OP 4.04 Natural Habitats	The WBG supports the protection, maintenance, and rehabilitation of natural habitats and their functions. The conservation of natural habitats is essential for long-term sustainable development. Natural habitats are land and water areas where (i) the ecosystems' biodiversity is formed largely by native plant and animal species, and (ii) human activity has not essentially modified the area's primary ecological functions. The Bank encourages borrowers to incorporate into their development and environmental	Considering the characteristics of the natural setting in Suriname, based on policy 4.04, mining activities may impact particularly sensitive receptors (natural habitats, local populations centers, and dispersed Indigenous communities). The SESA will also consider the WBG's EHS Guidelines (both general and mining specific) and the ESMF, which integrates natural habitats considerations in exclusion criteria, as appropriate for the selection of specific sites for interventions. The ESMF will be considered during the

Operational Policies (OPs)	Description	Applicability
	strategies analyses of any major natural habitat issues, including identification of important natural habitat sites, the ecological functions they perform, the degree of threat to the sites, priorities for conservation, and associated recurrent-funding and capacity-building needs.	SESA. An identification of (a) natural habitat issues and special needs for natural habitat conservation, including the degree of threat to identified natural habitats (particularly critical natural habitats), and (b) measures for protecting such areas in the context of the SCSD project will also be considered.
OP 4.09 Pest Management	This policy assists borrowers to manage pests that affect either agriculture or public health, and promotes and supports the safe, effective, and environmentally sound pest management, in particular the selection and use of recommended Classification of Pesticides by Hazard and Guidelines to Classification	This policy is triggered after considering the characteristics of the mining sector activities and local environmental conditions, which may affect public health and pest management for mining operations, like railway (use of herbicides) and mining camps (use of pesticides) for vector control. As part of this OP, it will assess the capacity of the Suriname regulatory framework and institutions to promote and support safe, effective, and environmentally sound pest management.
OP 4.10 Indigenous Peoples	These are defined to be a distinct, vulnerable, social and cultural group possessing a number of characteristics including collective attachment to geographically distinct habitats or ancestral territories in the project area and to the natural resources in these habitats and territories.	This policy is triggered given the presence of Indigenous peoples (IP) near areas where economical, agribusiness and mining activities occur in Suriname. Indigenous Peoples are closely tied to the land, forests, water, wildlife, and other natural resources, and therefore special considerations apply if the project affects such ties. Additional to, but not within the scope of the ESMF, a Social Assessment will be conducted, including consultation with IP representatives to inform about the preparation of the SESA and the IPPF (Indigenous Peoples Planning Framework) as part of the ESMF¹. (See also the ITPPF safeguard Document). These documents will be considered as part of the SESA. Projects should include measures to (a) avoid potentially adverse effects on the Indigenous Peoples' communities; or (b) when avoidance is not feasible,

¹ The SESA will include an impact assessment to estimate the total potential impact to Indigenous peoples and vulnerable groups in all the sectors identified, and will include detailed analysis of potential benefits, impacts and risks concerning indigenous peoples of potential sector specific investments

Operational Policies (OPs)	Description	Applicability
		minimize, mitigate, or compensate such effects.
OP 4.11 Physical Cultural Resources (PCR)	This policy addresses physical cultural resources, which are defined as movable or immovable objects, sites, structures, groups of structures, and natural features and landscapes that have archaeological, paleontological, historical, architectural, religious, aesthetic, or other cultural significance.	Although the project is not expected to have direct negative impacts on cultural property, 'chance finds' during the implementation of activities could be possible. As part of the environmental assessment (OP 4.01 Environmental Assessment) 'Chance finds' procedures are incorporated into the ESMF, including procedures to identify PCRs, and avoid or mitigate any adverse impacts on physical cultural resources, provisions for managing chance finds, any necessary measures for strengthening institutional capacity, and a monitoring system to track the progress of these activities.
OP 4.12 Involuntary Resettlement	This policy aims to address and mitigate risks of physical relocation, loss of land and other assets, sources of income and means of livelihood by local people due to proposed projects. The policy also applies to the involuntary restriction of access to legally designated parks and protected areas resulting in adverse impacts on the livelihoods of the displaced persons. This policy explicitly states that individuals without recognizable legal right or claim to the land or assets they occupy, or use must be provided resettlement assistance, as well as compensation for loss of assets other than land, if such would result from the project.	This policy is triggered, as social issues related to involuntary resettlement associated with mining projects, since these potential impacts are essentially covered by the SESA. The Resettlement Policy Framework (see RPF safeguard Document) of the SCSD as well as a resettlement plan will be considered during the SESA.
OP 4.36 Forests	The objective of this policy is to assist borrowers to harness the potential of forests in a sustainable manner, integrate forests effectively into sustainable economic development and protect the vital local and global environmental services of forests.	This policy is triggered given the potential of mining development to impact forests negatively, and/or the rights and welfare of local communities and their relationship with forests, as well as bring changes in the management, protection, or utilization of natural forests or plantations, whether they are publicly, privately, or communally owned. SESA will consider environmental assessments, poverty assessments, social analyses, Public Expenditure Reviews, and other economic and sector work to identify the economic, environmental, and social significance of forests in Suriname.

Operational Policies (OPs)	Description	Applicability
OP 4.37 Safety of Dams	This policy aims for ensuring that appropriate measures are taken, and sufficient resources are provided to ensure that dams are safely designed, and their construction supervised by experienced and competent professionals, including the adoption and implementation of dam safety measures for the design, bid tendering, construction, operation, and maintenance of the dam and associated works.	This policy is triggered for mines that require the construction of new slime or tailing dams. The ESMF considers the potential impacts on dam failure on the environment.

Source: (WBG, June 2019a) modified by RINA, 2022

In August 2016, the WBG adopted a new set of environment and social policies called the Environmental and Social Framework (ESF). As of October 1, 2018, the ESF applies to all new WBG investment project financing. These policies will be required by borrowing governments seeking financing by the WBG and will be referred to in the SESA report as a benchmark for good international practice.

3.2 IFC EHS GUIDELINES – GENERAL AND MINING

The following sections describe the different general and mining sector specific EHS guidelines, developed by the WBG and IFC, that are applicable to a mining project.

3.2.1 Air Quality Guidelines

The GOS has not established air quality levels standards in its legal or regulatory framework, therefore, the WBG's General ESH Guidelines will apply. As stated in the WBG's General EHS guidelines, for *"Projects with significant sources of air emissions, and potential for significant impacts to ambient air quality, should prevent or minimize impacts by ensuring that:*

- *Emissions do not result in pollutant concentrations that reach or exceed relevant ambient quality guidelines and standards² by applying national legislated standards, or in their absence, the current World Health Organization (WHO) Air Quality Guidelines (see Table 1.1.1), or other internationally recognized sources³;..."*

The following table presents the revised *New WHO 2021 ambient air quality levels and targets* (WHO, 2021) compared with the former *WHO 2005 Air Quality Guidelines (AQG) targets* listed in WBG's General EHS Guidelines.

² Ambient air quality standards are ambient air quality levels established and published through national legislative and regulatory processes, and ambient quality guidelines refer to ambient quality levels primarily developed through clinical, toxicological, and epidemiological evidence (such as those published by the World Health Organization).

³ For example, the United States National Ambient Air Quality Standards (NAAQS) (<http://www.epa.gov/air/criteria.html>) and the relevant European Council Directives (Council Directive 1999/30/EC of 22 April 1999 / Council Directive 2002/3/EC of February 12, 2002).

Table 3-2: Comparison of WHO Ambient Air Quality Levels and Targets (2005 vs 2021).

Pollutant	2005 Targets*		New 2021 AQG Levels & Targets**	
	Averaging Period	Guideline value in $\mu\text{g}/\text{m}^3$	Averaging Period	Guideline value in $\mu\text{g}/\text{m}^3$
Sulfur dioxide (SO_2)	10 minutes	50 (Interim target-2) 500 (guideline)	10 minutes***	500 (guideline)
	24-hour	125 (Interim target-1) 50 (Interim target-2) 20 (guideline)	24-hour	125 (Interim target-1) 50 (Interim target-2) 40 (Air Quality Guideline)
Nitrogen dioxide (NO_2)	1-hour	200 (guideline)	1-hour***	200 (guideline)
	1-year	40 (guideline)	1-year	40 (Interim target-1) 30 (Interim target-2) 20 (Interim target-3) 10 (Air Quality Guideline)
	N.A.	N.A.	24-hour	120 (Interim target-1) 50 (Interim target-2) 25 (Air Quality Guideline)
Particulate Matter (PM_{10})	1-year	70 (Interim target-1) 50 (Interim target-2) 30 (Interim target-3) 20 (guideline)	Annual	70 (Interim target-1) 50 (Interim target-2) 30 (Interim target-3) 20 (Interim target-4) 15 (Air Quality Guideline)
	24-hour	150 (Interim target-1) 100 (Interim target-2) 75 (Interim target-3) 50 (guideline)	24-hour	150 (Interim target-1) 100 (Interim target-2) 75 (Interim target-3) 50 (Interim target-4) 45 (Air Quality Guideline)
Particulate Matter ($\text{PM}_{2.5}$)	1-year	35 (Interim target-1) 25 (Interim target-2) 15 (Interim target-3) 10 (guideline)	Annual	35 (Interim target-1) 25 (Interim target-2) 15 (Interim target-3) 10 (Interim target-4) 5 (Air Quality Guideline)
	24-hour	75 (Interim target-1) 50 (Interim target-2) 37.5 (Interim target-3) 25 (guideline)	24-hour	75 (Interim target-1) 50 (Interim target-2) 37.5 (Interim target-3) 25 (Interim target-4) 15 (Air Quality Guideline)
Ozone (O_3) - Short term daily maximum 8 hr	8-hour daily maximum	160 (Interim target-1) 100 (guideline)	8-hour daily maximum	160 (Interim target-1) 120 (Interim target-2) 100 (Air Quality Guideline)
Ozone (O_3) - Peak Season	N.A.	N.A.	8-hour daily maximum	160 (Interim target-1) 70 (Interim target-2)

Pollutant	2005 Targets*		New 2021 AQG Levels & Targets**	
	Averaging Period	Guideline value in $\mu\text{g}/\text{m}^3$	Averaging Period	Guideline value in $\mu\text{g}/\text{m}^3$
				60 (Air Quality Guideline)
Carbon Monoxide (CO)	N.A.	N.A.	24-hour	7 (Interim target-1) 4 (Air Quality Guideline)
			8-hour***	10 (Air Quality Guideline)
			1-hour***	35 (Air Quality Guideline)
			15- minute***	100 (Air Quality Guideline)

Sources:

* Table 1.1.1. of the WBG's (2007) General Environmental, Health, and Safety (EHS) Guidelines (page. 4).

** WHO (2021) WHO global air quality guidelines: particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide.

*** short average periods still valid (Table 3.25 - page 135 in WHO (2021))

3.2.2 Noise and Vibration Guidelines

The WBG General EHS Guidelines requires that noise impacts should not exceed the Decibel A Scale (DbA) levels presented in Table 3-3, or result in a maximum increase in background levels of 3 decibels (dB) at the nearest receptor location off-site. The GOS still needs to establish noise levels standards in its legal or regulatory framework.

Table 3-3: WBG's Noise Level Guidelines

Noise Level Guidelines (a)		
Pollutants	One Hour LAeq* (dBA)	
	Daytime 07:00 - 22:00	Nighttime 22:00 - 07:00
Residential; institutional; educational (b)	55	45
Industrial; commercial	70	70

Source: WBG, 2007

Notes:

(a) Guidelines values are for noise levels measured out of doors. Source: Guidelines for Community Noise. (WHO, 1999).

(b) Noise monitoring should be carried out using a Type 1 or 2 sound level meter meeting all appropriate IEC standards

In addition, during construction and decommissioning activities, noise and vibration may be caused by the operation of pile drivers, earth moving and excavation equipment, concrete mixers, cranes and the transportation of equipment, materials, and people. Some recommended noise reduction and control strategies to consider in areas close to community areas include:

- Planning activities in consultation with local communities so that activities with the greatest potential to generate noise are planned during periods of the day that will result in least disturbance.
- Using noise control devices, such as temporary noise barriers and deflectors for impact and blasting activities, and exhaust muffling devices for combustion engines.

- Avoiding or minimizing project transportation through community areas.

3.2.2.1 Occupational Permissible Noise and Vibration Exposure Levels

The WBG General EHS Guidelines provide noise limits for various working environments (see Table 3-4). These guidelines require that:

- No employee should be exposed to a noise level greater than 85 dB(A) for a duration of more than 8 hours per day without hearing protection;
- No unprotected ear should be exposed to a peak sound pressure level (instantaneous) of more than 140 dB(C).
- The use of hearing protection should be enforced actively when the equivalent sound level over 8 hours reaches 85 dB(A), the peak sound levels reach 140 dB(C), or the average maximum sound level reaches 110dB(A).
- Prior to the issuance of hearing protective devices as the final control mechanism, use of acoustic insulating materials, isolation of the noise source, and other engineering controls should be investigated and implemented, where feasible.
- Periodic medical hearing checks should be performed on workers exposed to high noise levels.

Table 3-4: WBG's ESH Guidelines on Noise limits for Various Working Environments

Location/activity	Equivalent level LAeq,8h dB(A)	Maximum LAmax, fast dB(A)
Heavy Industry (no demand for oral communication)	85	110
Light industry (decreasing demand for oral communication)	50 - 65	110
Open offices, control rooms, service counters or similar	45 - 50	-
Individual offices (no disturbing noise)	40 - 45	-
Classrooms, lecture halls	35 - 40	-
Hospitals	30 - 35	40

Source: WBG, 2007

The WBG General EHS Guidelines defines vibration as the exposure to hand-arm vibration⁴ (from equipment such as hand and power tools) or whole-body vibrations⁵ (from surfaces on which the worker stands or sits), which should be controlled through wise selection of equipment, installation of vibration dampening pads or devices, and by limiting the duration of exposure. In addition, the limits for vibration and action values, (i.e., the level of exposure at which remediation should be initiated), are provided by

⁴ Vibration transmitted through hand: holding a vibrating tool, 5 - 1500 Hz, ISO 5349-1. (CCOHS, 2005) or vibration workers experience when using tools such as chain saws, jackhammers, grinders, drills, etc. Excessive Hand-Arm Vibration can lead to what is commonly referred to as Hand-Arm Vibration Syndrome (HAVS). (Larson Davis, 2022a)

⁵ Vibration transmitted through a supporting area to whole body: sitting in a chair, standing on vibrating ground 0.5 - 80 Hz; ISO 2631-2 Sea sickness 0.1- 0.5 Hz, ISO 2631-1. (CCOHS, 2005) or Whole-body vibration (WBV) is the vibration transmitted through the seat or feet of a worker while operating machinery or standing on vibrating platforms. Some work environments where WBV is of concern include mining, farming, construction, trucking, sailors, helicopter pilots, and industrial equipment. (Larson Davis, 2022b)

the ACGIH⁶. The exposure levels should be checked on the basis of daily exposure time and data provided by equipment manufacturers.

While the government of Suriname has not developed an occupational workplace exposure standard for WBV, it is useful to make a comparison to an accepted standard in order to work out how much HAV and WBV exposure is likely to cause a risk to workers. This is done by using daily vibration exposure A(8) or vibration dose value (VDV), as developed by the American Conference of Governmental Industrial Hygienists (ACGIH) and the European Union (EU). The ACGIH has developed Threshold Limit Values (TLV's) for vibration exposure and advises that these values be applied in conjunction with other protective measures, as shown in Table 3-5 and Source:

Table 3-6. The European Union (EU) Directive 2002/44/EC provides the TLV and Action Levels for HAV and WBV (see

Table 3-5: ACGIH TLVs for Hand-Arm Vibration (HAV) Exposure

Threshold Limit Values (TLV) for Hand-Arm Vibration (HAV) Exposure	
Total Daily Exposure Duration (hours)	Maximum value of frequency weighted acceleration (m/s ²)
4 to less than 8 hours	4
2 to less than 4 hours	6
1 to less than 2 hours	8
less than 1 hour	12

Source: (CCOHS, 2005)

Table 3-6: ACGIH TLVs for Whole Body Vibration (WBV) Exposure

TLV for Whole Body Vibration (WBV) Exposure	
Total Daily Exposure Duration (hours)	Value dominant RMS* acceleration component (m/s ²)
8 to less than 16	2.2
4 to less than 8	3.4
2.5 to less than 4	4.8
1 to less than 2.5	8.1
25 min to less than 1 hr	12.1
16 min to less than 25 min	14.4
1 min to less than 16 min	19.2
*RMS = root mean square	

Source: (CCOHS, 2005)

⁶ The American Conference of Governmental Industrial Hygienists - ACGIH, 2005

Table 3-7: EU Directive 2002/44/EC Limit and Action Levels for HAV and WBV

TLV for Exposure of HAV and WBV			
Exposure	Exposure	Daily Limit ^(a)	Daily Action Level ^(b)
HAV	Daily Vibration exposure A(8) - 8 hours	5 m/s ²	2.5 m/s ²
WBV	Daily Vibration exposure A(8) - 8 hours	1.15 m/s ²	0.5 m/s ²
	Vibration Dose Value (VDV)	21 m/s ^{1.75}	9.1 m/s ²
Notes: (a) controls must be put in place to ensure a worker is not exposed under any circumstances to a daily vibration exposure above. (b) action should be taken to reduce exposure to below this value.			

Source: European Union, 2019

3.2.3 Treated Wastewater Discharge Quality Guidelines

The GOS has not established treated wastewater discharge levels standards in its legal or regulatory framework, therefore, the WBG's General ESH Guidelines will apply. The WBG General EHS Guidelines provides treatment recommendations and requirements for wastewater prior of discharging into the environment (see

Table 3-8). These include the management of industrial wastewater (i.e., Treatment or process water, utilities operations and stormwater) and sanitary wastewater (i.e., from domestic sewage, food service, and laundry facilities serving site employees).

For sewage discharges from industrial facilities to surface water, the WBG's General EHS Guidelines specifies that:

- *"if sewage from the industrial facility is to be discharged to surface water, treatment to meet national or local standards for sanitary wastewater discharges or, in their absence, the indicative guideline values applicable to sanitary wastewater discharges shown in Table 3-8;"*
- *"If sewage from the industrial facility is to be discharged to either a septic system, or where land is used as part of the treatment system, treatment to meet applicable national or local standards for sanitary wastewater discharges is required."*

Table 3-8: WBG's EHS Guidelines for Treated Sanitary Sewage Discharges

Pollutants	Units	Guideline value
pH	pH	6 - 9
BOD	mg/l	30
COD	mg/l	125
Total Nitrogen	mg/l	10
Total Phosphorus	mg/l	2
Oil and Grease	mg/l	10
Total Suspended Solids	mg/l	50
Total coliform bacterial	MPN(b) / 100 ml	400(a)
Notes: (a) Not applicable to centralized, municipal, wastewater treatment systems, which are included in EHS Guidelines for Water and Sanitation. (b) MPN = Most Probable Number		

Source: IFC, 2007

3.3 FREE, PRIOR AND INFORMED CONSENT (FPIC)

FPIC is a standard protected by global human rights' legislation. It reaffirms the right to self-determination and the freedom to seek one's own economic, social, and cultural growth. The FPIC principle focuses on the Indigenous Peoples' right to self-determination over land and resources on the premise of safeguards that those rights cannot be affected or impacted or even diminished without consultation and Free, Prior, and Informed Consent. Considered an emerging standard in the dialogue on Indigenous Peoples rights, Free, Prior and Informed Consent (FPIC) is becoming part of customary international law. It is especially applicable for the safeguard of property. FPIC aims to ensure the inclusions, disclosure and respect for Indigenous and Tribal Peoples in any and all decision-making process that involves them, their land or any other property.

Elements of FPIC can be defined as follows:

- free: no coercion, intimidation or manipulation;
- prior: consent is obtained in advance of the activity associated with the decision being made, and includes the time necessary to allow Indigenous and Tribal Peoples to undertake their own decision-making processes;
- informed: Indigenous Peoples have been provided all information relating to the activity and that that information is objective, accurate and presented in a manner and form understandable to Indigenous and Tribal Peoples;
- consent: implies that Indigenous Peoples have agreed to the activity that is the subject of the relevant decision, which may also be subject to conditions."

Although Suriname has not ratified the Labour Organization Convention (ILO) 169, it has voted in favor of the United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP) which was adopted

by the General Assembly on Thursday, 13 September 2007 which specifically mentions FPIC in its Articles 10, 11, 19, 28 and 29, and the obligation to consult in its Articles 19 and 32

The requirement of consultation and FPIC procedures is clearly explained in the Environmental Framework Act that was published in the Official Gazette of the Republic of Suriname on May 07, 2020. The act clearly defines FPIC as:

an internationally used abbreviation for the Free Prior Informed Consent principle from the international law, in relation to Indigenous and tribal people, and comprises of the manner in which the government should engage Indigenous and tribal people in decision-making related to activities in their living area. Only based on proper information, common processes of decision-making and free will, can the Indigenous and tribal people take well-informed decisions regarding the allowance of any form of economic activities in their living area. In situations where a direct interest exists for the Indigenous and tribal people within their living area and whereby permits and or concessions are granted to persons or decisions are taken on the use of their living area, the approval of the Indigenous and tribal people is required before decisions are taken by the government. In cases these involve government projects included in policy programs approved by the people's representative body, the Indigenous and tribal people must be consulted in advance before any right is granted to third parties (p. 2)

The Act further includes the obligation for the National Environmental Authority that NMA procedures “ensure that the FPIC principle is applied in decision-making processes that regard the living area of the Indigenous and tribal people” *p.6 Article 3 Duties and competences of the NMA Paragraph 1.m.)

4 LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK IN SURINAME'S MINING SECTOR

4.1 INTRODUCTION

This section provides a description of the relevant regulations set by the GOS for social planning and environmental protection relevant and applicable to the mining sector. Therefore, the legislative, regulatory and institutional requirements guiding the mining sector development project, are discussed in more detail below.

In addition, Suriname is signatory to a number of international conventions that address environmental and social issues. These international conventions are the key drivers behind for the development of legislation in most countries, including Suriname.

The GOS have different ministries and institutions responsible for the management of the environment and natural resources, and social themes.

4.2 LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK

The legal and regulatory framework of Suriname is composed of the national Constitution, Laws or Acts of Parliament (also called Wet Landsverordening prior to 1975), Decrees (Decreet)⁷, Government State Orders (Staatsbesluiten). In addition, there are regulations namely, Presidential Resolutions, Presidential Orders, Ministerial Orders (beschikkingen) and District Ordinances (verordeningen). The regulatory framework is also supported by international agreements, treaties, conventions, etc. The following sections describe in detail the legal and regulatory framework of Suriname.

4.2.1 National constitution and laws

Suriname is governed in accordance with its National Constitution (GOS, 2000) established in 1987 and last amended in 1992. Article 41 states that the “natural riches and resources are the property of the nation and should be used in economic, social and cultural development. The nation has the inalienable right to take full possession of its natural resources in order to utilize them for the economic, social and cultural development of Suriname.”

The Suriname Constitution sets specific social objectives related to environmental aspects, which include:

- Article 6 (a) – “the identification of the potential for development of its own natural environment and the augmentation of the capacity to increasingly expand that potential”; and
- Article 6 (g) – “creating and improving the necessary conditions for the protection of nature and for the preservation of the ecological balance”.

In addition, the Constitution also states that all workers have the right to safe and healthy working conditions (Article 28), and everyone has a right to health (Article 36). Finally, Article 42 affirms that the law shall guarantee that the method in which trade and industry are conducted, shall not be contrary to the national objectives and the public's interest, notably public order, health, morality and state security.

⁷ Decrees date from the period of 1980 – 1986 and have the same status as past or present Laws

In 2020, the 'Milieu Raamwet S.B. 2020 no. 97 (Environmental Framework Act⁸ (EFA) S.B. 2020 no. 97)' was established for the protection and sustainable management of the environment in Suriname and the implementation and carrying into effect of obligations deriving from the membership of Suriname to international agreements, notably the UN Framework Convention on Climate Change, the UN Convention on Biological Diversity, the Paris Agreement, the Stockholm Convention (Conventions on persistent organic pollutants (POPs), and the Rotterdam Convention *Prior Informed Consent (PIC) Procedure for Certain Hazardous Chemicals and Pesticides in International Trade. Another important matter is the investigation, prosecution and trial of offences defined in this Act as environmental offences. (UNEP, 2022)

The Act is composed of 77 articles divided into 11 Chapters: General provisions (I); The National Environment Authority (II); Duty of Care (III); Environmental Strategy (IV); Activities and Environmental Consequences (V); Control of Pollution – Environmental Pollution and Standards (VI); Waste and Hazardous Substances (VII); Legal Protection Mechanisms (VIII); Environmental offences and Sanctions (IX); Provisions on Enforcement and further provisions (X); Transitional and Final Provisions (XI). (UNEP, 2022)

The Act establishes the National Environment Authority (NMA) as a statutory body. It shall be the successor to the Environmental Coordination Office of the Presidential cabinet. The National Institute for Environment and Development in Suriname (NIMOS) will transition into the NMA.

In July 2020, the institutional structure for environmental management changed with the change of Government. The structural change included the establishment of a new Ministry for Spatial Planning and Environment (ROM). The Ministry of ROM aims to coordinate all environmental activities in the country.

The legal positioning of the Ministry of ROM became a priority, and a formal working group was established to amend the Environmental Framework Act (EFA). The amendment proposes that the Ministry become primarily responsible for coordinating Environmental Policy while the NIMOS is being transformed into the National Environmental Authority. A Presidential Resolution (PB. 31) was promulgated to facilitate the transition period scheduled to start August 15, 2020. Although the Environmental Framework Act was adopted in 2020, the legislation that would allow for its regulations for further implementation of the Act has yet to be approved. As such, the National Environmental Authority still needs to be installed. In addition, a Bill to amend the EFA has now been prepared. The date for the Act to enter into force along with its provisions has yet to be confirmed but it is now expected that the amendments will be discussed in Parliament toward the end of 2022 or early 2023⁹.

It is stipulated in the EFA that the NMA shall prepare the National Environment Strategy and be the main focal point for environmental protection in Suriname. It shall also ensure that the FPIC principle is applied in decision-making processes affecting the habitat of the Indigenous and tribal peoples. Key chapters include (UNEP, 2022):

- Chapter III defines a duty of care for the environment. This duty means that everyone knows or can reasonably suspect that his or her actions or omissions shall have adverse consequences for the environment, must omit such action insofar as this can be reasonably required.
- Chapter IV requires the NMA to elaborate the environmental strategy as incorporated in the National Development Plan for Strategy into a national environmental policy plan. The Act sets out the priorities of such policy. One priority is the inventory, protection and regulation of endemic flora and fauna and traditional and medicinal knowledge. The national plan shall be elaborated in environmental programs, which shall also be established at district level. The NMA shall publish an environmental report every four years. The Act establishes an Environment

⁸ <https://leap.unep.org/countries/sr/national-legislation/environmental-framework-act-no-97-2020>

⁹ Information provided by the Minister of Spatial Planning and the Environment (ROM).

Fund. Activities for which an environmental impact assessment is required shall be defined by a State Decree. This Act sets out the procedures for such an environmental impact assessment.

- Chapter V, the NMA shall be informed about government activities that may affect the environment and shall carry out an environmental impact analyses for such activities or policies or a strategic impact analysis for economic sectors such as forestry, water and agriculture.
- Chapter VI is about environmental pollution and rehabilitation of the environment. The NMA shall define polluting activities and be responsible for environmental authorizations.
- Chapter VII indicates that the NMA shall establish standards and procedures for waste treatment, reuse and recycling as well as the processing of waste. The NMA establishes rules for handling with hazardous substances. It is also charged with monitoring activities in this area prevention of environmental pollution.
- Chapters VIII and IX are about environmental offences. Penal law shall be the ultimate resort for such offences. Article 47 defines environmental offences and for this purpose indicates various other laws including, amongst others the Agriculture Act, the Forestry Act and the Water Resources Act.

In the past years, NIMOS has done some preparatory work by drafting subsidiary legislation necessary for the implementation of the Environmental Framework Act. The following regulations have been prepared in this regard:

1. A draft regulation to determine the activities subject to an EIA

The implementation regulation gives an overview about the activities for which an EIA is or is not required. The list includes activities related to different kinds of projects and indicates to which category (A, B) they belong. According to the determined category, NMA (transitions from NIMOS) shall decide whether the execution of an EIA shall take place. The regulation further indicates that project activities, which fall under category C, do not require an EIA, but are still obliged to apply for an environmental license.

2. A draft regulation to implement Environmental Impact Assessment

This draft regulation has now been combined with the draft regulation to determine the activities subject to an EIA whereas, the proponent of an activity is obliged to ask compulsory advice from NMA for the submission of an application for a license whether an EIA is required. The regulation sets further provisions regarding the determination of the scope of the required EIA, specifications for the qualification of persons to conduct an EIA, minimum requirements for the Environmental Impact Reports (EIR) including the submission and assessment of the EIRs.

3. A draft regulation on Pollution Control

This implementation regulation lays down provisions concerning the prevention of environmental pollution by means of inter alia introducing a licensing system. For an effective monitoring of compliance with the provisions under or pursuant to the Environmental Act, the regulation also determines the pollutants, their values, quantities and other technical details in connection with the measurability of environmental pollution. Regarding existing sources of pollution, it is obliged to register these sources with NMA. The regulations further provide provisions regarding spills and accidental releases.

4. A draft regulation on Hazardous Substances

The implementation regulation lays down provisions concerning hazardous substances to prevent environmental pollution by means of inter alia introducing a licensing system. For an effective monitoring of compliance with the provisions under or pursuant to the Environmental Act, the regulation also determines the hazardous substances as well as the provisions concerning inter alia the license, the transport, the use, the treatment and the registration of such substances.

Additional regulations regarding administrative and institutional matters, as well on waste, hazardous substances and audits still need to be drafted.

Other relevant environmental legislation in Suriname includes:

- The 'Hinderwet G.B. 1930 no. 40 z.l.g. bij S.B. 2001 no. 63 (Nuisance Act G.B. 1930 no. 40 as amended by S.B.2001 no. 63)' defines the permit requirements to control noise and air pollution for industrial development projects. The permits are issued and enforced by local District Commissioners (Buursink, 2005; SRK, 2007);
- The 'Natuurbeschermingswet, 1954 no. 26 z.l.g. bij S.B. 1992 no. 80 (Nature Conservation Act 1954 G.B. 1954 no.26 as amended by S.B. 1992 no.80)' defines the procedures to establish and manage conservation areas. According to the Act, the only allowable activities in the protected areas are activities of scientific, educational or cultural importance. Conservation areas can be established by State Order and managed by the Forest Service (LBB) who delegated this responsibility to the Nature Conservation Division (NDC).
- The 'Jachtwet 1954 G.B. 1954 no. 25 z.l.g. bij S.B. 1997 no. 33 (Game Act 1954, G.B. 1954 no. 26 as lastly amended by S.B. 1997 no. 33)' provides rules for protecting fauna and the regulation of hunting. Under this Act, animals are designated as protected animals and receive legal protection.
- Jachtbesluit 2002 S.B.2002 no. 116 (Game State Order 2002 S.B. 2002 no. 116) provides for hunting seasons, bag limits and hunting permits.
- The 'Wet Bosbeheer S.B. 1992 no. 80 (Forest Management Act S.B.1992 no. 80)' provides a framework for forest management, exploitation and related sector activities to guarantee sustainable utilization of forest resources. The Act classifies forests in Suriname in three ways: permanent, temporary (maintained for the time being), or one-time exploitation forest (land intended for other future use where tree cover will be completely cleared). Resource exploitation in public forests (domain land) is allowable with a permit or forest concession granted by the Ministry of Land Policy and Forestry Management (GBB), which issued regulations for logging, deforestation, and processing-related activities.

The following Table 4-1 presents laws, decrees, acts related to mining, extractive industries, and associated infrastructure development.

Table 4-1: National Regulatory Framework

Decree/Act	Description(s)	Implementing Agency
MINERAL RESOURCES		
Decreet Mijnbouw S.B. 1986 no. 28 z.l.g. bij S.B. 2017 no. 41 (Mining Decree S.B. 1986 no. 28 as amended by S.B. 2017 no. as amended by S.B. 1997 no. 41)	1. Governs exploration and exploitation of mineral resources. 2. Article 2 stipulates that all raw materials in and above the ground, including the territorial sea, are property of the State. Provides legal expectations and requirements to exploit minerals nationally including consideration of environmental and social impacts.	MNR
Besluit Mijnbouwinstallaties S.B. 1989 no.38 (State Order on Mining Installations S.B. 1989 no.38)	Provides provisions for offshore mining installations. Provides legal expectations and requirements to exploit minerals.	MNR
Petroleumwet 1990 S.B. 1991 no. 7 z.l.g. bij S.B. 2001 no. 58	Provides provisions for the exploration and exploitation of hydrocarbons.	MNR

Decree/Act	Description(s)	Implementing Agency
(Petroleum Act 1990 S.B. 1981 no. 7 as amended by S.B. 2001 no. 58)	Provides specific requirements for the management of environmental impacts.	
State Order Registration Mining Rights S.B. 1989 no. 39	The Mortgage Depositary is obliged to keep a registry according to the model indicated in the annex provided, in which the writing takes place from deeds to establishment and transfer listed exploration and exploitation rights. Provides requirements for the registration of mining rights.	MNR
Concession Act (G.B. 1907 no. 84) as amended by G.B. 1944 no. 129	Contains rules concerning the exploitation of the public utilities. The President can grant concession for the use of domain land for the construction and operation of works of public utility. Provides requirements for the exploitation of public utilities.	MNR
Staatsolie's Concession Agreement; Decree E8-B, S.B. 1981 no. 59	The State grants authorization to Staatsolie Maatschappij Suriname N.V. through a concession agreement for research into the presence and exploitation of hydrocarbons. Concession agreement between the State and Staatsolie Maatschappij Suriname N.V.	MNR
Maritime Zones Act S.B. 2017 No.41	The Maritime Zones act was promulgated in April 2017 and has thereby repealed the act of 1978 (Wet Uitbreiding Territoriale Zee en instelling Economische Zone, S.B. 1978 no. 26). This Act defines the size of the maritime zones of the GOS and the authority that the GOS has with respect to these zones. Suriname's Exclusive Economic Zone (EEZ) begins at the outer limit of the territorial sea of 12 nautical miles and consists of the contiguous zone that is up to 24 nautical miles into the sea and the Continental Shelf that is up to 200 nautical miles into the sea. In the territorial sea, Suriname has sovereignty over the land and the sea area, as well as the air above and the soil below the territorial sea. This law gives the State the authority to use the powers that a coastal State has in the maritime zones as regulated in UNCLOS.	MAS
Boorwet G.B. 1952 no. 93 (Drilling Act G.B. 1952 no.93)	Provides provisions for drilling in Suriname. Provides specific requirements for drilling programs.	Head of MNR
Landsverordening van 25 januari 1958 houdende machtiging tot het aangaan van overeenkomsten met de Suriname Aluminum Company inzake de ontwikkeling van het waterkrachtpotentieel van Suriname G.B. 1958 no. 4	Agreement between the Government of Suriname and Suralco LLD concerning the development of hydropower potential. Provides context to pre-existing mineral agreements in Suriname.	MNR

Decree/Act	Description(s)	Implementing Agency
(Law of January 25, 1958, authorizing the conclusion of agreements with the Suriname Aluminum Company regarding the development of the hydropower potential of Suriname G.B. 1958 No. 4) (Brokopondo Agreement 1958B)		
Wet Wijziging Brokoponde - overeenkomst S.B. 2019 no.96 (Act Amendment Brokoponde Agreement S.B. 2019 no.96)	Provides for the purpose of its early termination of the agreement between the GOS and Suriname Aluminum Company LLC and to settle the consequences of that termination.	Government of Suriname
Gross Rosebel Agreement S.B. 1994 no. 22 as amended by S.B. 2002 no. 115	Agreement between the Government of Suriname, Grassalco NV and Golden Star Resources Limited for the exploration and exploitation of minerals. Provides context to pre-existing mineral agreements in Suriname.	MNR
WET van 6 mei 2013, houdende toestemming tot het aangaan van een overeenkomst tot nadere wijziging en aanvulling op Delfstoffenovereenkomst van 7 april 1994 met Grasshopper Aluminum Company N.V. en Cambior Inc. (S.B. 1994 no. 22, zoals gewijzigd bij S.B. 2002 no. 115) S.B. 2013 no. 92 (LAW of May 6, 2013, containing permission to enter into an agreement to further amend and supplement the Minerals Agreement of April 7, 1994 with Grasshopper Aluminum Company N.V. and Cambior Inc. (S.B. 1994 No. 22, as amended by S.B. 2002 No. 115) S.B. 2013 No. 92)	This Act authorizes the Government of Suriname to give permission to Grasshopper Aluminum Company N.V., Rosebel Goldmines N.V. and IAMGOLD Corporation to enter into an agreement that further amends the Minerals Agreement of 7 April 1994, such as amended by the agreement dated March 13, 2003.	Government of Suriname
Newmont (Surgold) Mining Corporation Mineral Agreement EX-10.2 3 d755143dex102.htm EX-10.2	The Mineral agreement between Surgold, a Subsidiary of Newmont Mining Corp. and the Ministry of Natural Resources of Suriname signed November 22, 2013, provides the terms of the agreement, the areas, the estimated resources and confirms that the GOS may hold a maximum share of 25% of the Suriname Mining Company.	Suriname government and Newmont Mining Corporation
LAND USE		
Planwet 1973 G.B. 1973 no. 89 (Planning Act 1973 GB. 1973 no. 89)	Provisions for national and regional planning e.g., land-use policy issues. Provides planning and land-use requirements.	Ministry of Finance and Planning-
Decreet uitgifte Domeingrond (S.B. 1982 no. 11) z.l.g. bij S.B. 2003 no.7 (Decree on the	Provides general regulations on the issuance of State-owned lands.	Ministry of GBB

Decree/Act	Description(s)	Implementing Agency
Issuance of State-Owned Lands (S.B. 1982 no. 11) as amended by S.B. 2003 no. 53	Provides regulations and requirements to acquire state land.	
Bouwwet G.B. 1956 no.30, z.l.g. bij S.B. 2002 no.72 (Building Act G.B. 1956 no.30 as amended by S.B. 2002 no.72)	Provides requirements for the construction of buildings. Provides regulations and requirements to construct buildings.	Ministry of Public Works
Bouwbesluit G.B. 1956 no. 108 z.l.g. bij S.B.2002 no.93 (Building State Order G.B. 1956 no.108 as amended by S.B. 2002 no.93)	The Building State Order sets specific drainage and sewerage requirements for construction of buildings. Provides regulations and requirements to construct buildings.	Ministry of Public Works
Stedebouwkundige wet (G.B. 1972 no. 96) z.l.g. bij S.B. 2002 no.72 (Urban Planning Act (G.B. 1972 no. 96) as amended by S.B. 2002 no. 72)	Provisions for urban development. Provides regulations and requirements to affect development.	Ministry of Public Works
Besluit aanwijzing woongebieden Commewijne, Wanica en Para en primaire wegen categorie a S.B. 2014 no.89 (State Order on designating residential areas Commewijne, Wanica and Para and primary roads category a S.B. 2014 no.89)	Through this State Order, the districts of Commewijne, Wanica en Para are designated as urban area as well as an area of five (5) kilometres on either side of the axis of designated primary roads category A as well as projected primary roads. It is stated in this State Order that the way of life of the communities living in the Interior will be respected. Designation of Commewijne, Wanica and Para as urban area.	Ministry of Public Works
Decreet ecologische omstandigheden in woongebieden (S.B. 1980 no. 68 zoals laatstelijk gewijzigd bij S.B. 1980 no. 121) (Law on Ecological Circumstances in Residential Areas S.B. 1980 no.68 as amended by S.B.1980 no.121)	To improve the ecological conditions in residential areas. Provides regulations for ecological circumstances in residential areas.	District Commissioners
Politie strafwet G.B. 1915 no. 77 z.l.g. bij S.B. 1990 no. 24 (Police Criminal Act G.B. 1915 no. 77 as amended by S.B. 1990 no. 24)	The Act punishes the lesser offences. Provides regulations and requirements for waste disposal.	Ministry of Justice and Police
Burgerlijk Wetboek (G.B. 1860. No.4) z.l.g. bij S.B. 2004 no. 25 ((Civil Code (G.B. 1860 no. 4) as amended by S.B. 2004 no. 25))	Regulates private law relating to contracts, property, family, and obligation. Provides regulations and requirements for expropriation.	
AIR QUALITY AND NOISE		
Hinderwet G.B.1930 no. 64 z.l.g. bij S.B. 2001 no. 63 (Nuisance Act G.B.1930 no. 64 as amended by S.B. 2001 no.63)	Controls industrial pollution (noise and air). Provides regulations and requirements control industrial pollution.	District Commissioners issue permits in consultation with Ministry of Health

Decree/Act	Description(s)	Implementing Agency
WATER/MARITIME		
Waterleidingswet G.B. 1938 no.33 (Water Supply Act G.B. 1938 no.33)	Contains prohibitions with respect to water wells, etc. that serve as water supply sources. Provides regulations and requirements for water pollution.	MNR, Ministry of Public Health
Decreet Havenwezen 1981 S.B. 1981 no.86 (Harbours Decree 1981 S.B. 1981 no.86)	Provisions for harbor activities. Provides regulations and requirements for the discharge of water and oil.	Maritime Authority Suriname and District Commissioners, assisted by the Prosecutors office, the Police and the Ministry of Trade and Industry
Wet Maritieme Beveiliging S.B. 2004 no. 90 (Maritime Security Act S.B. 2004 no.90)	Provisions for safety of ships and harbors. Provides regulations and requirements for ship movements.	Maritime Authority Suriname, Ministry of Trade and Industry, Maritime Safety Council
Wet Maritieme Autoriteit Suriname S.B. 1998 no. 37 (Maritime Authority Act S.B 1998 no.37)	This act establishes the Maritime Authority Suriname (MAS) as a corporation under Article 3 in the framework of privatization of public services. The MAS is responsible for safe and efficient maritime traffic to and from Suriname, and for control of water pollution by vessels, in accordance with international conventions ratified by Suriname. The MAS is responsible for the supervision and control of maritime navigation in accordance with the laws of Suriname. The MAS shall further render services to sea-going vessels, in particular with regard to exportation and importation of goods. Provides provisions for the management of maritime traffic.	Ministry of Transport, Communication and Tourism MAS
Wetboek van Strafrecht G.B. 1911 no.1 z.l.g bij S.B.2020 no. 42 (Penal Code G.B. 1911 no.1 as amended by S.B. 2020 no.42)	The Act regulates which offenses are considered a criminal offense, which offenses are regarded as an offense, and which types of penalties can be imposed. It provides regulations and requirements for water pollution.	Ministry of Justice and Police
Besluit Mijnbouwinstallaties S.B. 1989 no.38 (State Order on Mining Installations S.B. 1989 no.38)	Provisions for offshore mining installations. Provides regulations and requirements for releases to marine environments	MNR
Bestrijdingsmiddelen besluit 2005 S.B. 2005 no.21 (Pesticides State Order 2005 S.B 2005 no. 21)	To implement article 13 of the Pesticides Law Provides regulations and requirements for pesticide disposal.	LVV, Ministry of Public Health

Decree/Act	Description(s)	Implementing Agency
NATURAL ECOSYSTEMS (VEGETATION, FISH, AND WILDLIFE)		
Wet Bosbeheer S.B. 1992 no. 80 (Forest Management Act S.B. 1992 no. 80)	Provides a framework for forest management, exploitation, and related sector activities (e.g., primary processing and export) to guarantee sustainable utilization of forest resources. Provides for establishment of conservation forests. Provides regulations and requirements for forest management.	Ministry of GBB/SBB
Besluit Kabo speciaal beschermd bos S.B. 2012 no. 124 (State Order Special protected forest Kabo S.B. 2012 no. 124)	For the purpose of scientific activities, an area can be designated as special protected forest. Provides for the establishment of a special protected forest, which falls under the management of the Foundation Center for Agricultural Research in Suriname (CELOS).	Ministry of GBB
Besluit Mapane speciaal beschermd bos S.B. 2012 no. 123 (State Order Special protected forest Mapane S.B. 2012 no. 123)	For the purpose of scientific activities, an area can be designated as special protected forest. Provides for the establishment of a special protected forest, which falls under the management of the Foundation Center for Agricultural Research in Suriname (CELOS).	Ministry of GBB
Natuurbeschermingswet 1954, G.B. 1954 no. 26 z.l.g. bij S.B. 1992 no. 80 (Nature Conservation Act 1954, G.B. 1954 no. 26 as amended by S.B. 1992 no. 80)	Wildlife protection, establishment and management of protected areas. Prohibits any activities that may impact on protected areas, except for activities of scientific, educational or cultural importance. Provides regulations and requirements for wildlife protection.	LBB manages nature reserves with the Nature, Protection Commission in an advisory role.
Natuurbeschermingsbesluit 1966 van natuurreserveaat Wia Wia, Coppename monding en Brinkheuvel, G.B. 1966 no. 59 (Nature Conservation Order 1966 Wia Wia Nature Reserve Coppename Monding and Brinkheuvel G.B. 1966 no. 59)	Establishment of the Wia Wia Coppename monding en Brinkheuvel nature reserves. Relates to the provisions for designating the areas as natural reserves.	LBB is part of the Ministry of GBB. Delegated its task to NCD
Natuurbeschermingsbesluit 1969 van Galibi Natuur Reservaat, G.B. 1969 no. 47 (Nature Conservation State Order 1969 Galibi Nature Reserve G.B. 1969 no. 47).	Establishment of the Galibi nature reserve. Relates to the provisions for designating the areas as nature reserves.	LBB is part of the Ministry of GBB. Delegated its task to NCD
Natuurbeschermingsbesluit Hertenrits, G.B. 1972 no. 22 (Nature Conservation State Order 1972 Hertenrits G.B. 1972 no. 22).	Establishment of the Hertenrits nature reserve. Relates to the provisions for designating the areas as nature reserves.	LBB is part of the Ministry of GBB. Delegated its task to NCD
Natuurbeschermingsbesluit 1972 G.B. 1972 no. 25 Instelling Sipaliwini natuurreserveaat	Establishment of the Sipaliwini nature reserve. Relates to the provisions for designating the areas as nature reserves.	LBB is part of the Ministry of GBB. Delegated its task to NCD
Natuurbeschermingsbesluit 1986 van Peruvia, Wanekreek, Copie en Boven-Coesewijne	Establishment of the Peruvia, Wanekreek, Copie and Boven-Coesewijne Natuurreserve.	LBB is part of the Ministry of GBB.

Decree/Act	Description(s)	Implementing Agency
natuurreserveaat, S.B. 1986 no 52 (Nature Conservation State Order 1986 Peruvia, Wanekreek, Copie and Boven-Coesewijne Natuurreserve S.B. 1986 no. 52).	Relates to the provisions for designating the areas as nature reserves.	Delegated its task to NCD
Natuurbeschermingsbesluit 1998 S.B. 1986 no.52 Instelling Centraal Natuurreserveaat	Establishment of the Central Suriname Nature Reserve. Relates to the provisions for designating the areas as nature reserves.	LBB is part of the Ministry of GBB. Delegated its task to NCD
Balatawet (G.B. 1914 no. 51) (Balata Law G.B 1914 no 51)	Provides provisions for research on the availability of 'bolletrie' trees (rubber trees) and the exploitation of balata on state land as well as private land. Provides regulations and requirements for protection of rubber trees.	The President and Administrator of Finance (implementation), with State Council in an advisory role.
Politie strafwet G.B. 1915 no. 77 z.l.g. bij S.B. 1990 no. 24 (Police Criminal Act G.B. 1915 no. 77 as amended by S.B. 1990 no. 24)	Provides regulations and requirements for wildlife protection.	
Zeevisserijwet 1980 S.B. 1980 no. 144 z.l.g. bij S.B. 2001 no. 120 (Sea Fisheries Act 1980 S.B. 1980 no. 144 as lastly amended by S.B. 2001 no. 120)	1. Provides for registration, licensing, seaworthiness certification and other requirements for fishing vessels. 2. Provides for issuing of regulations for the protection of fish stocks. 3. Provides regulations and requirements for releases to marine environments.	LVV
Visstandbeschermingswet G.B. 1961 no. 44 z.l.g. bij S.B. 1981 No. 66 (Fishstock Protection Act G.B. 1961 no 44 as lastly amended by S.B. 1981 no. 66)	Protection of fish stocks in inland waters. Provides regulations and requirements for protection of fish stocks.	LVV
Jachtwet 1954 G.B. 1954 no. 25 z.l.g. bij S.B. 1997 no. 33 (Game Act 1954, G.B. 1954 no. 26 as lastly amended by S.B. 1997 no. 33).	This Act provides rules for protecting fauna and regulation of hunting. Under this Act, animals are designated as protected animals and receive legal protection. Hunting is only allowed on private land with the permission of the owner. However, hunting in built-up areas is prohibited. The Act provides regulations and requirements to protect the fauna and game.	LBB
Jachtbesluit 2002 S.B.2002 no. 116 (Game State Order 2002 S.B. 2002 no. 116)	Provisions for various animal species, including hunting seasons and numbers allowed. Provides regulations and requirements for protection of fauna and game.	LBB
Beschikking Instelling CITES Management Autoriteit S.B. 2016 no. 102 (Ministerial Order on the Establishment of the CITES Management Authority S.B. 2016 no. 102)	Each country that has signed up to the CITES Convention is required to establish a CITES Management Authority. Sets out the tasks and responsibilities of the CITES Management Authority.	LBB

Decree/Act	Description(s)	Implementing Agency
Beschikking Instelling CITES Wetenschappelijke Autoriteit S.B. 2016 no. 101 (Ministerial Order on Establishing the CITES Scientific Authority S.B. 2016 no. 101)	Each country that has signed up to the CITES Convention is required to establish a Scientific Authority, which has the task to advise the CITES Management Authority on different matters. Provides for the appointment of the members of the CITES Scientific Authority and lays down the tasks and responsibilities.	GBB
Ministeriele Beschikking benoeming leden van de CITES Wetenschappelijke Autoriteit no. 0699-22/Min GBB (Ministerial Order nominating the members of the CITES Scientific Authority no. 0699-22/mingbb Date 2 maart 2022)	The members of the CITES Scientific Authority are appointed. Per Ministerial Order the members of the CITES Scientific Authority are appointed.	GBB
Ministeriele Beschikking betreffende de instelling van het Noord Saramacca bijzonder beheersgebied S.B. 2001 no.88 (Ministerial Order to designate North Saramacca as special management area S.B. 2001 no.88);	Designation of special management area North Saramacca. Special management areas, by means of Ministerial Orders, are designated areas in the estuarine zone with a special ecological value. Provides for the geographical delimitation of the area.	Ministry of GBB
Ministeriele Beschikking betreffende de instelling van het Bigi Pan bijzonder beheersgebied S.B. 2002 no.34 (Ministerial Order to designate Bigi Pan as special management area S.B. 2002 no.34)	Designation of special management area Bigi Pan. Special management areas, by means of Ministerial Orders, are designated areas in the estuarine zone with a special ecological value. Provides for the geographical delimitation of the area.	Ministry of GBB
Ministeriele Beschikking betreffende de instelling van het Noord Coronie beheersgebied S.B. 2002 no.87 (Ministerial Order to designate North Coronie as special management area S.B. 2002 no.87); and	Designation of special management area North Coronie. Special management areas, by means of Ministerial Orders, are designated areas in the estuarine zone with a special ecological value. Provides for the geographical delimitation of the area.	Ministry of GBB
Ministeriele Beschikking betreffende de instelling van het Noord Commewijne/ Marowijne bijzonder beheersgebied S.B. 2002 no.94 (Ministerial Order to designate North Commewijne/Marowijne as special management area S.B. 2002 no. 94).	Designation of special management area North Commewijne/ Marowijne. Special management areas, by means of Ministerial Orders, are designated areas in the estuarine zone with a special ecological value. Provides for the geographical delimitation of the area.	Ministry of GBB
Richtlijnen Gronduitgifte Estuariene Beheersgebieden S.B. 2005 no. 16 (Ministerial Order "Guidelines for Land Issuance in the Estuarine Management Areas)	This Ministerial Order sets guidelines for the issuance and use of domain land within the estuarine zone. The considerations for setting the guidelines are to maintain the natural functions of the estuarine zones, such as coastal and shore protection function, hydrological function, and	Ministry of GBB

Decree/Act	Description(s)	Implementing Agency
	biological function, i.e., spawning and nursery area for fish, shrimp and birds. Provides guidelines for land issuance in special management areas.	
CULTURAL HERITAGE		
Grondwet van Suriname, G.B. 1987. Artikel 47 (Constitution of Suriname, Article 47).	States that the government of Suriname shall safeguard and protect the cultural heritage of the country, stimulate its preservation and promote the use of science and technology in the context of national development objectives. The provisions of Article 47 are broadly applicable to the goldmining industry if A) there are previously recorded cultural resources within a project concession and/or B) if cultural resources are identified during project activities.	Ministry of Education, Science, and Culture
Besluit Monumentenregistratie G.B. 1963 no.23, z.l.g. bij S.B. 2000 no.41 (State Order for Monuments Register, No. 41 S.B 2000)	The State Order sets out the procedure upon the registration of immovable property and parts thereof that have been designated by the Minister as objects of monument conservation. Further regulations concerning the design and management of the monument register must be established, in order to make registration as intended possible. Provides the procedures upon monuments registration.	Ministry of Education, Science and Culture
'Instelling bijzondere eisen aan bouwplannen voor de historische binnenstad en aangrenzende bufferzones S.B. 2003 no.34 (Ministerial Order on Plans for the Historic Downtown, No. 34 S.B. 2003)	This Order provides for special requirements for building plans for the historic inner city and adjacent zone. Since renovations of the historic inner city are highly susceptible to changes, this can result in the loss of the cultural historical character rules / requirements for the establishment of the construction plans. These requirements are also to maintain the cultural historic quality. Lays down the regulations and requirements for building plans in the historic inner city.	Ministry of Education, Science and Culture
Wet Behoud tot behoud van voorwerpen welke historische, culturele en wetenschappelijke waarde hebben G.B. 1952 no.14 (Preservation Act for the preservation of objects of historical, cultural and scientific value G.B. 1952 no. 14)	Controls the export of objects that have historical, cultural and scientific value. Provides regulations and requirements for protection of cultural heritage.	Ministry of Education, Science and Culture

Decree/Act	Description(s)	Implementing Agency
Stichting Bosbeheer 2011 (Forest Management Code of Practice).	Code requires project developers to establish a 50 m buffer around areas of cultural importance, settlements, fields, cultural, historical, spiritual and archaeological sites. If archaeological or cultural historical findings are made, relics and locations have to be reported immediately to Ministry of Education, Science, and Culture. The Code is applicable to the gold mining industry if cultural resources are identified during project activities.	Ministry of Education, Science, and Culture
Natuurbeschermingswet 1954, G.B. 1954 no. 26 z.l.g. bij S.B 1992 no. 80 (Nature Conservation Act 1954, G.B. 1954 no. 26 as amended by S.B. 1992 no. 80)	The President can designate land and waters, which are domain of the Country, as a nature reserve for protection and conservation of the natural resources present in Suriname after having heard the Council. To be designated as a nature reserve an area must meet the following requirements: the protection of the government is earned under alternately countryside and/or the presence of scientific or culturally important flora, fauna and geological objects. Sets out the criteria to be designated as a protected area.	LBB manages nature reserves with the Nature, Protection Commission in an advisory role.
OCCUPATIONAL HEALTH & SAFETY/PUBLIC HEALTH		
Veiligheidswet 1947 G.B. 1947 no. 142, z.l.g. bij SB. 1980 no.116 (Occupational Safety and Health Act 1947 G.B. 1947 no. 142, as amended by SB. 1980 no.116)	To advance safety and hygiene in enterprises so that the chance of accidents and occupational diseases can be reduced to a minimum. Provides regulations and requirements for health and safety in the workplace.	Ministry of Labor, Employment and Youth Affairs
Veiligheidsvoorschrift nr. 1, G.B. 1972 no. 95 (Safety regulation nr. 1, G.B. 1972 no. 95)	This regulation aims to prevent or diminish the risk of injuries in all enterprises. This regulation is important to factories, mechanical work, construction companies and companies working with machineries and cable systems. It provides technical procedures and measures when power drive machinery is in operation. To prevent or minimize injuries the project activities should be carried out in compliance with the safety regulation.	Ministry of Labor, Employment and Youth Affairs
Veiligheidsvoorschrift nr. 2, G.B. 1948 no. 104 (Safety regulation nr. 2, G.B. 1948 no. 104)	This regulation aims to promote hygiene (order and prevention of dust) in all enterprises. The regulation provides requirements for hygiene related to cleaning, drainage and washstands and requirements for washing facilities for employees. Project activities must be carried out taking into consideration the regulations regarding hygiene.	Ministry of Labor, Employment and Youth Affairs

Decree/Act	Description(s)	Implementing Agency
Veiligheidsvoorschrift nr. 3, G.B. 1948 no. 183 (Safety regulation nr. 3, G.B. 1948 no. 183)	This regulation prescribes some measures regarding first aid. These include the availability of an emergency compartment kit at site, certified persons entrusted with first aid and where the work involves the risk of drowning, efficient and visible means for rescue (swimming jackets, lifeboats, etc.) should be readily available. To prevent or minimize injuries the project activities should be carried out in compliance with the safety regulation.	Ministry of Labor, Employment and Youth Affairs
Veiligheidsvoorschrift nr. 4, G.B. 1948 no. 128; 1969, no 30 (Safety regulation nr. 4, G.B. 1948, no. 128; 1969, no. 30)	This regulation endeavors to prevent the production and spreading of hazardous or irritating gas or dust and to regulate its removal. The obligation is to avoid the use of white lead (ceruse), lead sulfate, and all products containing these pigments in the internal paintings of buildings and ships. In case project activities regarding the production and spreading of hazardous and irritating gas and dust, the regulation must be complied with.	Ministry of Labor, Employment and Youth Affairs
Veiligheidsvoorschrift nr. 5, G.B. 1950 no 121 (Safety regulation nr. 5, G.B. 1950 no. 121)	This regulation aims to protect workers against hazards caused by the weight of loads to be transported by sea or inland waterways. The sender of a package or object of 1.000 kg or more gross weight has the obligation to mark its gross weight plainly and durable on the outside before loading on a ship or vessel. The weight should be stated on the freight documents. For the construction, as well as the operational phases of a project, the transport of heavy load takes place, this regulation must be complied with.	Ministry of Labor, Employment and Youth Affairs
Veiligheidsvoorschrift nr. 6, S.B. 1981 no. 71 (Safety regulation nr. 6, S.B. 1981 no. 71). Dust Control State Order	This regulation is to prevent the occurrence of pneumoconiosis or other diseases caused by dust. This regulation is only applicable in case of activities regarding cleaning by streams with loose material (for example sandblasting).	Ministry of Labor, Employment and Youth Affairs
Veiligheidsvoorschrift nr. 7, S.B. 1981 no 72 (Safety regulation nr. 7, S.B. 1981 no. 72).	This State Order on working conditions, is to promote safe and comfortable working conditions related to inter alia hazardous and disturbing noises and vibrations, temperature, ergonomic aspects, excessive strenuous labor etc. In order to promote safe and comfortable working conditions, the project activities need to be implemented in compliance with this regulation.	Ministry of Labor, Employment and Youth Affairs

Decree/Act	Description(s)	Implementing Agency
Veiligheidsvoorschrift nr. 8, S.B. 1981 no.73 (Safety regulation nr. 8, S.B. 1981 no. 73) (State Order Ionizing Radiations)	The regulation aims to protect workers against ionizing radiation. It contains provisions on the prevention of exposure, permissible doses and level of radioactive exposure, visual and durable warning signs, medical examinations, hygiene, protective devices, etc. Special provisions regarding workspaces, lighting, hygiene, temperature and electrical installations are also included in this regulation. This regulation is only applicable in the case when workers are exposed to activities that release ionizing radiations.	Ministry of Labor, Employment and Youth Affairs
Veiligheidsvoorschrift nr. 9, S.B. 1981 no. 74 (Safety regulation nr. 9, S.B. 1981 no. 74) (State Order harmful gases and vapors).	The regulation intends to prevent hazardous risks to health by inhalation or exposure to noxious and irritating gasses and fumes. The measures relate to workspaces, clothing accommodation and changing rooms, ventilation and fresh air, maximum permissible amount of gas or fumes and the preventive measures. This regulation is applicable in case the project implementation contains areas belonging to associated canteens, dressing rooms, or sleeping quarters that are being set up where noxious and irritating gasses are released.	Ministry of Labor, Employment and Youth Affairs
Wet Arbeidsinspectie S.B. 1983 no. 42 z.l.g.b. S.B. 2017 no. 39 (Labor Inspection Act S.B. 1983 no. 42 as amended by S.B. 2017 no. 39)	Outlines the tasks and responsibilities of the Labor Inspector.	Ministry of Labor, Employment and Youth Affairs
Ongevallenwet G.B. 1947 z.l.g. bij S.B. 2007 no.26 (Industrial Accidents Act G.B. 1947 as amended by S.B. 2007 no.26)	To indemnify workers and their family against financial consequences of industrial accidents and occupational diseases. These accidents related to or in the course of employment include fatal injuries, but also the more gradual development of a sickness because of the performed labor. Provides regulations and requirements for health and safety in the workplace.	Ministry of Labor, Employment and Youth Affairs

Decree/Act	Description(s)	Implementing Agency
Rijwet 1971 G.B. 1917 no. 65 z.l.g. bij S.B. 2007 no. 92 (Driving Act 1971 G.B. 1917 no.65 as amended by S.B. 2007 no. 92)	The 'Rijwet 1971 G.B. 1917 no. 65 z.l.g. bij S.B. 2007 no. 92 (Driving Act 1971 G.B. 1917 no.65 as amended by S.B. 2007 no. 92)', as well as the 'Rijbesluit 1957 G.B. 1957 no. 103, z.l.g. bij S.B. 2020 no. 9 (State Order on Driving G.B. 1957 no.103 as amended by S.B. 2020 no. 9)' provides rules for participation in vehicular traffic. This legislation provides, amongst others, for rules regarding loading and unloading of cargo on public roads, transportation of large cargo and it prohibits. Prescribes rules regarding the regulation of traffic on the roads with regard to the use of vehicles. Nuisance caused by motor vehicles.	Ministry of Justice and Police
Bestrijdingsmiddelenwet (G.B. 1972 no. 151, zoals laatstelijk gewijzigd bij S.B.2005 no. 18) (Pesticide Act G.B. 1972 no. 151 as amended by S.B. 2005 no.18)	Regulates the trade and use of agrochemicals. Provides regulations and requirements for the use of agrochemicals.	LVV and the Ministry of Public Health
Muskietenbestrijdingswet GB 1952 no. 9 (The Mosquito's Act G.B. 1952 no. 9)	To combat mosquitoes and other insects judged to be destructive to the health of human beings and animals. Provides regulations and requirements for the control of mosquitos.	
Waterleidingswet G.B. 1938 no.33 (Water Supply Act G.B. 1938 no.33)	Establishes prohibitions with respect to water wells, etc. that serve as water supply sources. Provides regulations and requirements for protection of water wells.	MNR, Ministry of Public Health
Anchylostoomwet GB 1917 no. 83, geldende tekst GB 1937 No. 23 (Ancylostomiasis Act G.B 1917 no.83)	Contains rules for the protection of water wells against contamination from ancylostomiasis. This act prohibits the use of feces as fertilizer and protects wells against contamination with feces. Provides regulations and requirements for the protection of water wells.	Ministry of Public Health
Wetboek van Strafrecht G.B. 1911 no.1 z.l.g bij S.B.2020 no. 42 (Penal Code G.B. 1911 no.1 as amended by S.B. 2020 no.42)	Provides regulations and requirements for protection of human consumption of foodstuffs.	
Wet Goederenverkeer S.B. 2003 no. 58 (Act on the Movement of Goods SB. 2003 no. 74)	Provides general rules for international trade. Provides regulations and requirements for international trade.	Ministry of Economic Affairs, Entrepreneurship and Technological Innovation
Besluit Negatieve lijst S.B. 2003 no. 74 zoals laatstelijk gewijzigd bij S.B 2006 no. 100 State Order Negative List S.B. 2003 no.74 as amended by S.B. 2006 no. 100)	Regulates the international traffic of goods. Provides regulations and requirements for shipping of goods.	Ministry of Economic Affairs, Entrepreneurship and Technological Innovation

Decree/Act	Description(s)	Implementing Agency
Transportation		
Wet Wegenautoriteit S.B. 1995 no. 68 (Road Authority Act S.B. 1995 no. 68)	The tasks of the road authority are to manage and maintain primary roads and permanent connections therein; advise the government on the repair and maintenance of roads and fixed connections. Sets out the tasks of the Road Authority	Ministry of OW
Staatsbesluit Primaire Wegen S.B. 2002 no. 61 (State Order Primary Roads S.B. 2002 no. 61)	By State Order, roads are considered primary if they are a connection between an urban area on the one hand and a "production area of national significance or a regional residential core on the other. The primary roads are categorized and named. Sets out the primary roads.	Ministry of OW
Staatsbesluit Opcenten verbruiksbelasting S.B. 2000 no. 5 (State Order Surcharges on consumption tax S.B. 2000 no. 5)	The State Order lays down rules regarding the determination of surcharges on motor fuel consumption tax for road maintenance. Provides the surcharges on motor fuel consumption tax	Ministry of Finance and Planning
Staatsbesluit wijziging van het besluit Opcenten S.B. 2006 no. 63 (State Order amending the Opcenten Decree S.B. 2006 no. 63)	This State Order regards a rate change for Article 3 of the State Order Surcharges on consumption tax S.B. 2000 no. 5. Provides the surcharges on motor fuel consumption tax.	Ministry of Finance and Planning
Construction and Operation of Ports		
Decreet Havenwezen 1981 S.B. 1981 no.86 (Harbours Decree 1981 S.B. 1981 no.86)	Provisions for harbor activities. Provides regulations and requirements for the discharge of water and oil	Maritime Authority Suriname and District Commissioners, assisted by the Prosecutors office, the Police and the Ministry of Trade and Industry
Wet Maritieme Autoriteit Suriname S.B. 1998 no. 37 (Maritime Authority Act S.B. 1998 no.37)	This act establishes the Maritime Authority Suriname (MAS) as a corporation under Article 3 in the framework of privatization of public services. The MAS is responsible for safe and efficient maritime traffic to and from Suriname, and for control of water pollution by vessels, in accordance with international conventions ratified by Suriname. The MAS is responsible for the supervision and control of maritime navigation in accordance with the laws of Suriname. The MAS shall further render services to sea-going vessels, in particular with regard to exportation and importation of goods. Provides provisions for the management of maritime traffic	Ministry of Transport, Communication and Tourism MAS

Decree/Act	Description(s)	Implementing Agency
Wet Maritieme Beveiliging S.B. 2004 no. 90 (Maritime Security Act S.B. 2004 no.90)	Provisions for safety of ships and harbors. Provides regulations and requirements for ship movements.	Maritime Authority Suriname, Ministry of Trade and Industry, Maritime Safety Council
Bouwwet G.B. 1956 no. 30 z.l.g. bij S.B. 2002 no. 93 (Building Act G.B. 1965 no. 30 as amended by S.B. 2002 no.93)	Under the Building Act it is prohibited to build without a permit from the Director of the Ministry of Public Works. Under this Act, building is understood placing or erecting in whole or in part, renovate, alter and extend buildings or other structures to the fullest sense of the word, including hydraulic structures. Provides regulations and requirements to construct buildings.	Ministry of Public Works
Bouwbesluit G.B 1956 no. 108 z.l.g. bij S.B.2002 no.93 (Building State Order G.B. 1956 no.108 as amended by S.B. 2002 no.93)	The Building State Order sets specific drainage and sewerage requirements for construction of buildings. Provides regulations and requirements to construct buildings	Ministry of Public Works
Hinderwet G.B.1930 no. 64 z.l.g bij S.B. 2001 no. 63 (Nuisance Act G.B.1930 no. 64 as amended by S.B. 2001 no.63)	The 'Hinderwet G.B.1930 no. 64 z.l.g bij S.B. 2001 no. 63 (Nuisance Act G.B.1930 no. 64 as amended by S.B. 2001 no.63)' aims to prevent the cause of danger, damage or hindrance caused by undertakings (enterprises) to the outside fence surrounding environment. According to the Nuisance Act, all new undertakings need a written 'Nuisance Act Permit' ('Hinderwetvergunning') issued by the District Commissioner (DC) who should seek advice from the Bureau for Public Health, the Department of Labor Inspection, the Head of the Fire Department etc. In case of outside-fence hindrance caused by air pollution and noise, soil and water pollution and generation of solid or liquid waste, this act can be enforced. Provides regulations and requirements to control industrial pollution.	District Commissioners issue permits in consultation with Ministry of Health

Source: (ERM, 2011), (SWM, 2019) updated and modified by RINA, 2022

4.2.2 Court rulings from cases that affect directly or indirectly related to the mining sector and related stakeholders

Court rulings can be used as precedents for present and future mining legislation. It is therefore important to consider court rulings, whether in favor or against the claiming party. The following table shows court rulings against the State of Suriname that directly or indirectly affect stakeholders in the mining sector.

Table 4-2: Court cases and rulings

Court case	Description of case	Court ruling
The Moiwasa case, 2005	This case concerns the slaughtering of more than 40 N'djuka Maroon men, women and children by the Suriname army in the village of Moiwana. Those who escaped the attack supposedly fled into the surrounding forest, and then into exile or internal displacement. Although the Moiwana community members were not Indigenous to the region, the Village was settled by N'djuka clans late in the 19th Century and lived in the area with strict adherence to N'djuka custom until the 1986 attack. The community members did not possess the land as individuals, but as a community as a whole.	The Court ruled amongst others, that the State shall adopt such legislative, administrative, and other measures as are necessary to ensure the property rights of the members of the Moiwana community in relation to the traditional territories from which they were expelled and provide for the members' use and enjoyment of those territories. These measures shall include the creation of an effective mechanism for the delimitation, demarcation and titling of said traditional territories.
The Saramaka vs. Suriname case, 2005	The Saramaka people, descendants of self-liberated African slaves, have been living on their traditional territory in Suriname since the early 1700s. This non-Indigenous community lives in a traditional way by fishing, hunting and woodworking, and their relationship with the land is more than economic, but also spiritual and cultural. In 1986, Suriname adopted a new Constitution specifying that all non-titled lands and natural resources belonged to the State. In the 1990's, Suriname granted logging and mining concessions to private companies within the traditional Saramaka people's territory	The Court decided that although the Saramakas were not an Indigenous community, they had certain resemblances with traditional Indigenous communities and therefore enjoyed the same rights. As a consequence, they did not need a title in order to own the lands (possession was sufficient). The Court found a violation of articles 3 (Judicial Personality), 21 (Right to Property) and 25 (Right to Judicial Protection) of the American Convention of Human Rights in relation with articles 1.1 and 2 (obligations to respect, ensure and to give domestic legal effect of those rights). It recommended Suriname to "delimit, demarcate and grant a collective title over the territory of the members of the Saramaka people, in accordance with their customary laws, and through previous,

Court case	Description of case	Court ruling
	without consultation or their consent.	effective and fully informed consultations” and to abstain from acts that might affect the territory until such delimitation, demarcation and titling were completed. The Court also requested the State to review all concessions already granted. In addition, the State was required to give the Saramakas legal recognition of their collective judicial capacity and to adopt the legislative and other measures to recognize and ensure the community’s property rights, the right to be effectively consulted, and to provide effective recourses against violations of their rights.
Case of the Kaliña and Lokono peoples v. Suriname, 2015	Violations of the rights of the members of eight communities of the Kaliña and Lokono Indigenous peoples of the Lower Marowijne River in Suriname, owing to the absence at of a legal framework that recognizes the legal personality of the Indigenous peoples: no established legal and regulatory framework that would permit recognizing the right to collective ownership of the lands, territories and natural resources of the Kaliña and Lokono Indigenous peoples: issue of individual property titles to non-indigenous persons; the granting of concessions and licenses to carry out mining operations; and the establishment and continuation of three nature reserves in part of their ancestral territory... neither the granting of mining concessions and licenses nor the establishment and permanence to date of the nature reserves were subject to any consultation procedure aimed at obtaining the prior, free and informed consent of the Kaliña and Lokono peoples	The State violated the right to property recognized in Article 21 of the Convention, in relation to Articles 1(1) and 2 of this instrument, to the detriment of the Kaliña and Lokono peoples, by: (i) granting property titles to non-indigenous persons within the territory of the Kaliña and Lokono peoples; (ii) establishing and maintaining the Wia Wia, Galibi and Wane Kreek Nature Reserves, and (iii) granting a mining concession and authorizing mining operations within their traditional territory, all without conducting a consultation process aimed at obtaining their free, prior and informed consent in keeping with Inter-American standards. The Court ordered the State: 1. To grant the Kalina and Lokono peoples’ legal recognition of collective juridical personality, 2. Delimit, demarcate and grant collective title to the territory of the members of the Kalina and Lokono peoples. 3. Determine the territorial rights of the Kalina and Lokono peoples. 4. Take the appropriate measures to ensure the access, use and participation of the Kalina and Lokono peoples, etc.

4.2.3 Sectoral and cross-sectoral policies

The following sectoral and cross-sectoral policies are also considered relevant in relation to concerned environmental and socio-economic settings of the mining sector:

Suriname's National Development Plan 2021 – 2026 (NDP¹⁰): The general objective of the development plan is to build and maintain a national economy that is free from foreign domination with the participation of the citizens, and from which citizens benefit. The NDP looks to the future and identifies several policy areas as well as long- and short-term goals (GOS, 2022c). With regard to the gold sector (Chapter A6), the NDP mentions that forecasts indicate that the gold price will rise (significantly) in the future, thus making gold mining more lucrative. However, the danger is that unstructured small-scale gold mining will accelerate with disastrous consequences for the already severely degraded environment (GOS, 2022c).

For the gold mining sector, the NDP includes a strength, opportunities, weaknesses and threats (SWOT) analysis for the gold mining sector, which is presented in

A6.14.1: Informal gold sector must be reordered (planned in 2022).

1. A6.14.2: Allocate specific areas for small-scale gold mining (planned in 2023).
2. A6.14.3: Check which concession rights have been issued and see what the possibilities are to revoke them in the event of inactivity (planned in 2024).
3. A6.14.4: Establish and roll out a rehabilitation plan for degraded areas (planned in 2024).
4. A6.14.5: Exploring options for setting up Surinamese gold business/company (planned in 2024).
5. A6.14.6: Mining law amendment. (Amendment to Mining Act II). A new law should set clear rules for rehabilitation, those informal gold sectors and issuance policies (planned in 2025)¹¹.
6. A6.14.7: Implementation of gold mining without the use of toxic substances (the knowledge is already in Suriname), with in particular a phasing out of the use of mercury (planned in 2023).

For the later strategic action item A6.14.7, the NDP has set a plan to reach it, as shown in Figure 4-1.

¹⁰ Also known as Meerjaren Ontwikkelingsplan Suriname – MOPS.

¹¹ Consideration should be taken that amendments could be initiated with the installation of the Mineral Institute of Suriname planned for 2023.

Table 4-3: Gold mining sector SWOT Analysis

SWOT Analysis – Gold mining sector in Suriname	
Strengths	Weaknesses
• Large gold reserves in Suriname. • Demand for gold, however, prices fluctuate. • High wages and stable labor supply (but low labor rate). • Suriname intends to participate in the international Extractive Industries Transparency Initiative (EITI).	• Mining has profound effects on the environment such as: ○ Large scale deforestation ○ Mercury pollution (small scale mining) ○ Harmful substances used or released. • About environmental rehabilitation, there is little or no communication with the multinational companies. • Little use of “supply/local content” activities the industrial gold sector. • High health and safety risks for workers • Production takes place in remote areas making compliance checks difficult.
Opportunities ¹²	Threats
• Local production of inputs used in the gold sector. • Ratification of the Minamata Treaty by Suriname, creating momentum for the elimination of the use of mercury.	• Fluctuations in the world market price • An environmental disaster in case of major leakage of the cyanide storage occurs. • Further accumulation of mercury in the environment if the production methods of small-scale gold mining cannot be modernized, such as mercury poisoning of humans and animals. • Continuing and intensifying conflicts between small-scale and large-scale gold mining.

Source: (GOS, 2022c)

¹² An additional opportunity is that global internet access can now facilitate online access to information.

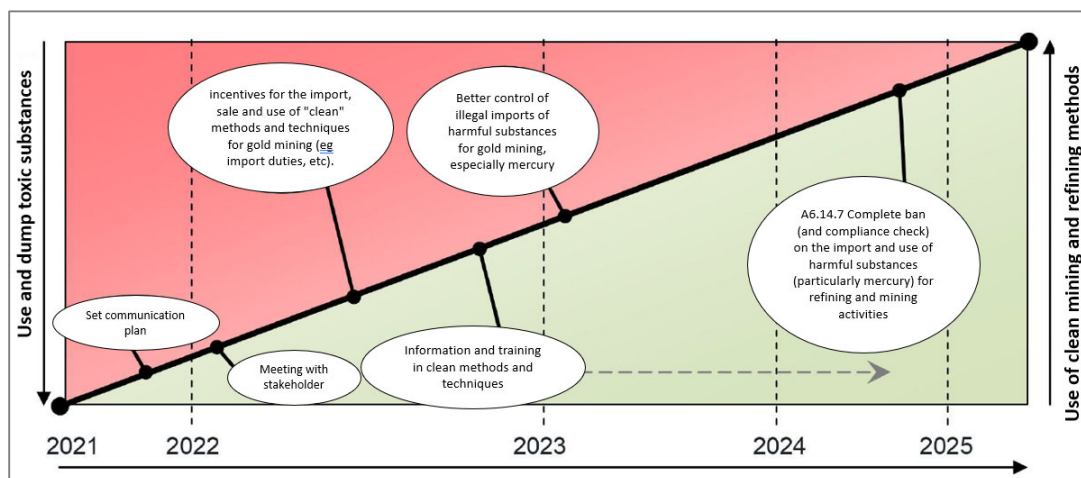


Figure 4-1: Development of strategic action item A6.14.7

Source: (GOS, 2022c) adapted by RINA, 2022

Based on Surinam's National Goals, in 2018 Surinam received funding from the Global Environment Facility (GEF)¹³ to improve the management of mining, especially artisanal and small-scale gold mining (ASGM), and promote environmentally responsible mining technologies (excluding the use of mercury), in order to reduce negative effects on biodiversity, forests, water and local communities, while reducing greenhouse gas emissions (known as the "EMSAGS Project") (European Commission, 2021). The key organizations involved in this initiative are the UNDP (implementing agency¹⁴), the MNR and NIMOS (executing agencies). The mid-term review report for the project carried out between August and December 2021 (UNDP, 2022), indicates that the project is offtrack and no real progress has been achieved on any of the main outcomes expected for this initiative, which include:

- Outcome 1, "Institutional capacity, inter-institutional coordination and availability of funding increased for improved management of ASGM";
- Outcome 2, "Policy and planning framework for the management of the environmental impacts of ASGM mining strengthened";
- Outcome 3 "Uptake of environmentally responsible artisanal small-scale gold mining practices increased"; and
- Outcome 4, "Knowledge availability and sharing increased at the national and regional scale on environmentally responsible ASGM".

1. **National Biodiversity Strategy:** Suriname developed a National Biodiversity Strategy (NBS), which establishes the national vision and strategic directions to conserve and sustain rich biodiversity and biological resources. Moreover, the NBS sets out its goals for sustainable management of the nation's natural resources and supports the equitable sharing of biodiversity related to services and benefits. The NBS provides a framework for the development of a Biodiversity Action Plan, which will identify the activities, tasks, outcomes, milestones, and implementation of a strategic program. The use and management of biodiversity remains a critical

¹³ GEF Project ID: 9288. <https://www.thegef.org/projects-operations/projects/9288>

¹⁴ UNDP - GEF Agency ID Project ID: 5627.

element in the maintenance and development of traditional societies for an emerging modern economy in Suriname.

2. **National Oil Spill Contingency Plan of Suriname (NOSCP) 2016:** The National Oil Spill Contingency Plan outlines Suriname's National Policy and Response Organization to a national level event as well as what is expected of the Responsible Party in ensuring a rapid, cooperative and effective response to an oil spill incident. The primary objective is to protect human health and safety, minimize social and environmental impacts, and ensure that the socio-economic and environmental conditions are restored. The Plan has been developed in conjunction with the Nationaal Coördinatie Centrum voor Rampenbeheersing (NCCR), the Maritime Authority Suriname (MAS), National Security, the National Institute for Environment and Development in Suriname (NIMOS), RAC/REMPEITC Caribe, Staatsolie Maatschappij Suriname N.V. (Staatsolie), and Oil Spill Response Ltd (OSRL). Following international and regional best practice, ISO 155441 and the International Maritime Organization (IMO) Manual on Assessment of Oil Spill Risk and Preparedness (2010 Edition), this plan establishes the roles, duties and responsibilities of the above-mentioned organizations in the response to a national level oil spill incident.

The NOSCP aims to ensure that any response operation recognizes the following as primary objectives:

- Protecting human health and safety
- Minimizing any social and environmental impacts
- Ensuring that the socio-economic and environmental conditions are restored, as near as is practicable, to pre-spill conditions.

4.2.4 International agreements

Suriname is signatory to a number of international agreements and conventions relating to environmental management, OHS, labor rights and Indigenous and Tribal Peoples. However, the international conventions are not always translated into national legislation. Some of the key agreements are listed in the following Table 4-4.

Table 4-4: International Agreements Relevant to Environmental and Social Issues

Agreement / Convention	Description	Status
CLIMATE CHANGE/AIR QUALITY		
Vienna Convention for the Protection of the Ozone Layer, 1985	Protection of the ozone layer came into force in 1988. Sets international standards for protection of the ozone layer; emissions from project potential to harm ozone layer.	Ratified in 1997.
Montreal Protocol on Substances that Deplete the Ozone Layer, 1988	Protection of the ozone layer. Came into force in 1988. As above.	Ratified in 1997.
United Nations Framework Convention on Climate Change (UNFCCC), 1994	Control of greenhouse gas emissions. Sets international guidelines on restrictions of GHG emissions in order to prevent climate change; Project will emit greenhouse gases from power generation through heavy fuel combustion.	Ratified in 1997.
Kyoto Protocol (Framework Convention on Climate Change), 1997	The Kyoto Protocol to the United Nations Framework Convention on Climate Change came into force in 2005. The Kyoto was superseded by the Paris Agreement in 2021.	Signed by Suriname on 22 April 2016

Agreement / Convention	Description	Status
	It applies the principle of common but differentiated responsibilities and sets binding targets for reducing greenhouse gas emissions for industrialized countries, recognizing them as those primarily responsible for the high levels of emissions currently present in the atmosphere.	
Paris Agreement (Framework Convention on Climate Change), 2015	The 21 st Conference of Parties to the United Nations Framework Convention on Climate Change (COP 21), held in Paris in 2015, concluded with the adoption of the Decision and the Paris Agreement. Its goal is to keep the global rise in temperature well below 2°C. It aims to achieve this by increasing adaptability to the adverse effects of climate change and promoting climate resilience and low carbon development. In order to achieve the agreed targets, the Paris Agreement establishes a reinforced transparency framework aimed at fostering mutual trust and promoting the effective implementation of the Agreement, increasing clarity and facilitating the monitoring of progress.	Ratified by Suriname on 13 Feb 2019
BIODIVERSITY/PROTECTED AREAS		
Convention on Nature Protection and Wildlife Preservation in the Western Hemisphere (Western Hemisphere Convention), 1942	Provisions to establish a set of protected areas; national parks to provide recreational and educational facilities; strict wilderness areas to be maintained inviolate; cooperation in the field of research between governments; species listed in annex to enjoy special protection and controls to be imposed on trade in protected fauna and flora and any parts thereof. Sets international requirements for the protection and preservation of nature/wildlife; project is in a high-biodiversity area and has potential to impact local wildlife and natural environment.	Signed by Suriname in 1985. Suriname has three Western Hemisphere Shorebird Reserves: - Coppenamemonding - Bigi Pan - Wia Wia
Convention on Wetlands of International Importance especially as Waterfowl Habitat (Ramsar Convention), 1971	Suriname acceded to this convention in 1985. Suriname has multiple Multi Use Management areas (MUMA) and protected areas hosting wetlands, including the following: • Bigi Pan MUMA • Noord Coronie MUMA • Noord Saramacca MUMA • Noord Commewijne/ Marowijne MUMA • Wanekreek Nature Reserve • Galibi Nature Reserve Sets international requirements for the protection of wetlands; project has potential to impact local wetland area.	Came into force for Suriname in 1985. The Coppenamemonding Nature Reserve is a Ramsar site.
The Cartagena Protocol on Biosafety to the Convention on Biological Diversity	The Cartagena Protocol, in force since September 2003, was adopted as a complementary agreement to the Convention on Biological Diversity. The Protocol seeks to protect biological diversity by managing the movements of Live Modified Organisms resulting from the application of modern technology	Accession since 27 March 2008 Ratified by Suriname in 2008

Agreement / Convention	Description	Status
	(LMOs) between countries. It establishes a procedure for prior informed agreement to ensure countries have the necessary information to make decisions about the importing of LMOs into their territory.	
Convention on the International Trade of Endangered Species of Wild Fauna and Flora (CITES), 1973	In force since 1975, to ensure that international trade in specimens of wild animals and plants does not threaten their survival and it accords varying degrees of protection to more than 33,000 species of animals and plants. Sets international restrictions/bans on trade of certain wild animals/plants. Project takes place in high biodiversity area.	Ratified by Suriname in 1981. New game regulations to comply with CITES requirements were passed in 2002.
United Nations Convention on Biological Diversity, 1992	Promotes development of national strategies for the conservation and sustainable use of biological diversity. Often seen as the key document regarding sustainable development. Sets guidelines for protection and promotion of biological diversity. Project takes place in high biodiversity area.	Ratified by Suriname in 1996. A National Biodiversity Strategy (2006) has been compiled as framework for a National Biodiversity Action Plan.
The International Plant Protection Convention, IPPC	First coming into force in 1952, the IPPC is an international plant protection agreement that aims to prevent the introduction and spread of plant pests and promote appropriate measures for their control. The purpose of this convention is to take action to prevent and introduce pests that are harmful to plants and plant products and to encourage appropriate measures against these pests. The Convention provides a framework and forum for international cooperation and the exchange of technical information in conjunction with regional and national plant protection organizations. It also plays an important role in trade, as it is recognized by the World Trade Organization (WTO) as the source for International Standards for Phytosanitary Measures (ISPMs) that affect trade.	Ratified by Suriname in 1977

Agreement / Convention	Description	Status
International Convention for the Regulation of Whaling	The International Convention for the Regulation of Whaling was signed in 1946 and it is the International Whaling Commission's founding document. The Convention includes a legally binding Schedule, which, amongst other things, sets out catch limits for commercial and Aboriginal subsistence whaling. In 1948 the International Convention for the Regulation of Whaling (ICRW) came into force. Establishing a system of international whaling regulations to promote conservation and development of whale species. An integral part of this treaty is the Schedule (The Schedule), which, among other things, indicates which whale species may not be hunted, how whaling will be carried out, and which areas are designated as whale sanctuary. The treaty also prohibits the capture of whale pups and their mothers.	Ratified by Suriname in 2004
United Nations Convention to Combat Desertification, 1994	In force since 1996, to combat desertification and mitigate the effects of drought through national action programs that incorporate long-term strategies supported by international cooperation and partnership arrangements. Sets guidelines to combat desertification. Project has potential to impact local water resources and quality and land use.	Ratified by Suriname in 2000. As part the implementation of this convention, a national report has been compiled. Currently discussions are underway on how to implement the convention in a more effective and efficient manner. The development of national action plan is prerequisite.
FOREST MANAGEMENT		
International Tropical Timber Agreement, 1994 (Tropical Timber 94)	To ensure that (by the year 2000) exports of tropical timber originate from sustainably managed sources, and to establish a fund to assist tropical timber producers in obtaining the resources necessary to reach this objective. Sets international targets for sustainable export of tropical timber; Project has potential to impact local forested area.	Suriname acceded in 1998.
HUMAN RIGHTS		
American Convention on Human Rights, 1969	International human rights instrument to establish a system of personal liberty and social justice based on respect for the essential rights of man, within the framework of democratic institutions. The bodies responsible for overseeing compliance are the Inter-American Commission on Human Rights and the Inter-American Court of Human Rights, both of which are organs of the Organization of American States.	Ratified by Suriname in 1987

Agreement / Convention	Description	Status
	Sets framework for protection of human rights; project has impact to affect lives of those living in the local community.	
International Convention on the Elimination of All Forms of Racial Discrimination, 1965	The Convention essentially resolves to “to adopt all necessary measures for speedily eliminating racial discrimination in all its forms and manifestations, and to prevent and combat racist doctrines and practices in order to promote understanding between races and to build an international community free from all forms of racial segregation and racial discrimination.”	Ratification/Accession , 1984
International Covenant on Economic, Social and Cultural Rights, 1966	It ensures the enjoyment of economic, social and cultural rights, including the rights to: education, fair and just conditions of work, an adequate standard of living, the highest attainable standard of health, social security.	Ratification/Accession ,1976
Convention on the Elimination of All Forms of Discrimination against Women, 1979	CEDAW is an international legal instrument that requires countries to eliminate discrimination against women and girls in all areas and promotes women’s and girls’ equal rights.	Ratification/Accession ,1993
Convention on the Rights of Persons with Disabilities, 2006	The Convention aims to enhance opportunities for people with disability to participate in all aspects of social and political life including access to employment, education, health care, information, justice, public transport and the built environment.	Ratification/Accession ,2017
Convention on the Rights of the Child, 1989	The United Nations Convention on the Rights of the Child (UNCRC) is a legally binding international agreement setting out the civil, political, economic, social and cultural rights of every child, regardless of their race, religion or abilities.	Ratification/Accession ,1993
CULTURAL HERITAGE		
UNESCO Convention Concerning the Protection of the World Cultural and Natural Heritage, 1972 (World Heritage Convention)	Promotes cooperation among nations to protect heritage from around the world that is of such outstanding universal value that its conservation is important for current and future generations. Sets guidelines for the conservation of significant heritage/cultural sites.	Ratified by Suriname in 1997. Suriname has two World Heritage Sites: - Central Suriname Nature Reserve; and - Historic Inner City of Paramaribo
INDIGENOUS PEOPLE		
UN formulation of Declaration on Rights of Indigenous People, 1994. UN approval of Declaration on Rights of Indigenous People, 2007.	Promotes the rights of Indigenous people. Sets standards for protection of Indigenous people.	Signed by Suriname

Agreement / Convention	Description	Status
REGIONAL DEVELOPMENT/COOPERATION		
Amazon Cooperation Treaty, 1978 (amended in 1998)	Aimed at the promotion of sustainable development of the Amazon Basin. Member states are Bolivia, Brazil, Colombia, Ecuador, Guyana, Peru, Suriname and Venezuela.	Suriname is a founding member,
LABOR / HEALTH & SAFETY		
Constitution of the International Labor Organization	Promotes opportunities for women and men to obtain decent and productive work, in conditions of freedom, equity, security and human dignity. Sets international labor standards; project will employ large workforce.	Suriname member since 1976.
Constitution of the Pan American Health Organization	To improve health and living standards of the countries of the Americas. Sets guidelines to improve health and living standards; project has potential to impact local health/living standards.	Suriname member since 1976.
POLLUTION PREVENTION AND CONTROL		
Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal	The agreement, in force since May 1992, seeks to protect the health of people and the environment from the harmful effects of hazardous wastes. The provisions of the Convention revolve around the reduction of hazardous waste generation and the promotion of the environmentally sound management of hazardous wastes, the restriction of transboundary movements of hazardous wastes, and the application of a regulatory system for permissible movements of hazardous waste.	Suriname acceded on 20 Sept 2011
Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade	In force since February 2004. It aims to promote shared responsibility and cooperation among Parties in addressing the international trade of certain hazardous chemicals, in order to protect human health and the environment from potential harm. The agreement establishes a prior informed consent (FPIC) procedure for the import of hazardous chemicals.	Suriname acceded on 20 May 2000
Stockholm Convention on Persistent Organic Pollutants	Came into force in 2004. It aims to protect human health and the environment from Persistent Organic Pollutants (POPs). The Convention requires Parties to take steps to eliminate or reduce the production, use, import, export and emissions into the environment of POPs and includes provisions on access to information, public awareness, training and participation in the development of implementation plans.	Ratified on 20 Sept 2011
Minamata Convention on Mercury	The Minamata Convention, in force since August 2017, was adopted at the 2013 Plenipotentiary Conference in Kumamoto, Japan. The aim of this global treaty is to protect human health and the environment from anthropogenic and	Suriname Acceded on 02 August 2018

Agreement / Convention	Description	Status
	other releases of mercury and mercury compounds. It includes provisions on public information, environmental education, promotion of participation and capacity building.	
The London Convention on Prevention of Pollution by Dumping of Wastes and Other Matter 1972	The London Convention entered into force in 1975 and is an agreement to control pollution of the sea by dumping and to encourage regional agreements supplementary to the convention. Its objective is to promote the effective control of all sources of marine pollution and to take all practicable steps to prevent pollution of the sea by dumping of wastes and other matters.	Suriname acceded on 21 October 1980
United Nations Convention on the Law of the Sea; UNCLOS	The Convention entered into force in accordance with its article 308 on 16 November 1994, 12 months after the date of deposit of the sixtieth instrument of ratification or accession. Today, it is the globally recognized regime dealing with all matters relating to the law of the sea. The United Nations Convention on the Law of the Sea lays down a comprehensive regime of law and order in the world's oceans and seas establishing rules governing all uses of the oceans and their resources. It enshrines the notion that all problems of ocean space are closely interrelated and need to be addressed as a whole. UNCLOS consists of 320 articles and 9 annexes and manages all aspects of the law of the sea. This is done by establishing rules on the management of all uses of the seas and their resources.	Ratified by Suriname in 1998
International Convention Relating to Intervention on the High Seas in Cases of Oil Pollution Casualties, 1969	The Convention entered into force on 6 May 1975. The 1969 Intervention Convention applied to casualties involving pollution by oil. To combat oil pollution from ships, by giving coastal states the right to take measures to prevent, reduce or eliminate pollution of the sea.	Suriname succeeded to the convention in 1976
International Convention for the Prevention of Pollution from Ships, 1973 as modified by the Protocol of 1978 relating thereto (MARPOL)	MARPOL came into force in 1983 and initially comprised Annex I (Regulations for the Prevention of Pollution by Oil Regulations) and Annex II (Regulations for the Control of Pollution by Noxious Liquid Substances in Bulk). Four more annexes have been added; parties must accept Annexes I and II, but the other four are voluntary. Suriname has ratified Annexes I through V. The MARPOL Convention is the main convention concerned with the prevention of pollution from ships. This is because the deliberate, negligent, accidental discharge of oil and other harmful substances from ships is a serious source of pollution. Harmful substances are understood to mean all substances, which, if they end up in the sea, could endanger human health; can cause damage to the marine flora and fauna, damage the recreational opportunities that the sea offers or can	Ratified by Suriname in 1988

Agreement / Convention	Description	Status
	have a disruptive effect on other lawful uses of the sea (Article 2 paragraph 2 MARPOL Convention). In addition to rules that serve to prevent and minimize pollution from ships, the convention also contains six technical annexes.	

Source: SRK, 2007 modified and updated by RINA, 2022

Since 1919, the International Labor Organization has maintained and developed a system of international labor standards aimed at promoting opportunities for women and men to obtain decent and productive work, in conditions of freedom, equity, security and dignity (ILO, 2022). The ILO Governing Body has identified eight “fundamental” Conventions, covering subjects that are considered to be fundamental principles and rights at work: freedom of association and the effective recognition of the right to collective bargaining; the elimination of all forms of forced or compulsory labor; the effective abolition of child labor; and the elimination of discrimination in respect of employment and occupation. The ILO has also designated another four Conventions as governance (or priority) instruments, thereby encouraging member States to ratify them because of their importance for the functioning of the international labor standards system. (ILO, 2019). Of these, Suriname has ratified eight (out of ten) fundamental conventions, three (out of four) governance conventions and 22 (out of 176) principal technical conventions. These and other relevant ILO conventions are presented in Table 4-5.

Table 4-5 Relevant ILO Conventions

Type	Category	#	Convention (C) - Protocol (P)	Date	Status
Fundamental	Freedom Association	1	C087 - Freedom of Association and Protection of the Right to Organize Convention, 1948 (No. 87)	15-Jun-76	In Force
		2	C098 - Right to Organize and Collective Bargaining Convention, 1949 (No. 98)	05-Jun-96	In Force
	Force labor	3	C029 - Forced Labour Convention, 1930 (No. 29) P029 - Protocol of 2014 to the Forced Labour Convention, 1930 ratified on 03 Jun 2019 (In Force)	15-Jun-76	In Force
		4	C105 - Abolition of Forced Labour Convention, 1957 (No. 105)	15-Jun-76	In Force
	Discrimination	5	C100 - Equal Remuneration Convention, 1951 (No. 100)	04 Jan 2017	In Force
		6	C111 - Discrimination (Employment and Occupation) Convention, 1958 (No. 111)	04 Jan 2017	In Force
	Child labor	7	C138 - Minimum Age Convention, 1973 (No. 138) Minimum age specified: 16 years	15 Jan 2018	In Force
		8	C182 - Worst Forms of Child Labour Convention, 1999 (No. 182)	12 Apr 2006	In Force

Type	Category	#	Convention (C) - Protocol (P)	Date	Status
	OHS	9	C155 - Occupational Safety and Health Convention, 1981 (No. 155)	/	not ratified
		10	C187 - Promotional Framework for Occupational Safety and Health Convention, 2006 (No. 187)	/	not ratified
Principal Governance	Labor	1	C081 - Labour Inspection Convention, 1947 (No. 81)	15-Jun-76	In Force
	Labor	2	C122 - Employment Policy Convention, 1964 (No. 122)	15-Jun-76	In Force
	Labor	3	C144 - Tripartite Consultation (International Labour Standards) Convention, 1976 (No. 144)	16-Nov-79	In Force
Principal Technical	Labor	1	C014 - Weekly Rest (Industry) Convention, 1921 (No. 14)	15-Jun-76	In Force
	OHS	2	C017 - Workmen's Compensation (Accidents) Convention, 1925 (No. 17)	15-Jun-76	In Force
	OHS	3	C019 - Equality of Treatment (Accident Compensation) Convention, 1925 (No. 19)	15-Jun-76	In Force
	OHS	4	C042 - Workmen's Compensation (Occupational Diseases) Convention (Revised), 1934 (No. 42)	15-Jun-76	In Force
	Labor	5	C095 - Protection of Wages Convention, 1949 (No. 95)	15-Jun-76	In Force
	Labor	6	C118 - Equality of Treatment (Social Security) Convention, 1962 (No. 118) Has accepted Branch (g)	15-Jun-76	In Force
	Freedom Association	7	C135 - Workers' Representatives Convention, 1971 (No. 135)	15-Jun-76	In Force
	Freedom Association	8	C154 - Collective Bargaining Convention, 1981 (No. 154)	05-Jun-96	In Force
	Labor	9	C097 - Migration for Employment Convention (Revised), 1949 (No. 97)	/	not ratified
	OHS	10	C102 - Social Security (Minimum Standards) Convention, 1952 (No. 102)	/	not ratified
	Labor	11	C143 - Migrant Workers (Supplementary Provisions) Convention, 1975 (No. 143)	/	not ratified
	OHS	12	C148 - Working Environment (Air Pollution, Noise and Vibration) Convention, 1977 (No. 148)	/	not ratified
	OHS	13	C152 - Occupational Safety and Health (Dock Work) Convention, 1979 (No. 152)	/	not ratified
	OHS	14	C161 - Occupational Health Services Convention, 1985 (No. 161)	/	not ratified

Type	Category	#	Convention (C) - Protocol (P)	Date	Status
	OHS	15	C162 - Asbestos Convention, 1986 (No. 162)	/	not ratified
	Labor	16	C168 - Employment Promotion and Protection against Unemployment Convention, 1988 (No. 168)	/	not ratified
	Indigenous People	17	C169 - Indigenous and Tribal Peoples Convention, 1989 (No. 169)	/	not ratified
	OHS	18	C170 - Chemicals Convention, 1990 (No. 170)	/	not ratified
	OHS	19	C176 - Safety and Health in Mines Convention, 1995 (No. 176)	/	not ratified
	Labor	20	C183 - Maternity Protection Convention, 2000 (No. 183)	/	not ratified
	OHS	21	P089 - Protocol of 1990 to the Night Work (Women) Convention (Revised), 1948	/	not ratified
	OHS	22	P155 - Protocol of 2002 to the Occupational Safety and Health Convention, 1981	/	not ratified

Source: (ILO, 2022)

4.2.5 National institutional framework for environmental and social management

Suriname's mining sector is governed by the Mining Decree (1986) and regulated by the Ministry of Natural Resources, the Ministry of Finance, the Ministry of Regional Development and the Ministry of Trade, Industry and Tourism. The legal and fiscal framework for the oil, gas and mining sectors is also supported by other legislation, including the Corporate Income Tax Act and Dividend Tax Act. (EITI, 2022). Applications for mining rights are processed by the Geological Mining Service (GMD) and are reviewed and approved by the Ministry of Natural Resources in accordance with the Mining Decree. Suriname does not have a public mining license register, however some information about issued licenses is included in EITI Reports (EITI, 2022). EITI Suriname is administered by the Suriname Multi-Stakeholder Group (MSG), which is hosted by the Ministry of Natural Resources. The MSG is chaired by the Minister of Natural Resources. (EITI, 2022)

On the other hand, the environmental approval process is to be administered under the new Environmental Framework Act 2022, through the yet to be operational National Environment Authority (in Dutch: Nationale Milieu Autoriteit, NMA), an institution transitioned from the National Institute for Environment and Development in Suriname (in Dutch: Nationaal Instituut voor Milieu en Ontwikkeling in Suriname, NIMOS), thus forming the new National Environment Structure in Suriname. Through this structure, the provisions included in the Environmental Framework Act will be implemented and applied in an efficient and effective manner. (NIMOS, 2022b).

The key institutions can be summarized as follows:

4.2.5.1 Ministry of Natural Resources - MNR

The aim of the Ministry of Natural Resources (MNR) is to ensure an integral sustainable and efficient management and development of the production of the natural resources present in Suriname, taking into account the environment, safety and rights of local communities. The organizational structure of the MNR has recently changed. The Ministry now consists of four (4) directorates, as further detailed in the organizational structure, namely (MNR, 2022a):

1. Directorate General Administration NH
2. Directorate of Water
3. Directorate of Energy
4. Mining Directorate

The organization has been expanded to include directorate and departmental tasks. The determining organizational objectives for this are:

1. Planning, monitoring and control of the spending of the budget and other resources.
2. Optimization of the operational execution of tasks.
3. Application of integrated management in the various policy areas.
4. Application of advancing technology within the various policy areas.
5. Continuous quality improvement and quality maintenance of the execution capacity.
6. Entry into regional and international partnerships.
7. Sustainable professional and personal development of staff.

Based on the objectives described above, the organization chart of the MNR is presented in Figure 4-2.

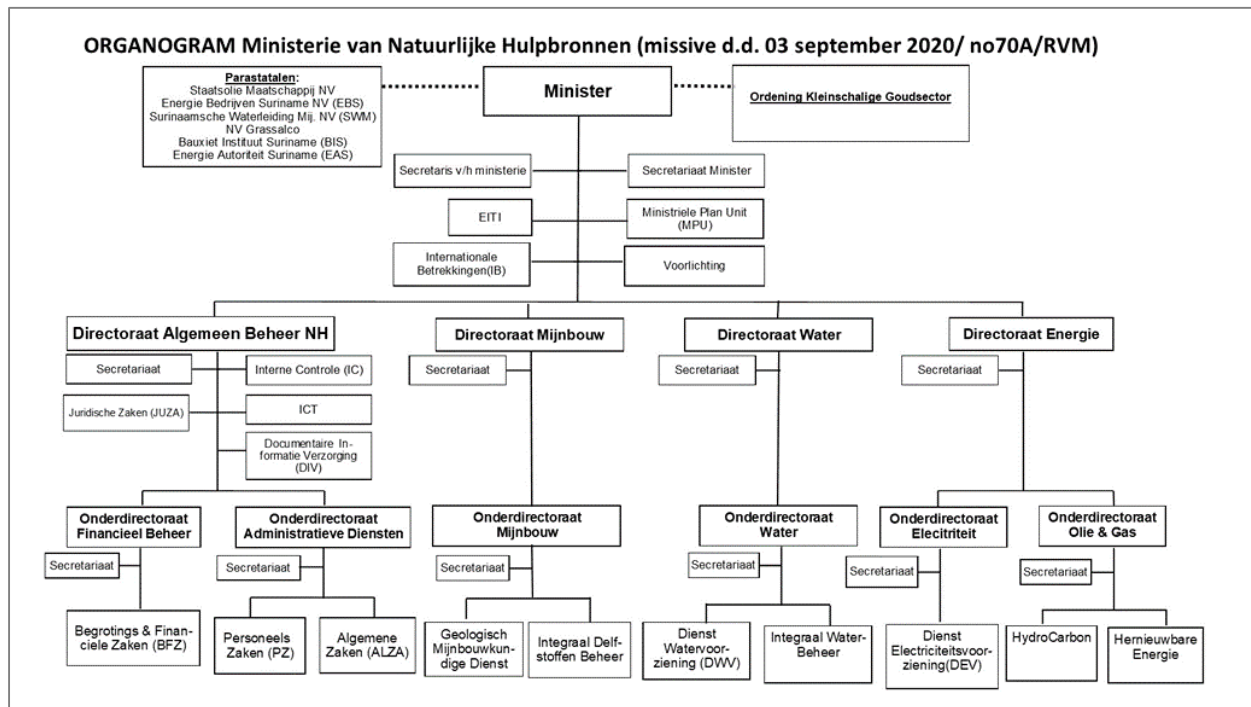


Figure 4-2: MNR organizational structure

Source: (MNR, 2022a)

4.2.5.2 Ministry of Land Policy and Forest Management

The Ministry of Land Policy and Forest management is responsible for the issuance of Domain Land, physical planning, nature conservation and forest management. The institutional coordinating function of the Ministry of Land Policy and Forest Management is, among others, related to: (i) proper land

allocation, in cooperation with relevant ministries, where necessary in an interdepartmental context; (ii) land use where necessary in an interdepartmental context; and (iii) monitoring the lawful and efficient use of issued land where necessary in an interdepartmental context.

4.2.5.3 Ministry of Regional Development and Sport

The Ministry of Regional Development and Sport is responsible for the development of rural areas and the provision services outside Paramaribo. Duties in each of the ten districts are managed by district admirations, headed by District Commissioners.

4.2.5.4 Ministry of Spatial Planning and Environment - ROM

In July 2020, the institutional structure for environmental management changed with the change of Government. The structural change included the establishment of a new Ministry for Spatial Planning and Environment (ROM). The Ministry of ROM aims to coordinate all environmental activities in the country. Legal positioning of the Ministry of ROM became a priority, and a formal working group was established for amending the Environmental Framework Act. The amendment proposes the Ministry to become primarily responsible for coordinating Environmental Policy while the NIMOS will transition into the National Environmental Authority (NMA). According to the Presidential Resolution (PB. 31), . Starting August 15, 2020, the Ministry of ROM and NMA transition into their new roles within 18 months, the Ministry of ROM and NIMOS had 18 months for the transition, which was scheduled to initiate August 15. . However, the transition has not yet been completed and it is expected that this period will be extended.

4.2.5.5 The Ministry of Transport, Communication and Tourism (MTCT)

The Ministry of Transport, Communication and Tourism (MTCT) is responsible for the transport sector (water, air and road transport), as well as public transport and its supervision. In addition, the ministry has the task to facilitate the water and air traffic sector, in particular with regard to the safe and efficient handling of persons and goods, pilotage and the management of all ports. Furthermore, the ministry is also in charge of the telecommunications sector and tourism in general as well as the further development of the tourism sector in particular.

4.2.5.6 Ministry of Public Works

The Ministry of Public Works is, among other things, responsible for the preparation and implementation of various Civil Engineering and Construction Works, traffic technical facilities, the issuance of construction and allotment permits, and also of green areas, street maintenance and waste processing.

The Road Authority (RA) is currently the only Parastatal organization under the Ministry of Public Works. The RA is known as the institution that has the necessary knowledge and experience with regard to, among other things, the rehabilitation of roads and the maintenance of verges. The Road Authority was established by the law of 25 July 1995, in which the following tasks are described:

1. Management and maintenance of primary roads and river crossings
2. Advising the government on the repair and maintenance of roads and fixed river crossings.

Currently, the RA maintains and manages 859.79 km of Primary roads. The activities of the RA are divided into the maintenance of the primary roads in and outside Paramaribo. Furthermore, the maintenance of verges and adjacent bridges that are connected to the primary roads, culverts, traffic facilities, road furniture, underground infrastructure and all other work that falls within the tasks of the institute established in 1995.

4.2.5.7 Bauxite Institute Suriname

The Bauxite Institute Suriname was established on February 17, 1981, by Decree E-9 S.B. 1981 no. 14 as a legal entity “sui generis” established to control the entire bauxite happening in Suriname in order to enable the Government to obtain a fair share from the proceeds of bauxite and its derivatives. The

core tasks of the Bauxite Institute Suriname are to advise the government on the rational development of the bauxite industry and to obtain a fair share of the revenues for Suriname, which can be categorized as follows.

- ✓ Collecting and processing data related to events in and around the bauxite industry nationally and internationally;
- ✓ Monitoring all activities of the bauxite companies in Suriname and checking them against the law and the various relevant agreements;
- ✓ Preparing and contributing to the implementation of the Government's bauxite policy; and
- ✓ Maintaining international contacts in this industry.

4.2.5.8 [National Coordination Centre for Disaster Management - NCCR](#)

NCCR is a division of the Ministry of Defense that develops national policies on disaster management and acts as coordinator and facilitator in crisis and disaster management during crises and disasters.

4.2.5.9 [Extractive Industries Transparency Initiative \(EITI\) Suriname](#)

EITI Suriname is administered by the Suriname Multi-Stakeholder Group (MSG), which is hosted by the Ministry of Natural Resources. The MSG is chaired by the Minister of Natural Resources. (EITI, 2022)

Suriname was found to have made *meaningful progress* in implementing the 2016 EITI Standard in March 2021, following its first Validation. The Validation identified 19 corrective actions to be addressed by the country's next Validation, expected to commence in April 2023. (EITI, 2022)

4.2.5.10 [National Institute for Environment and Development in Suriname \(NIMOS\)](#)

NIMOS was established on March 15, 1998, as the executive body of the National Environmental Council (which no longer exists). NIMOS is an autonomous government foundation, supervised by a Foundation Board and consists of the following 7 board members. The board is appointed for 2 years by the President of the GOS. The day-to-day management and overall coordination of NIMOS is undertaken by the General Director. Currently NIMOS has 27 staff members, of which 20 are employed on a permanent basis and the remaining on a project basis. (NIMOS, 2022b)

NIMOS consists of several offices, each tasked with specific responsibilities, namely (NIMOS, 2022b):

1. Administration
2. Environmental & Social Assessment
3. Environmental Monitoring & Enforcement
4. Environmental Legal Services
5. Environmental Education & Public Outreach
6. Environmental Planning & Information Management
7. Environmental Funding & Investments (not operational).
8. Environmental Research (not operational).

The mission of NIMOS is to "*initiate the development of a national legal and institutional framework for environmental policy and management in the interest of sustainable development*" that is to be driven by the following goals (NIMOS, 2022b) :

1. Achieving National Environmental Legislation in the broadest sense of the word.

2. Preparing and implementing Environmental Protection legislation.
3. Coordinating and monitoring of Compliance.

Under the new Environmental Framework Act, NIMOS will be responsible to develop the State of the Environment Report that will be used to inform policy and decision making of the NMA (NIMOS, 2022b). In addition, the new Act will require the development of Implementation Regulations to facilitate its execution. A total of 21 implementation regulations (i.e., Decrees) have been identified, however, the top 6 or most important ones were selected to be drafted first and currently under development, including (NIMOS, 2022b):

1. EIA Activities,
2. EIA Procedure,
3. Environmental Permit,
4. Hazardous Substances,
5. Environmental Fund,
6. Rules and Procedures for the Environmental Controllers.

NIMOS has also developed Generic Environmental Assessment Guidelines as well as specific Guidelines. These Guidelines carefully explain the procedures as how they should be carried out if the Environmental Framework Law and the Environmental Regulations were in place (NIMOS, 2022b).

The EIA process described by NIMOS is divided in 5 phases, namely (NIMOS, 2022b):

1. Screening phase: This is the phase in the EIA process, in which NIMOS decides whether an Environmental Impact Assessment is needed or not. If yes, NIMOS will indicate the nature and extent of the analysis. NIMOS acknowledges three categories of EIA:
 - a. Category A: an EIA is mandatory.
 - b. Category B, either:
 - i. An EIA is required (Category B, path 3);
 - ii. Another environmental document is needed (e.g., Environmental Management Plan) (Category B, path 2);
 - iii. No EIA is needed, but some environmental information is required before a decision can be taken (Category B, path 1).
 - c. Category C: No EIA is required, but the project proponent will have to keep with the minimal guidelines.
2. Scoping phase: In this phase the Terms of Reference (ToR), sometimes also called Scoping report, for the EIA-study has to be prepared.
3. Assessment phase: In the assessment phase, the EIA-study is to be carried out by a qualified consultant or consultancy firm by order of the project proponent.
4. Review phase: During the review process, the submitted Environmental Impact Statement (EIS) is reviewed by NIMOS.
5. Decision-making phase: After the EIS is reviewed, NIMOS should send the environmental advice regarding approval or denial of the project to the permitting agency.

Meanwhile NIMOS has the following sector specific EIA Guidelines in use, namely:

- ✓ Volume I: Generic (2009)
- ✓ Volume II: Mining (2005)

- ✓ Volume III: Forestry (2005)
- ✓ Volume IV: Social Impact Assessment (2005)
- ✓ Volume V: Power Generation and Transmission Projects (2005)
- ✓ Volume VI: Aquaculture Projects (2011)
- ✓ Volume VII: Agriculture Project (2013)
- ✓ Guidance Note (2017) (Supplement to the Generic Environmental Assessment Guidelines Volume 1)
- ✓ Volume IX: Part 1 - Offshore Oil and Gas Reconnaissance and Exploration Drilling 2020

4.2.5.11 [National Environment Authority– NMA](#)

With the approval of the Environmental Framework Act (EFA), the National Environmental Authority (in Dutch: Nationale Milieu Autoriteit) is established and responsible for enforcing this law. With this institution, the National Environmental Council (NMR), which has not been operational for many years, ceased to exist and the Environment Coordination within the President's Office has been renamed NMA, while NIMOS will cease to exist and will transition into the NMA. In practice, the situation has developed differently than prescribed by the EFA; the staff of Environmental Coordination is employed by the new Ministry of ROM and NIMOS is incorporated into the NMA, a process that is now underway.

The NMA will be responsible for administering the Environmental Impact Assessment process and Pollution Control. The NMA is to be responsible for the implementation and enforcement of the Environmental Framework Act.

4.2.5.12 [State Oil Company Suriname \(Staatsolie Maatschappij Suriname N.V.\)](#)

The State Oil Company Suriname is involved with the development of Suriname's hydrocarbon potential over the full value chain, to generate electricity, and to develop renewable sustainable energy resources. It is further engaged with assessments of hydrocarbon potential, promotion, contracting, and monitoring activities of other oil companies on behalf of the State.

4.2.5.13 [Staatsolie Hydrocarbon Institute N.V. \(SHI\)](#)

Since its establishment in December 1980, Staatsolie has acted as the state vehicle through which the government oil policy was executed and whereby foreign companies could only explore for and produce oil through service contracts with Staatsolie. The regulator's role was further developed through the establishment of the Petroleum Contracts division in 2000, followed by the establishment of SHI in 2020. SHI is an agent of the Government of the GOS and performs a regulatory role for the exploration and drilling activity (exploration and production)

4.2.5.14 [Minerals Institute](#)

The government of Suriname wants to promote a regulated and responsible expansion of the mining industry, including improved control over informal goldmining. However, weak institutional capacity in the public sector and the absence of an adequate legal and regulatory framework have inhibited prospects for growth. Moreover, weak corporate governance in the small-scale mining activity and the lack of social and environmental standards have diminished the sustainability and positive social impacts of the sector. The current mining policy aims to create a Minerals Institute – through the merger of the Geological Mining Service (GMD), the Bauxite Institute of Suriname (BIS) and the Suriname Gold Sector Regulation Commission (OGS¹⁵) - and to enact a new mining law since these currently are incapable

¹⁵ The Ordering Goudsector (OGS) was established in 2011 and it is a planning commission housed within the MNR and in charge of overseeing the small-scale gold mining sector.

of meeting legal deadlines for licensing, developing regulations and procedures for their activities, or enforcing the regulatory framework for artisanal and small-scale mining. (MNR, 2019)

The government of Suriname by State Decree plans to create the "Delfstoffen Instituut Suriname" (DIS) or Minerals Institute Suriname for the promotion, coordination, regulation, monitoring and inspection of the mining industry aimed at an optimal contribution to Suriname's sustainable development. Its creation is a response to the expected growth of sector activities in the country, the changing focus (from bauxite to gold and other minerals), the increasing complexity of financial, social and environmental challenges, and the increased demand for governance and accountability in managing the sector. (MNR, 2019)

In March 2022, the Minister for Natural Resources officially launched the mineral reform for the restructuring process of the mining sector. The Suriname Competitiveness and Sector Diversification Project (SCDS-Project) started in December 2021 and should ultimately form a good basis for setting up the Mineral Institute of Suriname. The Minister indicated that a legal and institutional framework will be needed to monitor mining areas and restructuring the small-scale gold sector. The Ministry of Natural Resources is holding workshops with stakeholders and a number of online discussions with various organizations. (MNR, 2022b)

The Minerals Institute Suriname will be given the functions of Land Registry, Geological Survey and Inspectorate in order to regulate the mining sector in an efficient and transparent manner through cooperative and transparent work processes in which technical and capacity issues are analyzed in a participatory manner, and with the digitization of the geological survey and capacity enhancement. (MNR, 2022b). Other land registration is to be executed by the Ministry of Spatial Planning and Environment.

5 ENVIRONMENTAL AND SOCIAL CHARACTERISTICS

This section describes the biophysical and socio-economic environment indicating area where possible impacts can occur from the mining sector.

5.1 INTRODUCTION

Suriname is endowed with many natural resources including soil, water, forests and mineral ores. The fertile soil of the young coastal plain, combined with the large freshwater swamps and rivers in the north are favorable for large scale agriculture. The largest part of the country (almost 88 % of total land area) consists of tropical rainforest, marginally used by local people for non-timber forest products. The cultivated land represents 8% of the total land area, whilst the remaining 4% consists of other non-forested natural areas (savannahs, swamps and wetlands). The cultivated area is used for forestry (3.7% of total land area), agriculture (3.2 %), and settlements, mining and hydropower reservoirs (1.3 %). Taking into account the low historical deforestation rate (between 0.03 and 0.04 % per year), Suriname is identified as a country with a High Forest Cover, and Low Deforestation rate (HFLD-country). (Berrenstein, 2016).

5.2 PHYSICAL ENVIRONMENT

5.2.1 Climatic Conditions

Suriname is located north of the equator on the northeastern coast of South America, between 40-60 north latitude and 540-580 west longitudes. Suriname has a tropical climate with abundant rainfall, uniform temperature, and high humidity. The country shows a smooth orography, with height ascending as we move southwards from the only coast in the north (Figure 5-1). This difference in altitude leads to diverging coastal and interior climates. (IADB, 2021)

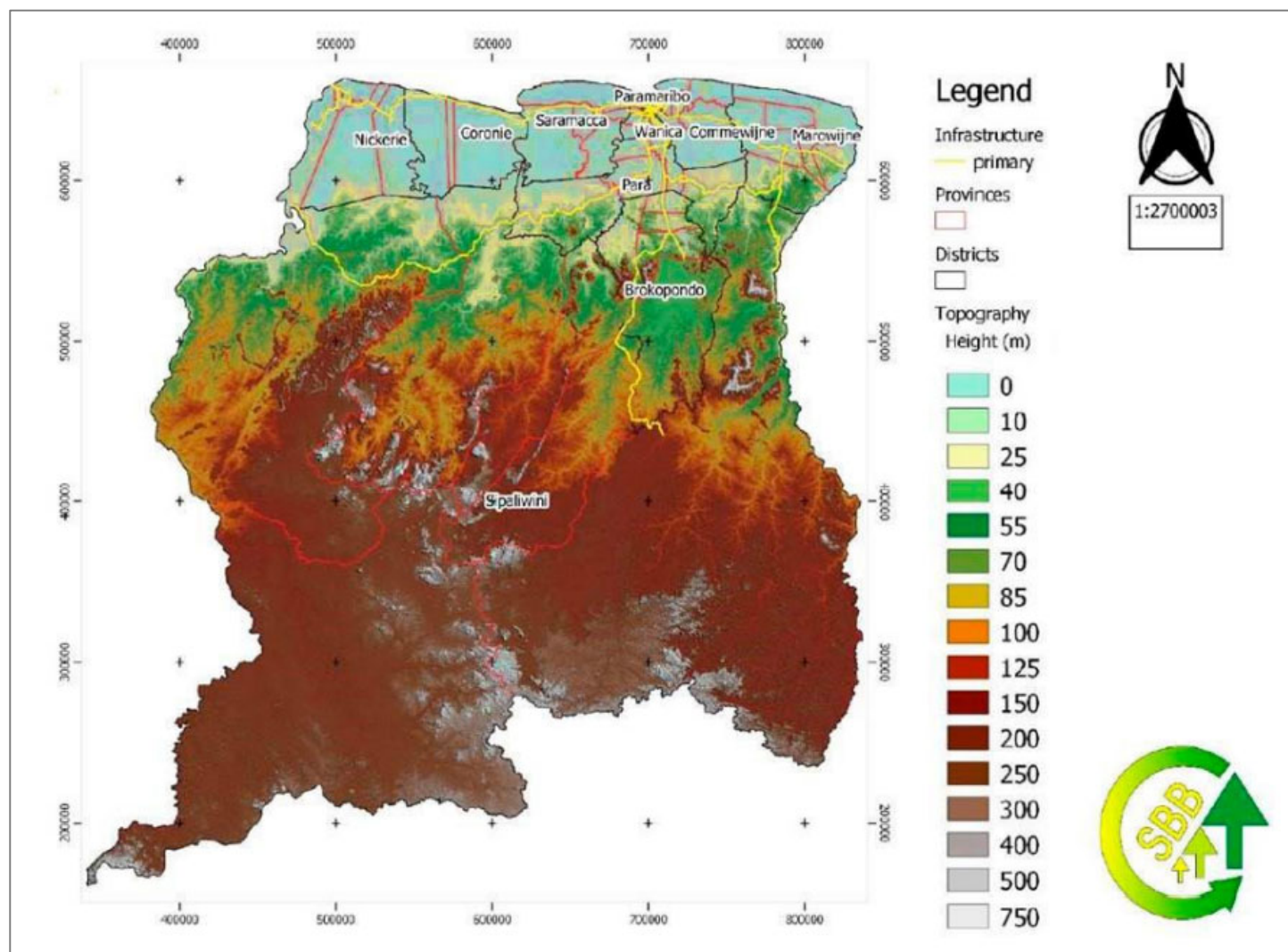


Figure 5-1: Orography of Suriname

Source: (IADB, 2021)

5.2.1.1 Temperature

The average daily temperature in the coastal region is 27.6°C, with an average daily variation of 4°C. There is relatively little variation in temperature between the seasons in the coastal region. January is the coldest month with an average temperature of 27.5°C, and October is the warmest with an average temperature of 29.2°C. The annual variation of the average temperature is 2-3°C. The interior has similar figures, with an even smaller annual variation of average temperatures (IADB, 2021). The average annual temperature in 2019 was 28.0 °C with an average minimum temperature of 23.9 °C and an average maximum temperature of 31.5 °C (GBS, March 2020).

5.2.1.2 Rainfall or Precipitation

Suriname has two wet and two dry seasons. The beginning of the seasons is not simultaneous for all areas. Most of the precipitation is measured during the rainy season. The Inter Tropical Convergence Zone (ITCZ) is responsible for precipitation (GBS, March 2020).

In 2019 the average yearly precipitation was 1,617.5 mm (GBS, March 2020). The average annual rainfall in Suriname is 2,200 mm; however, localized rainfall varies over the terrain where the coastal plains receive on average 1,500 -1,750 mm and the central region receives 2,500 - 3,000 mm. Suriname experiences two wet seasons: a major wet season between May and July when most of the country receives 250 - 400 mm per month, and a minor wet season from November to January, which brings around 150 - 200 mm of rainfall per month. The country also experiences two short dry seasons: a short dry season from February-April with mean monthly rainfall of 100 mm and a long dry season from August to December with less than 100 mm of rainfall per month. Suriname is south of the hurricane belt and therefore evades this threat. (SWRIS, 2022). The average temperature and precipitation per month are presented in Figure 5-2.

Precipitation amounts vary across the country. On average, Paramaribo receives 1,756 mm of rainfall annually, Bigi Pan in the northwestern coast receives 1,796 mm/year, Kwamalasamutu in southern Suriname receives 2,392 mm/year, and Tafelberg in central Suriname receives 1,851 mm/year. Variation in monthly rainfall results in two wet and two dry seasons in the northern part of Suriname. In the south, one wet and one dry season are distinguished (IADB, 2021).

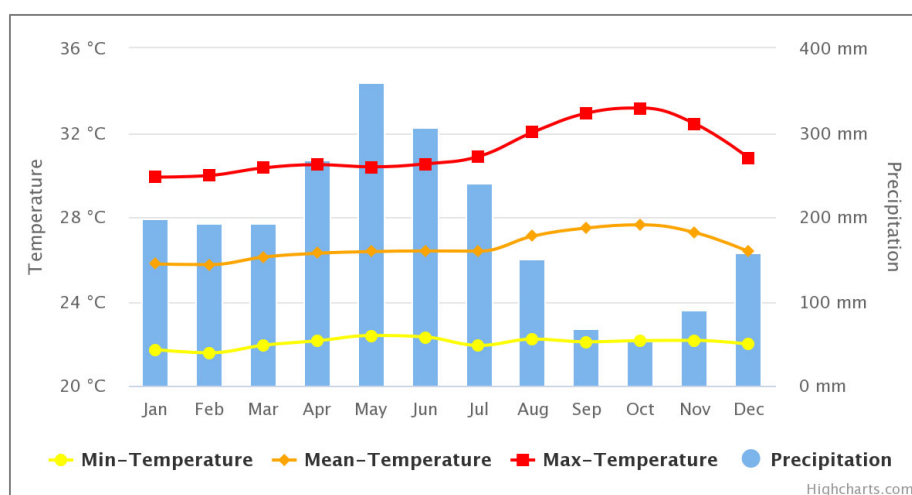


Figure 5-2: Monthly Climatology (Temperature and Precipitation) for Suriname between 1991 - 2020

Source: (WBG, 2022)

In the whole country, the rainfall varies from less than 1,750 to greater than 3,000 millimeters annually. The year can be roughly divided in two wet seasons (April to mid-August and December to February) and two dry seasons (February to April and mid-August to December) as it can be visualized in Figure 5-3 (SWRIS, 2022)

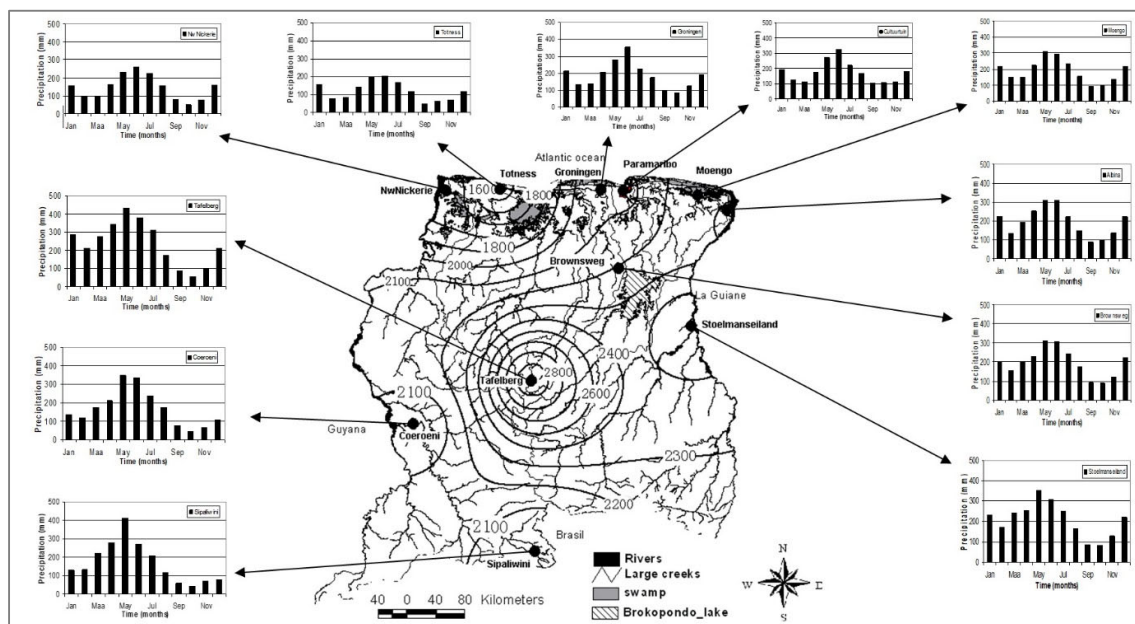


Figure 5-3: Annual and monthly rainfall distribution in Suriname (1961-1985)

Source: (SWRIS, 2022)

Inter-annual variations in climate are caused by the El Niño Southern Oscillation (ENSO)¹⁶. El Niño episodes bring dry conditions throughout the year, and bring warmer temperatures between June and August, while La Niña episodes bring wetter conditions throughout the year and cooler temperatures between June and August. The observed mean in relative humidity is 81%. (SWRIS, 2022)

5.2.1.3 Evaporation

The annual evaporation is about 1,700 mm (1931-1960). About 67% of the precipitated water on the Suriname rivers basins returns to the atmosphere by evapotranspiration and approximately 33% is drained off into the Atlantic ocean. (SWRIS, 2022)

5.2.1.4 Wind

Winds in Suriname generally move in a northeasterly direction. Maximum average wind speed ranges from 30-42 km/h on the coast to about 30 km/h in the interior. Wind is highest during the dry seasons, with up to 40 km/h in March, and a second stronger peak in September and October. Wind speeds are relatively high along the seashore and decrease as one moves inland. Wind speeds of 20 to 30 km/h generally occur during the day and drop dramatically during the evening and night, especially in the interior (IADB, 2021).

¹⁶ Usually occurs once every 2-7 years

Although Suriname falls outside of the hurricane belt, the weather is occasionally affected by tails of hurricanes. Local gales (in Surinamese called "sibibusi") appear before storms, generally at the end of the rainy seasons. During such gales maximum wind speeds of 20-30 m/s have been recorded. Such gales occur over the whole country and may destroy trees and houses. No information is available on the frequency of such gales. (GOS, 2016)

5.2.1.5 Humidity

Relative humidity is very high for all of Suriname, 80% or higher, and has a latitudinal gradient, with the maximum values found at the coast and slightly lower values further inland. Daily humidity is on average 80-90% in the coastal regions. It is lower in the central and southern regions of the country, with an average of 75%. In forested areas, air humidity depends, among other things, on the penetration of solar radiation. Variations of relative humidity are between 70-100% in forested areas and between 50-100% in open areas (IADB, 2021). In 2019 the average humidity was between 74% and 80% (GBS, March 2020).

5.2.1.6 Air quality

In general, the air quality in Suriname deteriorates during dry conditions and improves during the wet season. During rainy seasons, rain suppresses dust particles in the atmosphere and alleviates air pollution. Dust emissions are further reduced by damp soil conditions. During the dry season, dust emissions generally increase as the soils become desiccated (SWM, 2019).

5.2.2 Climate change adaptation

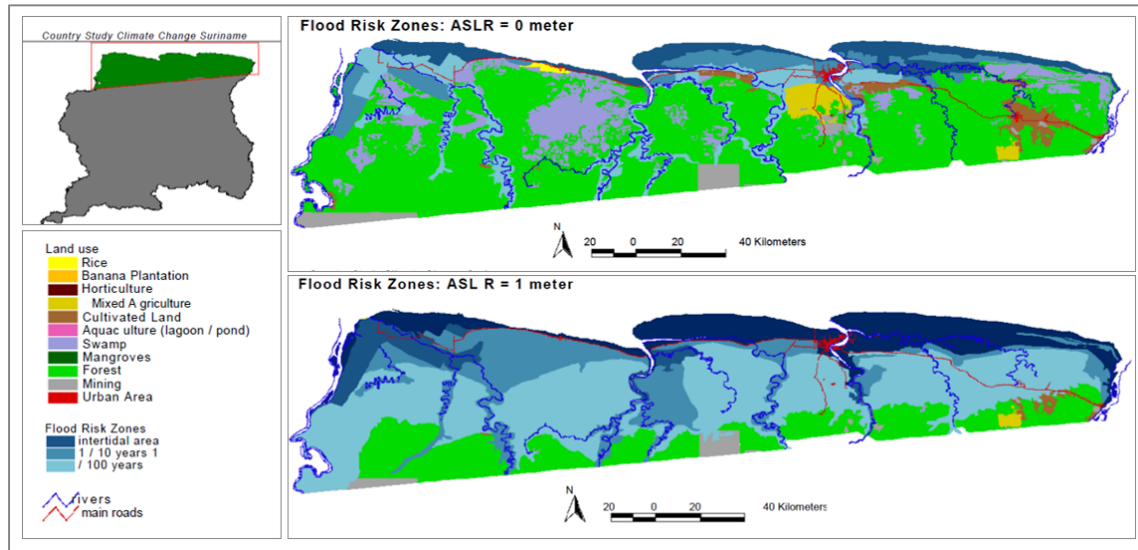
In the United Nations Framework Convention on Climate Change (UNFCCC) of 1992, climate change was defined as "a change of climate, which is attributed directly or indirectly to human activity, which changes the composition of the global atmosphere, and which is observed on top of the natural climate variability over comparable intervals of time." The negative effects of climate change are not limited to sea level rise, but affect all components of the environment: wind, precipitation and temperature. These three components of the climate can cause alone or jointly, disasters, but there are also natural disasters, such as earthquakes, infectious diseases and plagues, the risk of which must be assessed as adequately as possible (GOS, 2017).

5.2.2.1 Sea Level Rise

Rapidly rising global temperature, melting of polar ice caps, rising of the sea water level and changes in climate patterns and natural ecosystems are now becoming a daily concern (GBS, March 2020). Suriname is a country with a low-lying coast where over 80% of the population resides, and where the major economic activities and infrastructure are concentrated. These areas are and can suffer extensive losses and damages from the effects of climate change (UNDP, 2019).

For people, sea level rise is one of the major impacts of climate change, and although more detailed and accurate estimates of the impact of the rise in the sea level are needed (GOS, 2017), some studies indicate that a 1m sea level rise (SLR) would significantly impact 7% of the population of Suriname, particularly considering that most of this population is located in the coastal areas of the country (Dasgupta et al, 2007). Figure 5-4, shows the extension of ocean waters without sea level rise (ASLR = 0m) and sea level rise scenario (ASLR = 1m).

For a young nation with less than 600,000 inhabitants, of which the majority lives in flat coastal areas and heavily relying on natural ecosystems, the climate change impacts can be potentially catastrophic and is expected to be an enormous setback to the development of the country (Berrenstein, 2016). Considering that the coastal plain is vulnerable to sea level rise, and that Paramaribo is approximately between 0 and 3 m above sea level, the country has been included in the list of ten vulnerable countries with low-lying coastal plains that are threatened by sea level rise in this century (GBS, March 2020).



Source: (GOS, 2017) adapted by RINA, 2022

Figure 5-4: Modelled Sea Level Rise (+1 m) extent for Coastal Suriname

5.2.2.2 Droughts and Floods

Suriname is one of the most vulnerable countries in the world to the impact of flooding. The country is prone to periodic flooding due to heavy rainfall, especially when combined with spring tides. Approximately 87 percent of Suriname's population lives along the 386 km long coastal plain (around 67 percent in Paramaribo), and flooding affects most of the population and an estimated 90 percent of human activities. Flooding is exacerbated by poor drainage in the relatively highly populated urban areas on the coast, such as the capital city of Paramaribo. In addition, the coastal zone is also susceptible to erosion and some parts are prone to coastal flooding (WBG, 2022a).

Suriname has not experienced major natural disasters, and these have remained relatively limited, without large numbers of casualties. These natural disasters include the 2004 drought in the interior, where villages could no longer be reached via the river and had to get food aid, the floods of 2006 and 2008 in the interior, then in the coastal plain, and the wind gusts (strong local whirl winds) reporting 29 events between 2015 and 2019 (GBS, March 2020). Other flooding events in the country were recorded in 2021 with an estimated 10,100 people affected (OCHA, 2022).

The WBG is financing the Saramacca Canal System Rehabilitation Project¹⁷ (approved in 2019) which aims to support the country to reduce flood risk in the greater Paramaribo area and improve the Saramacca Canal System. It will improve the resilience against flooding by improving the ability of the Saramacca Canal to discharge water efficiently and safely, while improving vessel transport and strengthening the government's capacity to manage and operate the canal's drainage system.

5.2.2.3 Disaster Response and Relief

For disaster and emergency response, the government of Suriname has the established National Coordination Center for Disaster Relief (NCCR). The role of the NCCR is to coordinate the various services (Police, Fire department, National Army, NGOs and other parties) so that assistance can be

¹⁷ <https://projects.worldbank.org/en/projects-operations/project-detail/P165973>

provided in a very effective and organized manner for areas within the Surinamese territory affected or those involved in an incident (GBS, March 2020).

5.2.2.4 Carbon Sink Country

Suriname is a negative carbon-emitting country yet is committed to playing its part in the global fight against climate change. It provides a key ecosystem service to the world. About 94% of Suriname's territory remains forested, storing about 11 gigatons of carbon and absorbing (carbon sink country) more than 8.8 million tons of forest carbon annually (UNDP, 2019).

Despite its limited budget, the country has made progress towards climate-resilient and compatible development through a number of green policies and initiatives – in particular, actions to protect and use its forests sustainably (UNDP, 2019). These and others are included in its second Nationally Determined Contribution (NDC) of 2022 also known as the national climate pledge and were submitted in December 2019 in fulfilment of obligations under the Paris Agreement on climate change (GOS, 2019b).

5.2.3 Soil and Land Management

The current land degradation in Suriname is largely a result of mining activities. A distinction must be made between: a) land degradation accumulated over a long period of time as a result of the bauxite production, with little or no recovery of the environmental damage, particularly the land degradation and b) the continuing impact of the current mining activities in the gold sector. In 2017, Suriname planned to formulate a strategy and action plan for the limitation and restoration of land degradation based on research and consultations with stakeholders with the focus on mined areas. Regarding the large-scale agricultural activities while maintaining the environmental legislation to be adopted, strategies will be developed with stakeholders to reduce land degradation. Crucial in this respect is agreeing on standards and mechanisms for monitoring, controlling and reporting (GOS, 2017).

5.2.3.1 Geology, geography and soils

Suriname covers about one tenth of the Precambrian Guiana Shield¹⁸, a deeply weathered, rainforest-covered hill and mountain land stretching from the Amazon River in Brazil to the Orinoco in Venezuela (BOSMA, 1984) (see Figure 5-5). Geologically, the Guiana Shield is an ancient Precambrian land mass, 4 billion to 590 million years old, comprised of varied formations of sedimentary and igneous origin especially granites and gneisses (Huber, 1995). The age and history of the Guiana Shield uplift provide important clues to the origin and evolution of topographic formations such as drainage divides and waterfalls particularly relevant to the distribution patterns of aquatic fauna. For example, the Kaieteur Falls with an approximate height of 226m in Guyana isolates the only known habitats of the catfish species *Lithogenes villosus*, considered relict taxa. We also note that there are gaps in knowledge about the geology and biogeography of the Guyana Shield (LUJAN & ARMBRUSTER, 2011).

Eroded material from the Guiana Shield was deposited in the subsiding areas to the north. The oldest outcropping sediments in this coastal area have been deposited during three distinct periods resulting in three different formations (BOSMA, 1984):

- ✓ The Zanderij Formation, consisting of medium to coarse sands to sandy clays, which have been deposited by braided rivers at the foot of the Guiana Shield in the form of alluvial fans;
- ✓ The Coropina Formation, consisting of fine sandy to clayey marine sediments, which have been deposited north of (and on top of) the sediments of the Zanderij Formation. The marine clays were deposited as extensive clay flats and the marine sands as ridges, which were later severely eroded.

¹⁸ The Guiana Shield forms the northern part of the Amazonian Craton, the core of the South American continent, and is separated from its southern counterpart, the Central Brazilian shield, by the Amazon–Solimões basin.

The remaining surface deposits of this formation are found as “islands” surrounded by the sediments of the Mara Formation;

- ✓ The Mara and Coronie Formations. The Mara Formation consists of marine clays and peat that were deposited in the eroded areas in between the remnants of the Coropina Formation, and in gullies that had been cut into the former Old Coastal Plain. The Coronie Formation in the coastal plain is predominantly of marine origin, but along the main rivers' sediments have been deposited in fluvial and estuarine environments. The deposits consist of mudflats of marine clays and ridges of sands and shells. The finer sediments (clay, silt and fine sand) originate from the Amazon basin, while coarser sands come from the Hinterland.

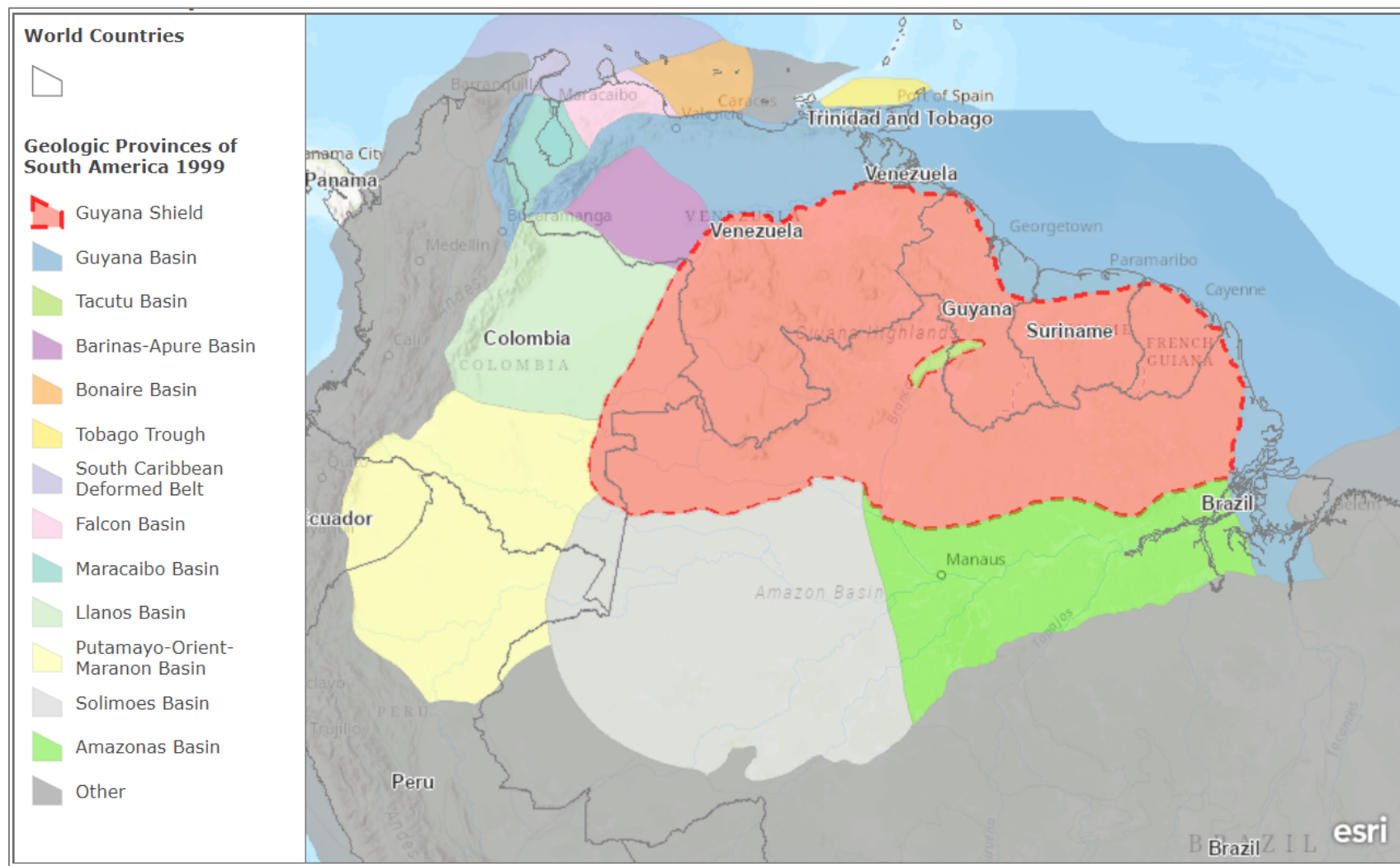


Figure 5-5: Geological provinces of South America

Source: (USGS, 1999)

The Suriname part of the shield consists of three distinct belts of low- and high-grade metamorphic rocks, separated by vast areas composed of granitoid and meta volcanic rocks. The basement is locally overlain by remnants of the sub-horizontal Precambrian Roraima sandstone formation and is cut by Proterozoic and Permo-Triassic dolerite dikes. In the coastal area to the north, it is covered by Mesozoic to Holocene sediments (BOSMA, 1984).

The Precambrian crystalline basement of Suriname consists of two high-grade metamorphic gneiss belts of Archean and Lower Proterozoic age, a likewise Lower Proterozoic volcanic -sedimentary greenstone belt, and a granitoid-volcanic complex in-between. The basement carries a few remnants of a once extensive cover of flat lying Middle Proterozoic and Permo Triassic dolerite dikes. The basement was essentially formed during the Trans-Amazonian Orogenic Cycle from about 1,800 to 2,000 million years ago (Ma). Younger, largely vertical movements culminated during the Nickerie Metamorphic Episode at about 1,200-1,300 Ma. This episode may be attributable to a continental collision along the westernmost margin of the Guiana Shield (at the present position of Colombia and Venezuela) (Vletter, 1984).

Suriname can mainly be distinguished by three landscapes (or geographic regions) as described below and shown in Figure 5-6 [(SWRIS, 2022); and (GOS, 2016)]:

1. The Coastal Plain: the coastal plains are about 21,000 km² and can be divided into:
 - a. The young coastal plain (lower plain): consists of soft fluvial clays and of sand ridges. This area is lower than 6 m above mean sea level. In this area percolation and surface drainage are very slow, because of the clay and peat soils. Most water is removed by evapotranspiration.
 - b. The old coastal plain (upper plain): consists of silt clays, loams, sandy loams and fine sand. This area lies between 6 and 12 m above mean sea level. In this area percolation and surface drainage are very slow.

The coastal plain area plays an important role in the maintenance of drinking water reserves because the greatest amount of rainfall percolates into the ground here, recharging the freshwater aquifers. The soils of the Young Coastal Plain are the most fertile zone of Suriname and has contributed to the development of large-scale agricultural activities. The soils of the Old Coastal Plain (generally composed of clay, sandy-clay or clayey-sand) offer good opportunities for agricultural development, in particular, horticulture. (Berrenstein, 2016)

2. The Cover Landscape or Savannah Belt consists of white coarse sands and coarse sandy clays. This area is about 8,500 km² and lies between 15 and 20 m above mean sea level. In this area, groundwater flow is the main contributory to discharge. The period of time to peak in this area is about 6.5 to 15 hours. This area reacts more quickly to rainfall. The savannah belt is an important source of recharge for the aquifers.
3. The Interior Highlands or Precambrian shield or basement. This occupies about 80% Suriname's landscape. Infiltration and percolation in this area is very small. Surface runoff is very high, and the drainage density is high. Time to peak period is about 13 to 28 hours with a concentration time of 51 to 72 hours.

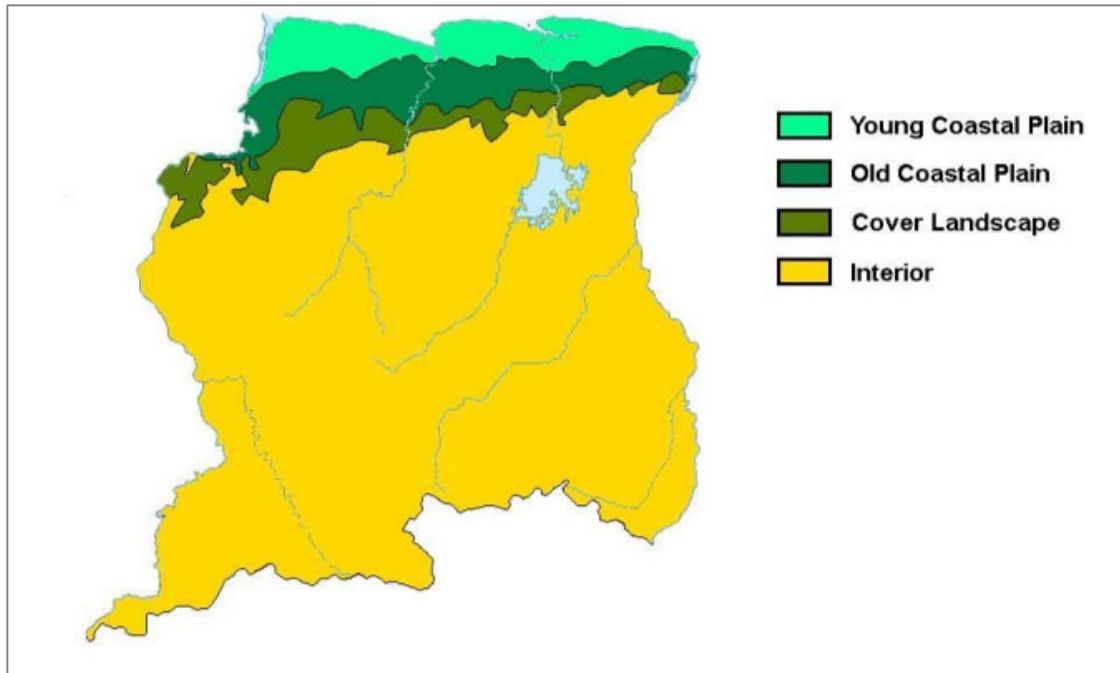


Figure 5-6: Geographic regions of Suriname

Source: (GOS, 2016)

The coast of Suriname is part of the coastal system, which extends from the mouth of the Amazon River in Brazil to the mouth of the Orinoco River in Venezuela. It is a muddy coastline where mud, including fine sands, from the Amazon Basin are being deposited. The coast is subjected to active geomorphological development, which is determined by a system of cyclic accretion and erosion. Both are linked to the presence of shoreface-attached mudflats and mud banks, which continuously migrate to the west (GOS, 2016).

The Global Lithological Map (GLiM)¹⁹ developed by the University of Hamburg was used to extract the soil types of Suriname (see Figure 5-7). The soils over the major area of Suriname consist of metamorphic rocks in the interior highland geographic region, followed by siliciclastic sedimentary rocks mainly along the coastal plain and savannah belt regions). Acid plutonic rocks, acid volcanic rocks, basic plutonic rocks, and other rocks are distributed in lower proportions around the interior highland geographic region (GLiM, 2022). The siliciclastic sedimentary rocks are clastic non-carbonate sedimentary rocks that are almost exclusively silica-bearing, therefore, mining activities in these silica bearing soils requires special protection measures for exposed personnel.

A more detailed graphical depiction of Suriname's geology and mineral resources is presented in Figure 5-8.

¹⁹ Global Lithological Map (GLiM) – available at: <https://www.geo.uni-hamburg.de/en/geologie/forschung/aquatische-geochemie/glim.html>

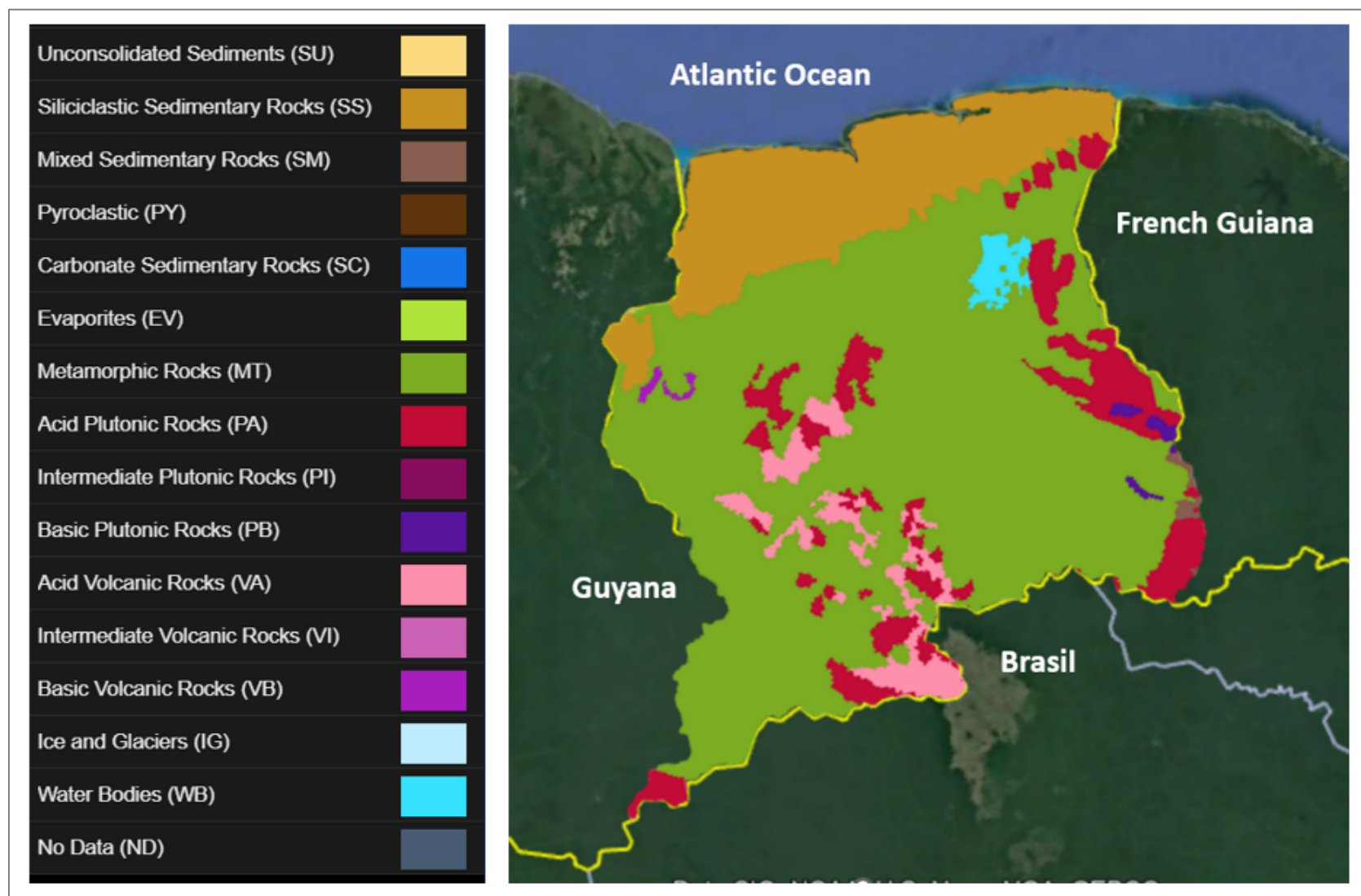


Figure 5-7: Lithological map of Suriname

Source: (GLiM, 2022).

Note: The average resolution of the GLiM mapping is 1: 3 750 000 and should be considered referential until soil sampling and testing are completed.

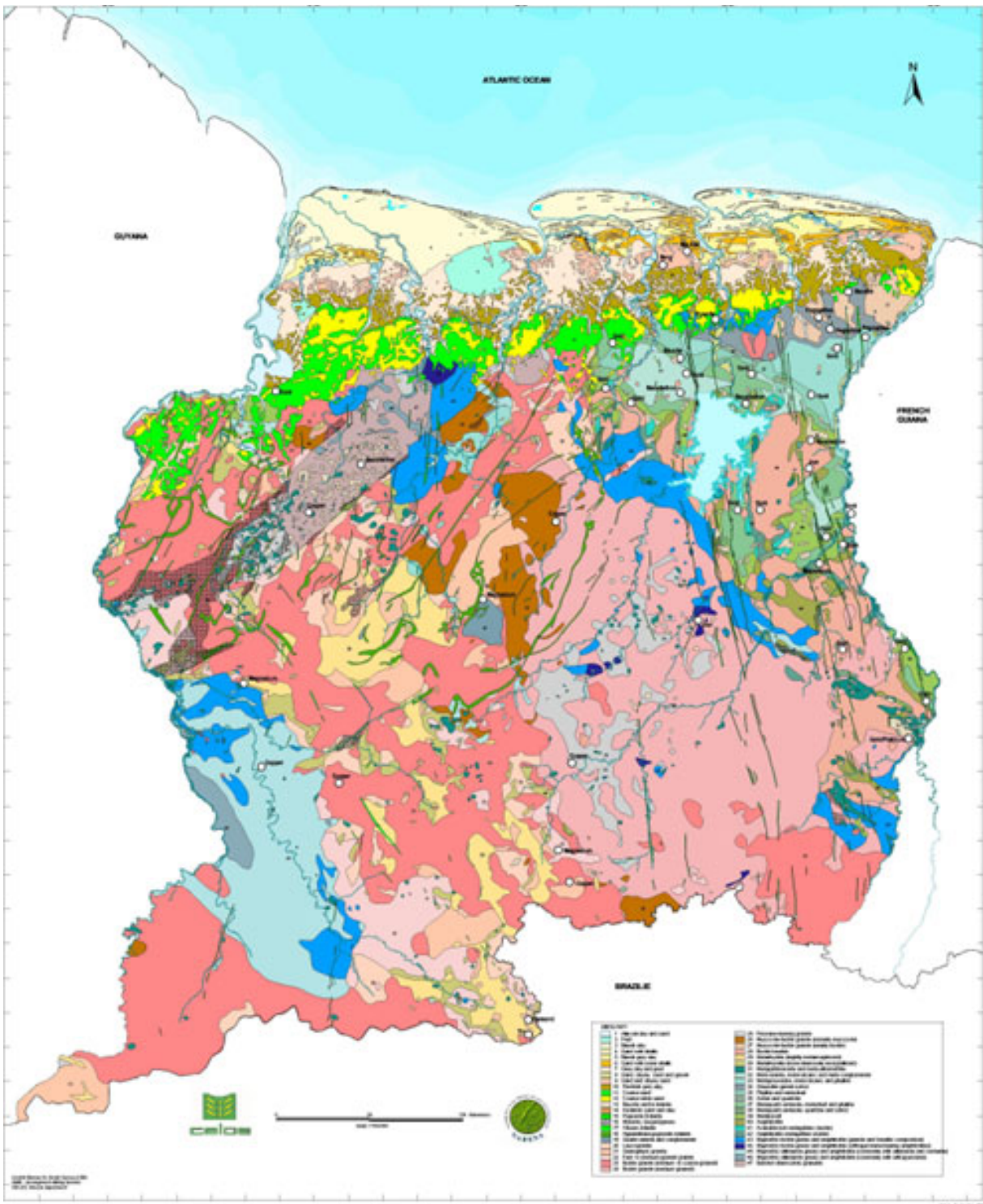


Figure 5-8: Map of Suriname's Geology and Mineral Deposits.

Source: (SWRIS, 2008)²⁰

²⁰ Available at: <https://www.swris.sr/map/general-maps/geology/>

5.2.4 Water Resources

Water is an important and valuable resource (raw material), and its value will increase in the future. It is also one of the important resources of Suriname. A compelling standard for regional and zoning plans will be the protection of the quality of both soil and surface water. Therefore, the mapping of existing water resources, and the planning of new ones is of utmost importance. The drainage of areas is a key issue in planning a livable and healthy environment, and in the development of regional and zoning plans. Storage of water will become a standard component of all regional plans. Laws and regulations regarding the use and extraction of water will be incorporated into the new planning legislation (GOS, 2017).

The pollution of Suriname's water resources is one of the most threatening issues in the development process: this relates to the pollution of waterways, such as rivers and creeks, often through illegal production activities such as gold mining (GOS, 2017).

5.2.4.1 [Water Sources and Quality](#)

In Suriname, water resources are abundant and can be found as surface or groundwater. The total outflow into the Atlantic ocean consist of 80% of the basement area runoff (126,500 km²), 8% of the savannah area and 8% of the coastal plain areas (21,000 km²). (SWRIS, 2022)

The main fresh water source is the abundant annual rainfall, which, together with the topography, soil types and land cover has resulted in many streams and large wetlands. Seven main rivers, originating in the Interior of the country, annually convey about 4,800 m³/s of fresh water into the Atlantic Ocean, which is about 30% of the annual rainfall. The Marowijne and the Corantijn Rivers contribute 70% to the total discharge. Of the remaining rainfall, the largest part evaporates and only a small part percolates to the aquifers, recharging ground water reserves. The hilly to mountainous interior has a dense network of streams, while the low-lying coastal area has extensive swamps and fewer streams. Some of these swamps have salt or brackish water. Surface water resources (rivers, creeks and swamps) are mainly used for irrigation (e.g., for rice and banana cultivation) and for the generation of hydropower. Groundwater is mainly used for potable purposes. However, excessive extraction in some areas results in increased intrusion of saltwater. (Berrenstein, 2016)

5.2.4.1.1 [Surface waters](#)

The basement area is the major source of surface water, and the savannah belt is the major source of groundwater runoff. (SWRIS, 2022) The main surface waters in Suriname are presented in Figure 5-9.

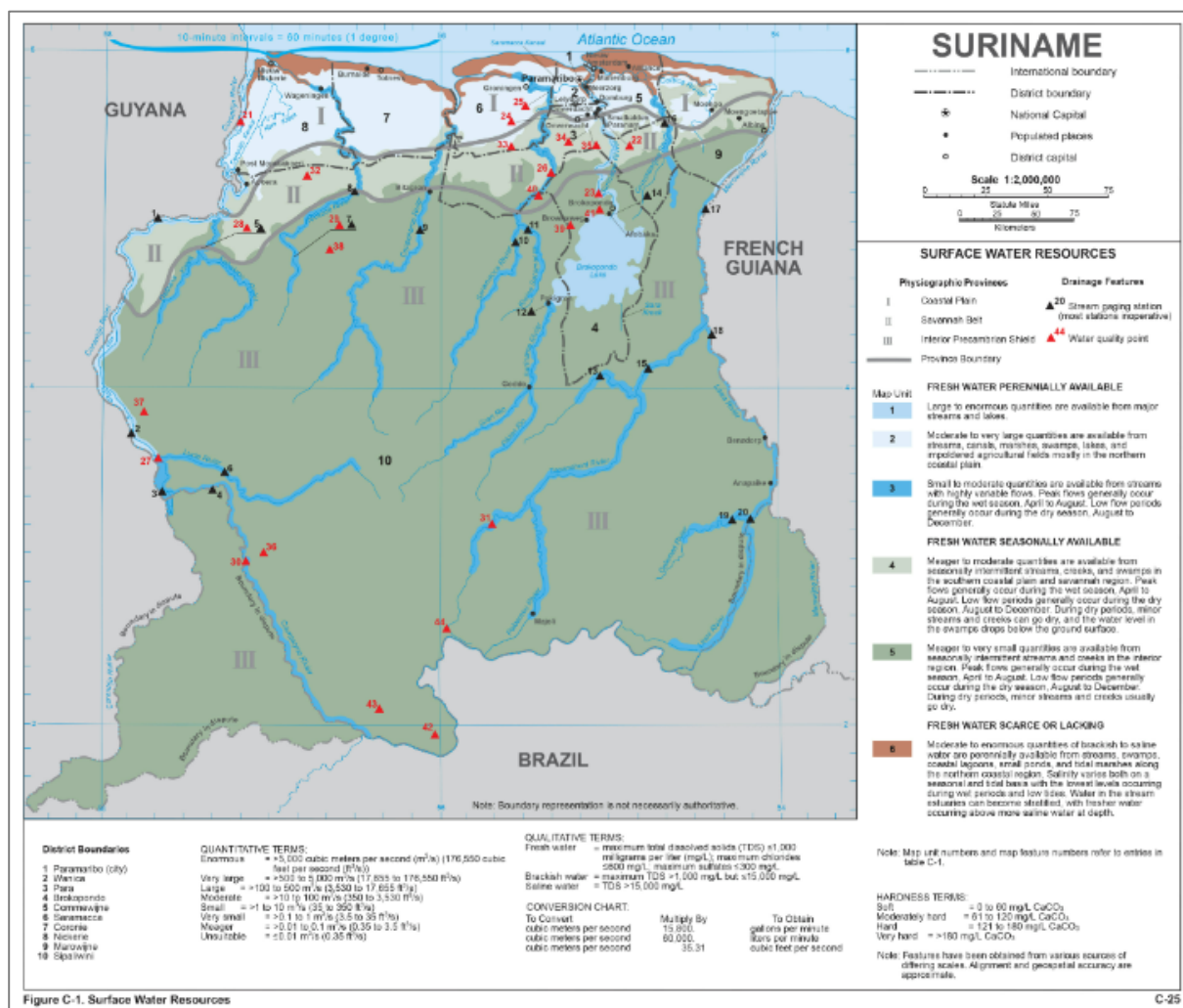


Figure 5-9: Surface waters of Suriname

Source: (SWRIS, 2022)²¹

²¹ Surface waters map of Suriname. Available at: <https://www.swris.sr/map/water-maps/surface-water/>

Rivers

Suriname has seven major streams draining toward the Atlantic Ocean (see Table 5-1). The largest are the Marowijne and Corantijn rivers that form the country's borders, respectively, in the east and the west. These two rivers flow northward and drain almost 58% of the country. The Coppename and the Suriname rivers also flow northward and together drain about 24% of the country. The smallest rivers are the Nickerie, Saramacca, and Commewijne rivers, which drain 16% of the country's waters. The final 2% of the country consists of coastal areas with direct drainage into the Atlantic Ocean. The Suriname river drains on average about 4,800 m³/s (756 mm/year) freshwater into the Atlantic ocean. The main rivers are tidal and contain brackish water in the coastal area. (SWRIS, 2022).

Table 5-1: Hydrological characteristics of the main rivers in Suriname

N	Main river	Catchment Area (km ²)	Average Annual Discharge (m ³ /s)	Extreme Low Discharge (m ³ /s)	Extreme High Annual Discharge (m ³ /s)	Specific discharge	Estimated Rainfall (mm)	Estimated Runoff (mm)	Estimated evapotranspiration (MM)
1	Marowijne	68,700	1,780	100	15,000	25.9	2,246	1,780	100
2	Commewijne	6,600	120	5	1,200	18.2	n/a	n/a	n/a
3	Suriname	16,500	426	20	4,000*	25.8	2,414	426	20
4	Saramacca	9,000	225	10	2,000	25.0	2,294	225	10
5	Coppename	21,700	500	25	4,200	23.0	2,349	500	25
6	Nickerie	10,100	178	10	1,800	17.6	2,272	178	10
7	Corantijn	67,600	1,570	100	15,000	23.2	2,271	1,570	100
*Before the construction of the dam									

Source: (SWRIS, 2022)

All four major rivers and their near-coastal subsidiaries have estuarine characteristics. The large volumes of tidal water and the strong tidal currents give these river estuaries a more or less funnel-shape and a wide mouth. They are characterized by mixing up of fresh and muddy saline water, and by relatively large tidal variations. In the river estuaries considerable quantities of marine mud are moved upriver with the intruding tide. Sedimentation of this mud occurs on the riverbed and as flats at the inside of river bends. The opposite outer bend, on the other hand, may be eroded by undercutting of the riverbank. (GOS, 2016)

Swamps and Wetlands

The coastal area of Suriname, in particular the young Coastal plain, is very low and flat with elevations of between the 0-10m. The "coastal belt" is about 5 km wide in the east and about 50 km wide in the west. Swamps cover about 60 percent of the coastal plain. In many areas, water is fresh during the wet seasons from precipitation falling directly on the swamps. During dry seasons, the swamps revert to brackish conditions. The four largest swamps are the Nani swamp, Coronie swamp, Coesewijne swamp and Surnau swamp. (SWRIS, 2022)

The Nanni swamp supplies irrigation water for about 18,000 ha of agricultural lands, mostly for rice crops. The total area of the swamp is about 1,550 km². The only source of water available for the swamp is rainfall and inflow from the Zanderij formation and the Old coastal plain. A water balance shows that in a year with average rainfall, the discharge is 12.5 m³/s. In an extreme wet year discharge increases to 16.8 m³/s., and in an extreme dry year the discharge can fall to 9.5 m³/s. (SWRIS, 2022).

The total area of the Coronie swamp is about 700 km². The area of the Coesewijne swamp depends on its water level and during the rainy season it may extend over 800-900 km². (SWRIS, 2022)

Reservoirs

The artificial lake, the Brokopondo reservoir²², is the largest artificial open freshwater resource in Suriname. It is located in the Suriname river at Afobaka, 194 km upriver. It covers an area of 1,560 km². The purpose of this reservoir is to provide electricity for the aluminum smelter at Paranam (operational in 1964). Controlled discharge at Brokopondo provides a mean yearly flow of approximately 250 m³/s. (SWRIS, 2022)

5.2.4.1.2 Groundwater and aquifers

The coastal plain area plays an important role in the maintenance of drinking water reserves because the greatest amount of rainfall percolates into the ground here, recharging the freshwater aquifers. These aquifers are the main resource for potable water in the coastal areas (mainly around the capital). Ninety-five percent of the country's total supply of potable water comes from ground water. The ground water resources of Suriname are used for public supply and to lesser extent industry. Suriname contains two hydrologically distinct provinces, the Interior Precambrian Shield of crystalline rocks, comprising 80 percent of the country, and the coastal plain basin, comprising the remaining 20 percent. There is an abundance of good quality groundwater contained in the coastal basin. Groundwater in the young coastal plain is not renewable, however, groundwater in the old coastal plain and savannah belt area is renewable. (SWRIS, 2022)

The aquifers of Suriname are in the coastal basin, which are built up of unconsolidated sediments consisting of a sequence of clay, sands, sandy clay, clayey sands, gravel, with more or less kaolin content, and thin bed organic compound. These aquifers, which have been classified based on the geological ages, are the Nickerie, Onverwacht, A-sand, Coesewijne, Zanderij, Coropina and Demerara.

The most important freshwater aquifers from where water is withdrawn are (SWRIS, 2022):

- ✓ A-sand aquifers: no recharge; depths from 130-190 m
- ✓ Coesewijne aquifers: no recharge; depths from 70-110 m, and
- ✓ Zanderij aquifers: recharge from Savannah area; depths from 15-60 m

²² Formerly known as Prof. Dr. Ir. Van Blommenstein reservoir.

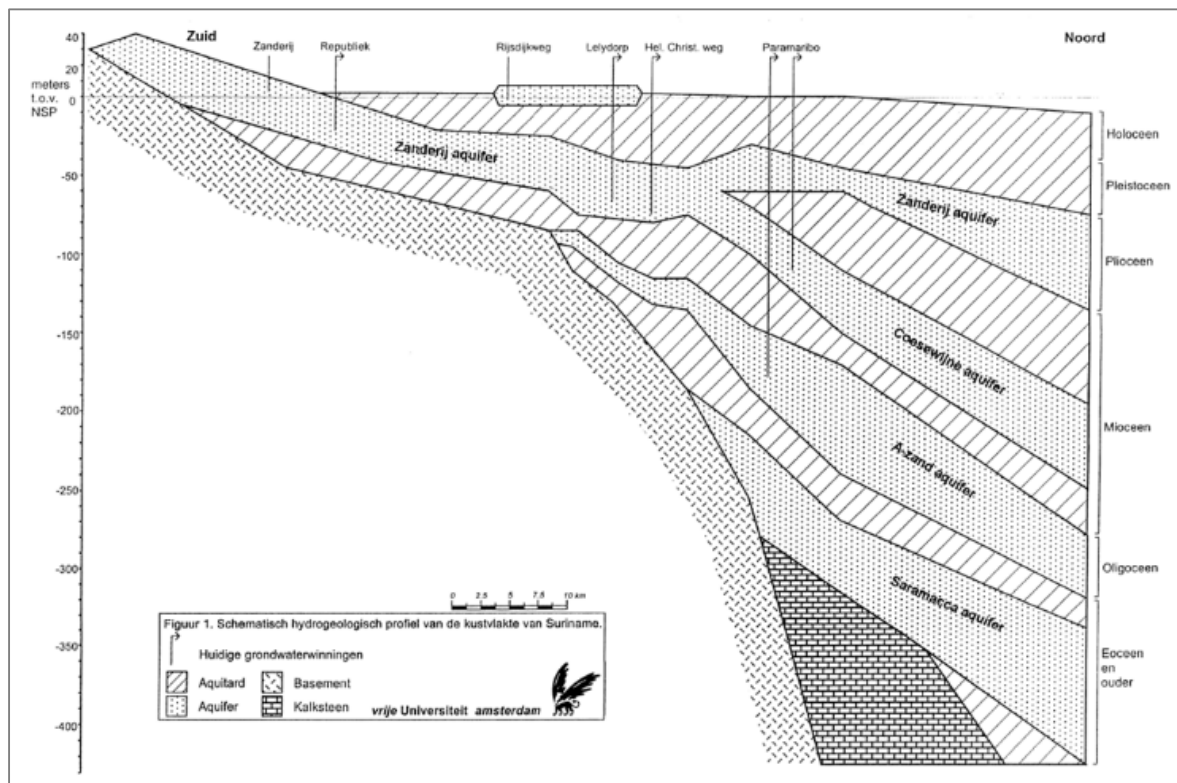


Figure 5-10: Groundwater Aquifers of Suriname

Source: (SWRIS, 2022)²³

²³ Groundwater aquifers of Suriname figure. Available at: <https://www.swris.sr/map/water-maps/groundwater-aquifers/>

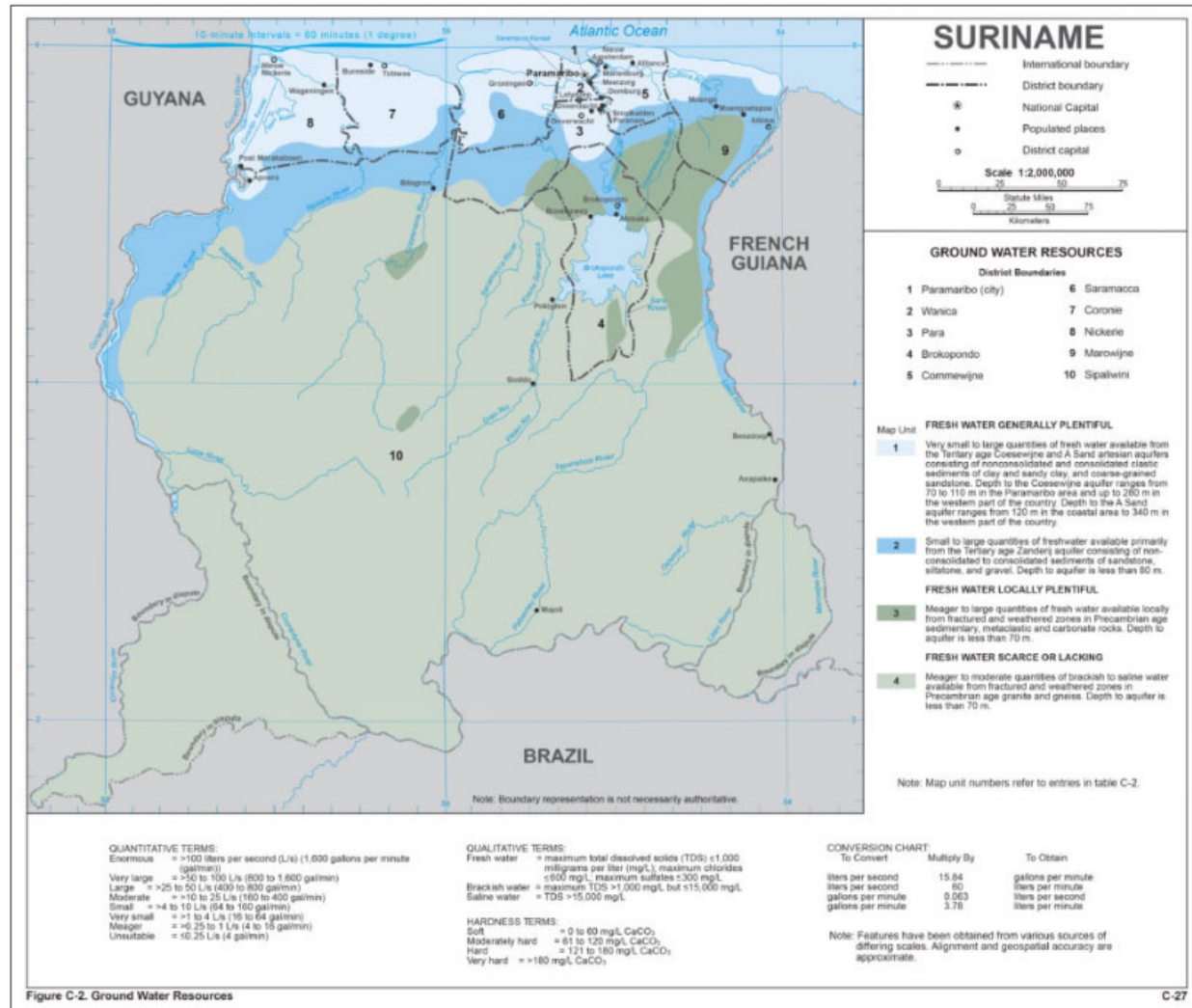


Figure 5-11: Groundwater

Source: (SWRIS, 2022)²⁴

²⁴ Groundwater map of Suriname. Available at: <https://www.swris.sr/map/water-maps/groundwater/>

Groundwater conditions in the northern Interior Precambrian Shield are generally unfavorable because the geological formations in this province have little or no primary permeability. The south, which is an active system, is recharged directly from rainfall and coincides with the savanna and old coastal plain. Generally, the salinity increases towards the coast. The water in the Zanderij aquifer is fresh²⁵ throughout the old coastal plain, and brackish²⁶ in the young coastal plain, particularly adjacent to the rivers. The change is abrupt. In the Coesewijne aquifers freshwater continues farthest north. Higher salinity²⁷ extends farthest inland along concealed lines in the A-Sand. (SWRIS, 2022)

The Suriname Water Supply Company (SWM²⁸) has currently 15 production stations. The number of groundwater wells is about 78. The annual water production is estimated to be 38,106 m³. (SWRIS, 2022). The characteristics of these stations are shown in the Table 5-2:

Table 5-2: Production stations of the SWM

Production station	In use*	Mean production (m3)	Storage capacity (m3)	Storage capacity (m3)	Aquifer	Cl-content (in mg/l)
Albina	1960	10	2000	2000	n/a	30
Benies	1990	n/a	2000	2000	n/a	n/a
Blauwgrond	1992	140	4000	10.000/4000	A-zand	340
Boskamp	n/a	n/a	n/a	n/a	n/a	11
Europolder	n/a	n/a	n/a	n/a	n/a	<200
Flora	1989	120	2000	2000	A-zand	250
Groningen	n/a	n/a	n/a	n/a	n/a	85
Groot Henar	n/a	n/a	n/a	n/a	n/a	70
Helena Christina	after 2002	300	1100	1100	A-zand/Coesewijne	160
Kampong Baroe	n/a	n/a	n/a	n/a	n/a	85
Koewarasan	n/a	n/a	n/a	n/a	n/a	135
La Vigilantia	n/a	n/a	n/a	n/a	n/a	11-200
Leiding 9a	n/a	n/a	n/a	n/a	n/a	165
Lelydorp	1982	400	1000	1000	Zanderij	75
Leysweg	1972	340	4000	4000	A-zand/Coesewijne	240
Livorno	1981	450	1700	1700	A-zand	100
Meerzorg	n/a	n/a	n/a	n/a	n/a	170
Moengo	after 1999	200	50	50	n/a	n/a
Nickerie West	n/a	58	n/a	n/a	n/a	350

²⁵ Salinity scale / Fresh water: total dissolved solids (TDS): 1-1,00 mg/l and Chloride: 0-250 mg/l.

²⁶ Salinity scale / Brackish water: total dissolved solids (TDS): 12,000-24,000 mg/l and Chloride: 7,000-14,000 mg/l.

²⁷ Salinity scale / Saline water: total dissolved solids (TDS): 24,000-34,000 mg/l and Chloride: 14,000-20,000 mg/l.

²⁸ Surinaamsche Waterleiding Maatschappij - SWM

Production station	In use*	Mean production (m3)	Storage capacity (m3)	Storage capacity (m3)	Aquifer	Cl-content (in mg/l)
Nw. Nickerie	1958	160	1000	1000	n/a	340
Paradise	n/a	n/a	n/a	n/a	n/a	15
Peperhol	n/a	n/a	n/a	n/a	n/a	102
Peperpot	n/a	n/a	n/a	n/a	n/a	270
Pont Buiten	n/a	n/a	n/a	n/a	n/a	91
Republiek	1933	250	400	400/4000	Zanderij	10
Sidoredjo	After 2002	150	600	600/650	n/a	30
Totness	n/a	n/a	n/a	n/a	n/a	60
Tourtonne	1990	100	2000	2000	A-zand	290
Uitkijk	n/a	n/a	n/a	n/a	n/a	170
Van Hattenweg	1997	1100	6100	6100	Zanderij	25
WK-Plein	1958	500	8400	8400	490	260
Wonoredjo	n/a	n/a	n/a	n/a	n/a	40

Source: (SWRIS, 2022)

Note: SWM water supply production stations in use

In terms of water quality information from official sources is not available. However, public sources indicate that water bodies downstream from small scale gold mining activities may be contaminated, as shown in Figure 5-12. Mercury is the key component used by small-scale mining operations to separate gold from soil. Designated by the World Health Organization as among the top 10 chemicals of major public health concern globally, exposure to mercury can result in severe neurological damage (Joshi, 2015).

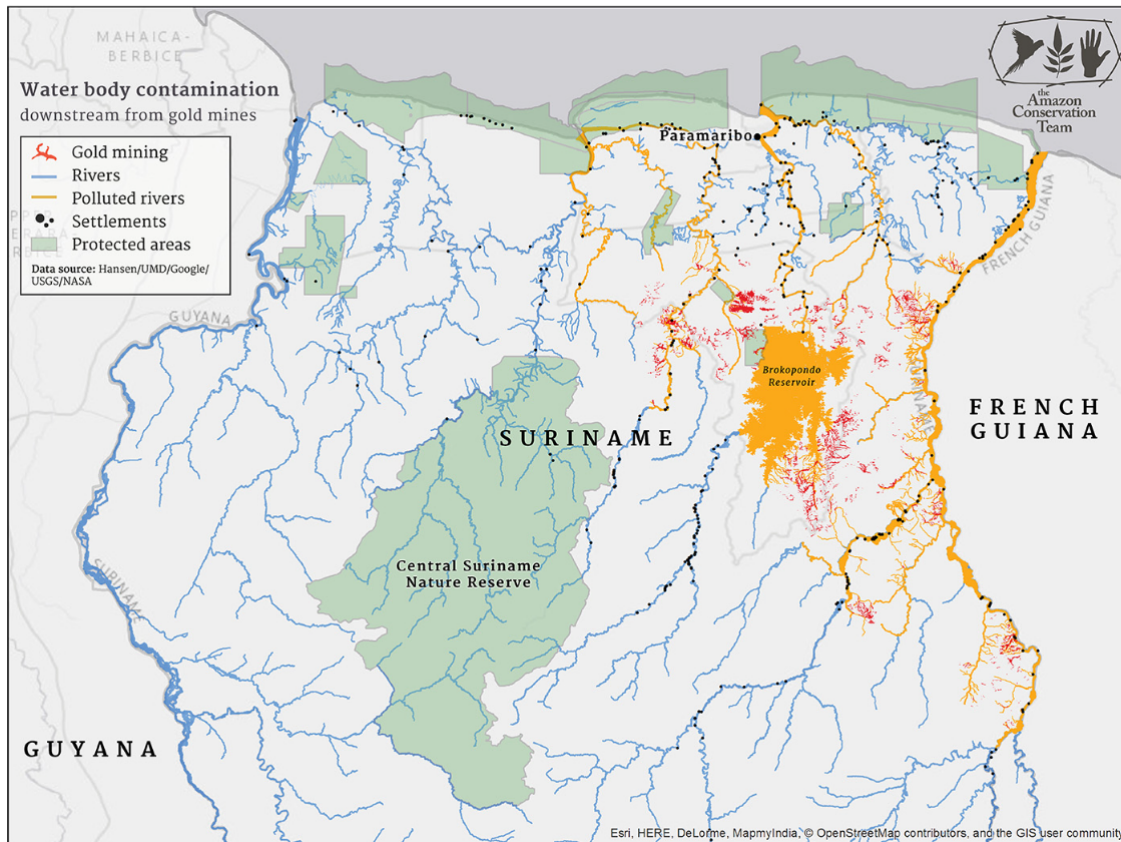


Figure 5-12: Potential Water Bodies Contamination from ASGM activities upstream

Source: Mongabay, 2015

5.2.4.2 Water Use and Accessibility

While improving water quality is critical to prevent disease, improving the accessibility and availability of drinking water is equally important, particularly for women and girls who usually bear the primary responsibility for carrying water, often over long distances. Overall, 98.2 % of the population in Suriname has access to improved drinking water sources, 99% in urban region, 98 % in rural coastal and 91% in the rural interior region. (GBS, March 2020).

The source of drinking water for the population varies by district. In Paramaribo, 88% of the population uses drinking water that is piped into their dwelling or into their yard or plot. For Nickerie this is 81% and for Wanica and Para this is 72%. The households in the districts of Commewijne (27%), Brokopondo (33%) and Sipaliwini (6%) have the lowest access to piped water into their dwelling. The second most important source of drinking water is rainwater collection. In Sipaliwini 73% of the households use rainwater. In Brokopondo and Sipaliwini, 8% of the households use surface water (rivers and ponds) as sources of drinking water, which is considered an unsafe source²⁹. (GBS, March 2020).

In general, the water quality in Suriname is good, but the quality of surface water in both urban and rural areas is under severe pressure by mining and industrial activities, poor waste processing and insufficient

²⁹ <https://statistics-suriname.org/wp-content/uploads/2019/08/Suriname-MICS-6-Survey-Findings-Report.pdf>

sanitation facilities. The Surinamese Water Company (SWM³⁰) laboratory and the Environmental Control Division of the Bureau for Public Health (BOG³¹) carry out daily water quality testing at the various distribution points (GBS, March 2020).

5.2.4.2.1 Water Distribution Network

Suriname, under SWM management has five water distribution/supply areas or branches. These are described in Table 5-3. In 2019, the average production for the entire SWM network was estimated at 6,276 m³/h resulting in an annual production of 50,084,000 m³ of water (GBS, March 2020).

Table 5-3: SWM water distribution / supply branches

#	Branch	Description / Comments	Average Production (m ³ /hr)	Storage Capacity
1	Central branch	In terms of area and of the number of connections, the Central Branch is considered the largest branch. This area covers the district of Paramaribo and large areas of Wanica and Para.	5,045	28,005
2	West branch	The West Branch area covers the following districts and subregions: 1. Nickerie District: • Nieuw Nickerie subregion • Sidoredjo subregion • Wageningen subregion * • Henar subregion * • Paradise subregion * 2. Coronie District: • Totness subregion* 3. Sipaliwini District: • Kabalebo subregion – Apoera area*	703	3,126
3	East branch	The East branch area covers the district of Marowijne, providing drinking water supply for Moengo, Wonoredjo and Albina.	107	1,388
4	Commewijne branch	The Commewijne branch* provides the drinking water supply for the subregions Meerzorg, Peperpot and Mariënborg.	341	1,390
5	Saramacca branch	Added to the SWM network in 2016.	409	918
TOTAL			6,276	40,817

Source: (GBS, March 2020)

Note: * added to the SWM network in 2016

³⁰ Surinaamsche Waterleiding Maatschappij - SWM

³¹ Bureau voor Openbare Gezondheidszorg - BOG

In 2019, the SWM's pipeline network extended along 4,799 km in Central, East, West, Commewijne and Saramacca branches. The districts that are not within the SWM network are provided with water through the Water Supply Service (DWV³²) of the Ministry of Natural Resources and the foundation Funding Development Interior (FOB³³), a technical arm of the Ministry of Regional Development and Sport³⁴. There are also NGOs and private companies that provide water for Surinamese households that are not part of the SWM, DWV or FOB network (GBS, March 2020).

In terms of water consumption, a total of 28,323,000 m³ were consumed through the SWM network in 2019 (see Table 5-4). The water consumption is dominated by "House Connections" with approximately 72% of the total consumption in 2019 (see Table 5-4). (GBS, March 2020).

Table 5-4: Water Consumption through SWM network in 2019

District / Branch	Production (m ³ x 1,000)	Consumption (m ³ x 1,000)
Paramaribo, Wanica and Para / Central	41,793	24,508
Nickerie / West	2,560	1,885
Marowijne / East	886	772
Commewijne	2,213	854
Saramacca	2,631	305
Total	50,084	28,323

Source: (GBS, March 2020).

Table 5-5: Water Consumption per type of connection

Connection	2019 (m ³)	2019 (%)
Yard connection	2,025,902	7%
House Connection	20,308,958	72%
House with Swimming pool	45,750	0%
Industrial & Commercial Connection	4,628,565	16%
Public Connection	1,313,751	5%
Total	28,321,926	100%

Source: (GBS, March 2020).

In the Central and West Branches groundwater is processed into drinking water and in the East Branch, river water (surface water) and groundwater are processed into drinking water.

³² Dienst Watervoorziening - DWV

³³ Fonds Ontwikkeling Binnenland

³⁴ Ministerie van Regionale Ontwikkeling en Sport.

5.3 BIOLOGICAL ENVIRONMENT AND BIODIVERSITY

5.3.1 Ecosystems

The natural terrestrial ecosystems of Suriname are shown in Figure 5-13 and can be characterized as follows (Berrenstein, 2016):

- ✓ **Mangrove forest and brackish herbaceous swamps** are estuarine and influenced by coastal changes and tidal action, consisting of mangrove forests, locally interrupted by salty to brackish lagoons and low vegetation of succulent salt plants and brackish herbaceous swamps. Mangrove forests protect the coastline and riverbanks against erosion and stimulate accretion, serve as nursery grounds for many species of marine fish and shrimps, and contain a high degree of biodiversity. The estuarine zone of Suriname is a wetland of international importance as it contains the feeding and nesting sites for Caribbean coastal birds and feeding grounds for migratory birds from North America. Along the coastline, mainly in the eastern part of the country, sand and shell beaches are found. These beaches are nesting places of international importance for four species of sea turtles.
- ✓ **Freshwater herbaceous swamps** dominate the Young Coastal Plain and part of the Old Coastal Plain. These wetlands play an important role in supplying freshwater to the estuarine zone, thus contributing to the maintenance of the brackish condition of these waters. The swampy areas are inundated throughout most of the year, allowing a peat layer to accumulate on top of the mineral soil.
- ✓ **Low swamp forest** is a low swamp forest, including palm swamp forest, covering about 3% of Suriname. If no forest fires occur, it develops into high swamp forest.
- ✓ **Swamp forests** are high swamp forests and creek forests. High swamp forests are found in the shallow freshwater swamps of the coastal plain where strong fluctuations in surface water levels occur, being the climax vegetation in the ecological succession of wet areas. Creek forests are found along creeks and are enriched by a large number of species from the surrounding high dryland forest.
- ✓ **Marsh forest** is a seasonal swamp forest found on poorly drained soils. During the rainy seasons, the soil may be totally inundated. In the dry seasons, the soil does not desiccate. Species diversity is intermediate between that of the high swamp forests, and the high dryland forests. Marsh forests are usually rich in palms.
- ✓ **High dryland forest** covers about 80% of Suriname, which is found at 400-600 meters above sea level. These are upland forests with the presence of a high and dense canopy at 25-45m with emergent trees up to 50-60m. Tree species diversity is high (between 100-300 species per hectare), it is an important habitat for large mammals and houses at least 500 bird species.
- ✓ **Mountain forests** are upland forests found above 400m. These forests are frequently covered in clouds, which reduces the solar radiation and plant productivity. Temperatures are lower and the average humidity is higher than in high dry land forest of the lowlands. Epiphytes (such as mosses, ferns, orchids and bromeliads) are abundant in both numbers and species. The flora and fauna of these montane forests is quite different from those of the lowland forests.
- ✓ **High and low xerophytic forests** are found in savannah areas. This type of forest commonly grows on excessively drained soils and on permeable soils resting on impermeable subsoils, hard pans and rocks. As a result, there is a water shortage in the soil during dry seasons. This vegetation type can also be found in the hilly/mountainous areas in the interior, especially where laterite caps, conglomerates, dolerite and granite rocks and sandstones are covered with a thin layer of soil, where they are also known as mountain savannah forest.
- ✓ **Open savannah** can be considered as remnants of the extensive Pleistocene climate savannah, that once covered Suriname almost entirely. Savannah ecosystems only survived where they were frequently burnt. At present, ten different types of savannahs have been distinguished in Suriname.

Most savannahs are found scattered over the savannah belt. In the absence of fire, savannah vegetation eventually develops into forest.

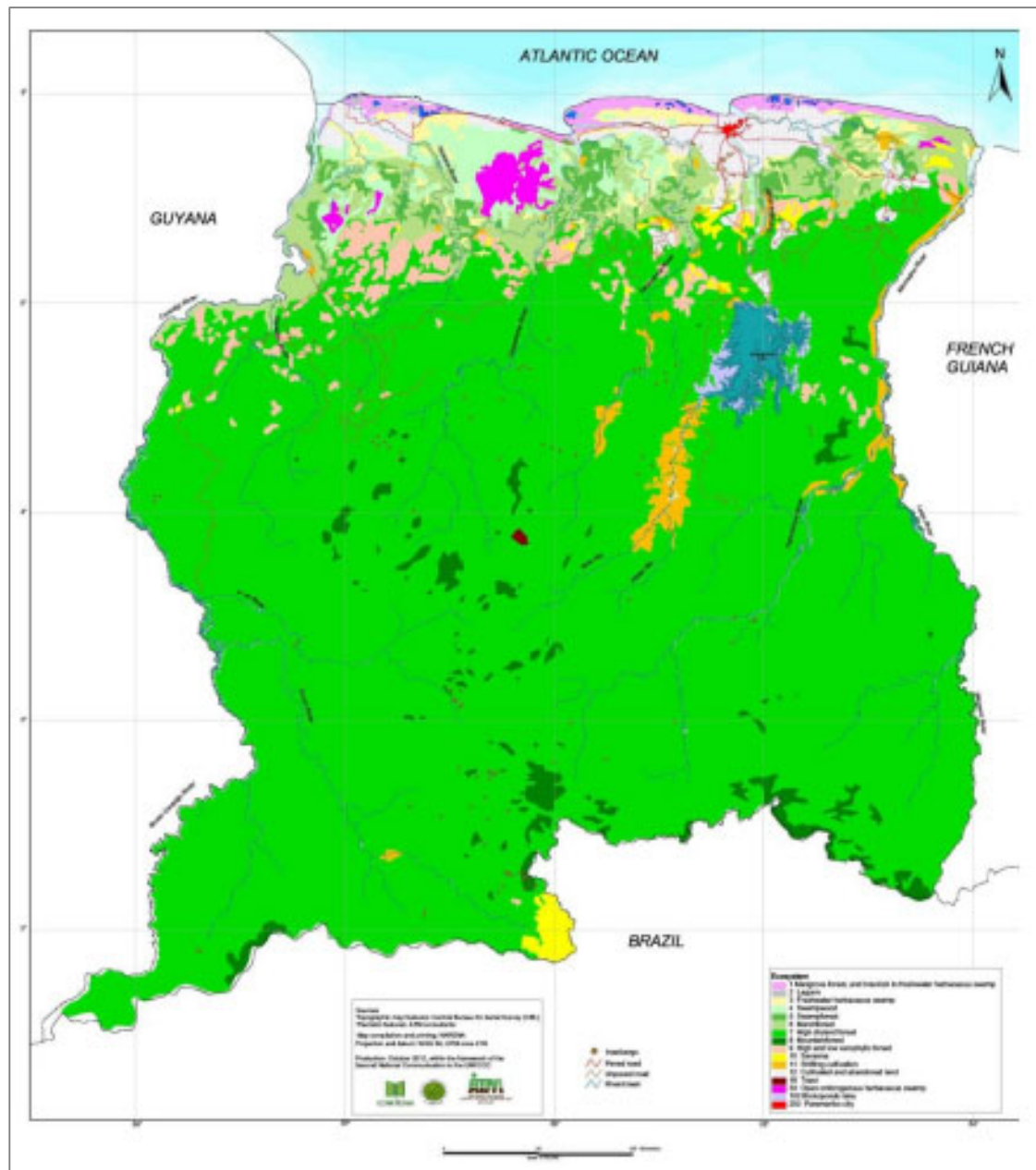


Figure 5-13: Ecosystems of Suriname

Source: (Berrenstein, 2016)

5.3.1.1 Coastal and Marine resources

The GOS reports that there are serious concerns related to coastal and marine resources, since these are being subject to depletion, including marine water, beaches and coastlines, marine protected areas, mangroves, fisheries and aquaculture, and other coastal and marine resources. Particularly marine water and coastal areas are under increasing environmental stress from pollution, overfishing, and degradation of coastlines (GBS, March 2020)

Suriname's marine area is subdivided in four zones (GBS, March 2020):

1. **The Continental Inner or Brown water Zone:** The water reaches a depth of about 30 m and is brown due to the heavy load of mud; along the coast, there are extensive mud banks that move slowly to the west, and sand banks that either also move along or have fixed locations near the mouths of major rivers;
2. **The Continental Mid or Green-water Zone:** The water is between 30 and 60 m deep and is still a bit muddy, but much less so than in the previous zone; the water has a greenish tint due to the abundance of algae, and is thus biologically very productive;
3. **The Continental Outer or Blue-water Zone:** The water is between 60 and 100 m deep and is clear, not muddy; the water has a blue tint due to the limited presence of algae, and is biologically less productive than the previous zone;
4. **The Deep Sea Blue-water Zone:** The water is blue and deeper than in the previous zone; this zone starts where the continental slope begins to drop off (this slope starts at about 100 m depth, and levels off at several thousand meters below the surface of the sea).

5.3.1.2 Sensitive ecosystems

Despite being quite resilient and fast growing, mangrove ecosystems are considered sensitive globally due to the key roles of the plants themselves in coastal ecological and geomorphological processes. The six mangrove species are present in Suriname (MBG, 2022), including red mangrove (*Rhizophora harrisonii*, *Rhizophora mangle*, *Rhizophora racemose*), white mangrove (*Laguncularia racemosa*) and black mangrove (*Avicennia germinans* and *Avicennia schaueriana*) species. The conservation of Suriname mangroves seems relatively ensured, as the vast majority of their coastal distribution is currently protected by four MUMAs and one Nature Reserve (see below) along the Suriname coast (Figure 5-15). In addition, GOS received international funding to setup a Mangrove Biodiversity Monitoring System in 2019 with 12 permanent monitoring stations, and this work was able to determine that mangrove cover in Suriname is estimated to have increased by approximately 1% over the period 1988-2018 (SBB, 2019).

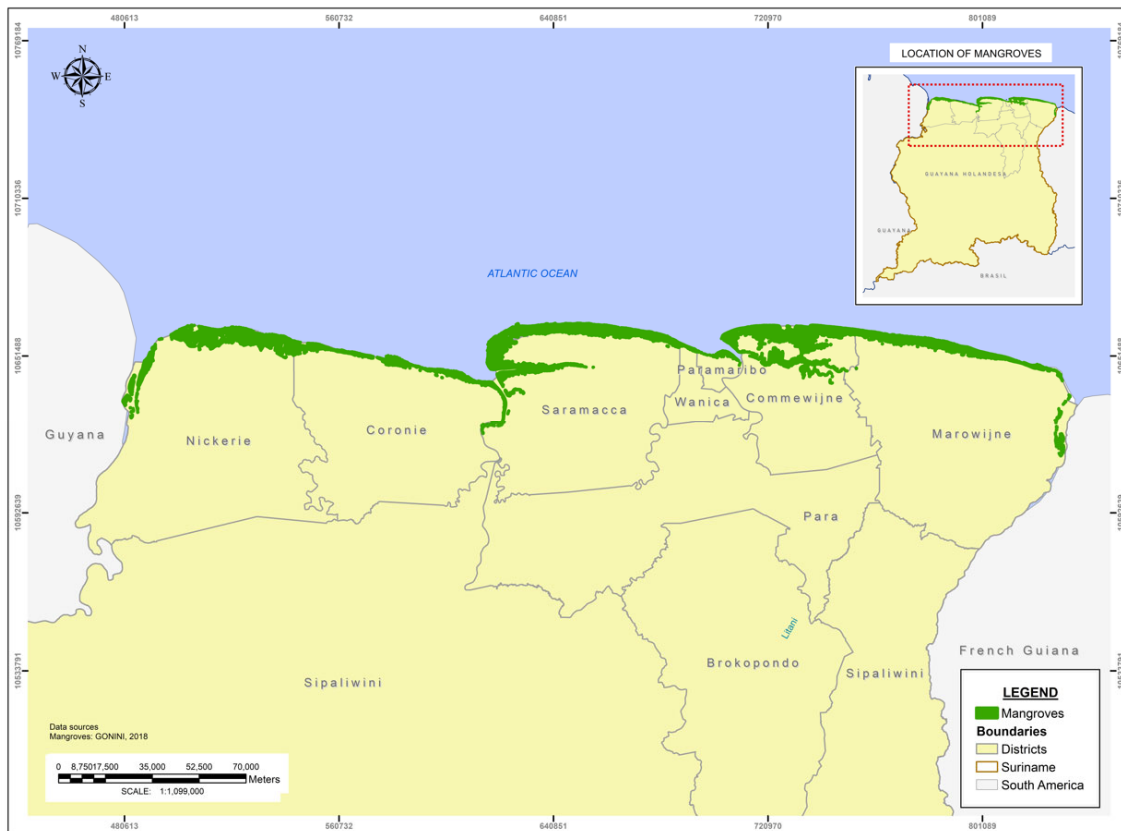


Figure 5-14: Mangrove Distribution along the Suriname coastline

Source: (GOS, 2022b) adapted by RINA, 2022

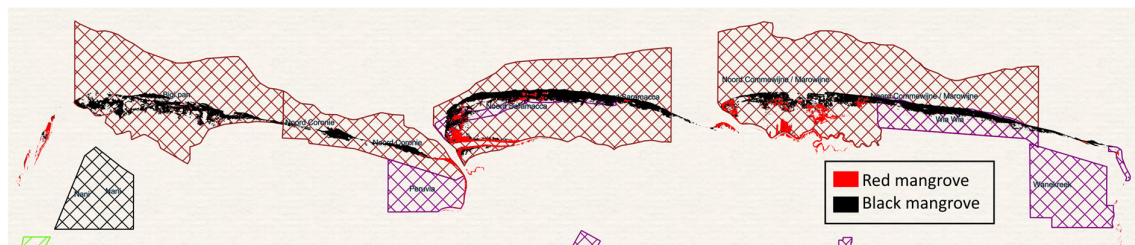


Figure 5-15: Distribution of red and black mangrove species dominated mangrove areas, MUMAs (red hatching) and Nature Reserves (purple hatching) along the Suriname coastline

Source: (SLMS, 2022)

5.3.2 Flora and Fauna

Being a country with extensive tropical lowland forests, diversity of biological species in Suriname is high and it was estimated in 2007 (ESS, 2007) that the country housed:

- ✓ approximately 5100 plant species (mosses, ferns and spermatophytes - seed producing plants);

- ✓ 318 freshwater fish species and approximately 360 marine fish species;
- ✓ 102 amphibian species (mainly frogs and toads);
- ✓ 175 reptile species (lizards, snakes, turtles and caimans);
- ✓ 715 bird species, from the coast, the savannahs and closed rainforests;
- ✓ 192 mammal species among marine (dolphins and whales), freshwater (manatee, otters), and terrestrial (deer, cats, monkeys, bats, etc.) environments.

A summary of species included in the Red List of Threatened species developed by the International Union for Conservation of Nature's (IUCN) for Suriname is presented in Table 5-6.

Table 5-6: Taxonomic groups occurring in Suriname with extinction risk assessments published on The IUCN Red List (Aug 2022)

Taxonomic group	Total assessed species	Total known threatened species (CR, EN & VU)	EX & EW	CR	EN	VU	NT	LR/CD	LC	DD
<i>ACTINOPTERYGII</i>	961	17	0	0	2	15	8	0	887	49
<i>AMPHIBIA</i>	103	1	0	0	0	1	0	0	96	6
<i>ARACHNIDA</i>	1	0	0	0	0	0	0	0	1	0
<i>AVES</i>	712	10	0	0	2	8	29	0	673	0
<i>BIVALVIA</i>	4	0	0	0	0	0	0	0	2	2
<i>CEPHALOPODA</i>	30	0	0	0	0	0	0	0	23	7
<i>CHAROPHYACEAE</i>	2	0	0	0	0	0	0	0	2	0
<i>CHONDRICHTHYES</i>	61	18	0	6	6	6	11	0	17	15
<i>GASTROPODA</i>	17	0	0	0	0	0	0	0	17	0
<i>HOLOTHUROIDEA</i>	5	0	0	0	0	0	0	0	4	1
<i>INSECTA</i>	107	1	0	0	1	0	0	0	97	9
<i>LILIOPSIDA</i>	100	2	0	0	0	2	1	0	92	4
<i>MAGNOLIOPSIDA</i>	834	27	0	1	3	23	1	0	778	13
<i>MALACOSTRACA</i>	35	0	0	0	0	0	0	0	31	4
<i>MAMMALIA</i>	211	9	0	0	1	8	8	0	183	11
<i>POLYPODIOPSIDA</i>	7	0	0	0	0	0	0	0	7	0
<i>REPTILIA</i>	124	7	0	0	2	5	0	0	114	3

Source: (IBAT, 2022)

Notes: Red List Categories: EX = Extinct; EW = Extinct in the Wild; CR = Critically Endangered; EN = Endangered; VU = Vulnerable; NT = Near Threatened; LR/CD = Lower Risk/Conservation Dependent; LC = Least Concern; DD = Data Deficient.

5.3.3 Forest Management

In 2017, the government identified that the pristine natural forest (14.8 million hectares of tropical forest), which is largely still in the 'heart' of the Suriname would be divided into different categories, each with its own economic potential. The wood production forest (4.5 million ha) accounts for 30.4 percent of the total tropical forest. Of the production forest, 62 percent had already been produced in 2015. The integration of the national development planning and the regional planning should result in structure and zoning plans, which should facilitate the sustainable use and management of forests: strengthening regulatory and supervisory institutions and the involvement of the local population are key principles in this approach. Suriname actively participates in REDD+. The program can play an important role in the activities that should be implemented to operationalize Suriname's sustainable development strategy. This international program focuses on the provision of technical assistance through the application of innovative technology, analysis and reporting methods, as well as by mobilizing financing (GOS, 2017).

5.3.4 Deforestation

Despite Suriname's wealth of forest cover, the negative impact from mining industries, the largest sector of the country's economy, has been evident, both from illegal (small- scale) and legal mining. Although this sector has attracted most of the investments (national and international) in the country, consequently raising the export trend and the development of related sectors³⁵, it is also the cause of almost 70% of the country's annual deforestation (see Figure 5-16). In addition, other change drivers include hydropower construction, agriculture (including slash and burn), and road construction. (Terra-i, 2014)

The Global Forest Resources Assessment 2020 report for Suriname (FAO, 2020) indicates that about 97% of the country is still covered by forest, thus being equal with French Guyana as the most forested country in the world. Indeed, Suriname has a very low deforestation rate, which does not show an increase in recent years, ranging from 0.05% to 0.1% over the last decade (see Figure 5-18). Suriname has a long history of sustainable forest management practices and forest protection. The Ministry of Physical Planning, Land and Forest Management (Min. of RGB) is the Ministry that is responsible for forest management in Suriname. Within this Ministry, the Foundation for Forest Management and Production Control (SBB) is the institution responsible for sustainable forest management in the country. As a forestry institution, SBB has the responsibility to report to the Global Forest Resources Assessment (FRA) 2020. (FAO, 2020)

The designation of 2.3 million ha of the country (14% of the total land area) as protected areas, and the development of the Celos Management System as a tool for the sustainable utilization of the 4 million ha forest designated for sustainable production of forest products, within the so called "Forest Belt", are evidence of the country's effort to keep the natural forest intact. (FAO, 2020)

In Suriname, the so-called Reduced Impact Logging (RIL) system is practiced and promoted in the production forest, minimizing the damages to the forest as a result that the forest cover remains intact even after timber has been harvested, by allowing natural regeneration. Nevertheless, the increased timber production has its challenges, to monitor the application of this RIL-guidelines as mentioned in the draft Code Of Practice for SFM, which might lead to unforeseen forest degradation (FAO, 2020).

Developments in sectors other than the forestry sector also leads to deforestation, for instance mainly due to planned and unplanned mining, mostly gold mining, a fabricated hydropower lake, agriculture, infrastructure construction etc. (FAO, 2020). The total area of deforestation due to ASGM for the period of 2000 to 2016 is around 700.28 km² (GOS, 2019a). The mining areas (gold and bauxite) are being rehabilitated with tree planting; approximately 1,150 ha of the mining areas has been rehabilitated (FAO, 2020).

³⁵ Conservation International Guyana, Projekt-Consult GmbH, WWF Guianas 2013. Guyana's Extractive Industry Sector (EIS) Online [<http://goo.gl/hRiXHV>]

The national statistics of Suriname indicate that the forest cover has decreased 1,669 km² (equivalent to 166,900 ha) from 1990 to 2019 (see Figure 5-17), which represents a 1.02% decrease if compared to the total land area of Suriname equivalent to 163,820 km².

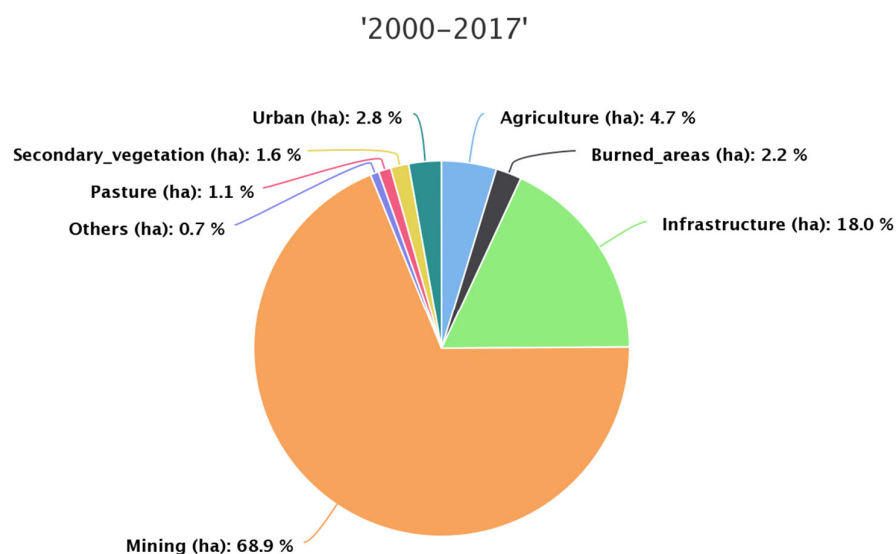
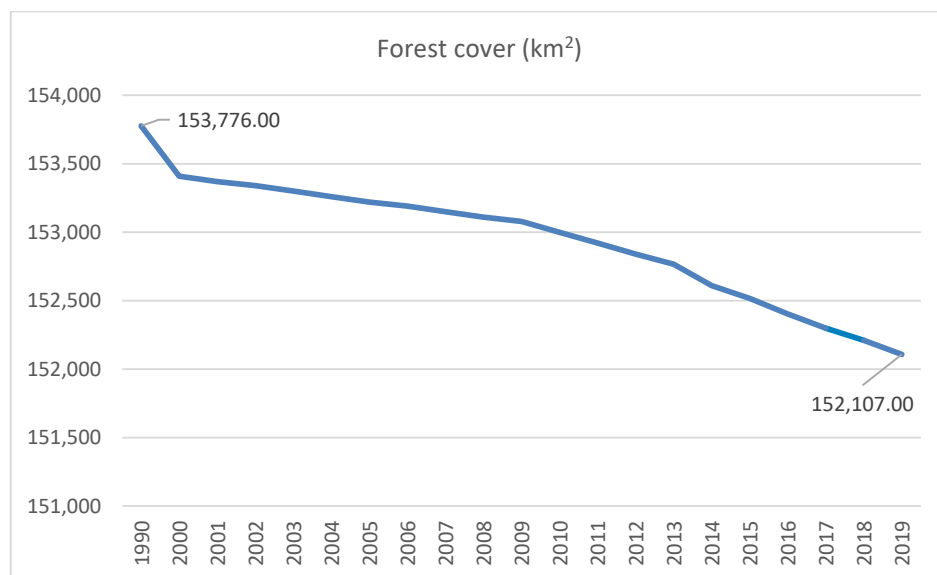


Figure 5-16: Sources of deforestation in Suriname 2000 – 2017

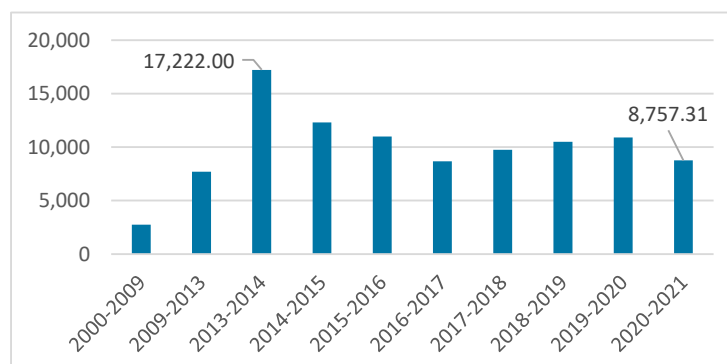
Source: SBB (2022)



Source: (KOPI, 2022) – Forest Cover

Figure 5-17: Forest Cover in Suriname since 1990 to 2019

Since 2014, the national statistics indicate that the annual deforestation area in Suriname averages 11,139.96 ha per year, ranging from its peak in 2013-14 from 7,222 ha and decreasing to 8,750 ha in 2020-21 (see Figure 5-18).



Source: (KOPI, 2022) - Deforestation³⁶

Figure 5-18: Annual Deforestation Area (2000 – 2021)

The national statistics indicate that between 2000 and 2017, the mining and infrastructure were the main drivers of deforestation in the country, accounting together for approximately 87% (91,106 ha) from 2000 until 2017. The GOS has mapped the extent of deforestation since 2000. Figure 5-19 shows the extent of deforestation per period until 2019.

Table 5-7: Post- Deforestation LULC information Deforestation Area (2000 – 2017)

#	Period	Agriculture		Burned Areas		Infrastructure		Mining		Others		Pasture		Secondary vegetation		Urban		Total
		ha	%	ha	%	ha	%	ha	%	ha	%	ha	%	ha	%	ha	%	Total
1	2000-2009	872	3.5	243	1.0	2,829	11.4	19,494	78.7	169	0.7	150	0.6	-	-	1,027	4.1	24,784
2	2000-2013	1,286	2.3	2,052	3.7	7,498	13.5	41,797	75.2	101	0.2	228	0.4	940	1.7	1,714	3.1	55,616
3	2000-2015	2,213	2.6	2,502	2.9	12,964	15.2	62,102	72.9	281	0.3	455	0.5	1,205	1.4	3,424	4.0	85,146
4	2000-2017	4,956	4.7	2,302	2.2	18,843	18.0	72,261	68.9	769	0.7	1,114	1.1	1,642	1.6	2,931	2.8	104,818

Source: (KOPI, 2022)

³⁶ <https://kopi.sbb.sr/index.php?r=deforestationinformation%2Findex>

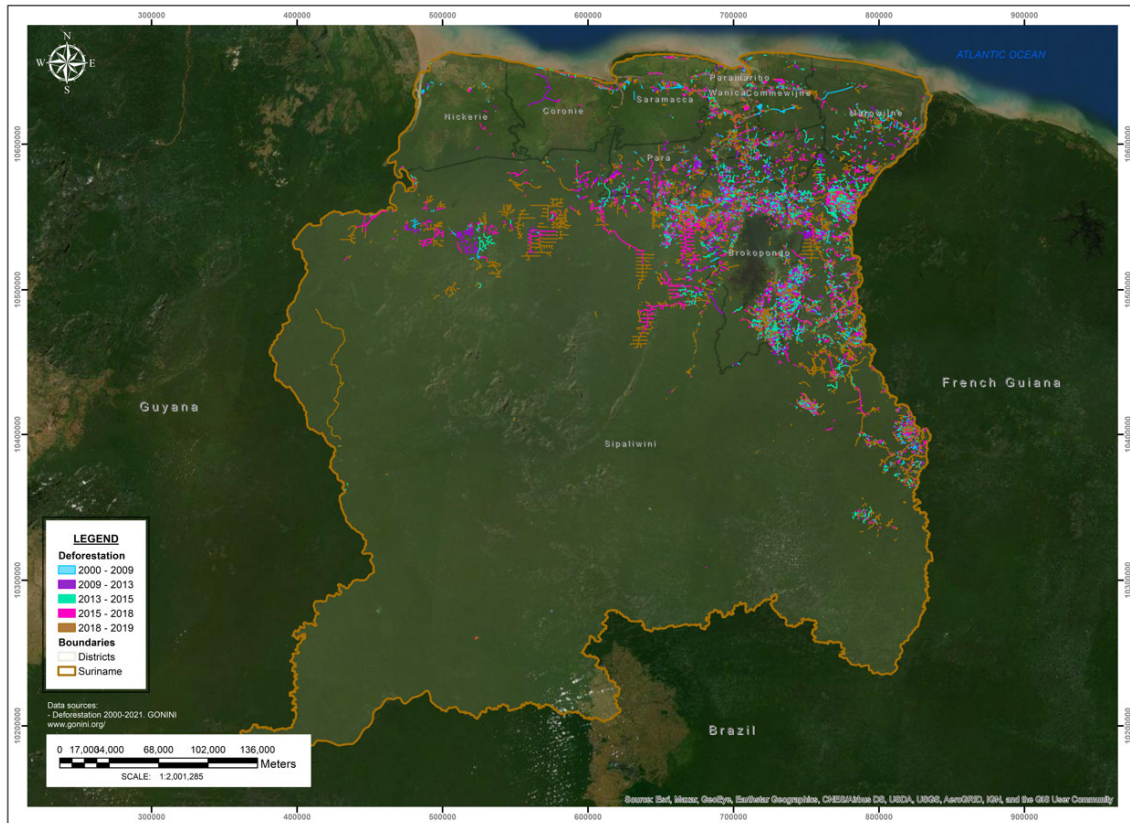


Figure 5-19: Deforestation extent (2000-2019)

Source: (GOS, 2022b) adapted by RINA, 2022

5.3.5 Biodiversity management

Suriname has already made a great investment in the preservation of its biodiversity by designating territories as nature-parks or protected areas. The strategy to protect and manage these environmental areas is the highest priority for biodiversity preservation. However, the government recognizes that it does not mean that the preservation of this would be unimportant outside these environmental areas. In particular, after considering the main factors threatening Suriname's high biodiversity:

- Mining (specifically gold mining);
- Forestry (in particular non-regulated logging);
- Both the legal and illegal hunting and fishing and trade in protected animals; and
- Mechanical agriculture and the use of agrochemicals.

5.3.6 Protected Areas

Protected areas have been established based on the following legislative products (Ouboter, 2002):

- ✓ Police Penal Code of 1915 (amended in 1942): Estuary of the Coppename River; became Coppename Monding Nature Reserve in 1961, based on the Nature Preservation Law of 1954.

- ✓ Nature Preservation Law (Government Bulletin 1954, No. 26), amended in 1992: All present Nature Reserves.
- ✓ “Agricultural Act” of 1936 and Degree L-2 of 15 June 1982: Brownsberg Nature Park was given on long-term lease to the Foundation for Nature Preservation (STINASU) based on this act.
- ✓ Planning Act (Government Bulletin 1973, No. 89): Bigi Pan Multiple-Use Management Area was created based on this act.
- ✓ Forest Management Act (Government Bulletin 1992, No. 80): Provides the possibility for maintaining forests that could have special scientific, educational, cultural, or recreational value.

The World Database on Protected Areas (WDPA) reports that Suriname has 22 Protected Areas, of which two are international (IBAT, 2022). However, four of these 20 national protected areas were only proposed and were never accepted by the GOS. The 16 national protected areas in existence are of three types and are shown in Figure 5-20, while there are also two international types (Table 5-8). The coverage of protected areas in Suriname has been calculated as 14% for terrestrial areas and 1.54% for marine areas (IBAT, 2022), though the figure for terrestrial areas should in fact be slightly lower due to the incorrect inclusion of proposed protected areas that do not exist. The coverage of protected areas in Suriname has been calculated as 14% for terrestrial areas and 1.54% for marine areas (IBAT, 2022), though the figure for terrestrial areas should in fact be slightly lower due to the incorrect inclusion of proposed protected areas that do not exist.

Table 5-8: Total number of Protected Area types designated at a national level and under international conventions and agreements

Type	Number of Protected Areas	Jurisdiction Name
Multiple Use Management Area	4	National
Nature Park	1	National
Nature Reserve	11	National
Ramsar Site, Wetland of International Importance	1	International
World Heritage Site (natural or mixed)	1	International

Source: (IBAT, 2022)

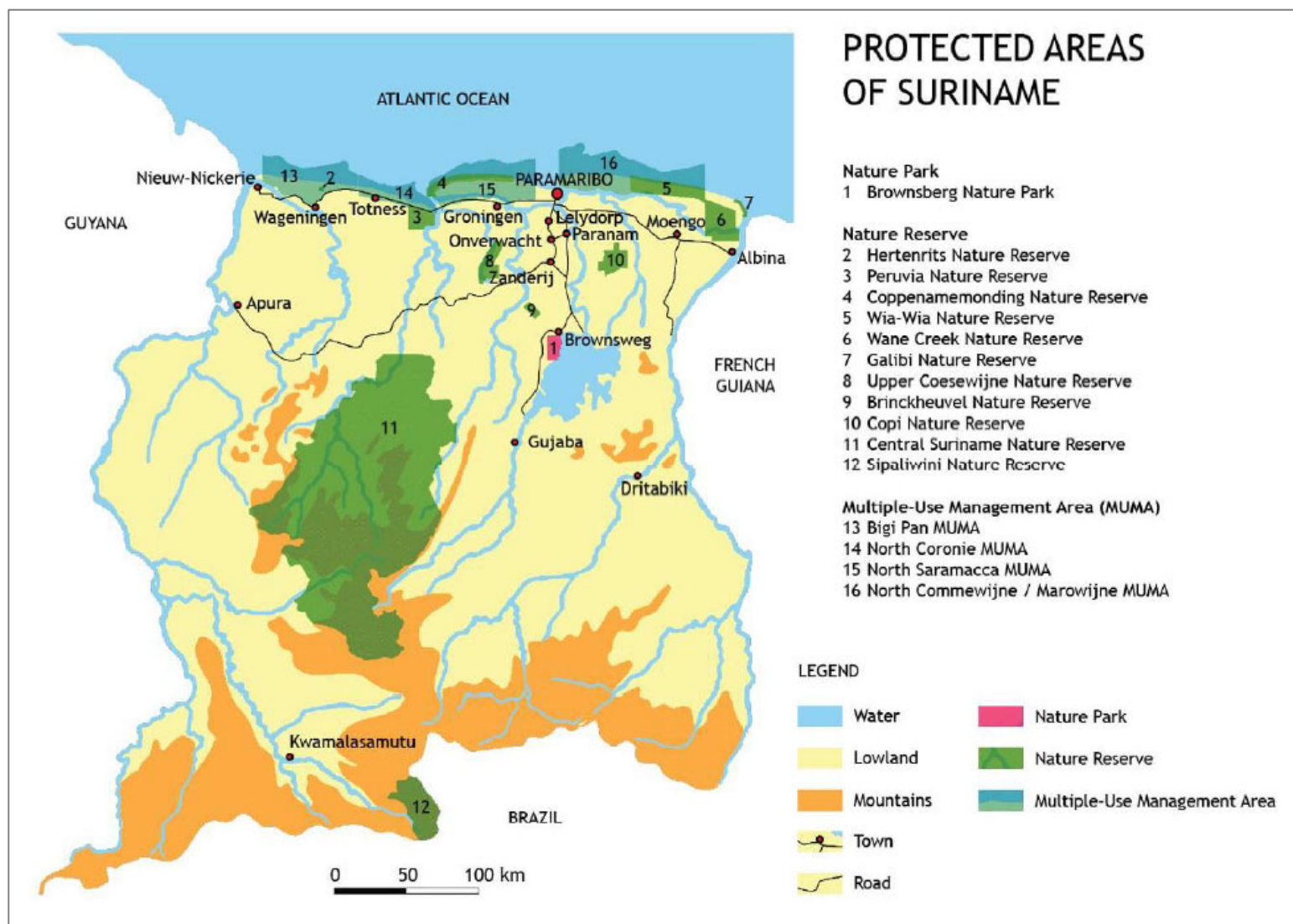


Figure 5-20: Protected areas of Suriname

Source: (Berrenstein, 2016)

The existing protected areas in Suriname are listed in Table 5-9. A further five protected areas had been proposed in the past but never adopted (Ouboter, 2002; Mohadin, 1993; SLMS, 2022), which include those listed below. The first four on this list are within the area of the North-West Suriname IBA (see following section).

- ✓ Kaboeri Kreek Nature Reserve (69,700 ha).
- ✓ Nani Nature Reserve (46,300 ha).
- ✓ Mac Clemen Forest Reserve (6,300 ha).
- ✓ Snake Kreek Forest Reserve (4,200 ha).
- ✓ Lelydorp Forest Reserve (500 ha).

Table 5-9: Suriname Protected Areas

#	Site name	Date established	Total terrestrial area (hectares)
Nature Reserve			
1	Coppename Monding Nature Reserve	1953/1961	17,000
2	Wia-Wia Nature Reserve	1961	39,200
3	Brinckheuvel Nature Reserve	1961	6,100
4	Galibi Nature Reserve	1969	2,000
5	Sipaliwini Nature Reserve	1972	82,200
6	Hertenrits Nature Reserve	1972	100
7	Boven-Coesewijne Nature Reserve	1986	26,200
8	Copi Nature Reserve	1986	28,300
9	Wane Kreek Nature Reserve	1986	55,500
10	Peruvia Nature Reserve	1989	30,900
11	Central Suriname Nature Reserve	1998	1,630,300
Nature Park			
12	Brownsberg Nature Park	1969	16,200
Multiple Use Management Area (MUMA)			
12	Bigi Pan MUMA	1987	160,200
14	Noord Coronie MUMA	2001	64,000
15	Noord Saramacca MUMA	2001	186,900
16	North Commewijne - Marowijne MUMA	2002	269,900
	Total		2,615,000

Source: modified from (Ouboter, 2002); (Berrenstein, 2016), and (SLMS, 2022)

5.3.7 Key Biodiversity Areas

Key Biodiversity Areas (KBAs) are sites that contribute significantly to the global persistence of biodiversity, on land, in freshwater or on the seas. These sites have been identified over the last four decades by an international network

of NGOs, academic institutions and government organizations using a series of complementary approaches. Two subsets of KBAs have been identified so far across virtually all countries worldwide (IBAT, 2022):

- ✓ Important Bird and Biodiversity Areas (IBAs; over 13,000 sites identified by the BirdLife International Partnership)
- ✓ Alliance for Zero Extinction Sites (AZEs; 853 sites identified by the Alliance for Zero Extinction)

KBAs have been identified for birds by the BirdLife International Partnership, for Critically Endangered or Endangered species restricted to single sites through the Alliance for Zero Extinction, and for other mammals, reptiles, amphibians, fish, plants and invertebrates through the Critical Ecosystem Partnership Fund (CEPF) hotspot profiling process. According to IBAT (2022), the Surinamese KBAs include no AZEs and 13 IBAs. Table 5-10 shows how about 50% of the Surinamese IBA surface area is already protected by the national protected areas (visualized in Figure 5-21), with 8 of 13 IBAs entirely or mostly covered by national protected areas.

Table 5-10: Summary of IBAs located in Suriname

IBA name	Area (ha)	IBA Criteria*	Final Code	Area Protected
Bigi Pan	136,000	A2, A4i, A4iii	SR001	yes
Northern Coronie	66,237	A2, A4i, A4iii	SR002	yes
Northern Saramacca	217,530	A2, A4i, A4iii	SR003	yes
Northern Commewijne/ Marowijne	333,423	A2, A4i, A4iii	SR004	yes
North-West Suriname	216,386	A2, A3	SR005	no
Boven Coesewijne Nature Reserve (BCNR)	27,000	A3	SR006	yes
Brownsberg Nature Park (BB)	13,000	A3	SR007	yes
Nassau mountain	28,850	A3	SR008	no
Bakhuys mountains	561,327	A1, A3	SR009	no
Kabalebo / Arapahu	1,683,697	A1, A3, A4i	SR010	no
Central Suriname Nature Reserve (CSNR)	1,600,000	A1, A2, A3	SR011	yes
Lely mountain	78,820	A3	SR012	no
Sipaliwini Nature Reserve	100,000	A1, A3	SR013	yes
All IBAs	5,062,270	A1, A2, A3, A4i, A4iii	-	2,493,190**

Source: (BirdLife International, 2022)

Notes: *A1 = Globally Threatened Species; A2 = Restricted Range Species; A3 = Bioregion-restricted assemblages; A4i = Globally important congregations - 1% or more of a biogeographic population; A4iii = Globally important congregations -. at least 20,000 waterbirds, or at least 10,000pairs of seabirds, of one or more species. **This is a crude estimate, actual official georeferenced boundaries will be required to make an accurate estimation.

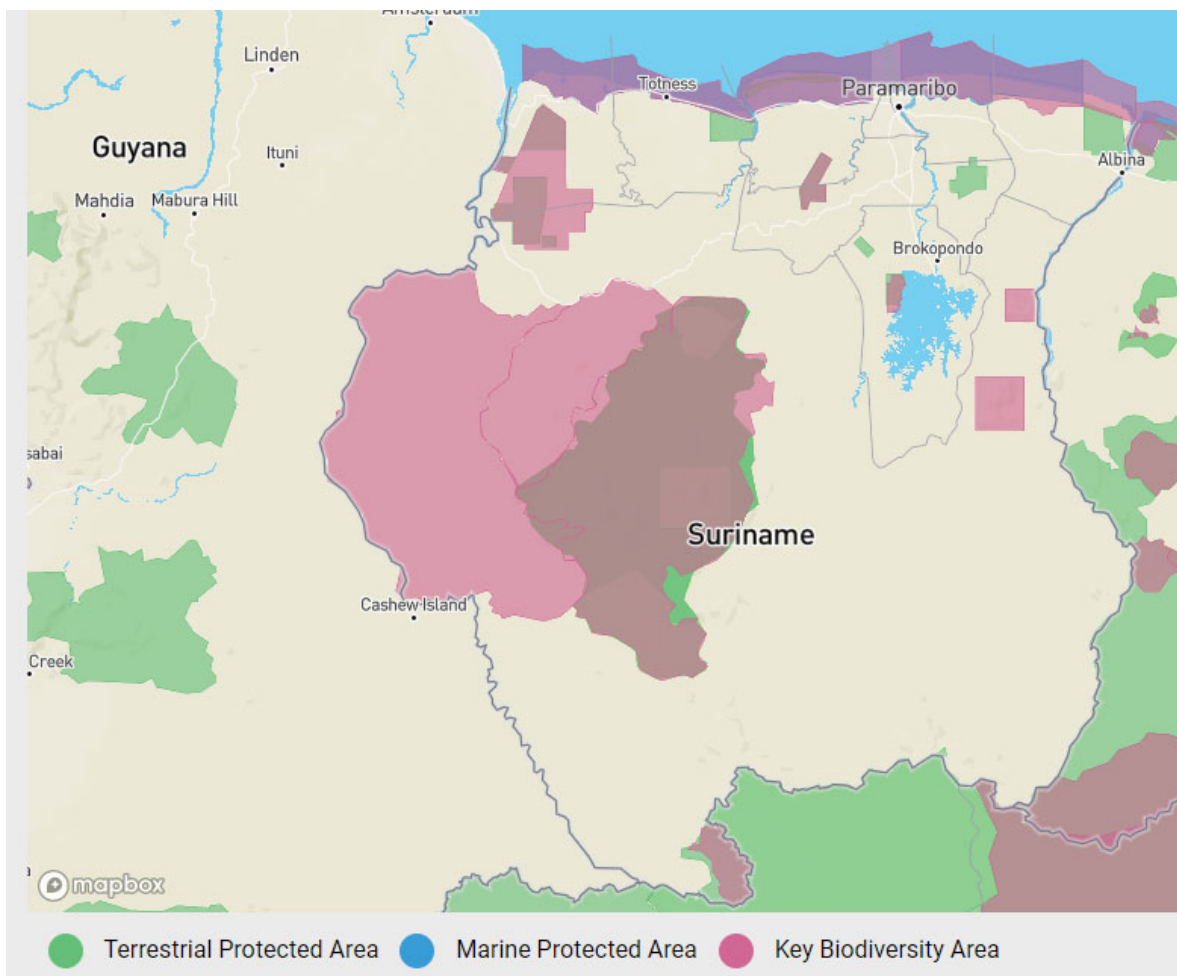


Figure 5-21: Overlap of Protected Areas with KBAs within Suriname

Source: modified from (IBAT, 2022)

5.4 SOCIO-ECONOMIC

This section presents the key and relevant socio-economic aspects of the current conditions in Suriname, in relation to mining development. These and other aspects will be addressed in more detail during the SESA.

5.4.1 Demographics and Population

The great diversity of the population of Suriname comprises Indigenous tribes³⁷ (Native Amerindians) and tribal people (descendants of African slaves), Maroon diaspora, the European diaspora (Dutch descendants), Javanese diaspora (descendants or Indonesia), Hindostani diaspora (descendants or India), creoles and others, while the greatest contribution of all these ethnic groups is the preservation of cultural values, norms and traditions. The population composition in percentage is as follows: Hindustani, 27.4%; Creoles, 17.7%; Maroons, 14.7%; Javanese, 14.6%; Mixed, 12.5%, others (Chinese, Indigenous people, Lebanese, European, etc.) The percentage of Indigenous population in Suriname is only 3.8%, mainly living in different parts of the country (GOS, 2020).

According to the geographical data sourced from the National Land Monitoring System of Suriname (GOS, 2022), the Maroon and Amerindian populations are geographically concentrated in the districts of Sipaliwini, Para and Brokopondo. All of them are in the interior zone of the country, far from the coastline. While the coastal strip of the Atlantic Ocean is a fertile and cultivated plain area where most of the population lives, the interior is sparsely populated and characterized by dense tropical forests.

Out of the 10 districts of Suriname, Sipaliwini hosts the largest number of Amerindians' villages (23) and Maroons' villages (138). This is followed by the districts of Para and Marowijne, which have 14 and 10 Amerindian villages respectively, and the district of Brokopondo that has 30 villages inhabited by Maroons. The remaining districts of Suriname host less than 5 villages inhabited by Amerindians and/or Maroons. A brief analysis of the number of villages classified by culture per district and resorts, is presented in Table 5-11.

It is worth mentioning that the data consulted and hosted by GOS (2022) does not include the population residing in the district of Paramaribo.

Table 5-11: Distribution of cultures by districts and ressorts.

Districts and Ressorts	# of Amerindian villages	# of Maroon villages	# of villages inhabited by other cultures	Total
Brokopondo				
Boven-Suriname	0	1	0	1
Brownsweeg	0	9	5	14
Centrum	1	6	1	8
Duwata	0	3	0	3
Klaaskreek	0	1	5	6
Kwakoegeon	0	1	2	3
Marschall	0	7	3	10
Sarakreek	0	5	0	5
<i>total absolute</i>	1	33	16	50
<i>total %</i>	2%	15%	5%	9%
Commewijne				
Bakkie	0	0	1	1
Meerzorg	2	0	5	7
Tamanredjo	0	0	3	3
<i>total absolute</i>	2	0	9	11
<i>total %</i>	3%	0%	3%	2%

³⁷ See Figure 5-33: Indigenous Territories

Districts and Ressorts	# of Amerindian villages	# of Maroon villages	# of villages inhabited by other cultures	Total
Coronie				
Johanna Maria	0	0	1	1
Welgelegen	0	0	2	2
<i>total absolute</i>	0	0	3	3
<i>total %</i>	0%	0%	1%	1%
Marowijne				
Albina	7	10	16	33
Galibi	2	0	1	3
Moengo	0	10	9	19
Moengotapoe	0	4	0	4
Patamacca	0	4	9	13
Wanhatti	1	7	8	16
<i>total absolute</i>	10	35	43	88
<i>total %</i>	17%	16%	14%	15%
Nickerie				
Wageningen	2	0	0	2
Westelijke Polders	1	0	1	2
N.A.	1	0	0	1
<i>total absolute</i>	4	0	1	5
<i>total %</i>	7%	0%	0%	1%
Para				
Bigi Poika	1	0	4	5
Boven-Coppename	1	0	0	1
Carolina	4	3	3	10
Kwakoegron	0	2	0	2
Para Noord	1	0	0	1
Para Zuid	5	0	1	6
Para-Oost	2	1	6	9
N.A.	0	1	0	1
<i>total absolute</i>	14	7	14	35
<i>total %</i>	23%	3%	5%	6%
Saramacca				
Calcutta	2	0	1	3
Groningen	1	0	0	1
Kampong Baroe	0	4	0	4
Paramakka	1	0	0	1
Wayamboweg	0	0	1	1
<i>total absolute</i>	4	4	2	10
<i>total %</i>	7%	2%	1%	2%
Sipaliwini				
Boven Saramacca	0	19	34	53
Boven-Coppename	2	5	4	11
Boven-Suriname	0	54	43	97
Coeroeni	8	0	0	8
Duwata	0	3	0	3
Kabalebo	4	0	2	6
Paramakka	0	11	0	11
Tapanahony	9	46	130	185
<i>total absolute</i>	23	138	213	374
<i>total %</i>	38%	62%	70%	64%
Wanica				
Lelydorp	1	4	0	5
N.A.	1	0	0	1

Districts and Ressorts	# of Amerindian villages	# of Maroon villages	# of villages inhabited by other cultures	Total
<i>total absolute</i>	2	4	0	6
<i>total %</i>	3%	2%	0%	1%
N.A.	0	1	3	4
%	0%	0%	1%	1%
Grand Total	60	222	304	586
	10.24%	37.88%	51.88%	100.00%

Source: (GOS, 2022b) adapted by RINA, 2022.

Dutch is the official language; however more than 16 other languages are spoken. Sranang Tongo (Surinamese, is the native language and is lingua franca among others), Caribbean Hindustani (a dialect of Hindi), Javanese (GOS, 2020).

In 2019, the estimated population of Suriname reached a total of 598,000 inhabitants over a land territory of 163,820 km² (approximately 3.65 inhabitants per km²). The sex distribution of the population accounted for 51.4% being female (307,372 inhabitants) and 48.6% being males (290,628 inhabitants). In 2019, 19.3% had 2 household³⁸ members, 17.3% had 3 household members and 17.5% had 4 household members. Only 3.9% had more than 9 household members. Most households in Paramaribo and Wanica have a household size of 2 to four household members (GBS, March 2020).

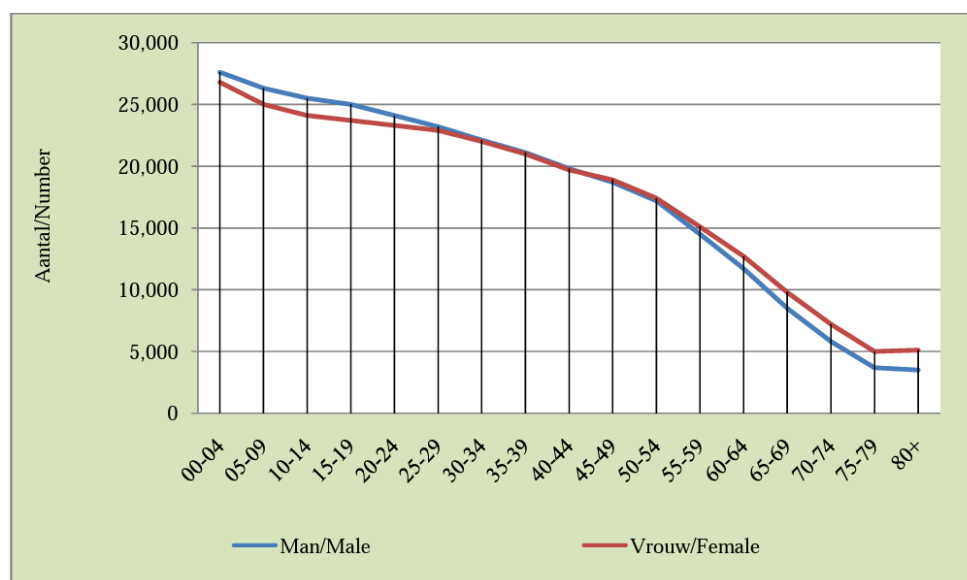


Figure 5-22: Estimated (mid-year) Population, 2019 by sex and age groups

Source: (GBS, March 2020)

The population is unequally distributed along the total land area of the country. About 66% of the population lives in the urban area while 20 % in the rural area and 14% in the interior (GOS, 2022a).

³⁸ A household is a person or group of persons, who have made arrangements individually or as a group, for providing themselves with food or other essentials for living (GBS Census, 2012).

5.4.2 Education

In Suriname, the education system consists of pre-primary, primary, junior and senior secondary schools, which also includes technical and vocational education as well as higher or tertiary (technical/ vocational) education (GOS, 2022a). Completion rates are 85 % at primary education level and progressively decreasing at the junior secondary and senior secondary level to respectively 49 % and 24% (GOS, 2022a). Completion rates are higher for children living in urban areas and those from the richest quintiles compared to their peers from the rural areas and the poorest quintile. The literacy rate in Suriname is about 98 percent, and on average, less than 10 percent never completed primary education.

The Covid-19 pandemic has adversely impacted learning due to school closures and limitations in rolling out distance learning. This learning loss will mainly weigh in on labor productivity going forward, putting the students from low-income households in a disadvantageous position (WBG, 2022a).

For the rural and tribal communities in Suriname, primary education is the highest accessible level in their own region and in some areas, there are secondary schools (Zanderij, Albina and Apoera). Only the Nickerie district provides senior secondary education in addition to the capital city of Paramaribo (VIDS, 2020).

5.4.3 Health and Sanitation

Access to safe drinking water, sanitation and hygiene (WASH) is essential for good health, welfare and productivity and is widely recognized as a human right³⁹. Inadequate WASH is primarily responsible for the transmission of diseases such as cholera, diarrhea, dysentery, hepatitis A, typhoid and polio. Diarrhea diseases exacerbate malnutrition and remain a leading global cause of child deaths. Drinking water may be contaminated with human or animal feces containing pathogens, or with chemical and physical contaminants with harmful effects on child health and development. While improving water quality is critical to prevent disease, improving the accessibility and availability of drinking water is equally important, particularly for women and girls who usually bear the primary responsibility for carrying water, often over long distances. (GBS, March 2020).

The most significant environment-related diseases in Suriname are malaria, dengue fever and Leptospirosis. These can be reduced through good public policies and making financial investments, expanding access to water, sanitation facilities and hygiene education, ensuring that garbage is collected properly, promoting the use of clean household fuels, and controlling the (insect and animal) vectors that transmit diseases. The total number of confirmed cases or hospitalized cases reported between 2015 and 2019 for Malaria, Dengue, Leptospirosis Chikungunya and Zika are presented in Table 5-12.

Table 5-12: Environment-related health cases recorded between 2015 and 2019

Type	Description	2015	2016	2017	2018	2019	2020*	2021*	Total***
Malaria	Suspected cases	9*	13*	5*	11	4	-	-	15
	Positively tested cases	3	10*	3	5	-	-	-	11
Dengue	Suspected cases	19*	19*	19*	17	44	235	9	305
	Positively tested cases	-	-	2	1	-	58	-	61
Leptospirosis	Suspected cases	53*	74	58*	71	25	15	18	203
	Positively tested cases	-	-	2	1	-	-	-	3
Chikungunya	Non-hospitalized cases	830	30	10	-	-	-	-	870

³⁹ The human rights to water and sanitation were explicitly recognized by the UN General Assembly and Human Rights Council in 2010 and in 2015.

Type	Description	2015	2016	2017	2018	2019	2020*	2021*	Total***
	Hospitalized cases	41	2	1	-	-	1	-	45
Zika	Non-hospitalized cases	-	1,144	15	-	-	**	**	1159
	Hospitalized cases	2	22	1	-	-	**	**	25

Source: (GBS, March 2020); and (GBS, 2022)

Notes: * data updated and sourced from Table 11.6 of (GBS, 2022)

** data not available (GBS, 2022)

In terms of sanitation, the national statistics for 2018 indicate that a high proportion of the population (94.5% of the population) in Suriname have access to improved sanitation facilities per district showing a 3.8% increase when compared with 2010 (see Table 5-13). The same statistics also indicate that although the districts of Brokopondo and Sipaliwini have shown a significant improvement (29% and 38%, respectively), a good proportion of the population does not have access to improved sanitation facilities, which is particularly important for Sipaliwini where half of the population is without such facilities.

Table 5-13: Proportion of the population in Suriname that have access to improved sanitation facilities per district (2010-2018)

Region / District	2010	2018	Change (+/-)
Paramaribo	97.9	98.5	0.6
Wanica	98.3	98.9	0.6
Nickerie	99.4	99.8	0.4
Coronie	98.0	99.6	1.6
Saramacca	83.4	97.3	16.7
Commewijne	94.2	97.9	3.9
Marowijne	88.4	85.5	-3.3
Para	91.6	91.1	-0.5
Brokopondo	55.5	71.7	29.2
Sipaliwini	37.0	51.1	38.1
Suriname	91.0	94.5	3.8

Source: (GBS, March 2020).

Suriname does not have wastewater treatment systems for municipal households nor treatment facilities for Industrial facilities. The amount of wastewater is estimated to be equal to the amount of water delivered (GOS, 2019a). Considering that an estimated that terms of water consumption consumed through the SWM network in 2019 (see Table 5-4 in Section 5.2.4.2.1), the wastewater discharged without treatment in the districts supplied by the SWM is estimated to be 28,323,000 m³. Part of the domestic sewage (feces and urine) from urban areas, is treated in septic tanks (in 2010, 86% of homes in Paramaribo had septic tanks). The effluent of the septic tanks is collected in the street sewers and drains indirectly to the river through the Saramacca canal or is pumped into the Atlantic Ocean through pumping stations (GEF-CReW, 2010). The situation in the regional or rural areas follows a similar pattern to that of Paramaribo and the greater Paramaribo area, with no existing infrastructure or treatment facilities for household or industrial facilities.

Health and gold mining activities

Gold mining goes together with health threats due to water pollution from mercury and cyanide, poor working conditions and housing, and lack of health facilities. Some problems that occur due to the small scale and illegal gold mining are degradation of forest lands by wild, scattered excavation, inefficient mining practices, social disruption and destabilization of local communities (GBS, March 2020). This is of particular concern, given that “*relatively high levels of mercury have been found in hair samples from women and children in interior areas of Suriname*”... “*levels similar to those found in previous cohort studies that have found negative neurodevelopmental impacts on children exposed during gestation and possibly during early childhood*” (Ouboter, et al., 2018).

In terms of mercury, the GOS calculated the releases of mercury into the environment for each source category. The primary metal production accounts for 97% of the releases, as seen in Table 5-14 (GOS, 2019a). The GOS identified that the individual sub-categories contributing with the highest mercury releases to the atmosphere were:

1. Gold and silver extraction with the mercury-amalgamation process (89%)
2. Informal waste burning and Informal dumping of general waste (6%)
3. Gold extraction and initial processing by other processes than mercury amalgamation (3%)

The individual mercury releases sub-categories contributing with the highest inputs of new mercury were:

1. Gold and silver extraction with the mercury-amalgamation process (70%)
2. Gold extraction and initial processing by other processes than mercury amalgamation (27%).

Community health and mercury

Mercury releases to the environment from mining are described below:

- Air: Diffuse sources as small-scale gold mining from informally burned waste with fluorescent lamps, batteries thermometers, and during the amalgamation process of gold with mercury.
- Water: Surface run-off and leachate from mercury contaminated soil and waste dumps
- Land: Uncollected waste products dumped or buried informally, local un-confined releases from industry such as on-site hazardous waste storage/burial or spreading of sewage sludge with mercury content on agricultural land (sludge used as fertilizer).
- By-products and impurities: Metal mercury or calomel as by-product from non-ferrous metal mining (high mercury concentrations)
- Sector specific waste treatment / disposal: Hazardous industrial waste with high mercury content that is deposited in dedicated safe sites, and confined deposition of tailings and high-volume rock/waste from extraction of non-ferrous metals.

Table 5-14: Estimated mercury releases into the environment

Source category	Calculated Mercury (Hg) output Kg/year							
	Air	Water	Land	By-products and impurities	General waste	Sector specific treatment / disposal	Total releases by source category	% total releases *3 *4
5.1: extraction and use of fuels/ energy sources	22.30	0.70	-	0.30	-	0.40	24	0.0%
5.2: Primary (virgin) metal production	16,869.30	23,829.70	44,563.40	962.60	-	-	86,225	97.0%
5.3: Production of other minerals and materials with mercury impurities *1	-	-	-	-	-	-	-	0.0%
5.4: Intentional use of mercury in industrial processes	-	-	-	-	-	-	-	0.0%

Source category	Calculated Mercury (Hg) output Kg/year							
	Air	Water	Land	By-products and impurities	General waste	Sector specific treatment / disposal	Total releases by source category	% total releases *3 *4
5.5: Consumer products with intentional use of mercury (whole life cycle)	274.90	291.30	284.50	-	430.90	-	1,282	1.4%
5.6: Other intentional product/ process use *2	0.50	16.40	2.60	1.60	10.90	10.40	42	0.0%
5.7: Production of recycled metals	-	-	-	-	-	-	-	0.0%
5.8: Waste incineration and burning	867.10	-	-	-	-	-	867	1.0%
5.9: Waste deposition/landfilling and wastewater treatment *3 *4	207.70	350.70	1,661.40	-	-	-	2,220	0.9%
5.10: Crematoria and cemeteries	2.10	-	7.10	-	-	-	9	0.0%
Sum of Quantified Releases *3 *4	18,244.00	24,346.00	44,858.00	965.00	442.00	11.00	88,864	100.0%
Notes: * 1 Includes production of cement, pulp and paper, lime and light weight aggregates. * 2 Includes dental amalgam fillings, manometers and gauges, lab chemicals and equipment, Hg use in religious rituals and folklore medicine, and miscellaneous product uses. * 3 The estimated quantities include mercury in products, which has also been accounted for under each product category. To avoid double counting, the release to land from informal dumping of general waste has been subtracted automatically in the TOTALS. *4 The estimated input and release to water include mercury amounts, which have also been accounted for under each source category. To avoid double counting, releases to water from wastewater system/treatment have been subtracted automatically in the TOTALS Source: (GOS, 2019a)								

It was estimated that approximately 62,159 kg of mercury is released into the environment each year due to ASGM, of which 16,267 kg was released to air, leaving 45,892 kg mercury released to land and water. Thus, mercury pollution from ASGM in Suriname can be linked to the fact that there is often no separate collection or treatment of tailings before release into the environment. (GOS, 2019a). In addition, some hot spots of mercury contamination related to mining may include:

- Tailings/residue deposits from artisanal and large-scale gold mining.
 - In the ASGM areas, tailings are not controlled and are directly deposited into the environment. Some companies have taken control measures before waste is discharged into the environment.
 - For the large-scale companies, the waste is collected into a controlled tailing pond, the overflow is discharged into the environment in a controlled matter.
- Tailings/residue deposits from other non-ferrous metal extraction.

These hot spots need to be evaluated, in terms of location and extent, contamination path analysis, potential affection of communities and sensitive ecosystems.

5.4.4 Labor market

The labor market is divided into three components, which, although connected, are very distinct because of their nature, size and challenges. There are two main institutions responsible for the production of labor market information in Suriname. 1) The Labor Market Directorate of the MOL collects and analyses data based on jobseekers who have registered at the Department for Direct Mediation in that Ministry. 2) The General Bureau for Statistics (ABS) collects data on the workforce during regular household surveys, as well as the population census carried out every 10 years. The most recent nationwide data are from the population census of 2012. More recent data, (available until 2016), have been derived from the household survey. However, data from the household surveys are only gathered for the districts of Paramaribo, Wanica, Commewijne and Nickerie. The data collected during the census are sex disaggregated.

In 2018, the Gold Sector Ordinance (OGS⁴⁰) identified that around 40,000 persons are directly and indirectly involved with ASGM activities in Suriname, and 7342 persons are registered by OGS as employees directly involved with ASGM activities. Nevertheless, the number of illegal ASGM miners and whether they hold a concessions rights issued by the government under the Mining Decree are unknown and would require a special study. In addition, the potential exposure of ASGM miners to mercury through different methods, such as mercury amalgamation of gold is unknown, (GOS, 2019a). Considering that the majority of ASGM taking place in Suriname's Greenstone Belt, where most gold deposits are located, is being undertaken by overall Brazilians (Garimpeiros), and to a lesser extent by the Surinamese Maroons, (GEF, 2016), the potential Occupational Health and Safety issues from these activities have not been considered in this sector and therefore need to be evaluated.

5.4.5 Economic Sectors

In 2019 the National Income per capita was Surinamese dollars SRD40,916 (US\$ 1,282.39). The largest private sector contributors to Suriname's GDP in 2019 were (GBS, March 2020):

1. Manufacturing
2. Wholesale and Retail trade
3. Financial Intermediation
4. Transport, Storage & Communication
5. Construction
6. Mining and Quarrying
7. Agriculture, Hunting and Forestry.

In terms of merchandise exports during 2021 (see Table 5-15), mining products are on the top 2 of the list (precious stones or metals and mineral products) and accounting to approximately 90.9% of the exports with a combined value of 2,004 US\$Mn. The composition of exports over the last 10 years is presented in Figure 5-24.

Table 5-15: Merchandise Exports 2021: Ten Most Important Products

#	Exports by Product	in US\$Mn	in % of total
1	Pearls, precious stones or metals	1,793	81.3
2	Mineral Products	211	9.6
3	Wood & Wood Products	72	3.3
4	Vegetable Products	38	1.7
5	Animal & Animal Products	34	1.5
6	Machinery / Electrical	16	0.7
7	Metals	11	0.5
8	Foodstuffs	9	0.4
9	Chemicals & Allied Industries	7	0.3
10	Transportation	5	0.2
11	Other	9	0.4

⁴⁰ <https://gov.sr/thema/ordering-goudsector-ogs/#>

	Total	2,204	100.0
--	-------	-------	-------

Source: (CBS, 2022)

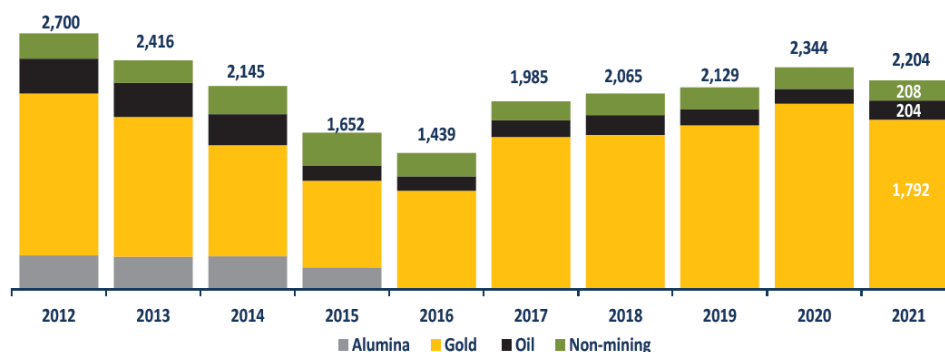


Figure 5-23: Composition of Exports (US\$Mn)

Source: (CBS, 2022)

The total government revenue by sector over the last 10 years (2012 -2021) is presented in Figure 5-24. The government mining revenue in 2021 represented 13.3% of GDP. Mining Sector Government Revenue by Major Industry is presented in Figure 5-25, with 55% sourced from gold mining and 45% from oil. In 2021, the formal gold mining sector employed 3,046 people, oil sector employed 1,073 people and the bauxite sector 25 people (see Figure 5-26).



Source: (CBS, 2022)

Figure 5-24: Total Government Revenue - by Sector

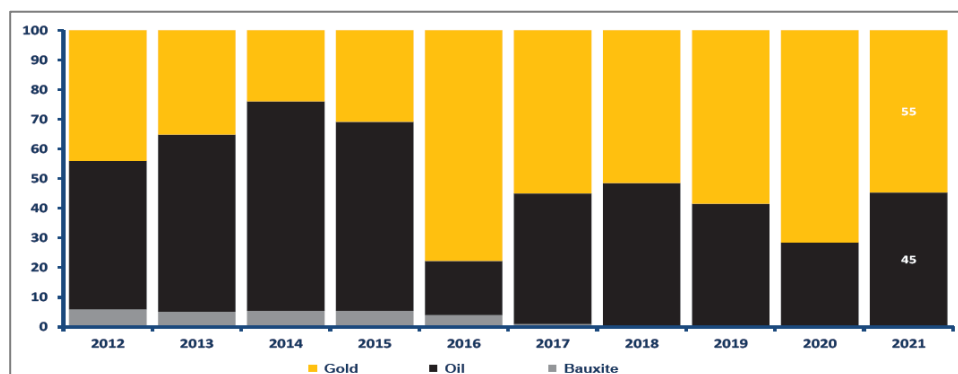


Figure 5-25: Mining Sector Government Revenue - by Major Industry (%)

Source: (CBS, 2022)

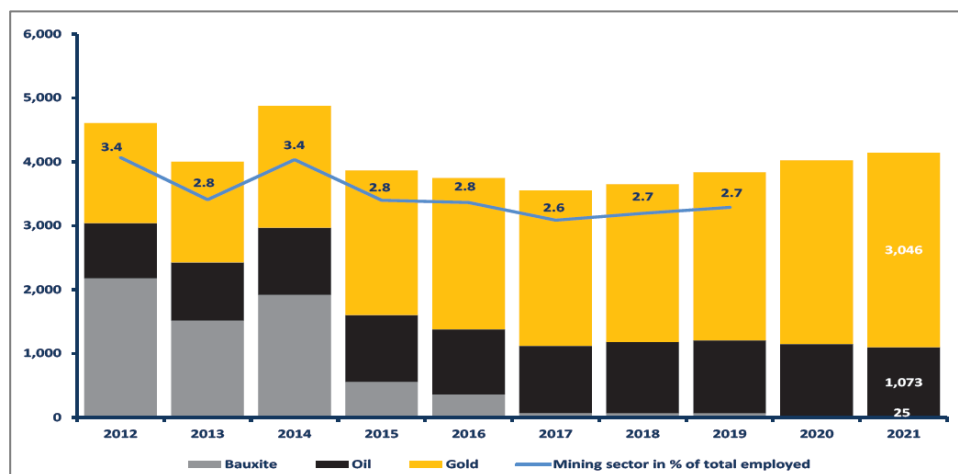


Figure 5-26: Mining Sector Employees

Source: (CBS, 2022)

The economy is driven by its abundant natural resources, with mining accounting for nearly half of public sector revenue and gold representing more than 80 % of total exports, what makes Suriname extremely vulnerable to external shocks (WBG, 2022a).

5.4.6 Poverty

About one in four people in Suriname, representing 26 % of the population, live on less than US\$5.50 a day. The poverty rate is much higher in the interior region where more than half the population live in poverty. Extreme poverty, defined as the percentage of the population with an income below US\$1.9 a day was estimated at 5 percent (WBG, 2022a).

Suriname was severely impacted by the COVID 19 pandemic, increasing the existing domestic vulnerabilities. Critical sectors, such as mining and oil, took major hits and many employees in the services and sales sectors experienced temporary job losses (WBG, 2022a).

The Indigenous and Maroon tribal communities in the interior are (materially) the poorest groups in Suriname. For the Maroon population this represented 74.2% and for the Indigenous population it reached 65.4%, of which another 18.1% was considered semi-poor (VIDS, 2020).

5.4.7 Indigenous and Tribal Peoples

5.4.7.1 Land acquisition

Indigenous villages in the Amazon forest are being affected by deforestation and mercury contamination caused from small-scale gold mining. Such contamination is not only affecting their health but also the environment, contaminating rivers and fish⁴¹. Mercury contamination, whether from legal or illegal mining, comes mainly from small-scale mining practices. Large mining companies in Suriname are currently phasing out the use of mercury seeking more environmentally friendly alternatives. Meanwhile, mercury contamination affects the health of community members and has a direct impact on their way of life. Fish is the main source of proteins for many Indigenous and tribal communities and therefore, they are obligated to seek alternative protein sources. In addition, Suriname is alleged to be selling Indigenous land to gold-mining organizations⁴².

Along with Guyana and French Guiana, Suriname is the only Amazonian country or territory not to have ratified Convention 169, a 1989 treaty from the International Labour Organization (ILO) that legally recognizes Indigenous and tribal peoples' right to self-determination (Lyons & Espinosa, 2022).

The impacts of gold mining will be reviewed in detail during the Environmental Impact Assessment to be conducted during the SESA study,

5.4.8 Archaeological and Cultural Heritage

The archaeological record in Suriname extends back at least 10,000 years. In the western hemisphere, archaeological resources are typically divided into two broad chronological categories. Pre-Contact (also known as Pre-Columbian) resources are archaeological resources that pre-date the arrival of European explorers and colonizers while post-Contact resources are Indigenous and/or Euro-American resources that postdate European arrival. The following sections provide a summary of the types of archaeological resources known to occur in Suriname. Both pre-Contact and post-Contact archaeological resources can be found in urban and rural settings, with sites in rural and undeveloped areas as the most susceptible to be impacted from mining projects.

Pre-contact archaeological sites have been identified across Suriname (Figure 5-28). A 2003 synthesis of the archaeological record identified approximately 109 Pre-Columbian archaeological sites in the country (Versteeg A. H., 2003). While the distribution of previously recorded sites suggests a greater concentration of Pre-Columbian sites along the coast plain, this is more likely a result of 20th century agricultural, industrial, infrastructure, and other activities uncovering sites rather than a lack of or lower number of sites being present in the interior.

⁴¹ Submission by the University of Oklahoma; and <https://www.undp.org/suriname/news/effectively-managing-gold-mining-suriname>

⁴² Submission by the University of Washington School of Public Health. <https://www.ohchr.org/sites/default/files/2022-05/university-of-washington-school-of-public-health.pdf>



Figure 5-27: The largest mound in Suriname, Hertenerits (left) and Pre-Columbian raised fields visible in a flooded field

Source: (Versteeg A. H., 2008) (Rostain, 2010)

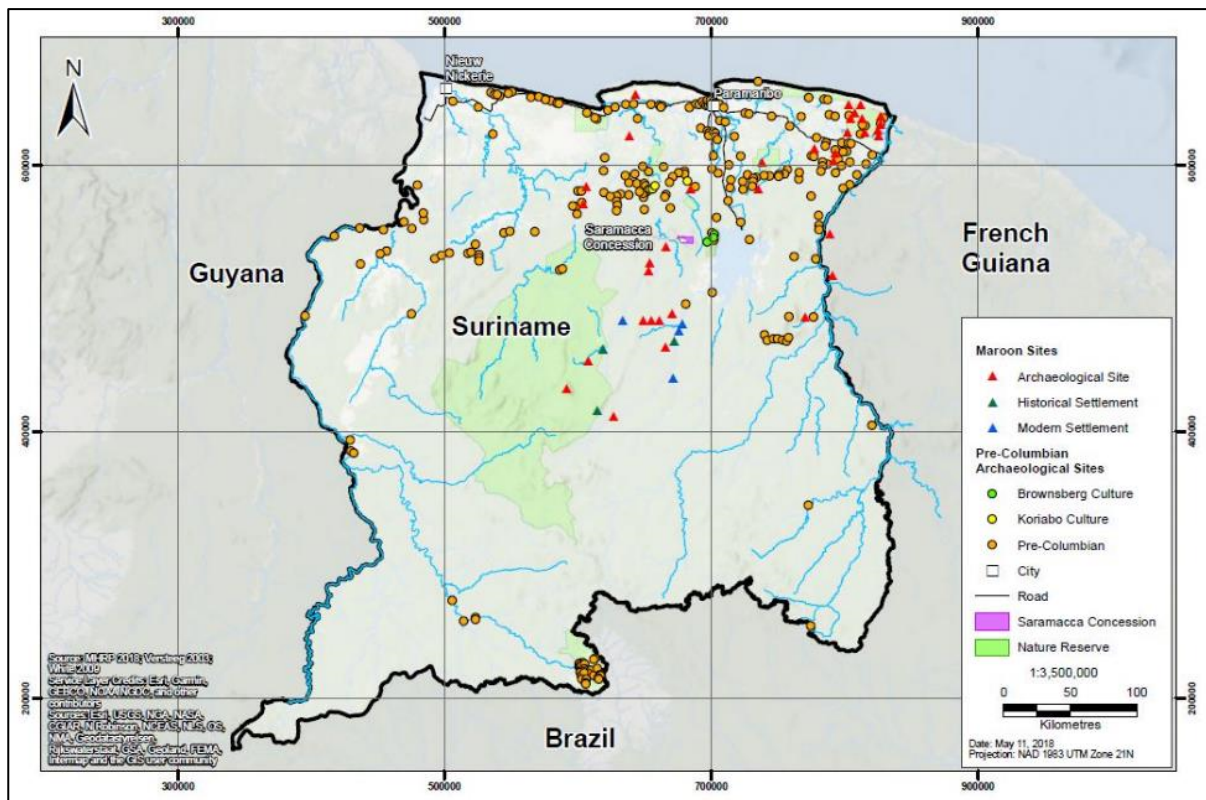


Figure 5-28: Previously Recorded Archaeological Sites in Suriname

Source: (ERM, 2018)

These sites include Sipaliwini Period (8,000 B.C. – A.D. 1,000) small open air and rock shelter camp sites in the Sipaliwini Savanna across the southern third of the country and isolate stone tool finds along interior rivers; Brownsberg and Koriabo period village sites located on high, sandy banks of rivers and creeks in the middle of the

country; and Buckleburg (A.D. 300-700) and Hertenrits (A.D. 700-1,250) villages built on artificial mounds and raised agricultural fields along the coastal plain. (Rostain, 2008) (Versteeg A. H., 2003) (Versteeg A. H., 2008).

During the 19th and early 20th century, the industrialization of the agricultural and mining sectors resulted in the establishment of towns, industrial facilities, mines, and associated infrastructure into the interior of Suriname. The ruins of the Marenburg Sugar Factory provide an example of a late 19th century through early 20th century archaeological site consisting of the ruined sugar factory and the plantation railway, the oldest railway in Suriname (Kautzor, 2014a); see Figure 5-29.



Figure 5-29: Abandoned sugar mill (left) and train engine (right) - Marenburg Sugar Factory

Source: (Kautzor, 2014a)

The most significant built heritage resource in the country is the Historic Inner City of Paramaribo, which is inscribed on United Nations Educational, Scientific and Cultural Organization (UNESCO) World Heritage List. This built heritage resource consists of mainly wooden buildings, many dating from the 17th through 18th centuries, which stylistically blend European architectural, North American, and local Creole culture architecture (Figure 5-30). Individual buildings within the district include the Fort Zeelandia Area (18th century military fort), the current Presidential Palace, St. Petrus en Paulus Roman Catholic Cathedral, and several historic homes (UNESCO, 2002). The listing of these resources by the Surinamese government demonstrates the historic and architectural significance of colonial period-built heritage in the country (SEARCH, 2022).



Figure 5-30: Example of Built Heritage Resources in Paramaribo - World Heritage Site

Source: (UNESCO, 2022a)

In addition to the Historic Inner City of Paramaribo, the Surinamese government has nominated the settlement of Joden Savanne and Cassipora Cemetery for listing on the UNESCO World Heritage List. The site is located south of

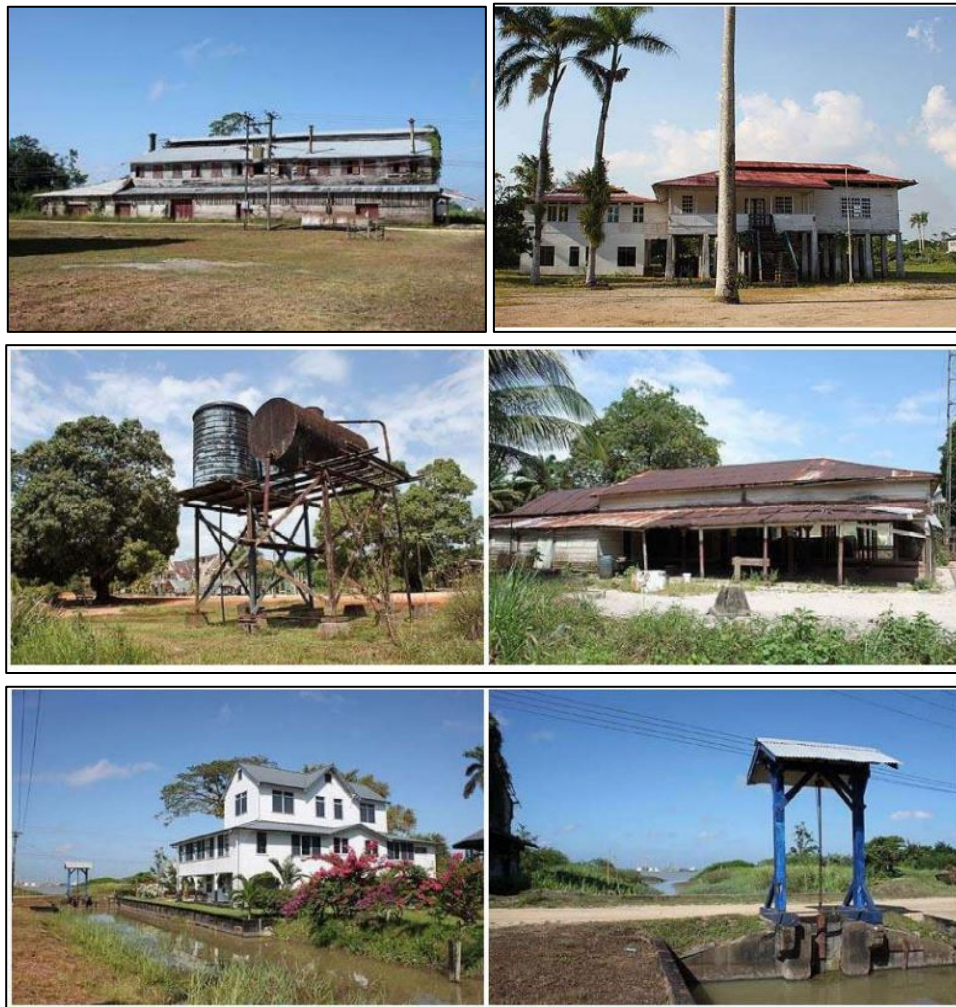
Paramaribo on the Suriname River and is the location of the earliest Jewish community in the western hemisphere. Jews fleeing the Spanish Inquisition founded the settlement in 1665, a flourishing agricultural community, and built a synagogue in 1685, owning multiple plantations throughout the 17th and 18th centuries. The site was abandoned in the early 19th century when the Jewish community relocated to Paramaribo. Today the site consists of the ruins of the synagogue, several buildings, and two cemeteries (see Figure 5-31) containing 450 marked graves (UNESCO, 2022b).



Figure 5-31: Ruins at Joden Savanne and Cassipora Cemetery. Synagogue ruins (left) and cemetery (right)

Source: (Jodensavanne Foundation, 2022)

In addition to these two internationally significant sites, the built heritage of Suriname includes diverse statues and monuments associated with historic events as well as the diverse ethnic groups who emigrated to the country at various times. While historic buildings, structures, and monuments in urban settings are unlikely to be subject to impacts from mining development, built heritage resources associated with historic economic and social activity in rural or areas are more likely to be impacted by mining projects. (SEARCH, 2022). Examples of these types of resources include historic sugar refineries, historic plantation homes and offices, and historic transportation infrastructure such as railways and canals, all of which played a significant role in the history and development of Suriname, and are significant for the historic, architectural, and aesthetic value (Figure 5-32).



Notes: Sugar refinery (upper left), Plantation House (upper right), Railway Water Tanks (middle left), Railway Station (middle right), Plantation House and Canal (lower left), and Canal (lower right)

Figure 5-32: Examples of 19th and 20th century Built Heritage in Suriname.

Source: (Kautzor, 2014b) (Kautzor, 2014a)

Other cultural heritage aspect are the living heritage resources, which include buildings, structures, places, and/or natural landscape features actively used by stakeholders as part of a living cultural tradition, often associated with religious beliefs, practices, ceremonies, or rituals; folklore or oral history; and/or mythical or historic events. Living heritage resources associated with emigrant and Indigenous populations can be found across Suriname. In urban areas, religious structures such as churches, mosques, synagogues, temples, and cemeteries used by the resident Christian, Muslim, Jewish, and Hindu populations are an important component of Suriname's Built Heritage. These resources are typically located in urban centers or villages, away from mining developments. Cemeteries, however, can be found in rural settings and typically are significant living heritage resources for local communities, providing a visible link to their ancestors and providing a sense of belonging to a location or place (SEARCH, 2022).

Living cultural heritage

Living heritage resources of the Indigenous and Maroon populations of Suriname are more likely to be found in rural areas that could be potentially impacted by mining activities. The Indigenous population of Suriname is typically divided into five Indigenous groups. The Lokono (Arawaks), Kalinya (Caribs), and Warao peoples are found in the coastal plain while the Trio (Tiryo, Tareno) and Wayana reside in southern Suriname (ACT, 2015). There are also descendants of other Indigenous peoples who are no longer present today, among others the Akoerio, the Apalai, the Wai-Wai, the Okomoyana, the Mawayana, the Katuena and the Tunayana. Of these peoples, only a few remain in Suriname (VIDS, 2020).

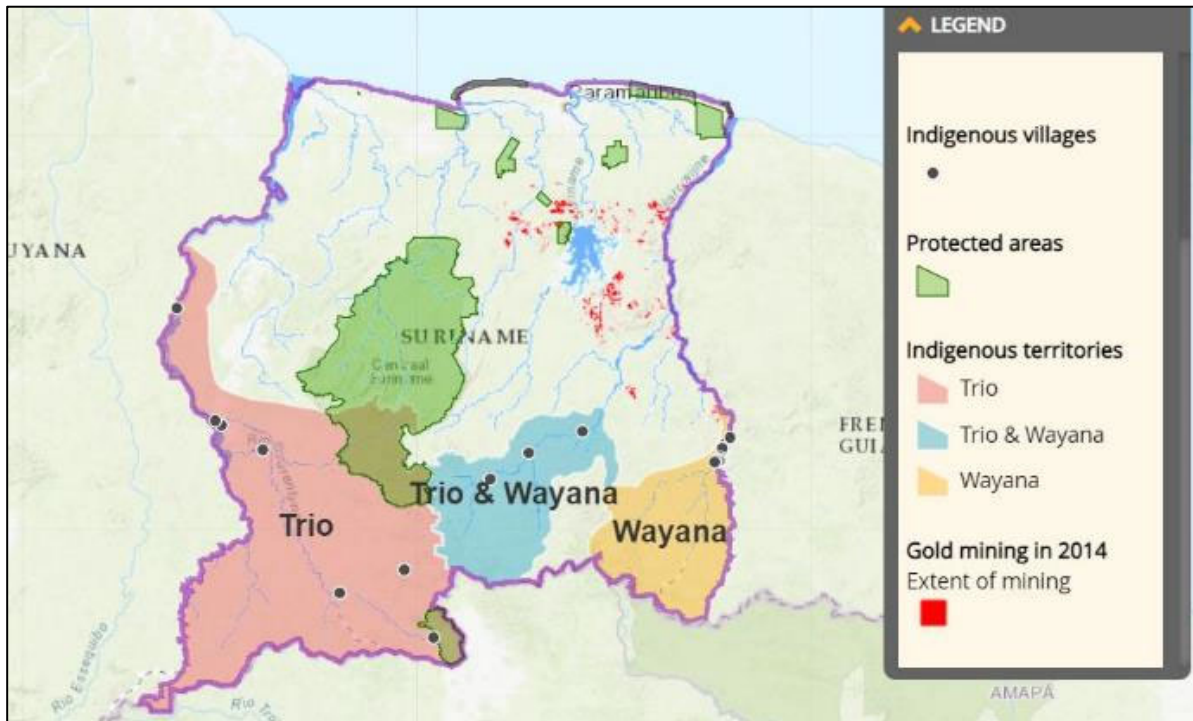


Figure 5-33: Indigenous Territories

Source: (ACT, 2015)

The Indigenous groups living in the coastal plain are located outside of the areas targeted for goldmine development, although living heritage sites associated with these peoples could be impacted by associated infrastructure development. The Wayana peoples in the southeast of Suriname, however, are located within or near potential goldmining areas. Wayana communities live in small remote villages often lead by a piyai or shaman centered around a community house referred to as “tukusipan” (Figure 5-34). These traditional community houses are important living heritage resources that play a central role in community life (Heemskerk et al. 2007). In addition to the tukusipan, local cemeteries associated with Wayana communities should be considered locally significant/important living heritage resources (SEARCH, 2022).



Figure 5-34: Traditional Wayana Tukusipan

Source: (Heemskerk et al. 2007).

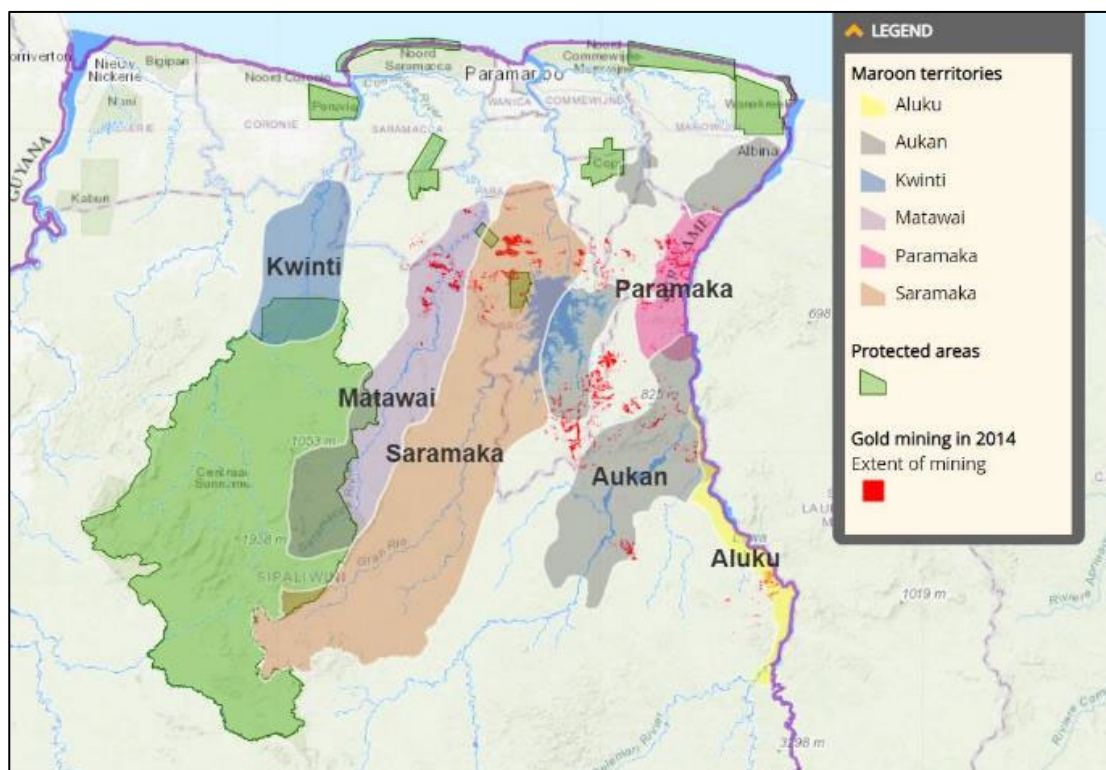


Figure 5-35: Maroon Communities in Suriname

Source: (ACT, 2015).

Note: Extent of mining: Evaluation of deforestation caused by gold mining using remote sensing - Amazon Gold Rush: Gold Mining in Suriname (amazonteam.org)

The Maroon population of Suriname is divided into six communities: Aluku, Aukan, Kwinti, Matawai, Paramak, and Saramaka. Figure 5-35 depicts the traditional territorial affiliations of these six communities relative to the extent of the goldmining industry in 2014, demonstrating that much of the modern mining industry is found in the traditional territories of the Matawai, Saramaka, Paramaka, Aluku, and Aukan communities (ACT, 2015). Natural landscape features such as mountains, creeks, rivers, and forests play a significant role in Maroon oral history, (as locations of former villages or battlefields during colonial period conflicts), as well as areas of ritual practice and belief. These places and features function as important living heritage resources as part of the living cultural tradition of these communities (ACT, 2016).

Based on the above descriptions of Cultural and Archaeological heritage of Suriname, in the rural areas these are at risk from the impacts of potential mining developments. A critical component of the SESA will be to develop policies and frameworks for the identification of physical cultural resources that are targeted and scaled to the risks presented by the development of the gold mining industry in Suriname, and the development of standard approaches to avoiding, minimizing, and/or mitigating potential impacts that are informed through stakeholder consultations.

5.4.9 Gender

According to the Global Gender Gap Index for 2018 (WEF, 2018), Suriname scores well in the field of education with a gender parity of 0.991, and in the field of health with a gender parity 0.973. But in the economic participation and opportunities the gender parity the score is only 0.638 (GOS, 2022a). Some of the areas in which women are lagging behind compared to their male peers are: payment, labor market participation, unemployment, child labor, engagement in sectors with higher wages and completion of vocational training (GOS, 2022a).

Results from the Living Conditions Survey in Suriname carried out in 2016-2017 show that the women's labor force participation is 43% compared to 68% of men within an urban setting. Only 1% of the men's labor force (those who have the potential to work, aged 15-64) are engaged in unpaid care and domestic work, while this represents 19% for their female counterparts. Additionally, about 98% of those engaged in unpaid care and domestic work are women (GOS, 2022a).

6 OVERVIEW OF THE MINING SECTOR OF SURINAME

The extractive industries play a central role in Suriname's economy. Bauxite has been the dominant industry for most of the past century, although extraction and exports have come to a halt recently as accessible reserves were exhausted. Gold has emerged as the primary extractive industry, including private industrial gold mines and substantial artisanal and small-scale gold mining (ASGM) operations that represent a significant source of employment. The state oil company Staatsolie has been successfully extracting low-cost onshore oil for decades, with new prospects for offshore oil recently identified. (WBG, 2017). In Suriname, the mining of minerals includes (EITI, 2019b):

- Gold and silver based on the Mining Decree 1986 and Mineral Agreements between Government and companies;
- Diamond mining (still in the stages of reconnaissance and therefore not commercialized yet);
- Bauxite (production of bauxite ended in 2015);
- Quarry aggregates for building purposes;
- Other ores (such as kaolin, manganese, chromium, tin, phosphate and silica sand).

The extractives sector drives Suriname's economy, however, it is highly concentrated with potential for increased revenue generation from mining. Suriname is located on the Guiana Shield geological formation that has significant mineral resources. One main bauxite extraction operation dominated the sector for the past 100 years, but operations ended in 2015 after accessible reserves were exhausted. Two new significant gold mine operations were established in the past decade (2009-2019). The mining sector also includes significant informal artisanal and small-scale operations and domestic firms supplying goods and services to extractive operations. Despite its economic importance, however, there are significant opportunities for the GOS to improve sector governance and revenue generation from mining, given limitations in sector revenue controls and traceability. (IBRD, 2019)

The majority of mining is taking place in Suriname's Greenstone Belt, which covers 24,000 km² of Central-East Suriname (SS, 2017). The location of the Greenstone Belt is shown in Figure 6-1 with other key spatial features such as the location of the industrial gold mining sites (Rosebel and Merian mines).

The interior of the country is mineral-rich containing a number of important deposits of bauxite, beryl, copper, diamonds, gold, platinum, iron ore, manganese, pegmatite and tin; exploitation of minerals mainly concentrates on bauxite and gold. Historically, extraction activities of mineral ores mainly occurred in the northern parts of the Interior, but mining activities have increasingly been moving further south over the last decennia. (Berrenstein, 2016)

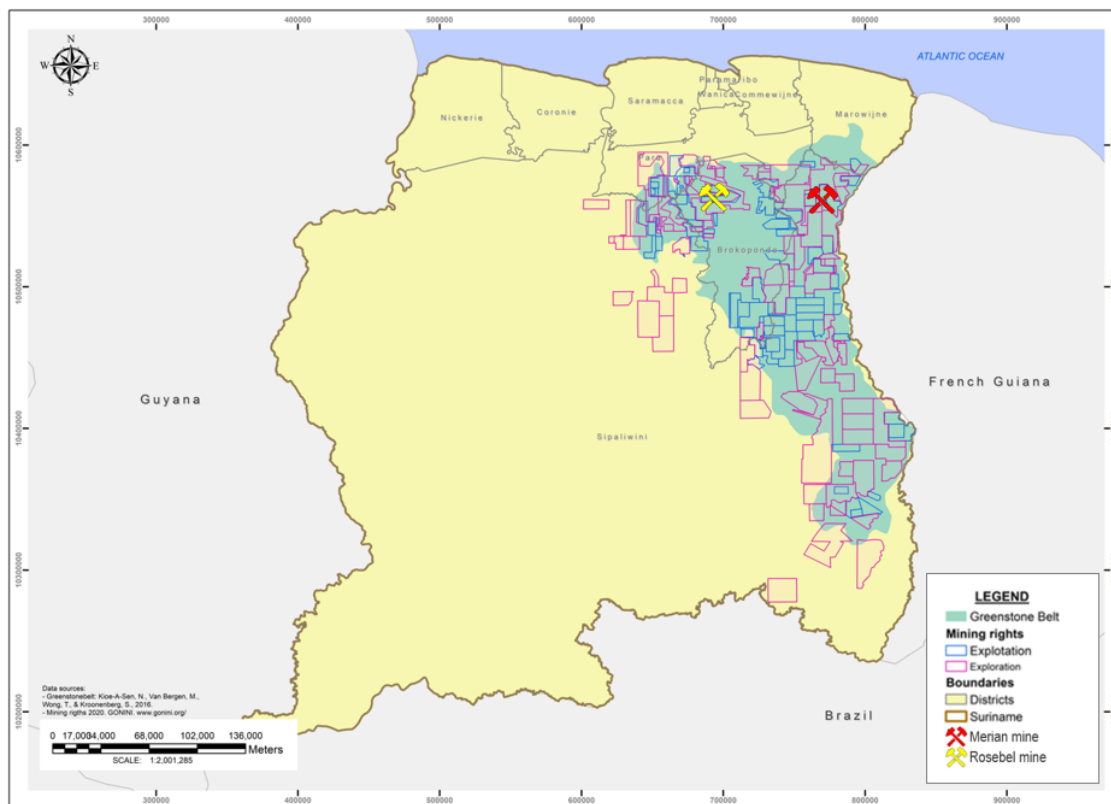


Figure 6-1: Greenstone belt, mining titles and current Industrial Mining sites operating

Source: (GOS, 2022b) and (ACT, 2015) adapted by RINA, 2022

Weak legal and regulatory frameworks and limited institutional capacity hinder economic development, diversification, and sector governance. In mining, the current 1986 decree governing the sector has not been complemented with implementing regulations. Management of mining titles and licensing procedures are far from good practice, resulting in lack of clarity, weak transparency, and risks to property rights. Institutional capacity to monitor the sector and enforce laws and regulations is weak; the Geological Mining Service responsible for monitoring mining title holders, for example, has almost no masters-trained geologists. These challenges in sector regulation and monitoring contribute to limited GOS revenue generation from the sector. (IBRD, 2019)

There are also gaps in the regulatory framework to manage environmental and social impacts of economic activities, creating risks for sustainable development and for new private investors. Within mining, this creates risks for deforestation of the country's vast forest coverage (more than 90 percent of national territory) due to illegal gold mining, as well as widespread contaminative usage of mercury for gold extraction by artisanal and small-scale miners. There are Indigenous and tribal populations living in areas with mineral potential, which often have land rights conflicts with informal (and formal) mining operations. Lack of transparency and capacity to monitor and enforce environmental and social regulations can encourage a race to the bottom across sectors, to the detriment of formally established firms seeking to comply with global standards such as Responsible Agricultural Investment, Global Sustainable Tourism Criteria, and IFC Performance Standards on Social & Environmental Sustainability (IBRD, 2019).

A general analysis of macro-economic development outlook of the mining sector in Suriname is presented in Appendix B. In economic terms the importance of the mining sector in Suriname is highlighted, providing some economic and business environment data of the country, such as imports and exports, gross domestic product (GDP), national integration planning and development mechanisms. In addition, the appendix provides an overview of the resources sectors with focus on mining and the key main production clusters (industrial and ASM gold mining), and finally a general analysis of current status and development or relevant infrastructure to the sector (transport and energy). It is important to note that Appendix B provides the background for the following sections of this report.

6.1 FUTURE MINING DEVELOPMENT SCENARIOS

Scenarios are stories that may represent the future. Good scenarios are plausible, challenging and rigorously constructed to address the most critical questions that decision-makers need to face.

One cannot expect any given scenario to come true as it stands. They are not predictions, preferences, or forecasts. Rather the process of developing and using scenarios is intended to help participants learn and generate insights, both from exploring each scenario individually and from comparing and contrasting them.

Based on this precept the following forecast of possible mining scenarios is put forward as a starting point. There are three main strands to any analysis.

1. The first is the external, or global, environment faced by Suriname, its mining sector, and the different minerals in the world context. These external factors are largely outside of the influence of Suriname, and include commodity prices, substitution, climate change, and technological developments.
2. The second is the internal, or Suriname, environment and how it pertains to the mining industry and, ultimately, to a particular scenario. These internal factors should be within the ability of Suriname to change, or at least modify, given political will and/or a clear action plan as an outcome of the strategic options chosen out of the analysis of the potential scenarios. These include, political, economic, social, technological, environmental and legal.
3. The third strand is the integration and use of outcomes, conclusions and analysis from previous studies and reports put together over the past two decades on Suriname. These sources are an important resource that gives detail, depth and source variety to the analysis from the point of view of the geology, economics, policy, legal, environmental, social, and other contexts.

6.1.1 Sector 1: Bauxite

Restart of the bauxite industry that closed in 2015. This scenario has significant upsides if it is done in conjunction with the opportunities presented in the Oil and Gas sector in Suriname or by the significant gas finds in neighboring Guyana to produce low-cost electricity for the bauxite-alumina-aluminum processing and smelting. The recent closure of the bauxite industry after more than 100 years of mining and processing should have left a significant technological legacy in Suriname that could be reengaged and, since the international aluminum players have left the country, new contracts can be negotiated with them, if required.

There are significant challenges to this scenario. Startup costs will be high, with new infrastructure such as a gas-fired power station, refurbishing of railroad infrastructure and the Paranam plant. Environmental concerns based on the bauxite deposits being in heavily forested areas, will be a significant concern. And the definition of bauxite reserves and consequent mine feasibility studies and their ultimate construction will be high cost and will require many years before production could resume.

Reasons for closure in 2015

- Exhaustion of coastal bauxite reserves

- High cost of electricity required to produce alumina, and smelting to produce aluminum
- High cost of shipping bulky bauxite
- Decline in world aluminum prices.

Challenges to restarting bauxite industry:

- Exploration would have to be restarted and reserves would have to be defined and restated
- Bauxite deposits are inland from the coastal plain with significant potential for environmental impacts (forest, biodiversity, water quality)
- Bauxite deposits are of lower grade
- Infrastructure dedicated to the bauxite industry through 2015 will have to be re-evaluated and recommissioned (railways, Paranam plant)
- Cost for restart of industry.

Opportunities in restarting bauxite industry

- Suriname has significant bauxite reserves
- Potential for cheaper electricity from gas because of potentially large offshore gas finds, based on finds in neighboring Guyana, hence lower refining and smelting costs
- Suriname has a stated intention to restart the bauxite industry
- Suriname has over 100 years of experience in bauxite mining with the potential to restart without major aluminum mining companies
- New contracts can be negotiated with mining companies
- Suriname has deep coastal waters, which is favorable for bulk shipping.

6.1.2 Sector 2: Large-scale Gold Mining

Large-scale mining is currently undertaken by two mining companies, namely, IAMGOLD (Rosebel mine) and Newmont (Merian mine). Gold discoveries in Greenstone belts are typically low tonnage, with small reserves, although they can have a long mining life, a result of the geology of these deposits and complex exploration required to define ore reserves. Although they are typically mines with good grades, these characteristics make them difficult financing propositions for their owners, and they are typically owned by mid-tier and small-tier mining companies. The potential for further discoveries of this type of mines typically requires diligent exploration over a number of years by dedicated exploration teams.

Their compliance with environmental, social and closure, as well as being regulated by the national legal code, are generally also guided by international reporting guidelines and requirements, frequently driven by financing or stock exchange disclosure requirements.

These two large-scale mining companies have both been involved with litigation over land rights and land use issues with local inhabitants (see section 5.3), which highlights a significant challenge for the government over land use and rights.

Another challenge is the implementation of adequate environmental legislation, including legislation covering the closure of these facilities at the end of their commercial lives, which is seen as a legacy of the bauxite mining industry, for example.

The use of gravity circuits instead of mercury potentially holds valuable lessons and requires skills' development that might well cross over to the small-scale mining sector. Nevertheless, this will likely require enabling legislation or tax breaks to achieve. The sharing of knowledge and skills in the

management of deforestation, sediments and soils, and better management of water quality may eventually trickle down to small-scale mining.

This scenario is less guided by external (global) issues than by the implementation of adequate legislation and its enforcement, and possibly reforms that look to encourage exploration and retain the geological knowledge base in-country when a company withdraws from its prospecting areas.

This scenario therefore needs to look at its interface with political, economic, social, technological, environmental and legal factors internal to Suriname.

6.1.3 Sector 3: Formalization of Small-scale and Artisanal Mining

This scenario entails a detailed evaluation of the small-scale and artisanal mining sector from the perspective of the small-scale miners. As noted by Heemskerk and van der Kooye, 2002, “the socio-cultural and economic lives of small-scale gold miners merit more attention. Without incorporating the opinions, beliefs, and life realities of local miners, policy interventions aimed at changing the behavior of these miners are doomed to fail.”

From this perspective a full business-type analysis should be undertaken to determine the required strategy for the sector, with the intention -not of achieving perfection- but achieving meaningful progress in the particularly important social and environmental challenges around this industry sector.

The significant volume of in-country studies and reports, and the international library of case studies and knowledge base should be utilized for this purpose.

Success will need to be measured on the basis of policy implementation and meaningful key performance indicators (KPIs). It should be noted that formalization and legality are two different concepts that need to be clarified and well understood.

6.2 MINING DEVELOPMENT SCENARIOS

6.2.1 Introduction

As stated in the Inception Report, the proposal is that the analysis of the sector development be based on two scenarios, a base case (conservative) and accelerated (optimistic) scenarios. The analysis will be undertaken using assumptions (i.e., discoveries, resource studies, exploration results, Suriname's macroeconomic development, ease of doing business/export environment, transport infrastructure and services) and supporting factors (i.e., global demand for minerals, carbon credit opportunities). Once the scenarios are agreed by the client and validated by the stakeholders, the identification of environmental and social risks will be developed around these scenarios. These risks will include cumulative and indirect impacts by studying the possible location of large- and small-scale mining sites, and transport infrastructure and services:

- Current developments in the infrastructure sector (ports, highways, local roads, possible domestic/national aviation);
- Current constraints and opportunities; and
- Infrastructure needed to support the mining development program (capacity, investment costs, timing).

RINA has been reviewing existing policies, programs, plans, studies, and public information material, such as:

- National and sectoral laws and regulations, national or regional plans, future mining development;
- Existing public or private programs or initiatives, current projects, plans and trends;

- Contextual information at the country, sub-national or sector levels (based on screening and targeted online searches);
- Available remote sensing imagery and geographic data to understand the environmental and social constraints. These sites are but not limited to:
 - Integrated Biodiversity Assessment Tool (IBAT) - <https://www.ibat-alliance.org>
 - KOPI – statistical data for the National Forest Monitoring System – managed by the Foundation for Forest Management and Production Control (SBB) <https://kopi.sbb.sr/>
 - Suriname Water Resources Information System (WSRIS) - <https://www.swris.sr/>
 - Global Lithological Map (GLiM) - <https://projekte.cen.uni-hamburg.de/lithomap/>
 - National Land Monitoring System of Suriname - <https://www.gonini.org/>
- Cultural Heritage;
- Scientific literature; and
- International mining standards (ICMM, Cyanide Code).

6.2.2 Definition and Assumptions

The **Base Case Scenario** is a conservative scenario where timing is not considered to be a driving factor, potential outcomes can therefore be more fully considered and associated risks can be managed with the extra time available.

The **Accelerated Case Scenario** is where timing is a significant consideration and the integration of parts of the subsectors is not contingent on the alignment of all aspects before investment decisions are made. This scenario appears to be contingent on the early availability of cross funding from the Oil and Gas industry.

6.2.3 Preliminary Analysis

The preliminary analysis for both scenarios is presented in Table 6-1 for the Base Case (Conservative) Scenario and Table 6-2 for the Accelerated (Optimistic) Scenario.

Table 6-1: Base Case Scenario – Mining Sector Development

Sector	Constraints	Opportunities
Large Scale Bauxite and Alumina (mining, smelting/refining)	1. Infrastructure*: • Several constraints need to be considered, including access to surface land for development of new infrastructure (mine areas, road and rail infrastructure, port facilities, power plant, processing facilities), access to energy and or course financing.	• Development of Oil and Gas sector goes hand in hand with development of bauxite sector: Cheap electricity, cross funding from fiscal returns. • The national Oil and Gas company Staatsolie Maatschappij Suriname N.V. is involved in from exploration to extraction, production and refining. It is constantly increasing its production, which in 2022 shows a growth of over 50% compared to 2021
	2. Capacity: • Development of related legal and fiscal frameworks, including environmental legal framework, closure	

Sector	Constraints	Opportunities
	legislation, and legislation on land ownership and rights. 3. Investment Cost: - Costs of restarting the bauxite subsector would be high (> USD 3 billion). 4. Timing: - Time required to restart the subsector would be in the order of 10 years.	(Staatsolie Maatschappij Suriname N.V. , 2022) - Suriname has a stated intention to restart the bauxite industry. - Potential to impact other development goals in Suriname. - Suriname has significant bauxite reserves. - Over 100 years of experience in bauxite mining with the potential to restart without major aluminum mining companies. - Deep coastal waters (offshore) that are favorable for bulk shipping. - New contracts could be negotiated with mining companies and financial risks shared with other players.
Large Scale Gold Mining Sector	1. Infrastructure*: - minimal additional regional infrastructure requirements necessary, however some improvement in transport services may be required. - Litigation over land rights and land use issues with local inhabitants. 2. Capacity: - Increased interest and participation in the sector would require assessment and possible changes to legal, fiscal and land ownership frameworks. 3. Investment Cost: - Investment costs borne by mining companies might require legislation to internalize costs associated with mine closure and other potential environmental and social impacts. 4. Timing:	- Adequate time available for exploration and discovery of new deposits in Greenstone belt geological environment – continuation of current situation. - Chance to implement cross over of lessons, skills and legislation to ASGM sector, including environmental issues such as decreasing the use of mercury, and better management of soil, sediment, water, deforestation and biodiversity impacts.

Sector	Constraints	Opportunities
	• Ongoing, no real time constraints.	
Artisanal and Small-scale Gold Mining (ASGM)	1. Infrastructure*: • Infrastructure required in the areas where mining is taking place, including provision of basic services such as sanitation, power, health services and other development requirements. • Transport infrastructure improvements (rivers, airfields, roads) required as well as potential improvements in transport services (rural airstrip services) and river vessel services/stellings.	• Ability to affect some of Suriname's development goals. • Chance to implement cross over of lessons, skills and legislation to ASGM sector, including environmental issues such as decreasing the use of mercury, and better management of soil, sediment, water, deforestation and biodiversity impacts.
	2. Capacity: • Enabling legislation around land reform. • Local capacity building • Attention to environmental issues – phasing-out and eliminating the use of mercury, better management of soil, sediment, water, deforestation and biodiversity impacts.	
	3. Investment Cost: • Investment costs can be seen as economic and social development costs, potentially constrained by budget and policy.	
	4. Timing: • Timing associated with development goals, enabling legislation and budgetary availability.	

* Infrastructure needed to support the mining development program which can have environmental and social impacts

Source: RINA, 2022

Table 6-2: Accelerated (Optimistic) Case Scenario – Mining Sector Development

Sector	Constraints	Opportunities
Large Scale Bauxite and Alumina (mining, smelting/refining)	1. Infrastructure*: - Infrastructure dedicated to the bauxite industry through 2015 will have to be re-evaluated and recommissioned, on accelerated schedule (railways, Paranam plant). - Port infrastructure will need to be evaluated, designed and built, on accelerated schedule. - Energy supply for alumina and possibly aluminum production will have to be evaluated, designed and commissioned.	- Oil and Gas sector is set to take off rapidly with the discovery in neighboring Guyana of significant offshore gas reserves. Development of Oil and Gas sector takes place hand in hand with development of bauxite sector with accelerated cross funding from O&G fiscal returns. - Cheap electricity generation from gas-fired power station. - Suriname has a stated intention to restart the bauxite industry, potentially without the participation of international mining companies. - Potential to impact other development goals in Suriname in the medium term. - Suriname has significant bauxite reserves. - Over 100 years of experience in bauxite mining with the potential to restart without major aluminum mining companies. - Deep coastal waters (offshore) that are favorable for bulk shipping.
	2. Capacity: Accelerated development of related legal and fiscal frameworks, including environmental legal framework, closure legislation, and legislation on land ownership and rights would be required.	
	3. Investment Cost: - New contracts would not necessarily be negotiated with mining companies while financial risks would not be shared with other players but taken on by the Suriname government. Nevertheless, the GOS does not have the financial capacity to undertake such projects on its own. Therefore, a feasible option is that the GOS work in a joint venture with multinationals for new bauxite exploration and exploitation contracts as to restart the bauxite industry. - Costs of restarting the bauxite subsector would be high (>USD 3 billion) and would be disbursed over a relatively short time frame.	

Sector	Constraints	Opportunities
	4. Timing: • Exploration would have to be restarted and reserves would have to be defined and restated (> 2 years). Lower standards of reserve definition might be required, with associated risks. • Engineering feasibility studies for new mines and environmental permitting would take a considerable time to be completed (> 5 years). Permitting would have to be streamlined without affecting environmental and social protections.	
Large Scale Gold Mining Sector	1. Infrastructure*: • minimal additional regional infrastructure requirements necessary, however some improvement in transport services may be required. • Litigation and conflict over land rights and land use issues with local inhabitants associated with the development of mining infrastructure, adds to country risk and could negatively influence mining company investment decisions and entry of new players.	• Opportunity to accelerate implementation of crossover of lessons, skills and legislation to ASGM sector, including environmental issues such as decreasing the use of mercury, and better management of soil, sediment, water, deforestation and biodiversity impacts. This would likely require enabling legislation.
	2. Capacity: • Increased interest and participation in the sector would require assessment and possible changes to legal, fiscal and land ownership frameworks in the short term.	
	3. Investment Cost: • Investment costs borne by mining companies require legislation to ensure that costs associated with mine closure and other potential environmental and social impacts are internalized.	

Sector	Constraints	Opportunities
	4. Timing: Timing constraints are associated with the time needed to enable environmental, mine closure and land reform legislation and potentially changes to the tax regime.	
Artisanal and Small-scale Gold Mining (ASGM)	1. Infrastructure*: - Infrastructure required in the areas where mining is taking place, including provision of basic services such as sanitation, power, health services and other development requirements would require access to budget. - Transport infrastructure improvements (rivers, airfields, roads) required improvements in transport services (rural airstrip services) and river vessel services/stellings.	- Cross funding in the short term could be available from potential revenues from the Oil and Gas sector. - Ability to affect some of Suriname's development goals in short term. - Chance to accelerate the implementation of the crossover of lessons, skills and legislation to ASGM sector, including environmental issues such as decreasing the use of mercury, and better management of soil, sediment, water, deforestation and biodiversity impacts.
	2. Capacity: - Enabling legislation around land reform would have to be accelerated. - Local capacity building. - Attention to environmental issues – eliminating the use of mercury, better management of soil, sediment, water, deforestation and biodiversity impacts.	
	3. Investment Cost: Costs would require early access to the necessary budget, which might not be available if potential Oil and Gas revenues are not realized in the short term.	
	4. Timing: Timing would be constrained by enabling legislation and access to budget.	

Sector	Constraints	Opportunities
* Infrastructure needed to support the mining development program which can have environmental and social impacts		

Source: RINA, 2022

6.3 POTENTIAL CONFLICTS AROUND MINING

Suriname's forests provide housing sustenance to local Indigenous and tribal peoples. These peoples can be classified as Amerindians and Maroons, respectively (Kambel and MacKay 1999, in Heemskerk and van der Kooye, 2002). Maroons are the descendants of runaway African slaves, who established independent communities in the rainforest in the 16th and 17th centuries. These Indigenous and tribal peoples have maintained relative political, social and economic sovereignty within the nation state. However, their access to land, natural resources and self-determination is vulnerable because the government has refused to recognize their individual or group property rights (Kambel and MacKay 1999, Heemskerk and van der Kooye, 2002). Nevertheless, Article 41 of the Constitution of the Republic of Suriname (GOS, 2000) clearly states that, "Natural riches and resources are the property of the nation and shall be used to promote economic, social and cultural development. The nation has the inalienable right to take complete possession of its natural resources in order to utilize them to the benefit of the economic, social and cultural development of Suriname."

Conflicts around mining are likely to revolve around some of the same issues that have proved conflictive in the past. Examples of conflicts around the issue of mining concessions and the rights of traditional communities are the cases of IAMGOLD and the Ndyuka Maroon community of Nieuw Koffiekamp, and the Merian and Paramaka traditional territory (WBG, June 2019c).

In general, some conflicts around gold mining include (GOMIAN, 2011):

- Between LSM and SSM: Multinationals have acquired concessions in the areas of Gros Rosebel (Iam Gold) and Meriam (Newmont), where SSM have been –and still are- working first. The SSM have been removed with force and threats of force.
- Among local authorities: Some village chiefs allow miners to work on their village lands against payment of a fee or other form of material compensation. Not all traditional leaders agree with these practices. In the Ndyuka territory, village heads (kapiteins) were reprimanded by the paramount chief (granman) for allowing mining in the Mama Ndjuka creek, an important ceremonial ancestral site.
- Between local authorities and community members: Local authorities do not equally distribute the compensation they receive from miners working on village land to all villagers who are affected. This creates tension.
- Among SSM (local-local; local-migrant; migrant-migrant): Non-gold related conflicts (i.e., over women) seem to be the most likely causes of physical fights. Yet conflicts also exist over mining rights, access to land, payment, etc. For example, in some areas tribal leaders have decided that all migrant miners need to leave to allow local men to earn a living.
- Between SSM and park management (Brownsberg national park): Miners are working in the national park, which is not in agreement with the park management guidelines: Between SSM and national government: related to all of the above, and primarily to access to land and mining rights.

6.3.1 IAMGOLD and Ndyuka Maroon community of Nieuw Koffiekamp

In 1992, the Suriname state granted the Canadian exploration firm Golden Star Resources (GSR), exclusive rights to the 17,000 ha Rosebel gold concession in Brokopondo district, 85 km south of Paramaribo city. The Ndyuka Maroon community of Nieuw Koffiekamp and the lands used by villages

for subsistence activities were situated within this concession. The villagers were never consulted and only informed after the fact. Neither was the community of Nieuwe Koffiekamp mentioned in the application of Golden Star for an exploration concession, as required by law (IBP US, 2009 in (WBG, June 2019c)).

It was proposed that the community would be relocated, but for the people from Nieuw Koffiekamp this was unacceptable. The community had already been relocated from its original community of Koffiekamp in the early 1960s, to make place for a hydropower dam. That incidence had been very traumatic, leading to the loss of traditional lands, livelihood, and important cultural heritage sites. The government constructed “trans-migration villages” that were not constructed in line with the culture and customs of the Maroon people, and lacked basic infrastructure (drinking water, electricity).

Complicating the matter was that small-scale gold miners were already active in the area, many of whom from Nieuw Koffiekamp. These gold miners found –and find- it extremely insulting that they are considered “illegals” in their own home community. After the Rosebel concession was granted to GSR, even more small-scale miners moved to the area. GSR threatened to abandon the project if the miners were not removed and in response, the Minister of Justice and Police threatened with violence if the miners did not immediately vacate the area. Negotiations between the government and the community did not generate satisfactory results. According to human rights organization Moiwana '86, at least eight violations of the American Convention on Human Rights were committed due to company action and government inaction (MacKay, 2002, in (WBG, June 2019c)).

In 2001, the Canadian mining company Cambior acquired the Rosebel gold mine and in 2006, IamGold took over Cambior and thereby became the owner of Rosebel Gold Mines N.V. Since the commencement of negotiations between the multinational mining companies, the Suriname government and the community, periods of relative calm have alternated with fierce protests, roadblocks and violent outbursts by community members –often small-scale gold miners- against Rosebel Gold Mines N.V. (Theije, et al., 2014). Apart from the fact that people’s livelihoods had been affected, complaints included that a community forest area that was used to collect wood and Non-Timber Forest Products was confiscated; dynamite blasting damaged homes and disturbed elders; and the only access road to the village was regularly blocked by the firm (e.g., during blasting). Grievance mechanisms should have been set into place.

In late 2014, in an effort to finally resolve the conflict in Nieuw Koffiekamp, the Commission for Regulation of the Gold Sector (OGS), ordered small-scale gold miners to leave the IAMGOLD concession by Christmas. As an alternative, the OGS established a “small-scale gold mining reserve” several km from the village and advised the Koffiekamp miners to apply for a legal concession. However, the Koffiekamp gold miners soon discovered that the available concession area was about half the size of the area that had initially been promised, and after prospecting, they were convinced that the area did not have economically viable gold deposits (Plein, Makambo board member, pers. com. 24-02-2015). Moreover, they found the area too far away. As a result, the gold miners did not move to the small-scale gold mining reserve.

In following years, the cat and mouse game between Koffiekamp small-scale gold miners and Rosebel Gold Mines N.V. continued. Inhabitants of the village still experience nuisance and damage as a result of the company’s activities, including dynamite blasting, ever closer to the village. At the same time, the company faces difficult exploration conditions, with small-scale gold miners running into the exploration pit after every dynamite blast – creating an extremely hazardous situation.

In November 2017, IamGold and Makambo signed an agreement that allowed a selected group of Makambo members to mine for gold in a specific section of the Rosebel Concession, named Roma pit. This agreement resolved problems for a small group of small-scale gold miners, but many others continue to venture onto other areas of the RGM concession. Moreover, the villagers still suffer from blasting near their community and severe restrictions on their access to the forest and its resources, including areas for subsistence plots, hunting, and collecting (e.g., medicinal plants) (WBG, June 2019c)

6.3.2 Merian and the Paamaka traditional territory

The only clause in the Mining Code that refers to people living in tribal communities is Article 25, which states that applications for exploitation licenses must include a list of all tribal villages located in or near the requested concession (subsection 1). The inhabitants of the mentioned villages are “obliged to allow the holder of [a mining] right to carry out mining operations ... on land owned or occupied by them: (a) provided that they have been notified on time ... (b) and have been compensated in advance or been given assurance for such compensation.” (Art. 47). It is not required that affected communities be consulted about, participate in, or consent to, the allocation of the concession in question (IDB 2004). Even these minimal provisions are often ignored; mining concessions are frequently granted on tribal lands without prior “notification.” Compensation is insufficient or non-existent. The limited legal protection of traditional communities vis-à-vis external mining interests has caused conflict in several locations, most notably in Nieuw Koffiekamp and the Paamaka traditional territory/Merian.

The Ministry of Natural Resources has indicated that cultural sites are now taken into account in concession allocations. And advice from the Ministry of Spatial Planning, Land and Forest Management (ROGB), the Ministry of Regional Development (RO), and the District commissioners (DC), is now requested prior to any allocation. The DC also conducts a field orientation and delivers a report. Without these advisory documents, mining concessions are not granted (WBG, June 2019c).

7 IDENTIFICATION OF ENVIRONMENTAL AND SOCIAL RISKS AND VULNERABILITIES

This section provides a general description of potential impacts centered in themes (water, air, land use, geology soils, biodiversity, cultural heritage, etc.) and preliminary mitigation measures to be considered during the SESA.

Typical environmental impacts or effects of a mining projects include (WBG, 2017):

- landscape alteration (erosion, formation of sinkholes, and so forth) and pollution of air, soil and water (surface and groundwater),
- development and/or building of related Infrastructure such as roads, railways, port facilities, opening of land areas, construction of buildings, etc.
- mining is a water intensive industry, and the utilization of this resource can be particularly critical in areas where water is scarce or is heavily used by other local activities, such as agriculture or drinking, or is needed by ecosystems.
- decommissioning and closure of mines can also be a source of negative environmental impacts.

In terms of Artisanal and Small-scale Mining (ASM) communities often operate without any environmental safeguards, due to the country's lack of resources to manage environmental (and social) impacts. Therefore, ASM activities contribute either directly or indirectly to environmental degradation, through deforestation, loss of biodiversity, wildlife poaching for bush meat, soil erosion, dust, noise, habitat destruction, introduction of invasive species, siltation, and pollution of water bodies or alterations to rivers' regimes (WBG, 2017). In terms of ASGM, the activity has associated environmental and occupational health issues, particularly when practiced informally or with limited technical and material resources. The health and well-being of miners, their family members as well as nearby communities is often adversely affected (WHO, 2016). The WHO classifies the main hazards associated with ASGM as:

- Chemical hazards associated with the use and/or exposure to mercury, cyanide, exposure to chemicals contained in dust like silica, and toxic gases from blasting.
- Biological hazards associated with the infections from waterborne and vector borne diseases, sexually transmitted infections (HIV/AIDS) and tuberculosis.
- Biomechanical and physical hazards from carrying heavy loads (workers developing musculoskeletal disorders) and repetitive tasks (suffering from overexertion, exposure to vibrations, loud noises, heat, humidity, and radiation).
- Exposure to other hazards like drug and alcohol abuse, violence, nutritional deficits.

The WHO highlights that woman and children in ASGM must be given special consideration since women perform many tasks in and around the ASGM process and are subsequently exposed to many of the environmental and occupational health hazards described previously. The woman can transfer that risk to children during gestation, not to mention the exposure of woman and kids to other hazards such as violence from drug and alcohol abuse (WHO, 2016).

The following table provides a general summary of key preliminary environmental and social risks, impacts and vulnerabilities identified during the Scoping process and literature review. The purpose of the table is to compile most risks and impacts and discuss these in the workshops and during the continuous consultation process of the SESA.

Table 7-1: Preliminary Risks and Impacts and Strategic Objectives

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
Water	• Surface and groundwater pollution from toxic chemicals (e.g., mercury). • Surface and groundwater pollution from toxic materials – e.g., acid mine drainage (AMD) from waste dumps and failure of tailing dams. • Surface and groundwater contamination due to the generation of acid rock drainage (ARD) and metals leaching (ML) containing runoff / leachate, sedimentation of drainage networks. • Mercury contamination in freshwater ecosystems: water, sediments and fish. • Water scarcity and competition with other users and ecosystems. • Climate change increases the risks of droughts and floods, affecting water resources. • Earth-moving operations may mobilize sediment that can enter watercourses and disrupt water quality and quantity. • Sedimentation of natural water bodies.	• Strengthen the institutional framework and capacity of institutions. • Establish and implement sustainable gold mining, without the use of toxic substances such as mercury. • Protect surface water and aquifer resources. • Ensure sustainable water resources management. • Protect and preserve marine water quality. • Prevent conflicts (including transboundary) over water use, quality and quantity. • Enhance and/or maintain the quality and quantity of water resources. • Control water use and avoid potential pollution of water resources to international standards. • Establish acceptable water use and wastewater discharge quality levels. • Establish a water balance (including probable climatic events) for the mine and related process plant circuit and use this to inform infrastructure design. • Develop a Sustainable Water Supply Management Plan to minimize impact to natural systems by managing water use, avoiding depletion of aquifers, and minimizing impacts to water users. • Minimize the amount of make-up water. • Consider reuse, recycling, and treatment of process water where feasible (e.g., return of supernatant from tailings pond to process plant). • Consider the potential impact to the water balance prior to commencing any dewatering activities. • Consult with key stakeholders (e.g., government, civil society, and potentially affected communities) to understand any conflicting water use demands and the communities' dependency on water resources and/or conservation requirements that may exist in the area. • Consider zero discharge tailings facilities and completion of a full water balance and risk assessment for the mine process circuit including storage reservoirs and tailings dams.

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
		- Conduct a comprehensive series of accelerated leaching tests from feasibility study stage onwards, to evaluate the potential for ARD in all formations foreseen, to be disturbed or otherwise exposed by the mine according to internationally recognized methodologies. - Conduct comprehensive ARD / metals leaching (ML) testing / mapping on an ongoing basis with decreasing block size as formations are transferred from long- to medium- and short-term mining plans.
Air, Noise	- Air quality pollution. - Fugitive dust from blasting, exposed surfaces such as tailings facilities, stockpiles, waste dumps, haul roads and infrastructure. - Mercury emissions to air. - GHG emissions. - Noise and vibrations (from blasting, excavation, and transportation).	- Strengthen the institutional framework and capacity of institutions. - Enhance and/or maintain air quality (noise, dust emissions and vibrations management). - Control GHG and pollutant emissions to acceptable standards. - Implement dust suppression techniques (e.g., wetting down, use of all-weather surfaces, use of agglomeration additives) for roads and work areas, optimization of traffic patterns, and reduction of travel speeds. - Revegetate exposed soils and other erodible materials or cover promptly. - Clear new areas and open-up only when absolutely necessary. - Revegetate surfaces or otherwise render non-dust forming when inactive. - Plan activities in consultation with local communities so that activities with the greatest potential to generate noise are planned during periods of the day that will result in least disturbance. - Use noise control devices, such as temporary noise barriers and deflectors for impact and blasting activities, and exhaust muffling devices for combustion engines. - Avoid or minimize project transportation through community areas. - Use mechanical ripping where possible, to avoid or minimize the use of explosives. - Use of specific blasting plans, correct charging procedures and blasting ratios, delayed/micro delayed or electronic detonators, and specific in-

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
		situ blasting tests (the use of downhole initiation with short-delay detonators improves fragmentation and reduces ground vibrations). • Develop blast design, including a blasting-surfaces survey, to avoid over confined charges, and a drill-hole survey to check for deviation and consequent blasting recalculations. • Implement ground vibration and overpressure control with appropriate drilling grids. • Re-locating noise sources to less sensitive areas to take advantage of distance and shielding. • Design the foundations of primary crushers adequately, and to other significant sources of vibrations.
Geology and soils	• Soil and subsoil contamination from toxic chemicals and material – e.g., use of mercury, acid mine drainage (AMD) from waste dumps and failure of tailing dams.	• Strengthen the institutional framework and capacity of institutions – in particular, the regulation and control of hazardous waste generation registries, safe handling and transport, and disposal from industrial production sites. • Protect quality of seabed and sediments. • Preserve the hydrogeological environment. • Maintain / recover / improve soil quality. • Design of tailings storage facilities should take into account the specific risks / hazards associated with geotechnical stability or hydraulic failure and the associated risks to downstream economic assets, ecosystems and human health and safety. • Implement soil conservation measures (e.g., segregation, proper placement and stockpiling of clean soils and overburden material for existing site remediation); key factors such as placement, location, design, duration, coverage, reuse, and single handling should be considered. • Consider potential change of geotechnical properties in dumps due to chemical or biologically catalyzed weathering. Design new facilities as to provide for such potential deterioration of geotechnical properties with higher factors of safety. Stability / safety assessments of existing facilities should take these potential changes into account.

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
		• Conduct accurate assessment of worksite safety from rockfall and/or landslide. Particular attention should be given after heavy rainfall, seismic events and after blasting activities. Risks should be minimized by appropriate bench and pit slope design, blast pattern design, rock scaling, protective berms and minimizing traffic. • Assess the natural topography around the mine site, as well as mine related infrastructure such as cut slopes. Road alignments should be included in geotechnical stability analyses, especially in tropical climates or seismic zones with deeply weathered soils and high precipitation, natural geotechnical where risks may exist even before the start of mining activities. These conditions can be especially hazardous for settlements / housing related to mining activities. Especially underground, but also for surface features, modern topographical 3D deformation measurements and related specific processing and evaluation software should be the standard method for stability monitoring.

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
Flora and fauna	- Loss of biodiversity. - Physical damage to biotopes. - Fragmentation of habitats. - Removal of wildlife and forest cover. - Loss of swamp habitat through polder development. - Disturbance of fish and marine mammals due to seismic surveys.	- Preserve terrestrial and river ecosystems and their biodiversity. - Preserve critical habitats and protected or endangered species. - Ensure that protected areas (e.g., parks) are not affected by mining development. - Contribute to conservation of wildlife and wildlife habitats. - Plan and avoid sensitive areas and implement buffer zones. - Implement an effective monitoring program. - Implement mitigation measures appropriate for the type of habitat and potential impacts including, for example, post-operation restoration (which may include baseline inventories, evaluations, and eventual rescue of species), offset of losses, or compensation of direct users. - Avoid or minimize the creation of barriers to wildlife movement, or threats to migratory species (such as birds) and provide alternative migration routes when the creation of barriers cannot be avoided.
Energy use	The most significant energy consuming activities in mining are transport, exploration activities, drilling, excavation, extraction, grinding, crushing, milling, pumping, and ventilation processes.	- Strengthen the institutional framework and capacity of institutions. - Develop and implement an Energy Management Program including as minimum: (i) Reporting of principal energy flows within a facility at unit process level. (ii) Preparation of mass and energy balance. (iii) Definition and regular review of energy performance targets, which are adjusted to account for changes in major influencing factors on energy use. (iv) Regular comparison and monitoring of energy flows with performance targets to identify where action should be taken to reduce energy use. (v) Regular review of targets, which may include comparison with benchmark data, to confirm that targets are set at appropriate levels. - Perform a systematic analysis of energy efficiency improvements and cost reduction opportunities.
Waste	Inadequate management and disposal of general wastes generated by mining	Develop and implement strategies to manage the impact of inadequate management and disposal of wastes, including hazardous wastes, such

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
	operations (formal and informal). Formal and large mining operations generate large volumes of mined waste material or overburden that it is normally stored or used to build large structures such as waste dumps, tailings impoundments / dams, and containment facilities.	as training, actions plans, surveillance and monitoring. - Large mining operations as part of their Life of Mine planning process must include proper planning and operations for the design, construction and O&M phases for large structures so that geotechnical risks and environmental impacts are appropriately assessed. - Design, operation, and maintenance of structures according to specifications of ICOLD3 and ANCOLD4, or other internationally recognized standards based on a risk assessment strategy. - Waste dumps: Plan and design waste dumps to local geological considerations and the management of Potential Acid Generating (PAG) wastes (i.e., acid rock drainage – ARD) for which Geochemical Characterization methods need to be developed and implemented, - Tailings dams: Require tailings design standards and design management strategies that consider the varying local constraints and the nature / type of the tailings, and how these will be handled and disposed of during operation until permanent storage after decommissioning. - Hazardous wastes: guidance and requirements for management of hazardous materials and wastes, should be developed and implemented (including collection by specialized service providers, where possible).
Socio - economic	- Water conflicts. - Job creation and revenue sharing. - Disturbance to family networks, community structures and cultural and ethnic identities. - Increase in drug trade and prostitution. - Temporary restrictions on fishing activities in exploration areas. - Damage to local infrastructure due to blasting. - Increase in armed robbery, including illegal possession of weapons.	- Preserve social structure integrity. - Sustain and enhance agricultural and cattle economic activities. - Enhance community development. - Ensure benefits from mining development to local communities. - Avoid resource conflicts. - Minimize expectations over benefits of mining development. - Consult with key stakeholders (e.g., government, civil society, and potentially affected communities) to understand any conflicting land use demands and the communities' dependency on natural resources and / or conservation requirements that may exist in the area.

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
	- Irregular migration and overstay. - Change of navigation channels in the rivers (sand banks due to the dumping of tailings into the river).	
Land use and land tenure, and resettlement	- Allocation of land for mining activities. - Land rights. - Landscape alteration. - Economic and physical displacement. - Disruption of livelihoods and economic displacement. - Forced evictions. Population relocation and resettlement.	- Strengthen the institutional framework and capacity of institutions. - Consult with key stakeholders (e.g., government, civil society, and potentially affected communities) to understand any conflicting land use demands and the communities' dependency on natural resources and / or conservation requirements that may exist in the area. - Avoid, mitigate and/or compensate land acquisition in accordance with best practices.
Visual Impact	Mining operations, and in particular surface mining activities, may result in negative visual impacts, including: highwalls, erosion, discolored water, haul roads, waste dumps, slurry ponds, abandoned mining equipment and structures, garbage and refuse dumps, open pits, and deforestation.	- Consult with local communities about potential post-closure land use. - Incorporate visual impact assessment into the mine reclamation process. - Ensure that reclaimed lands, to the extent feasible, conform to the visual aspects of the surrounding landscape. - Include strategic placement of screening materials to mitigation measures including trees, and the use of appropriate plant species in the reclamation phase as well as modification in the placement of ancillary facilities and access roads.
Indigenous Peoples (IP) Maroons,	Involuntary resettlement of IP Livelihood impacts of IP	- Strengthen the institutional framework and capacity of institutions. - Protect Indigenous People and Tribal People and other vulnerable groups. - Recognize cultural diversity.
Cultural	Damage/disturbance to archeological	Strengthen the institutional framework and capacity of institutions.

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
	sites.	- Retain competent professionals to assist in the identification and protection of cultural heritage. - Develop provisions for managing chance finds through a chance find procedure, which will be applied in the event that cultural heritage is subsequently discovered. - Consult with the Affected Communities to identify cultural heritage of importance, and to incorporate into the decision-making process the views of the Affected Communities on such cultural heritage. Consultation will also involve the relevant national or local regulatory agencies that are entrusted with the protection of cultural heritage. - Apply mitigation measures that favor avoidance for sites where there is tangible cultural heritage that is replicable and not critical. - Apply the following measures where avoidance is not feasible.: - Minimize adverse impacts and implement restoration measures, in situ, which ensure maintenance of the value and functionality of the cultural heritage, including maintaining or restoring any ecosystem processes needed to support it. - Where restoration in situ is not possible, restore the functionality of the cultural heritage, in a different location, including the ecosystem processes needed to support it.
Health and safety	- Potential for significant effects on human health. - Malaria and other diseases spread (including sexually transmitted diseases). - Waterborne diseases (typhoid fever, cholera, leptospirosis and hepatitis A). - Vector-borne diseases (dengue fever, zika fever, chikungunya, malaria and leishmaniasis). - Exposure to mercury, cyanide and other chemicals - Elevated mercury levels in people.	- Prepare emergency response plans specifically applicable to exploration and production activities (considering the often geographically isolated nature of mining sites). - Prevent disruption, disturbance and nuisance to communities. - Ensure public and community safety in areas of mining operations. - Implement strategies to manage the impact of diseases through assessment, surveillance, actions plans, and monitoring. - Include a workplace program to prevent new HIV infections and provide care and support for infected and affected employees. - Undertake health awareness and education initiatives. - Train health workers in disease treatment. - Provide treatment through standard case management in on-site or

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
	- Heavy workloads, repetitive tasks, long working hours and unsafe equipment. - Alcohol and drugs use. Pressure into the access to health care.	community health care facilities (e.g., immunization programs). - Implement an integrated vector control program. - Engineer design reviews including careful scrutiny of roads, water storage and control facilities and surface water management strategies. - Collaborate and exchange in-kind services with other control programs in the project area to maximize beneficial effects, particularly distribution of treated bed net. - Apply outreach activities within the community, sector and / or broader society. - Develop an effective short and long-term monitoring and evaluation program for both workers and potentially affected communities. - Ensure a safe work environment for mining operators and employees. - Make sure cyanide use is consistent with the principles and standards of practice of the International Cyanide Management Code. The Cyanide Code includes principles and standards applicable to several aspects of cyanide use including its purchase (sourcing), transport, handling / storage, use, facilities decommissioning, worker safety, emergency response, training, and public consultation and disclosure.
Gender inequality and discrimination risks	- Unequal access to employment in mines or a loss of male support for household work. - Increased dis-advantages for women (ownership and possession of land, mineral rights, capital, and equipment). - Women left out of community decision-making processes and benefits received. Representation of woman.	- Strengthen the institutional framework and capacity of institutions. - Include the representation of women in the mining sector by promoting organizational representation through cooperatives, unions, federations, and trade associations should be a key policy focus moving forward.
Human Rights HDD	- Force labor and Child labor. - Destruction of ritually or culturally significant sites without compensation. - Violation of the right to a clean	- Use strategic context analysis of Suriname of international human rights framework (gap analysis) and apply to the mining sector. - Determine the risk and extent of labor rights violations, forced labor and child labor.

Theme	Risks & Impacts from mining (informal and formal)	Proposed Mitigation Measure
	environment and health affectations. Labor rights violations.	- Strengthen the institutional framework and capacity of institutions.
Vulnerability risks	Increasing negative living conditions of disadvantaged and the most vulnerable sections in the community (e.g., women, youth, children, and the elderly)	- Use vulnerability analysis to identify impacts on living conditions of disadvantage and vulnerable people and/or communities associated with mining projects or activities. - Develop and define measures to reduce exposure to vulnerabilities of these populations. - Strengthen the institutional framework and capacity of institutions
Mine Closure risks	- Liability to the government for rehabilitation of abandoned mining sites. - Affectation to the environment and communities. - Red mud lakes polluted with caustic soda and other metal and minerals. - Landfills with batteries, Spent Pot Lining (SPL) and other industrial waste. - Abandoned pits full of water, with high pH and possibly heavy metals. - Presence of settling ponds with chemical compounds. - Risk of impact on aquifers that serve to collect drinking water. Communities that are overly dependent on mining projects can suffer significant impacts after projects are closed.	- Include mine decommissioning and closure plans and cost estimates for rehabilitation. - Verify that mining contracts consider requirements for additional financial guarantees. - Verify that closure and post closure plans include appropriate aftercare and continued monitoring of the site, pollutant emissions, and related potential impacts. The duration of post closure monitoring should be defined on a risk basis; however, site conditions typically require a minimum period of five years after closure or longer. - Ensure that the mine decommissioning and closure plans address beneficial future land use (this should be determined using a multi-stakeholder process that includes regulatory agencies, local communities, traditional land users, adjacent leaseholders, civil society and other impacted parties), be previously approved by the relevant national authorities, and be the result of consultation and dialogue with local communities and their government representatives. - Determine the impact on communities from its dependence of mining projects and mining activities. - Develop community resilience and just transition of mining projects prior closure. - Strengthen the institutional framework and capacity of institutions.

Source: EHS General Guidelines, IFC, 2007; EHS Guidelines for Mining, IFC, 2007; Socio-economic, health and environmental impacts of mining in Suriname, with a focus on Artisanal and Small-scale gold Mining, Social Solutions, 2017, and modified by RINA, 2002.

8 STAKEHOLDER ENGAGEMENT

During the development of the Inception Report, stakeholders were identified in collaboration with MNR and based on the literature reviewed.

Key stakeholders include government authorities such as MNR, NIMOS, MRDS, donor agencies (WB, IDB, USAID and others), local and international NGOs (CI, WWF), CSOs, private companies, mining companies, and mining communities with special attention to Indigenous and Tribal communities involved in mining. The Inception Report describing the stakeholder engagement plan is presented as an Appendix A to this report. In addition, a general display map to be used as reference for the consultation process is presented in Appendix C. Please note that this map is for reference purposes and the sources of information used for its development include available spatial information sourced the National Land Monitoring System of Suriname (GONINI) and other maps identified during the information review process. The data is not official and, in some instances, due to the lack of available or updated information, some maps were used and roughly digitized until key reference information such as mining rights blocks, Indigenous and tribal peoples' territories, until an official GIS data is provided by the GOS.

9 PROPOSED CONTENTS OF THE SESA REPORT

Below is the proposed table of contents for the SESA report:

- 0. Executive Summary
 - 0.1. Project Context
 - 0.2. SESA Objectives
 - 0.3. SESA Methodology
 - 0.4. SESA Consultation Process
 - 0.5. Key Findings
 - 0.5.1. Institutional, Legal and Regulatory Framework
 - 0.5.2. Mining Sector Development Context
 - 0.6. SESA Action Plan
 - 0.6.1. Immediate Action
 - 0.6.2. Short Term Action
 - 0.6.3. Medium to Longer Term Action
- 1. Introduction
- 2. Project Background
 - 2.1. Background and context
 - 2.2. SESA ToR
 - 2.3. SESA Objectives
 - 2.4. Vision for Sustainable Management of Suriname Mining Sector
 - 2.5. Public Consultation (including IP approach)
 - 2.6. SESA Methodology
 - 2.7. Monitoring and Follow-Up
 - 2.8. Reporting
 - 2.9. Organization of the SESA Report
- 3. Mining Potential Trend Scenarios
- 4. Critical Environmental and Social Impacts and Risks
 - 4.1. Identification of General Strategic E&S consequences.
 - 4.2. Proposed mitigations and monitoring measures for E&S impacts
- 5. Key SESA Findings
 - 5.1. Institutional Capacity
 - 5.2. Policy, Legal and Regulatory Framework
 - 5.3. Mining Development Context
 - 5.4. Concluding Remarks
- 6. SESA Recommendations and Action Plan
 - 6.1. Proposed Mitigation, Monitoring and Evaluation Measures
 - 6.2. SESA Action Plan
- 7. SESA Implementation and Work Plan
- 8. SESA Monitoring Mechanism
- 9. Appendices
- 10. References

10 MECHANISM TO MONITOR IMPLEMENTATION OF SESA

Based on the analysis conducted, and the recommendations developed, an Action Plan will be developed by RINA to facilitate the implementation of the SESA. Priorities will be structured into immediate, short term (1-2 years) and medium term (3-5 years), including the Action Plan Cost Estimate.

The objectives of the SESA Action Plan will be:

- To present a series of recommendations arising from key findings of the SESA;
- To suggest a priority timeline for implementation of the recommendations;
- To identify the GOS Agency responsible for the implementation of the SESA;
- To provide some preliminary costs that will require confirmation and adjustment by applicable GOS Authorities.

The Action Plan will contain the preliminary related recommendations organized as follows:

- Findings: refers to a summary of the SESA key finding and forms the basis for the Recommended Actions.
- Priorities: organized in three priorities: immediate (2023-2024), short-term (2024-2025) and medium to long term (2026-2028). These priorities will define the ideal time span required to follow up on the Action Plan, and therefore the implementation of the recommendations.
- Justification: represents the justification for implementing the actions.
- Recommended Actions: a series of action(s) that should be undertaken to implement the recommendations.
 - Anticipated Outcome: describes the anticipated outcome if the action is fully implemented.
 - Responsibility: assigns the responsibility for the implementation of the action.
 - Cost: presents a preliminary cost estimate that should be re-determined by the responsible Suriname authority.

This action plan will include a suggested prioritization of activities or reforms that could be directly financed or supported by the Suriname Competitiveness and Sector Diversification (SCSD) Project during its implementation. This would include potential support to MNR/NIMOS or other GOS agencies, as well as to representatives of social or environmental groups to strengthen their capacity to engage in future discussions and planning about the mining sector's development.

REFERENCES

- ACT. (2015, October 12). *Amazon Gold Rush: Gold Mining in Suriname - Map Journal*. (The Amazon Conservation Team (ACT)) Retrieved October 10, 2022, from The Amazon Conservation Team (ACT): <https://www.amazonteam.org/maps/suriname-gold/>
- ACT. (2016). *Mapping the traditional lands of the Matawai Maroons in Suriname, one creek at a time*. Amazon Conservation Team (ACT). Retrieved October 05, 2022, from <https://www.amazonteam.org/mapping-the-traditional-lands-of-the-matawai-maroons-in-suriname-one-creek-at-a-time/>
- AGC. (2014). *A practical guide: Using Retorts to Reduce Mercury Use, Emissions, and Exposures in Artisanal and Small-Scale Gold Mining*. Artisanal Gold Council (AGC). Retrieved October 05, 2022, from http://www.artisanalgold.org/wordpress/wp-content/uploads/2017/01/Retort_guide_Oct2014_lowQ_version1.0_Eng.pdf
- Alcoa. (2019). *2019 Alcoa Sustainability Report*. Alcoa. Retrieved from <https://www.alcoa.com/sustainability/en/pdf/2019-Sustainability-Report.pdf>
- Berrenstein, H. a.-S. (2016). *Republic of Suriname - Second National Communication to the United Nations Framework Convention on Climate Change 2008-2012*. Paramaribo: Office of the President of the Republic of Suriname.
- BID. (2020). *SURINAME CONSOLIDATING A SUSTAINABLE ENERGY SECTOR (SU-L1055) LOAN PROPOSAL*. Interamerican Development Bank (BID). Retrieved from <https://www.gtai.de/PRO202001295028>
- BirdLife International. (2022, October 4). *Country profile: Suriname*. Retrieved from BirdLife International Data Zone: <http://datazone.birdlife.org/country/suriname>
- BOSMA, W. e. (1984). AN EXPLANATION TO THE GEOWGY OF SURINAME. *Contributions to the Geology of Suriname* 8, pp. 31-82.
- CBS. (2022). *Country Profile - Suriname*. Central Bank of Suriname (CBS). Retrieved October 10, 2022, from <https://www.cbvs.sr/images/content/statistieken/CP/SurinameCountryProfile.pdf>
- CCOHS. (2005). Occupational Vibration Exposure. *IOHA 2005 6th International Scientific Conference* (p. 84). Pilanesberg, South Africa: Canadian Centre for Occupational Health and Safety (CCOHS). Retrieved September 26, 2022, from https://www.ccohs.ca/events/ioha/vibration_ioha_2005.pdf
- Dasgupta et all. (2007). *The Impact of Sea Level Rise on Developing Countries: A Comparative Analysis*. World Bank Group (WBG).
- EITI. (2017). *Extractive Industries Transparency Initiative (EITI) Report of the Republic of Suriname for the Fiscal Year 2017*. Extractive Industries Transparency Initiative (EITI). Retrieved Sept 06, 2022, from Suriname 2017 EITI Report: <https://eiti.org/documents/suriname-2017-eiti-report>
- EITI. (2019). *Suriname - Fact Sheet*. Extractive Industries Transparency Initiative (EITI) of Suriname. Retrieved Sept 06, 2022, from <https://eitisuriname.org/wp-content/uploads/2019/04/seiti-factsheet-eng.pdf>
- EITI. (2019b). *EITI Suriname - History of Mining Industry*. Retrieved from EITI Suriname: <https://eitisuriname.org/en/about-suriname/history-of-mining-industry/>
- EITI. (2022, Sept 06). *Extractive Industries Transparency Initiative (EITI)*. Retrieved Sept 06, 2022, from Suriname: <https://eiti.org/countries/suriname>
- ERM. (2011). *SCOPING REPORT AND TERMS OF REFERENCE: Environmental and Social Impact Assessment Study of Surgold's Merian Project in Suriname*. Environmental Resources Management (ERM). Retrieved September 29, 2022, from https://s24.q4cdn.com/382246808/files/doc_downloads/operations_projects/south_america/documents/5-Merian_TOR-Final-Merian-ESIA-TOR_27Sept11.pdf
- ERM. (2016). *DRAFT-Environmental Impact Assessment Addendum*. Retrieved September 29, 2022, from <https://paperzz.com/doc/9382693/draft-environmental-impact-assessment-addendum>
- ERM. (2018). *Saramacca Satellite Mine Project: Environmental and Social Impact Assessment. Prepared for IAMGOLD Corporation by ERM Consultants Canada, Ltd*. Toronto, Ontario.
- ESS. (2007). *Nationaal Biodiversiteit Actie Plan (NBAP)*. Paramaribo, Suriname.
- European Commission. (2021). *Multiannual Indicative Programme 2021-2027*. European Comission - International Partnerships. Retrieved Jan 17, 2022, from https://international-partnerships.ec.europa.eu/countries/suriname_en
- European Union. (2019, 07 26). *EUR-Lex*. Retrieved September 26, 2022, from DIRECTIVE 2002/44/EC OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02002L0044-20190726>
- FAO. (2020). *Suriname - Global Forest Resources Assessment 2020*. Rome: Food and Agricultural Organization (FAO). Retrieved 09 16, 2022, from <https://www.fao.org/3/cb0068en/cb0068en.pdf>
- GBS. (2022). *Data on Diseases, Immunisation data and other relevant health statistics, 2015-2021*. General Statistical Office (GBS). Retrieved from <https://statistics-suriname.org/wp-content/uploads/2022/07/final->

- ODIN-5.1-5.2-Diseases-prevalence-Immunization-data-and-other-relevant-Health-statistics-2015-2021-EG.docx
- GBS. (June 2022). *Environmental Statistics - Summary 2022*. General Bureau of Statistics of Suriname. Retrieved September 05, 2022, from https://statistics-suriname.org/wp-content/uploads/2022/06/ENVIRONMENT_ODIN_Indicators-EG-dd-300622.xlsx
- GBS. (March 2020). *9th Environment Statistics Publication*. General Bureau of Statistics of Suriname (GBS). Retrieved from <https://statistics-suriname.org/wp-content/uploads/2021/03/Final-9th-environment-pub-2020.pdf>
- GEF. (2016). *Artisanal and Small-Scale Gold Mining (ASGM) National Action Plan (NAP) for Suriname*. Retrieved October 05, 2022, from Global Environmental Fund (GEF): <https://www.thegef.org/projects-operations/projects/9489>
- GEF. (2017). *Artisanal and Small-Scale Gold Mining (ASGM) National Action Plan (NAP) for Suriname*.
- GEF. (2017). *Artisanal and Small-Scale Gold Mining (ASGM) National Action Plan (NAP) for Suriname*. Retrieved October 05, 2022, from Global Environmental Fund (GEF): <https://www.thegef.org/projects-operations/projects/9489>
- GEF-CReW. (2010). *Suriname*. (U.-C. R. Analysis, Producer, & GEF Caribbean Regional Fund for Waste Management (GEF CReW)) Retrieved October 05, 2020, from GEF CReW: <https://www.gefcrew.org/index.php/participating-countries/suriname>
- GLiM. (2022). *University of Hamburg, Global Lithological Map*.
- GOMIAN. (2011). *Fact sheet Small scale gold mining in Suriname*. GOMIAN Small Scale Gold Mining in the Amazon. Retrieved September 26, 2022, from <https://www.gomiam.org/wp-content/uploads/2014/11/factsheet-suriname.pdf>
- GOS. (2000). *THE CONSTITUTION OF THE REPUBLIC OF SURINAME (BULLETIN OF ACTS AND DECREES 1987 NO. 166)*. Paramaribo: The Waterfront Press.
- GOS. (2013). *National Biodiversity Action Plan (NBAP) 2012-2016*. Government of Suriname (GOS), Ministry of Labour, Technological Development and Environment. Paramaribo, Suriname: Republic of Suriname. Retrieved September 29, 2022, from <https://www.fao.org/faolex/results/details/es/c/LEX-FAOC163417/>
- GOS. (2016). *Second National Communication to the United Nations Framework Convention on Climate Change*. PARAMARIBO, SURINAME: Office of the President of the Republic of Suriname. Retrieved August 30, 2022, from <https://unfccc.int/documents/144630>
- GOS. (2017). *Policy Development Plan 2017 - 2021*. Paramaribo: Government of the Republic of Suriname. Retrieved September 29, 2022, from <https://observatorioplanificacion.cepal.org/en/plans/policy-development-plan-2017-2021-suriname>
- GOS. (2018). *Macroeconomic outlook, investment and trade opportunities, September 2018*. Government of Suriname (GOS), Ministry of Finance. Retrieved October 05, 2022, from <https://statistics-suriname.org/wp-content/uploads/2021/05/Macroeconomic-outlook-investment-and-trade-opportunities-September-2018.pdf>
- GOS. (2019a). *National Inventory of Mercury Releases in Suriname 2019*. Republic of Suriname. Government of Suriname (GOS). Retrieved October 05, 2022, from <https://suriname.un.org/sites/default/files/2020-10/Report%20L2Mercury-FNweb.pdf>
- GOS. (2019b). *NATIONALLY DETERMINED CONTRIBUTION OF THE REPUBLIC OF SURINAME 2020 - 2030*. Retrieved September 05, 2022, from <https://unfccc.int/sites/default/files/NDC/2022-06/Suriname%20Second%20NDC.pdf>
- GOS. (2019c). *Suriname Decent Work Country Programme (DWCP) 2019 to 2021*. Government of Suriname (GOS) and International Labor Organization (ILO). International Labor Organization (ILO). Retrieved October 05, 2022, from https://www.ilo.org/wcmsp5/groups/public/---americas/---ro-lima/---sro-port_of_spain/documents/publication/wcms_727269.pdf
- GOS. (2020, June). *Suriname Minamata Initial Assessment Report 2020*. Ministry of Foreign Affairs, International Business and International Cooperation. Government of Suriname (GOS). Retrieved from <https://hlpf.un.org/sites/default/files/vnrs/2022/VNR%202022%20Suriname%20Report.pdf>
- GOS. (2020). *Suriname Minamata Initial Assessment Report 2020*. Government of Suriname (GOS). Retrieved October 09, 2022, from <https://suriname.un.org/en/download/51398/94234>
- GOS. (2022b). *Gonini - Geographical Information System (GIS) layers*. Retrieved from Gonini - National Land Monitoring System of Suriname: <https://www.gonini.org/>
- GOS. (2022c). *Surinam Multi-Year Development Plan 2022 - 2026*. Paramaribo, Suriname: Republic of Suriname. Retrieved from <https://observatorioplanificacion.cepal.org/en/plans/multi-year-development-plan-2022-2026-suriname>
- Grassalco. (2022). *Grasshopper Gold - Maripaston Gold Mine*. Retrieved from <https://grassalco.com/grasshopper-gold>

- GTR. (2015, March 09). *Suriname opens first gold refinery*. Retrieved October 9, 2022, from Global Trade Review (GTR): <https://www.gtreview.com/news/americas/suriname-opens-first-gold-refinery/>
- Huber, O. (1995). Geographical and physical features. In M. B. Garden.. Portland, USA: Timber Press.
- IADB. (2021). *State of the Climate Report: Suriname*. Inter-American Development Bank (IADB), Climate Change Division. Inter-American Development Bank (IADB). Retrieved August 30, 2022, from <https://publications.iadb.org/en/state-climate-report-suriname>
- IAMGOLD. (2022). *Rosebel Gold Mine, Suriname*. Retrieved October 09, 2022, from IAMGOLD OPERATIONS: <https://www.iamgold.com/English/operations/rosebel/default.aspx>
- IBAT. (2022). *Integrated Biodiversity Assessment Tool (IBAT)*. Retrieved August 31, 2022, from <https://www.ibat-alliance.org>
- IBAT. (2022, 10 4). *Suriname*. Retrieved August 31, 2022, from Integrated Biodiversity Assessment Tool (IBAT): https://www.ibat-alliance.org/country_profiles/SUR
- IBRD. (2019). *PROJECT APPRAISAL DOCUMENT ON A PROPOSED LOAN IN THE AMOUNT OF US\$23 MILLION TO THE REPUBLIC OF SURINAME FOR A COMPETITIVENESS AND SECTOR DIVERSIFICATION PROJECT*. WORLD BANK GROUP - INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT (IBRD). INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT.
- IBRD. (2019). *PROJECT APPRAISAL DOCUMENT ON A PROPOSED LOAN TO THE REPUBLIC OF SURINAME FOR A CSD PROJECT*. World Bank Group (WBG).
- IFC - International Finance Corporation. (2007). *Environmental, Health and Safety Guidelines for Mining*.
- IFC - International Finance Corporation. (2007). *Environmental, Health, and Safety General Guidelines*.
- ILACO. (2019). *Environmental and Social Impact Assessment (ESIA) of Energy Infrastructure Projects*. Retrieved September 29, 2022, from https://ewsdata.rightsindevelopment.org/files/documents/55/IADB-SU-L1055_wnAREte.pdf
- ILO. (2019). *Rules of the Game: An introduction to the standards-related work of the International Labour Organization*. International Labour Organization (ILO). Retrieved October 05, 2022, from https://www.ilo.org/global/standards/information-resources-and-publications/publications/WCMS_672549/lang--en/index.htm
- ILO. (2022). *International Labor Organization (ILO)*. Retrieved October 03, 2022, from Suriname Ratifications: https://www.ilo.org/dyn/normlex/es/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103287
- ITA. (2022, Aug 06). *Suriname - Country Commercial Guide. Mining and Minerals*. Retrieved Sept 06, 2022, from International Trade Administration (ITA): <https://www.trade.gov/country-commercial-guides/suriname-mining-and-minerals>
- Jodensavanne Foundation. (2022). *Our Gallery*. Retrieved from odensavanne Foundation: <https://www.jodensavanne.org/gallery>
- Joshi, A. (2015, October 30). *Gold mining explodes in Suriname, puts forests and people at risk*. Retrieved from Mongabay News: <https://news.mongabay.com/2015/10/gold-mining-explodes-in-suriname-puts-forests-and-people-at-risk/>
- Kautzor, T. (2014a). *The Railways of Suriname: Plantation Railways*. Retrieved October 05, 2022, from <https://www.internationalsteam.co.uk/trains/surinam04.htm>
- Kautzor, T. (2014b). *The Railways of Suriname, Suriname Landspoorweg*. Retrieved October 05, 2022, from <https://www.internationalsteam.co.uk/trains/surinam02.htm>
- KOPI. (2022). *Kennis, Onderzoek, Planning, and Informatie (KOPI)*. Retrieved August 30, 2022, from Foundation for Forest Management and Production Control (SBB) and National Forest Monitoring System (NFMS): <https://kopi.sbb.sr/index.php?r=deforestationinformation%2Findex>
- Larson Davis. (2022a). *Larson Davis*. (Larson Davis) Retrieved September 26, 2022, from Hand-Arm Vibration Syndrome: <http://www.larsondavis.com/applications/human-vibration-exposure/hand-arm-vibration-syndrome>
- Larson Davis. (2022b). *Larson Davis*. (Larson Davis) Retrieved September 26, 2022, from Whole-Body Human Vibration Exposure: <http://www.larsondavis.com/applications/human-vibration-exposure/whole-body-vibration>
- LUJAN, N. K., & ARMBRUSTER, J. (2011). The Guiana Shield. In J. S. (Eds.), & J. S. REIS (Ed.), *Historical Biogeography of Neotropical Freshwater Fishes*. (1st ed., pp. pp. 211-224). California, USA: University of California Press. Retrieved Sept 01, 2022, from <http://www.jstor.org/stable/10.1525/j.ctt1pp29c.17>
- Lyons & Espinosa. (2022, September 14). *Can Two New Bills Reshape Indigenous Rights and Illegal Gold Mining in Suriname?* Retrieved September 09, 2022, from Mongabay: <https://news.mongabay.com/2022/09/can-two-new-bills-reshape-indigenous-rights-and-illegal-gold-mining-in-suriname/>
- MAS. (2020). *National Report Suriname 2021 - 22nd Meso-American-Caribbean Sea Hydrographic Commission Meeting*. Paramaribo: Maritime Authority Suriname (MAS). Retrieved from <https://iho.int/uploads/user/Inter->

- Regional%20Coordination/RHC/MACHC/MACHC22/MACHC22_2021_03.11_EN_National.Report_Suriname_v1.pdf
- MBG. (2022, 10 4). *Tropicos.org. Missouri Botanical Garden*. Retrieved from <https://tropicos.org>
- MNR. (2019). *Note on Minerals Institute Suriname*. Ministry of Natural Resources (MONR) of Suriname, Republic of Suriname. Retrieved Sept 05, 2022, from <https://documents1.worldbank.org/curated/zh/715551576654339922/Note-on-Minerals-Institute-Suriname.docx>
- MNR. (2022a). *Ministry of Natural Resources*. Retrieved September 29, 2022, from About us: <https://gov.sr/ministeries/ministerie-van-natuurlijke-hulpbronnen/over-ons/>
- MNR. (2022b, March 09). *Ministry of Natural Resources*. (Republiek Suriname) Retrieved Sept 06, 2022, from Officiële doorstart Suriname competitive and sector diversification project; duurzame ontwikkeling mijnbouwsector, naar delfstoffen instituut: <https://gov.sr/officiële-doorstart-suriname-competitive-and-sector-diversification-project-duurzame-ontwikkeling-mijnbouwsector-naar-delfstoffen-instituut/>
- MNRE. (1984). *Mededeling Geologisch Mijnbouwkundige Dienst van Suriname Mededeling 27-1984, Contributions to the Geology of Suriname 8*. Ministry of Natural Resources and Energy of Suriname (MNRE).
- Mohadin, K. (1993). Conservation of freshwater ecosystems in Suriname. In P. Ouboter, *MONOGRAPHIAE BIOLOGICAE. Volume 70. The Freshwater Ecosystems of Suriname* (pp. 275-284). Kluwer Academic Publishers.
- Monsels, D. (2016). Bauxitedeposits in Suriname: Geological context and resource development. *Netherlands Journal of Geosciences — Geologie en Mijnbouw*, 95(4), 405–418. doi:10.1017/njg.2015.28
- N.V., S. M. (2022, December 30). Staatsolie Exceeds Production Targets. *Press Release*, p. 4.
- Newmont. (2019). *2019 SUSTAINABILITY HIGHLIGHTS - Beyond the Mine - Newmont Suriname Operations*. Retrieved from <https://www.newmont.com/operations-and-projects/global-presence/south-america/merian-suriname/default.aspx>
- Newmont. (2020). *Merian 2019 Economic Impact Report*. Newmont and Steward Redqueen. Retrieved from https://s24.q4cdn.com/382246808/files/document_library/economic_impact_reports/Suriname-Merian-2019-Economic-Impact-Report_National-Level.pdf
- NIMOS. (2022). Retrieved from THE APPROVAL PROCESS OF THE ENVIRONMENTAL FRAMEWORK ACT: <https://nimos.org/en/portfolio/revise-approve-env-frmwk-act/>
- NIMOS. (2022a). *NAP Project*. Retrieved October 10, 2022, from National Institute for Environment and Development in Suriname (NIMOS): <http://nimos.org/en/nap/>
- NIMOS. (2022b). *National Institute for Environment and Development in Suriname (NIMOS)*. Retrieved September 29, 2022, from About Us: <http://nimos.org/en/about-us/>
- OCHA. (2022). *Suriname: Country Profile (as of May 2022)*. Retrieved from UN Office for the Coordination of Humanitarian Affairs (OCHA): <https://reliefweb.int/report/suriname/suriname-country-profile-may-2022>
- Ouboter, et al. (2018, May 17). Mercury Levels in Women and Children from Interior Villages in Suriname, South America. *International Journal of Environmental Research and Public Health*, 15(5), 1007. doi:<https://doi.org/10.3390%2Fijerph15051007>
- Ouboter, P. (. (2002, February). *Directory of protected areas of Suriname*. Conservation International Suriname, Paramaribo. Retrieved September 02, 2022, from https://www.researchgate.net/publication/262972705_Directory_of_protected_areas_of_Suriname
- RG. (2022). *IAMGOLD ROSEBEL GOLD MINES*. Retrieved from IAMGOLD ROSEBEL GOLD MINES N.V.: <https://www.rosebelgoldmines.sr/>
- Rostain, S. (2008). *The Archaeology of the Guianas: An Overview*. . (e. H. Silverman and W. Isbell, Ed.) Springer, New York: In Handbook of South American Archaeology. .
- Rostain, S. (2010). *Pre-Columbian Earthworks in Coastal Amazonia. Diversity 2010 (2), pg. 331-352*.
- SBB. (2022, October 4). *Post- Deforestation Land Use Land Cover (LULC) information*. Retrieved from KOPI - Statistical data of the National Forest Monitoring System (NFMS): <https://kopi.sbb.sr/index.php?r=deforestationlanduselandscover2%2Findex>
- SBB, B. C. (2019). *GCCA+ Suriname Adaptation Project. Setting up a mangrove biodiversity Monitoring System. Final Report*. Paramaribo, Suriname: SBB.
- SEARCH. (2022). *Suriname Cultural Heritage Scoping Report*. SEARCH Inc.
- SITA. (2022). *Natural Resources*. Retrieved from Suriname Investment and Trade Agency (SITA): <https://sita.sr/sectors/natural-resources/>
- SLMS. (2022, 10 4). *GONONI - National Land Monitoring System of Suriname*. Retrieved from <https://www.gonini.org/>
- SS. (2017). *Socio-economic, health and environmental impacts of mining in Suriname, with a focus on Artisanal and Small-scale gold Mining*. Produced by Social Solution (SS) for United Nations Development Program (UNDP) and Global Environmental Facility (GEF).

- SWM. (2019). *Environmental and Social Analysis Water Supply Modernization Program SU-L1058*. Suriname Water Supply Company (SWM). Retrieved from <https://www.swm.sr/wp-content/uploads/2019/11/Fit-for-Disclosure-ESA-SU-L1058-1.pdf>
- SWRIS. (2022). *Suriname Water Resources Information System (SWRIS)*. Retrieved August 26, 2022, from Suriname Data - General: <https://www.swris.sr/data/general/>
- SWRIS, S. W. (2008, September). *Suriname Water Resources Information System - SWRIS*. Retrieved from Geology: Suriname's Geology and Mineral Deposits: <https://www.swris.sr/wp-content/uploads/2018/05/geology.pdf>
- Terra-i. (2014, 09 08). *Terra-i*. Retrieved 09 16, 2022, from SURINAME, A FOREST WEALTH COUNTRY AFFECTED BY AN ENVIRONMENTAL ISSUE: <http://www.terra-i.org/news/news/Suriname--a-forest-wealth-country-affected-by-an-environmental-issue.html>
- TMP. (2022). *About us - Moengo Port History*. Retrieved from Traymore Moengo Port (TMP): <https://traymorenv.com/en/about-us/moengo-port-history>
- UN. (2022). *General Assembly - Mercury, small-scale gold mining and human rights - Report of the Special Rapporteur on the implications for human rights of the environmentally sound management and disposal of hazardous substances and wastes, Marcos Orellana/HRC/51/35*. United Nations (UN) General Assembly. Retrieved October 07, 2022
- UNDP. (2019, July 30). *INDC Project Actions and Impacts: Suriname*. Retrieved October 05, 2022, from UNDP Climate Promise: <https://climatepromise.undp.org/research-and-reports/indc-project-actions-and-impacts-suriname>
- UNDP. (2022). *Midterm Review, Improving Environmental Management in the Mining Sector of Suriname, with Emphasis on Artisanal and Small Scale Gold Mining (ASGM)*. United Nations Development Program (UNDP). Retrieved Jan 17, 2022, from <https://erc.undp.org/evaluation/documents/download/19813>
- UNEP. (2022). *Environmental Framework Act (No. 97 of 2020)*. FAO, FAOLEX. Retrieved September 29, 2022, from <https://leap.unep.org/countries/sr/national-legislation/environmental-framework-act-no-97-2020>
- UNESCO. (2002). *Nomination of the Historic Inner City of Paramaribo for the World Heritage List of UNESCO*. Retrieved October 05, 2022, from <https://whc.unesco.org/en/list/940>
- UNESCO. (2022a). *Historic Inner City of Paramaribo: Photograph Gallery*. Retrieved October 05, 2022, from UNESCO: <https://whc.unesco.org/en/list/940/gallery/>. Accessed on October 5, 2022.
- UNESCO. (2022b). *The settlement of Joden Savanne and Cassipora cemetery. Tentative World Heritage List*. Retrieved October 05, 2022, from <https://whc.unesco.org/en/tentativelists/1083/>
- USGS. (1999). *Geologic Provinces of South America 1999*. Schenk, C.J, Viger, R.J., and Anderson, C.P., 1999, *Maps showing geology, oil and gas fields and geologic provinces of the South America region: U.S. Geological Survey Open-File Report 97-470-D*, 12 p. Retrieved August 30, 2022, from <https://services3.arcgis.com/6ivv17yGXadPbqNc/arcgis/rest/services/Geologic%20Provinces%20of%20South%20America%201999/FeatureServer>
- Versteeg, A. H. (2003). *Suriname Before Columbus. Libri Musei Surinamensis 1. Stitching Surinaams Museum, Paramaribo*. .
- Versteeg, A. H. (2008). *"Barrancoid and Arauquinoid Mound Builders in Coastal Suriname."* (e. H. Silverman and W. Isbell, Ed.) Springer, New York.: In *Handbook of South American Archaeology*. .
- VIDS. (2020). . *Baseline Report of the situation of the Indigenous people in Suriname 2020*. Available at: Association of Indigenous Village Leaders in Suriname (VIDS). Retrieved from https://indigenousexplorer.org/files/media/document/BaselineReport_ENG_final_10May2021.pdf
- Vletter, R. d. (1984). SYNTHESIS OF THE PRECAMBRIAN OF SURINAME AND REVIEW OF SOME OUTSTANDING PROBLEMS. *CONTRIBUTIONS TO THE GEOLOGY OF SURINAME* 8, pp. 11-30.
- WBG. (2017). *Oil, Gas and Mining: A sourcebook for understanding the extractive industries*. World Bank Group (WBG).
- WBG. (2017). *Suriname Sector Competitiveness Analysis: Identifying Opportunities and Constraints to Investment and Diversification in the Agribusiness and Extractives Sectors*. Washington, USA.: World Bank Group (WBG). Retrieved 09 16, 2022, from <https://openknowledge.worldbank.org/handle/10986/26205>
- WBG. (2022). *World Bank Group (WBG) Climate Change Knowledge Portal For Development Practitioners and Policy Makers*. Retrieved August 30, 2022, from Suriname - Climatology: <https://climateknowledgeportal.worldbank.org/country/suriname/climate-data-historical#:~:text=Suriname's%20geographical%20location%20just%20C%2C%20in%20the%20south%20ern%20regions>.
- WBG. (2022a). *Overview*. (World Bank Group (WBG)) Retrieved October 13, 2022, from The World Bank In Suriname: <https://www.worldbank.org/en/country/suriname/overview#1>
- WBG. (April 2015). *INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT INTERNATIONAL FINANCE CORPORATION AND MULTILATERAL INVESTMENT AGENCY COUNTRY PARTNERSHIP STRATEGY FOR SURINAME FOR THE PERIOD FY15-FY19*. World Bank Group (WBG). Retrieved

- October 04, 2022, from <http://documents1.worldbank.org/curated/en/753311468311980671/pdf/91238-REPLACEMENT-FILE-Box391426B-PUBLIC-disclosed-4-29-15.pdf>
- WBG. (February 2019b). *SCSD Indigenous and Tribal Peoples Planning Framework – ITPPF*. World Bank Group (WBG).
- WBG. (June 2019a). *SCSD Environmental and Social Management Framework*. World Bank Group (WBG).
- WBG. (June 2019c). *SCSD Rapid-Social-Assessment - RSA*. World Bank Group (WBG).
- WBG. (June 2019d). *SCSD Resettlement Policy Framework*. World Bank Group (WBG).
- WEF. (2018). *Global Gender Gap Report 2018*. World Economic Forum (WEF).
- WHO. (2016). *Environmental and occupational health hazards associated with artisanal and small-scale gold mining*. World Health Organization (WHO). Retrieved from <https://apps.who.int/iris/handle/10665/247195>
- WWF. (2005). *SITUATION ANALYSIS OF THE SMALL-SCALE GOLD MINING IN SURINAME - Reforming the sub-sector to promote sound management*. World Wildlife Fund (WWF).

Appendix A

Inception Report – SESA for the Mining Sector

Doc. No. P0032241-SR Rev. 2 –
January 2023





Ministry of Natural Resources Paramaribo, Suriname

Strategic Environmental and Social Assessment (SESA) for the Mining Sector of Suriname

Inception Report

Doc. No. P0032241-IR Rev. 4 – December 5, 2022

Rev.	Description	Prepared by	Controlled by	Approved by	Date
4	Final report	Johanne Hanco	NV	Julie Evans	12/05/2022

All rights, including translation, reserved. No part of this document may be disclosed to any third party, for purposes other than the original, without written consent of RINA Consulting Inc.

TABLE OF CONTENTS

	Page
LIST OF TABLES	4
LIST OF FIGURES	4
ABBREVIATIONS AND ACRONYMS	5
1 INTRODUCTION	6
1.1 GENERAL INFORMATION	6
1.2 BACKGROUND	7
1.2.1 PROJECT DESCRIPTION COMPONENT 1: FOUNDATIONS FOR PRIVATE INVESTMENT IN EXTRACTIVE INDUSTRIES	7
2 SCOPE AND OBJECTIVE OF THE SESA STUDY	9
2.1 EXPECTED OUTCOMES OF THE SESA STUDY	9
2.2 PRIMARY FORESEEN CHALLENGES AND CONCERNS	10
2.3 RINA TEAM SAFETY PROTOCOL	10
3 SESA METHODOLOGY	11
4 APPLICABLE STANDARDS	12
4.1 INTERNATIONAL STANDARDS	12
4.2 LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK	13
4.2.1 International Treaties	14
5 STAGES OF THE SESA	16
5.1 STAGE 1. SCOPING	16
5.1.1 TASK 1. Kick-off call virtual meeting	16
5.1.2 TASK 2. Desktop Review and Scoping Report	25
5.1.3 TASK 3. Stakeholders Mapping	27
5.1.4 TASK 4. Establishment of SESA Goals and Objectives	38
5.1.5 TASK 5. Conducting the validation workshops	41
5.2 STAGE 2. DESCRIPTION OF THE REGULATORY AND INSTITUTIONAL FRAMEWORK OF THE GOS CAPACITY FOR ENVIRONMENTAL, SOCIAL AND OCCUPATION HEALTH AND SAFETY ASSESSMENT AND MANAGEMENT	42
5.2.1 TASK 6. Development of an Institutional, Legal and Regulatory Framework and GOS Capacity Assessment	42
5.3 STAGE 3. IDENTIFICATION OF THE GENERAL, STRATEGIC AND SOCIAL CONSEQUENCES RESULTING FROM FORECASTED FUTURE DEVELOPMENT OF MINING	44
5.3.1 TASK 7. Analysis of Future Developments in Mining Environment and Social Consequences	44
5.4 STAGE 4. PROPOSED MITIGATION AND MONITORING MEASURES	45
5.4.1 TASK 8. Submit Mitigation Measures Including Indicators for Monitoring and Evaluation	45
5.5 STAGE 5. FORMULATION OF RECOMMENDATIONS AND A SESA ACTION PLAN	45
5.5.1 TASK 9. Development of the SESA Action Plan	45
5.5.2 TASK 10. Realization of the National Consultative Workshop and Validation of SESA Action Plan and Report	46
5.6 STAGE 6. FINAL CONSULTATIONS, REVIEW AND APPROVAL	47
5.6.1 TASK 11. Preparation of the Final Report	47
5.7 CONCURRENT STAGE - CAPACITY BUILDING AND ADVISORY SERVICES	47
5.7.1 TASK 12. Capacity building and advisory services	47
6 PROPOSED TOC FOR THE SESA REPORT	50
7 TIMELINE	52

8	REFERENCES	58
4	LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK IN SURINAME'S MINING SECTOR	2
4.1	INTRODUCTION	2
4.2	LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK	2
4.2.1	NATIONAL CONSTITUTION AND LAWS	2
4.2.2	INTERNATIONAL AGREEMENTS	11
4.2.3	INSTITUTIONAL FRAMEWORK	15
5	REFERENCES	20

APPENDIX A: RINA HEALTH AND SAFETY PLAN

APPENDIX B: AGENDA FOR VALIDATION WORKSHOP AND SITE VISITS

APPENDIX C: LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK IN SURINAME'S MINING SECTOR

LIST OF TABLES

Table 4.1:	Applicable standards, instruments, Suriname regulations and treaties	12
Table 4.2:	Treaties Signed by the GOS	14
Table 5.1:	Virtual Kick-off call participants	16
Table 5.2:	RINA Points of Contact	17
Table 5.3:	MNR Points of Contact	18
Table 5.4:	RINA Team Roles and Responsibilities	21
Table 5.5:	Stakeholders	27
Table 5.6:	Indigenous and Maroon Organizations and representations	35
Table 5.7:	Preliminary Strategic Objectives for Main Risks and Impacts from Mining	38
Table 5.8:	Capacity Building Program	49

LIST OF FIGURES

Figure 3.1:	Technical Approach and Methodology	11
Figure 5.1:	RINA's Team Composition	20
Figure 5.2:	Map of various indigenous and Maroon Groups in Suriname	34

ABBREVIATIONS AND ACRONYMS

ASM	Artisanal and Small-scale Mining
CPS	Country Partnership Strategy
CSO	Civil Society Organization
EIA	Environmental Impact Assessment
EITI	Extractive Industries Transparency Initiative
FPIC	Free Prior Informed Consent
GMS	Geological and Mining Service
GOS	Government of Suriname
IDB	Inter-American Development Bank
IFC	International Finance Corporation
ILO	International Labor Organization
ISO	International Organization for Standardization
ITPPF	Indigenous and Tribal Peoples Planning Framework
MNR	Ministry of Natural Resources
MRDS	Ministry of Regional Development and Sport
NGO	Non-Government Organization
NIMOS	National Institute for Environment and Development in Suriname
OECD	Organization for Economic Co-operation and Development
OSHA	Occupational Safety and Health Administration
SCSD	Suriname Competitiveness and Sector Diversification Project
SEA	Strategic environmental assessment
SESA	Strategic Environmental and Social Assessment
WB	World Bank
WBG	World Bank Guidelines
WTO	World Trade Organization

1 INTRODUCTION

1.1 GENERAL INFORMATION

RINA Consulting, Inc. (RINA) has been retained by the Ministry of Natural Resources (MNR) of the Government of Suriname (GOS) to conduct a Strategic Environmental and Social Assessment (SESA) of the Mining Sector financed by the World Bank's International Bank for Reconstruction and Development (IBRD). The development of this SESA is part of an overarching goal to strengthen the environmental and social standards and impact management under the Country Partnership Strategy (CPS) between the World Bank and the GOS.

As part of this consultancy, RINA will prepare the following:

Deliverable	Deliverable/Output	Main Objective
1	Inception Report and PPT (Methodology and Approach and Stakeholder list)	To include the workplan for consultations that will ensure that the stakeholders' concerns and advice are considered in the SESA.
2	Scoping Report	Frame the context and methodology for the SESA. This will include the Table of Contents for the SESA report.
3	Regulatory and Institutional Framework and Assessment of the GOS Capacity	Assess the adequacy of the framework, based on international experience, and the capacity of public agencies at national and local levels, and governmental organizations at national and community levels to address the potential impacts of developments in mining, especially for licensing and enforcement of legal ESHS requirements.
4	Identification of the general, strategic environmental and social consequences - and - Proposed Mitigation and Monitoring Measures report	Identify and evaluate the environmental and social consequences and risks, including climate change risks associated with existing developments and plans for investment in mining, and safety and occupational health risks and issues relevant to mining, considering the forecasts and scenarios developed. This will include mapping to identify overlapping areas of natural resources / forests/ other areas of high biodiversity value, with potential and actual extraction sites, and identify potential mitigation measures and monitoring requirements for each potential strategic impact on mining. Mitigation measures for strategic impacts may include, among others: implementation of appropriate legislation, including legal environmental discharge standards for emissions into air, water and soil; measures specifically targeting engagement with indigenous and tribal communities, drawing on initial mitigation measures to enable effective benefit sharing with such communities identified in the Indigenous and Tribal peoples planning Framework (ITPPF) and Rapid Social Assessment; revised institutional organization, streamlining environmental responsibility, creation of new GOS agency/ies;

Deliverable	Deliverable/Output	Main Objective
		strengthening of specific GOS Agencies; review or creation of environmental and social licensing agencies or mechanisms; enforcement of international best practices and operational standards guidelines for each sector; creation of protected areas or exclusion zones (No Go zones, buffer zones); etc. Specific indicators may be selected to allow for follow-up of the effectiveness of proposed mitigation measures. Indicators may be individual/stand-alone or combination (complex index), such as established indices, e.g., representing poverty reduction, migratory bird species, water quality, etc.
5	Draft SESA Report and Action Plan	Based on the analysis conducted, and the recommendations developed, an Action Plan will be developed to facilitate their implementation. Priorities will be structured into immediate, short term (1-2 years), and medium term (3-5 years), including the Action Plan Cost Estimate.
6	Final SESA Report and SESA Action Plan, including Stakeholder Consultation Report	Final SESA report will be presented after validation with stakeholders. The SESA will be accompanied by the Stakeholder Consultation Report.

This Inception Report will be reviewed by the client and later validated by the stakeholders in a workshop in Paramaribo and the interior. It is structured as follows:

- Introduction
- Scope and Objective of the SESA Study
- SESA Methodology
- Applicable Standards
- Stages of the SESA
- TOC for the SESA
- Timeline
- References

1.2 BACKGROUND

The Suriname Competitiveness and Sector Diversification Project (SCSD) has strong and compelling links with Suriname's National Development Plan (Nationaal Ontwikkelings Plan - NOP) 2022-2027 (NDP), which outlines the development trajectory for Suriname for upcoming years.

The NDP presents a vision and a strategy aimed at establishing "a diversified Surinamese economy, which is competitive in the competitive global market, has significantly more sustainable development, generates employment and equality, and keeps the environment liveable." The development vision in the NDP related to private investment in extractive industries is described below:

1.2.1 PROJECT DESCRIPTION COMPONENT 1: FOUNDATIONS FOR PRIVATE INVESTMENT IN EXTRACTIVE INDUSTRIES

This Project Component aims to strengthen foundations for private investment in extractive industries, by financing a geophysical data collection campaign and a geodata management system. A second

objective is to improve the regulatory and institutional framework for managing new investments in extractives. Foreseen activities under Component 1 include:

- Geological data program.
 - Airborne geophysical data collection campaign.
 - Financing of a geodata management system that can be used to manage and promote the data and the associated implications for new mining potential. The geodata management system will include geospatial information on Indigenous people's areas and natural habitats and other protected areas to enhance management of future social and environmental impact.
- Improving the regulatory framework for new investments in extractives.
 - GOS plans are underway to establish a Minerals Institute that will integrate and strengthen mining cadaster, geological data management, and mining inspectorate functions. GOS also has plans to establish an independent oil regulator to separate production and regulatory functions. The project will support the establishment and operationalization of these institutions.
 - The project will also support the development of regulatory frameworks to be implemented by these institutions that provide appropriate incentives for future private investments including clear mining titles, a level playing field, and compliance with transparent social and environmental regulations.
- Strengthening institutional capacity to manage social and environmental impacts of future private investments.
 - Finance a SESA covering mining and oil as well as the agriculture sector. This report is the Inception Report for the SESA for the mining sector.
 - Finance technical assistance for implementation of some of the recommendations of the SESA. This is expected to include training, advisory support, and other technical assistance to strengthen the capacity of environmental and Indigenous peoples' representative institutions to engage more effectively in managing social and environmental impacts of future investment activities.

2 SCOPE AND OBJECTIVE OF THE SESA STUDY

The SESA will entail a comprehensive mining sector-wide examination of potential impacts, both positive and negative, of future potential investments in the mining sector, and identify gaps in regulations, institutional capacity, and public consultation mechanisms that can be strengthened. This will help address the weaknesses in the current legal and regulatory framework and enable the GOS to integrate principles of sustainable development both upstream in sector planning efforts and downstream when specific projects are being prepared or evaluated.

The SESA will also contribute to a more transparent understanding of environmental and social regulations, creating a more level playing field for future private investors.

Specific objectives of the SESA are:

- ✓ To identify the social and environmental impacts that could be generated by the mining sector development, evaluating the scope and probability of these impacts due to increased activities in the sector and related development.
- ✓ To put forward recommendations to avoid, manage and/or attenuate these impacts.
- ✓ To facilitate the integration of these measures into a coherent policy and ensure their application; and
- ✓ To help in capacity building and training of the GOS officials in the management of the mining sector impacts, in particular, the Ministry of Natural Resources (MNR) and the National Institute for Environment and Development (NIMOS).

Because this SESA concerns the mining sector and requires a strategic look at environmental and social aspects, MNR and NIMOS will be particularly important to provide governmental guidance, advice and input to the expert team that will carry out the SESA:

When completed, the SESA will provide guidance to the MNR and NIMOS on systematically integrating environmental, socio-economic, health and safety concerns in policy, regulation and planning, project development, operations, maintenance, and decommissioning of mining sector activities.

The SESA can be the catalyst for the following outcomes:

- ✓ Increased attention to environmental, Indigenous and tribal community engagement, labor and resettlement, health, safety and security priorities associated with mining and development resulting in more responsible mining operations;
- ✓ Strengthened environmental constituencies;
- ✓ Improved social accountability by making the mining policy process more transparent, especially related to Indigenous and tribal communities; and
- ✓ Enhanced sector capacity for managing environmental, health, safety and socio-political risks associated with the development and operation of the mining sector.

Furthermore, the SESA will provide a benchmark for cumulative environmental and social effects with other ongoing sector programs and plans. The SESA will ensure that the concerns and expectations of different stakeholders are taken into account in the decision-making process for equity and sustainable development of Suriname's mining sector.

2.1 EXPECTED OUTCOMES OF THE SESA STUDY

The expected outcomes of the SESA study are a series of concrete measures for the mining sector to meet the needs of environmental protection, ensure the well-being of local populations, and identify clear institutional responsibilities for government, private sector, and civil society. These outcomes should lead to the growth of the mining sector driven by a more sustainable development process.

2.2 PRIMARY FORESEEN CHALLENGES AND CONCERNS

The substantial number of stakeholders whose participation and contributions to the SESA will be sought is one of its greatest strengths and the source of a significant challenge given the number of anticipated groups, diversity of locations, cultural diversity, economic and educational level differences.

The SESA will need to address the specific concerns and expectations raised by these different stakeholders, which will include the systematic integration of environmental, socio-economic, health and safety in policy, regulation and planning, project development, maintenance and decommissioning of mining sector activities, decision-making equity and sustainable development of the mining sector in Suriname. Other concerns may be raised during consultation with the stakeholders and added to this anticipated list. All concerns, observations and recommendations will be considered and addressed in the SESA report to provide transparency and clear information to all stakeholders on the overall process of the SESA implementation.

2.3 RINA TEAM SAFETY PROTOCOL

RINA will be guided by its approved H&S protocol while in Suriname to ensure the safety of the RINA team members and that of the stakeholders during meetings and workshops. Likewise, RINA's travel procedures will be implemented and followed while traveling to the country interior.

The H&S plan is provided in Appendix A of this Inception Report.

3 SESA METHODOLOGY

The methodology was designed to be consultative and participatory to strengthen institutional capacity, introduce good global practices, identify and inform specific reform efforts, and raise public awareness to integrate environmental and social concerns into the mining sector, as to develop the country's long-term sustainable development and in alignment with the NDP.

The methodology will ensure the safeguards for the protection of Indigenous and Maroon Peoples' rights, and their fair and transparent consultation and participation in the SESA in line with the ITPPF and following the Indigenous and Tribal People the FPIC procedures.

The methodology to be employed in the SESA process was derived from the TOR and from best and good SEA practice references (Dalal-Clayton and Sadler 2005)¹; (OECD 2006)²; (Partidário 2007; Partidário n.d.)³ and RINA's experience implementing similar projects.

The overall SESA study will contain three main components:

- ✓ A technical component underpinning the definition of objectives, targets, and indicators, which identify the relevant studies for each of the critical factors for decision-making, allowing necessary and sufficient information to be collected from within the available data.
- ✓ A process component that ensures the linking of a SEA process and the decision-making, planning, and programming processes, establishing the governance rules for the integration of the processes. This articulation between a SEA process and the strategic decision-making processes is what makes the SEA a flexible process adaptable to each scenario.
- ✓ A communication process, which is crucial for public participation and involvement, which assures the exchange of information and the cross-referencing of the multiple perspectives, the opinion making, an integrated vision and participative processes suited to the problem and to the critical decision moments. The communication component is adjusted to the characteristics of the target groups.

The SESA is to be used as a process of sector institutional and governance strengthening that materializes along with the implementation of mining sector reforms. The development of the SESA will follow 6 stages and 12 main tasks. The general approach to be used is presented in the Figure below

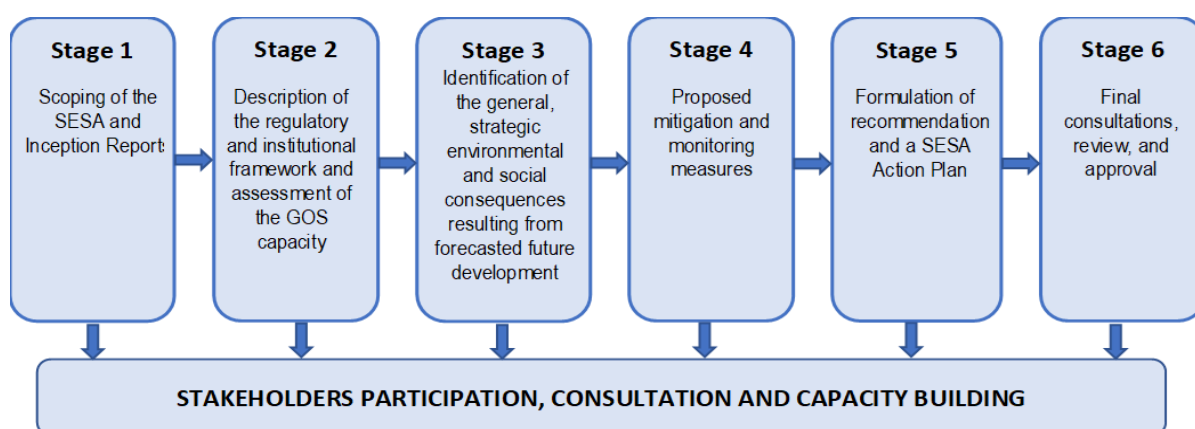


Figure 3.1: Technical Approach and Methodology

Source: RINA 2022

¹ <https://www.tandfonline.com/doi/pdf/10.1080/14615517.2017.1322811?needAccess=true>

² https://www.oecd-ilibrary.org/development/applying-strategic-environmental-assessment_9789264026582-en

³ http://content-ext.undp.org/aplaws_publications/1703425/SEA_guide_Portugal.pdf

4 APPLICABLE STANDARDS

4.1 INTERNATIONAL STANDARDS

Applicable international standards will be reviewed and used as reference for the SESA development. The applicable standards considered for the SESA will include:

Table 4.1: Applicable standards, instruments, Suriname regulations and treaties

Author(s)	Title	Date
World Bank	Operational Policy OP 4.01 - Environmental Assessment	April 2013
World Bank	Operational Policy OP 4.01 - Annex A	February 2011
World Bank	World Bank Operational Policy OP 4.01 - Annex B	January 1999
World Bank	World Bank Operational Policy OP 4.01 - Annex C	January 1999
World Bank	World Bank Operational Policy OP 4.04 Natural Habitats	April 2013
World Bank	World Bank Operational Policy OP 4.09 Pest Management	December 1998
World Bank	World Bank Operational Policy OP 4.10 Indigenous Peoples	April 2013
World Bank	World Bank Operational Policy OP 4.10 Indigenous Peoples Annex A	July 2005
World Bank	World Bank Operational Policy OP 4.10 Indigenous Peoples Annex B	July 2005
World Bank	World Bank Operational Policy OP 4.10 Indigenous Peoples Annex C	July 2005
World Bank	World Bank Operational Policy OP 4.11 Physical Cultural Resources	April 2013
World Bank	World Bank Operational Policy OP 4.12 Involuntary Resettlement	April 2013
World Bank	World Bank Operational Policy OP 4.12 Involuntary Resettlement Annex A	February 2011
World Bank	World Bank Operational Policy OP 4.36 Forests	April 2013
Instruments prepared as part of the SCSD		
GOS	SCSD Project A Assessment Document PAD	June-2019
GOS	SCSD Rapid-Social-Assessment	June2019
GOS	SCSD Resettlement Policy Framework (RPF)	June2019
GOS	SCSD- ENVIRONMENTAL AND SOCIAL MANAGEMENT FRAMEWORK (ESMF)	June-2019

GOS	SCSD TOR Strategic Environmental and Social Assessment (SESA)	June 2019
GOS	SCSD Indigenous and Tribal Peoples Planning Framework ITPPF	February 8, 2019
GOS	Country Partnership Strategy	2014
IFC Performance Standards		
IFC (World Bank)	E&S Performance Standard 1 Assessment and Management of Environmental and Social Risks and Impacts	2012
IFC (World Bank)	E&S Performance Standard 2 Labor and Working Conditions	2012
IFC (World Bank)	E&S Performance Standard 3 Resource Efficiency and Pollution Prevention	2012
IFC (World Bank)	E&S Performance Standard 4 Community Health, Safety, and Security	2012
IFC (World Bank)	E&S Performance Standard 5 Land Acquisition and Involuntary Resettlement	2012
IFC (World Bank)	E&S Performance Standard 6 Biodiversity Conservation and Sustainable Management of Living Natural Resources	2012
IFC (World Bank)	E&S Performance Standard 7 Indigenous Peoples	2012
IFC (World Bank)	E&S Performance Standard 8 Cultural Heritage	2012
IFC (World Bank)	EHS Environmental, Health, and Safety General Guidelines	April 2007
IFC (World Bank)	EHS Environmental, Health and Safety Guidelines for Mining	December 2007
Other International Standards		
ISO	ISO 45001 Occupational Health and Safety Management Systems	2018
ISO	ISO 26000:2010 Social Responsibility	2010
International Mining Standards		
ICMM	Principles of the International Council of Mining and Metals	2001
The Cyanide Code	The International Cyanide Management Code for the Manufacture, Transport, and Use of Cyanide in the Production of Gold	2022

Source: RINA, 2022 Note: ToR indicated to use OHSAS 18001; however, this standard had been withdrawn. Instead, ISO 45001 is in use.

4.2 LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK

Suriname's national legal and regulatory framework is based on civil law and derived from several sources, including a national constitution, laws, decrees, regulations, policies, strategies and plans, as well as international agreements, treaties, and conventions to which Suriname is a party. A complete

review of the legal, regulatory and institutional framework will be conducted and delivered as an output (Deliverable 3) of this consultancy as stated in the introduction of this Inception Report. The following Table describes the international treaties.

4.2.1 International Treaties

Suriname is signatory to a number of international agreements and conventions relating to environmental management, OHS, labor rights and Indigenous Peoples. The international conventions are not always translated into national legislation. Some of the key agreements are listed in the Table below.

Table 4.2: Treaties Signed by the GOS

Ratified treaties	Date of ratification
Basel Convention - Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal	02/09/2011
Cartagena Protocol - The Cartagena Protocol on Biosafety to the Convention on Biological Diversity	27/03/2008
CBD - Convention on Biological Diversity	12/01/1996
CEDAW Convention on the Elimination of All Forms of Discrimination against Women	03/01/1993
CERD International Convention on the Elimination of All Forms of Racial Discrimination	03/15/1984
CESCR international Covenant on Economic, Social and Cultural Rights	12/28/1976
CITES - Convention on International Trade in Endangered Species of Wild Fauna and Flora	17/11/1980
CRC Convention on the Rights of the Child	03/01/1993
CRPD Convention on the Rights of Persons with Disabilities	03/29/2017
CSICH - Convention for the Safeguarding of the Intangible Cultural Heritage	09/05/2017
Kyoto Protocol - Framework Convention on Climate Change	25/09/2006
Minamata Convention - Minamata Convention on Mercury	02/08/2018
Montreal Protocol - Montreal Protocol on Substances that Deplete the Ozone Layer	14/10/1997
Paris Agreement - Framework Convention on Climate Change	13/02/2019
Rotterdam Convention - Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade	30/05/2000
Stockholm Convention - Stockholm Convention on Persistent Organic Pollutants	20/09/2011
UNCCD - United Nations Convention to Combat Desertification in those Countries Experiencing Serious Drought and/or Desertification, particularly in Africa	01/06/2000

Ratified treaties	Date of ratification
UNFCCC - United Nations Framework Convention on Climate Change	14/10/1997
Vienna Convention - Vienna Convention for the protection of the Ozone Layer	14/10/1997

Source: UN ECLAC <https://observatoriop10.cepal.org/en/countries/42/treaties>

5 STAGES OF THE SESA

The following sections describe the stages of the SESA and associated tasks.

5.1 STAGE 1. SCOPING

The main objective of this stage is to set up the context for the SESA with extensive and substantial stakeholder inputs. The scoping will be validated by stakeholders through a validation workshop. This stage is comprised of the following tasks:

- Task 1: Kick-off Call
- Task 2: Desktop Review
- Task 3: Stakeholder Mapping
- Task 4: Establishment of SESA Goals and Objectives
- Task 5: Completion of the consultative and validation workshop and meetings with the stakeholders

5.1.1 TASK 1. Kick-off call virtual meeting

The Kick-off call was held virtually on Friday, September 02, 2022. The objective of the Kick-off call was to introduce the team, set up the communication protocol and the dates for the site visit. Meeting participants are identified in the Table below:

Table 5.1: Virtual Kick-off call participants

Participants	Institution
Aradhna Mathur	World Bank
Christopher Mays Johnson	World Bank
Isaac Alberto Mera Chehin	World Bank
Clyde Griffith	MNR
Susan Bansropansingh	MNR
Reana Burke	MNR
Jean Van Eenaeme	MNR
Lucia Dwarkasing	MNR
Tanja	MNR
Diana Vyent	MNR
Karishma Bhageloe	MNR
Reenuska Mahabier	MNR
Julie Evans	RINA

Participants	Institution
Johanne Hanco	RINA
Pedro Bastidas	RINA
Maria Paz Covolo	RINA

Source: RINA, 2022

During the call, RINA shared a Power Point presentation conveying the following:

- RINA's experience working with international development financial institutions and SESA studies for the World Bank and IDB.
- Requests for information pertaining to the mining sector and related to the SCSDC, particularly copies of the contracts awarded to mining companies. To date, RINA has not received those contracts.
- RINA points of contact, also summarized in the Table below:

Table 5.2: RINA Points of Contact

Name	Position	Responsibility	Communication
Julie Evans Julie.evans@rina.org +1 786 451-1380	SESA team leader / Contract Administrator	Oversee the development and accuracy of the SESA. Provide technical Oversight/Management for each of the stages comprised in the SESA.	Should be copied in all communications regarding the contract and Project implementation.
Johanne Hanco johanne.hanco@rina.org +1 514 512-2879	Project Coordinator	Facilitate communications, coordinate the implementation of the project, and with stakeholders and experts.	Should be copied in all communications regarding the contract and Project implementation.
Pedro Bastidas pedro.bastidas@rina.org +1 202 509-7426	Capacity Building	Prepare capacity building program and advisory services for GOS.	Should be copied in all communications regarding the contract and Project implementation
Salomon Emanuels asoebie51@gmail.com +597 8553324	Local coordinator / Public Participation	Coordinate local logistics and follow-up with stakeholders including Indigenous Peoples. Validation coordinator	Should be contacted for any issue regarding reaching out to stakeholders, sending invitations, field visits and workshop logistics

Source: RINA, 2022

- MNR updated the PPT to indicate its points of contact:

Table 5.3: MNR Points of Contact

Name	Position	Responsibility	Communication
Clyde Griffith clydegriffith1977@gmail.com	SCSDC 1 Component Coordinator	Managing and technically overseeing the implementation of all Component 1 project activities. Supervise PIU staff and consultants within MNR as well as supervise contractors and external consultants hired to implement project activities.	Should be copied on all communications.
Reenuska Mahabier reenuska.a.mahabier@gmail.com	PIU Technical Assistant	Assist the component coordinator with planning, coordinating, implementation and monitoring of project activities. Stimulating communication between different beneficiaries as well share project information among relevant stakeholders.	Should be copied on all communications
Karishma Bhageloe karishma.bhageloe@hotmail.com	PIU Administrative Assistant	Assist the Component coordinator with administration, planning, organizing and coordinating project activities (meetings, minutes).	Should be copied on all communications
Diana Vyent dladydie@hotmail.com	PIU Financial Assistant	Assist the component coordinator with planning, coordinating and carrying out financial activities within the project (prepares and controls payments).	Should be copied on all communications regarding invoicing and financial management
Lucia Dwarkasing lucia.dwarkasing@bdo.sr	Financial Specialist	Advise Component coordinator on all financial and control considerations of the project. Prepare project interim financial reports and annual financial statements.	Should be copied on all communications regarding invoicing and financial management
Jean Van Eenaeme jean.van.eenaeme@scarlet.be	Procurement Specialist	Assist the Component Coordinator with close out of all project processes and activities. Ensure that goods/ services/ works are procured in	Should be copied on all communications regarding the contract

Name	Position	Responsibility	Communication
		accordance with the approved Procurement Plan, with respect to timelines and in keeping with estimated costs.	and Project implementation.

Source: MNR, 2022

- RINA created a SharePoint space to share information gathered throughout the development of the SESA among MNR, World Bank and RINA, including maps and other data produced as part of the study. The location of the SharePoint is: <https://rinagroup.sharepoint.com/sites/85633>
- RINA presented its SESA project team as shown in the Figure below:

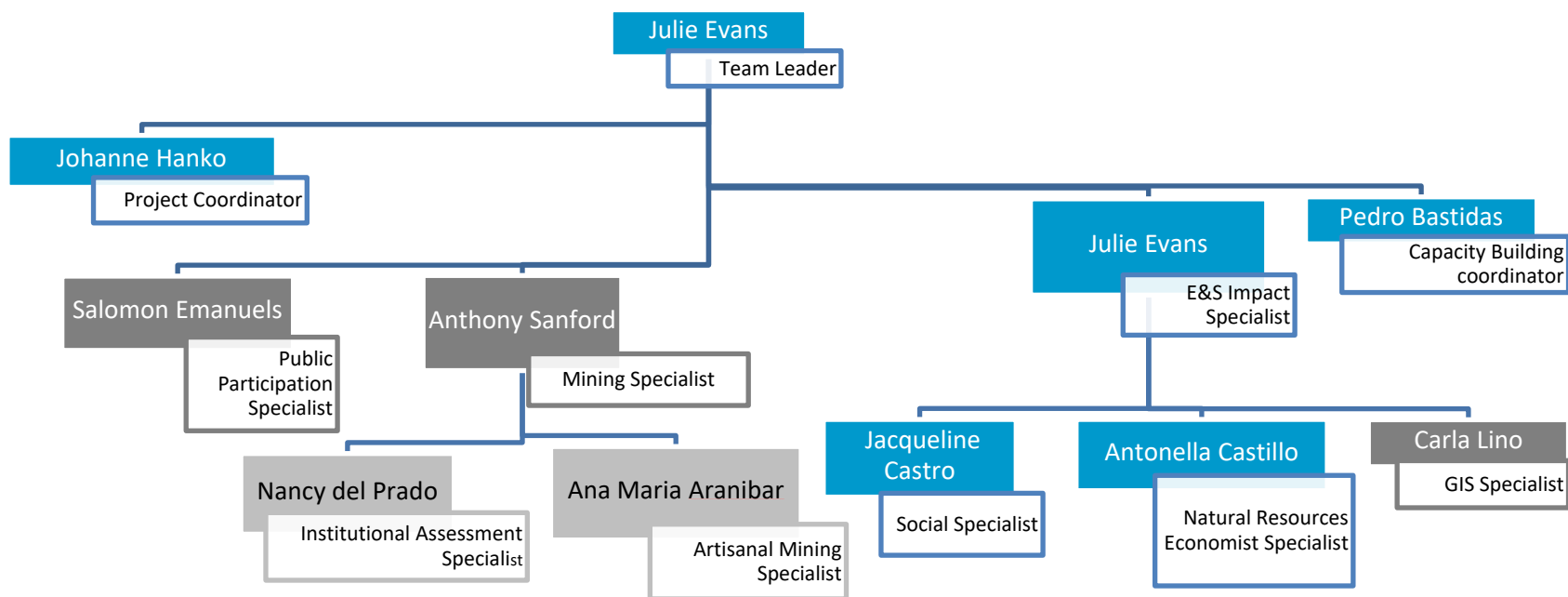


Figure 5.1: RINA's Team Composition

Source: RINA, 2022

Since the Kick-off call RINA has further developed and delineated its team roles and responsibilities as presented in the Table below.

Table 5.4: RINA Team Roles and Responsibilities

Name	Role	Responsibilities
Julie Evans	SESA Team Leader / EHS Specialist/Contract Administrator	– Oversee the development and accuracy of the SESA. – Provide technical Oversight/Management for each of the stages comprised in the SESA. – Ensure the compliance of the terms and conditions of the contract. – Provide technical Oversight/Management for each of the stages comprised in the SESA. – Coordinate all technical aspects of the SESA. – Review Environmental, Health and Safety protocols. – Propose mitigation, monitoring and good EH&S measures for the mining sector in Suriname. – Participate in the development of capacity building workshops for MNR and NIMOS.
Johanne Hanko	Project Coordinator	– Facilitate communications with client. – Coordinate the activities of the local consultants. – Keep track of stakeholder's participation (invitation and participation). – Assemble validation reports in coordination with local consultant.
Pedro Bastidas	Capacity Building Coordinator	– Prepare capacity building program and advisory services for GOS. – Prepare the agenda for the continuous capacity building workshops. – Prepare and disseminate materials to be used for the project and workshops. – Coordinate the attendance at the capacity building workshops.

Name	Role	Responsibilities
Salomon Emanuels	Local coordinator/ Public Participation Expert	– Support the RINA's international team with logistics for mobilization to travel to Suriname. – Responsible for the engagement with Indigenous Peoples and Maroons and Tribal communities according to the schedule and assist in the translation when required. Accompany RINA team to meetings in Paramaribo when required. – Coordinate all logistics related to the consultation/validation workshops including but not limited to invitations, venue and attendance, – Coordinate logistics for local capacity building during RINA visit to Suriname. – Collaborate in the mapping and identification of stakeholders and the preparation of the Stakeholder list. – Provide comments on the inception and scoping reports, and on the Draft SESA report.
Anthony Sanford	Mining Specialist	– Assess existing capacity of the relevant regulatory institutions. – Participate in the calibration of capacity assessment tools to be applied to the staff at the ministries. – Prepare a compilation of relevant policies, laws, regulations, guidelines, and contracts, including environmental licensing requirements (including EIA processes). – Participate in the adequacy assessment of the framework, based on international experience, and the capacity of public agencies at national and local levels, governmental organizations at national and community levels to address the potential impacts of developments in mining, especially for licensing and enforcement of legal ESHS requirement. – Assess mandates, capacity, incentives and transparency in licensing/permitting, Capacity of GOS for monitoring and enforcing environmental, social and occupational health and safety regulations. – Assess the adequacy of the existing grievance mechanisms to protect landowners and affected people, in particular Indigenous and tribal communities; and if applicable, provide recommendations to improve it.

Name	Role	Responsibilities
		- Describe the existing mechanisms for the social distribution of benefits to affected people; If applicable, provide recommendations to improve it. - Identify potential gaps regarding the international good practice, not only limited to the WB policies and safeguards but also practices from ILO, like ILO 169 and others.
Nancy del Prado	Institutional Assessment Specialist, Suriname regulatory framework specialist	- Identify the institutions responsible for policy, planning, regulating and enforcing legislation in the mining sectors. - Present the environmental, social, safety and occupational health legal, regulatory and institutional framework. - Participate in the assessment of existing capacity of the relevant regulatory institutions. - Prepare a compilation of relevant policies, laws, regulations, guidelines, and contracts, including environmental licensing requirements. - Participate in the adequacy assessment of the framework, based on international experience, and the capacity of public agencies at national and local levels, governmental organizations at national and community levels to address the potential impacts of developments in mining, especially for licensing and enforcement of legal ESHS requirement. - Assess mandates, capacity, incentives and transparency in licensing/permitting, Capacity of GOS for monitoring and enforcing environmental, social and occupational health and safety regulations in the above-mentioned sectors. - Collaborate in the assessment of adequacy of the existing grievance mechanisms. - Describe the existing mechanisms for the social distribution of benefits to affected people; If applicable, provide recommendations to improve it. - Identify potential gaps regarding the international good practice. - Provide recommendations for inclusion in the SESA report to strengthen the legal context to sustainably develop the mining sector.

Name	Role	Responsibilities
Ana Maria Aranibar	Artisanal Small Mining Expert	– Present an overview of the legal frameworks, best practices and initiatives in Artisanal and Small-Scale Mining (ASM) of selected five Latin American countries where small-scale mining is of high impact. For example, the Green Gold, Fair Trade, Fairmined initiatives or others. – Examine specific environmental, social and occupational health and safety (ESOHS) issues related to ASM in Suriname and present strategies to minimize their negative impacts in Suriname. – Collect and process information on ASM in Suriname, including interviews with relevant NGOs and/or stakeholder groups (e.g., Garimpeiros). – Analyze aspects to be considered in the formalization of ASM at the local, regional and national levels, and their possible impact. Explain how the informal sector may be impacted by a more formalized set of procedures and approaches. Summarize findings and proposed recommendations on formalization for inclusion in the SESA report to strengthen the sustainability of ASM in Suriname.
Jacqueline Castro	Social Specialist	– Scoping – Description of the regulatory and institutional framework and assessment of the GOS capacity for environmental, social, and occupational health and safety assessment and management, including opportunities to introduce good global practices; – Identification of the general, strategic environmental and social consequences resulting from forecasted future development of mining; – Proposed mitigation and monitoring measures, drawing on good global practices; Formulation of recommendations and a SESA Action Plan;

Name	Role	Responsibilities
Antonella Castillo	Natural Resources Economist	- Identify and evaluate the environmental and social consequences and risks, relevant to mining developed. - Estimate/model the potential impacts to ecosystems and vulnerable populations such as indigenous and tribal communities from expansion of the mining sector over a defined period of time. - Identify potential mitigation measures and monitoring requirements for each potential strategic impact on mining. - Develop specific indicators to allow for the follow-up of the effectiveness of proposed mitigation measures. - Consider good practice international standards, like the World Bank Safeguards and ILO policies, when developing the mitigation and monitoring measures. - Contemplate resettlement good practices to help enhance the Environmental and Social Framework.
Carla Lino	GIS Specialist	- Collect GIS information in open format or raster to develop maps that depict land use, natural protected areas, tribal land, etc. - Develop mapping as required.

Source: RINA, 2022

5.1.2 TASK 2. Desktop Review and Scoping Report

The purpose of this task is to assemble preliminary information relevant to the scope of the SESA applicable to the mining sector including artisanal and small-scale mining (ASM), forecast future mining development scenarios to be developed subsequently, identification of the environmental and social risks and vulnerabilities to be considered and assessed (e.g., potential indigenous and tribal community land use, water quality, air quality, geology and soils, biodiversity, marine resources, socio-economic, coastal activities, etc.), and identify the environmental and social characteristics of the areas that are most likely to be affected by future mining activities, identifying sensitive ecosystems, protected areas, areas of high biodiversity value, areas inhabited by indigenous peoples, maroons and fishermen, and the like. The identification will include small, medium and large scales mining activities, along with mined-out areas.

Two anticipated mining development scenarios in the upcoming years will be presented during our first meeting (virtual or in person) with the MNR:

- Base Case Scenario (Conservative)

- Accelerated Case Scenario

The analysis of the sector development under these two scenarios will be undertaken using assumptions (i.e., discoveries, resource studies, exploration results, Suriname's macroeconomic development, ease of doing business/export environment, transport infrastructure and services) and supporting factors (i.e., global demand for minerals, carbon credit opportunities). Once the scenarios are agreed by the client and validated by the stakeholders, the identification of environmental and social risk will be developed around these scenarios. These risks will include cumulative and indirect impacts by studying the possible location of large and small mining sites and transport infrastructure and services:

- Current developments in the infrastructure sector (ports, highways, local roads, possible domestic/national aviation);
- Current constraints and opportunities; and
- Infrastructure needed to support the mining development program (capacity, investment costs, timing).

RINA has been reviewing existing policies, programs, plans, studies, and public information material, such as:

- National and sectoral laws and regulations, national or regional plans, future mining development;
- Existing public or private programs or initiatives, current projects, plans and trends;
- Contextual information at the country, sub-national or sector levels (based on screening and targeted online searches);
- Available remote sensing imagery and geographic data to understand the environmental and social constraints. These sites are but not limited to:
 - Integrated Biodiversity Assessment Tool (IBAT) - <https://www.ibat-alliance.org>
 - KOPI – statistical data for the National Forest Monitoring System – managed by the Foundation for Forest Management and Production Control (SBB) <https://kopi.sbb.sr/>
 - Suriname Water Resources Information System (WSRIS) - <https://www.swris.sr/>
 - Global Lithological Map (GLiM) - <https://projekte.cen.uni-hamburg.de/lithomap/>
 - National Land Monitoring System of Suriname - <https://www.gonini.org/>
- Cultural Heritage;
- Scientific literature; and
- International mining standards (ICMM, Cyanide Code).

The references collected to date are listed at the end of this report as an Appendix. It should be noted that a Request for Information (RFI) was place to MNR during the Kick-off call but to date we have not been provided with the complete information. Likewise, we have requested GIS information from GONINI, but it has not been provided.

5.1.2.1 [Scoping Report](#)

The Scoping Report is the second output of the SESA study and will be presented under separate cover to be approved by the client and later validated by the stakeholders in conjunction with this Inception Report. This Inception Report and Scoping Report will be submitted for the consultation and validation process in the events to take place between October 24 and November 5, 2022. The agenda for the validation workshop and site visits is included as Appendix B to this Inception Report.

The Scoping Report will use the information from the desktop review and collected during Stage 1 of the SESA. Preliminary findings based on the document review will be presented, related but not limited to current project plans and trends in mining, forecast of future mining development scenarios to be developed subsequently, conflicts around mining, identification of environmental and social risks and vulnerability to be considered and assessed, Indigenous and tribal community land use, water quality, air quality, geology and soils. biodiversity, marine resources, socio economics, coastal activities, identification of environmental and social characteristics of the area's most likely to be affected by mining activities, sensitive ecosystems, protected areas, areas of high diversity and areas inhabited by Indigenous peoples, Maroons and fishermen.

The Scoping Report also will include a stakeholder analysis and present key strategic options, key restrictions, major interest, and will detail the stakeholder participation process and consultation plan, the common vision on environmental, social, occupational safety and health risks, and present a proposed monitoring mechanism.

5.1.3 TASK 3. Stakeholders Mapping

During the development of this Inception Report, stakeholders were identified in collaboration with MNR. Key stakeholders include government authorities such as MNR, NIMOS, MRDS, donor agencies (WB, IDB, USAID and others), local and international NGOs (CI, WWF), CSOs, private companies, mining companies, and mining communities with special attention to Indigenous People and Maroon and Tribal communities involved in mining. This latter group of stakeholders will be described in a separate item.

The following Table sets forth the identified stakeholders and their relevance to the SESA study. However, as we continue to define stakes and relevant parties, the following list may be subject to change. If additional stakeholders are identified, they will be included and consulted.

Table 5.5: Stakeholders

Sector	Name of Agency/Company	Relevance
Donor, development and aid agencies, UN and International organizations	United Nations Framework Convention on Climate Change (UNFCCC)	Suriname is located in the Amazon Forest and has a rich biodiversity. As mining has been the largest driver of deforestation in the country, UNFCCC has a key role as it defines forest reference (emission) levels to evaluate the country's forest status through REDDD+. Building capacity, estimating emission factors and activity data for the country.
Donor, development and aid agencies, UN and International organizations	World Health Organization of the United Nations (WHO)	One of WHO's country cooperation priority strategies for Suriname is to address social determinants of health, including "Strengthening national response to environmental health threats" and "Strengthening capacity and coordination to address workers' health." Mining activities generate direct negative impacts on health, especially affecting the lungs. This problem is compounded for gold mining activities where the use of mercury contaminated water used for consumption by whole families and where miners often use mercury for gold amalgamation without any protective equipment.
Donor, development and aid agencies, UN and International organizations	Forest Stewardship Council (FSC)	The FSC was established in 1993, as a follow-up to the United Nations Conference on Environment and Development (the Earth Summit at Rio de Janeiro, 1992) with the mission to promote international environmentally appropriate, socially beneficial, and economically viable management of the world's forests. FSC provides a system for voluntary accreditation and independent third-party certification, including in Suriname.

Sector	Name of Agency/Company	Relevance
Government	Ministry of Finance	The Ministry of Finance monitors the revenues and expenditures of the State and consists of three directorates, namely the Finance Directorate which takes care of the payment of the expenditures of the State and the Taxation Directorate where a large part of the revenues of the State (Government) enters. The third Directorate, Planning and Development Financing is responsible for planning and raising funds for development projects.
Government	The Ministry of Foreign Affairs, International Business and International Cooperation (BIBIS)	The Ministry of Foreign Affairs, International Business and International Cooperation (BIBIS) establishes and maintains relationships with the various countries and organizations in the world. The Surinamese diaspora are also involved in the efforts for a better Suriname. The Ministry's policy focuses on various areas among which, Border countries politics and the Development Cooperation and International Trade.
Government	Ministry of Economic Affairs, Entrepreneurship and Technological Innovation	The Ministry of Economic Affairs, Entrepreneurship and Technological Innovation is responsible for the promotion of domestic and foreign trade, including import and export policy, in collaboration with the relevant Ministries; matters of a commercial policy nature and the granting of import, export and exchange licenses, as far as these do not conflict with trade liberalization, all this in collaboration with the relevant Ministries.
Government	Ministry of Health	The activities of the Ministry of Health are determined by the policy it determines will best create better access to and quality of health care in Suriname's on the basis of research and consultations with impacted populations.
Government	Ministry of Labor, Employment and Youth Affairs	The mission of the Ministry of Labor, Employment and Youth Affairs is to realize sustainable development by establishing a healthy productive, development-oriented and safe labor market within the framework of the democratic constitutional state, whereby employers, employees and their organizations nationally and internationally, can fully contribute and fulfill their role.
Government	Ministry of Land and Forest Management	The Ministry of Land and Forest Management is responsible must ensure, among other responsibilities, sound planning, topography, cartography, geodesy, soil investigation and soil mapping; land use, allocation of land, land registry and the public registers, inventory,

Sector	Name of Agency/Company	Relevance
		exploration, optimal exploitation and management of the resource, flora and fauna; the management of nature and conservation.
Government	Ministry of Natural Resources (MNR)	MNR's mission is to ensure an integral sustainable and efficient management and development of the production of the natural resources present in Suriname, taking into account the environment, safety and rights of local communities. The Ministry is responsible for the development and execution of Suriname's energy policy.
Government	Ministry of Public Works	The Ministry of Public Works develops, builds and maintains the public good for a livable, clean and beautiful Suriname. The Ministry consists of four Departments: the Civil Engineering Works Department; Department of Construction and Urban Planning; the Public Green and Waste Management Department; and the Research and Services Department.
Government	Ministry of Regional Development and Sports	The Ministry of Regional Development and Sports consists of four Directorates and twelve Sub-directorates and facilitates sustainable regional development in cooperation with local authorities and traditional authorities on the basis of legislation, policy development, task coordination and optimal citizen participation.
Government	Ministry of Spatial Planning and the Environment (ROM)	ROM was established by presidential decree in response to the call for more attention to spatial planning and a more decisive environmental policy. Its mission is to ensure improvement in the quality of life in the living environment and an orderly country where development is approached in a responsible manner.
Government	Geology and Mining Service (GMD)	GMD is responsible for four main tasks: producing geological maps, making an inventory of minerals, advising the Minister on mining legislation, exploration permits and concessions, monitoring the resulting activities of third parties (mine inspection), and providing services to third parties in the field of geological exploration and mining inventory.

Sector	Name of Agency/Company	Relevance
Government	National Institute for Environment and Development in Suriname (NIMOS)	NIMOS is an autonomous government foundation that functions as an independent, technically competent, professional and progressive partner in the regulation and structuring of the various sectors in charge of the environment. Its mission is to initiate the development of a national legal and institutional framework for environmental policy and management in the interest of sustainable development.
Government	Bauxite Institute Suriname (BIS)	The BIS was founded in 1981 with the aim to preserve and expand the bauxite industry, to obtain a fair share of the proceeds from bauxite and its derivatives, and the acquisition of specific knowledge and expertise in the industry.
Government Foundation	The Management Institute for Land Registration and Land Information System (MI-GLIS)	MI-GLIS is responsible for the management of Land Registration and Land Information Systems in Suriname and was established by the Land Registration and Land Information System Act.
Mining Companies - Mixed	Large-scale: Newmont Suriname LLC (Newmont Suriname)	Newmont Suriname (previously known as Suriname Gold Company, LLC), is a fully- owned subsidiary of Newmont Goldcorp Corporation, and manages and operates the Merian mine on behalf of Suriname Gold Project CV, a Suriname limited partnership. Suriname Gold Project CV is jointly owned by Newmont (75%) and Staatsolie Maatschappij Suriname N.V., Suriname's State-owned oil company (25 %).
Mining Companies- Private company	Large-scale: IAMGOLD (Rosebel in Suriname)	IAMGOLD is a mid-tier international gold mining company operating in through Rosebel in Suriname. The Rosebel property lies approximately 85 kilometers south of Paramaribo. The mining concession covers 170 square kilometers, and gold production at Rosebel in 2022 is expected to be in the range of 155,000 to 180,000 ounces.
Mining Companies - Government owned	Large-scale: Grasshopper Aluminium Company (Grassalco)	Grassalco falls under the Ministry of Natural Resources and is a 100% government-owned mining company, active in the development of the country's mineral wealth. It was founded by government decree in 1971 to develop bauxite deposits in Suriname. Transport, trade, and industry are included in its Statutes of Incorporation alongside mining.
Mining Companies	Small scale: Sarafina NV.	Sarafina is a small-scale gold mining company that is working with a valid exploitation license facilitating small-scale mining operations.

Sector	Name of Agency/Company	Relevance
Mining Companies	Gold buyers/exporters: Century Mining Company NV	Century Mining Company NV is a Surinamese Gold Buying and Exporting Company with its head office at Wilhelminastraat.
Mining Companies	Gold buyers/exporters: Goldex NV.	Goldex NV. is a Surinamese gold buying and selling company with its head office at Paramaribo.
Private	Energy Authority Suriname (EAS)	The EAS is an autonomous institution, established by State decree 2016, no. 41. EAS performs supervisory and steering tasks and duties aimed at establishing and promoting optimal availability, affordability and sustainability in Suriname's energy sector.
Private	VSF: Suriname Business Association	The VSB was founded in 1950 with the purposes of promoting the economic and social well-being of the community. The VSB derives its legitimacy from Surinamese Constitution and is recognized by the ILO as the most representative Employers Business Member Organization in Suriname.
NGO/CSO/Association/Foundation.	Association of Surinamese Manufacturers (ASFA)	The ASFA, is an interest organization and represents the Surinamese production sector. Together with the government, it tries to create favorable preconditions to promote manufacturing in Suriname, represents the leading Surinamese manufacturers and aims to create a production-friendly environment.
NGO/CSO/Association/Foundation.	Stichting Business Support Organization Suriname (BSO)	The BSO organizes activities conducive to the promotion of businesses that focus on promoting Suriname, its products and services through activities such as trade fairs for the Chamber of Trade, Commerce and Industry, and activities that promote production and trade between local and foreign companies.
NGO/CSO/Association/Foundation.	World Resources Institute (WRI)	WRI works internationally to develop practical solutions that improve people's lives and ensure nature can thrive; in Suriname it works on the development of stronger climate plans to help enforce the 2015 Paris Agreement.
NGO/CSO/Association/Foundation.	WWF Guianas	WWF Guianas works together with the people of Suriname and Guyana to conserve their natural heritage for human well-being for now and for generations to come.

Sector	Name of Agency/Company	Relevance
NGO/CSO/Association/Foundation.	Conservation International - Suriname	Conservation International - Suriname works across the country to protect nature to conserve natural capital, and promote socio-economic development in Suriname, the greenest country in the world with unrivaled natural resources and globally significant biodiversity.
NGO/CSO/Association/Foundation.	Tropenbos Suriname	Tropenbos Suriname works to achieve sustainable management of tropical forestlands for the benefit of people, conservation, and sustainable development.
NGO/CSO/Association/Foundation.	Green Growth Suriname (GGS)	GGS builds projects designed for growth and driven by local needs & knowledge to assure sustainable solutions to further advance the people of Suriname's well-being and welfare by conserving the biological and cultural richness of the country. GGS envisions.
NGO/CSO/Association/Foundation.	Green Heritage Fund Suriname	Green Heritage Fund Suriname was established in 2005 with a guiding mission of encouraging wise decisions for the sustainable use of Suriname's natural resources.
NGO/CSO/Association/Foundation.	Extractive Industries Transparency Initiative (EITI)	EITI is an international nonprofit organization that promotes understanding of natural resource management, strengthening of public and corporate governance and accountability, and provides the data to inform policymaking and multi-stakeholder dialogue in the extractive sector.
NGO/CSO/Association/Foundation.	Spatial Planning Association Suriname (SPASU)	SPASU is a nonprofit organization that aims to promote spatial and urban planning in Suriname by increasing awareness and knowledge.
NGO/CSO/Association/Foundation.	Foundation for Forest Management and Forest Control (Stichting voor Bosbeheer en Bostoezicht -SBB)	SBB is a so-called Responsible Party within the REDD+ Program in Suriname. Their responsibilities encompass having a leading role in supporting the technical implementation of pillar 3 "Implementation Framework and tools", which includes the Historical deforestation mapping, National Forest Monitoring System and Forest Reference Emissions Levels.
NGO/CSO/Association/Foundation.	Stichting Suriname Environmental and Mining Foundation (SEMIF)	To promote, support and foster environmentally sound and responsible development of natural resources in Suriname, provide scholarships to Surinamese nationals in studies related to the development of natural resources; Providing support and assistance to enterprises of Surinamese nationals in their efforts to promote and develop natural resources.

Sector	Name of Agency/Company	Relevance
NGO/CSO/Association/ Foundation.	UTGS: Foundation United Tour guides Suriname	ensure preservation of guide work in Suriname by uniting guides and jointly building bridges to various tourism stakeholders.
NGO/CSO/Association/ Foundation.	Suriname Conservation Foundation (SCF)	The Suriname Conservation Foundation is the sustainable Surinamese environmental fund which is committed to the protection of biodiversity in general and the protected areas of Suriname in particular.
NGO/CSO/Association/ Foundation.	Water Forum Suriname (WFS)	Water Forum Suriname aims to promote sustainable water management in Suriname.
NGO/CSO/Association/ Foundation.	Mangrove Forum Suriname (MAFOSUR)	MAFOSUR aims to promote, sustainably manage, protect, conserve, rehabilitate and expand the mangrove areas and the mangrove ecosystem. We encourage the sustainable use of the services and goods produced by the mangrove ecosystem.
NGO/CSO/Association/ Foundation.	Amazon Conservation Team Suriname (ACT Suriname)	The Amazon Conservation Team (ACT) Suriname Foundation is committed to the protection of the Amazon rainforest and nature through participation with local Indigenous peoples in the south of Suriname.
NGO/CSO/Association/ Foundation.	Foundation for Sustainable Development of Pamaka (SDOP)	SDOP founded as a local development organization for the Pamaka communities because of the relationship between Newmont and the Pamaka people. SDOP aims to negotiate with Newmont and coordinate the implementation of development projects in the Pamaka communities.
NGO/CSO/Association/ Foundation.	PACT	Pact is supporting Indigenous people's organizations to increase their influence in the governance of the Amazon region, bringing together government, industry, miners and mining communities to make ASM safer, formal and more productive.

Source: RINA, 2022

Continuous engagement of any and all stakeholders who have access to the internet will be ensured through email communications and virtual meetings through Teams or Zoom platforms. This will allow whoever attends the meetings or receives information by email to be updated on the development of the SESA study and give them the opportunity to share their ideas and concerns either in writing or verbally during the meeting.

5.1.3.1 IP, Maroons Tribal Communities

Based on the ITPPF SCSD (2019), the area of influence for Component 1 (Extractive Industries), will be the area underlain by the Guiana Shield mainly referred to as “the interior” in Suriname. In this area, the ITPPF report identified the following IP and Maroon groups:

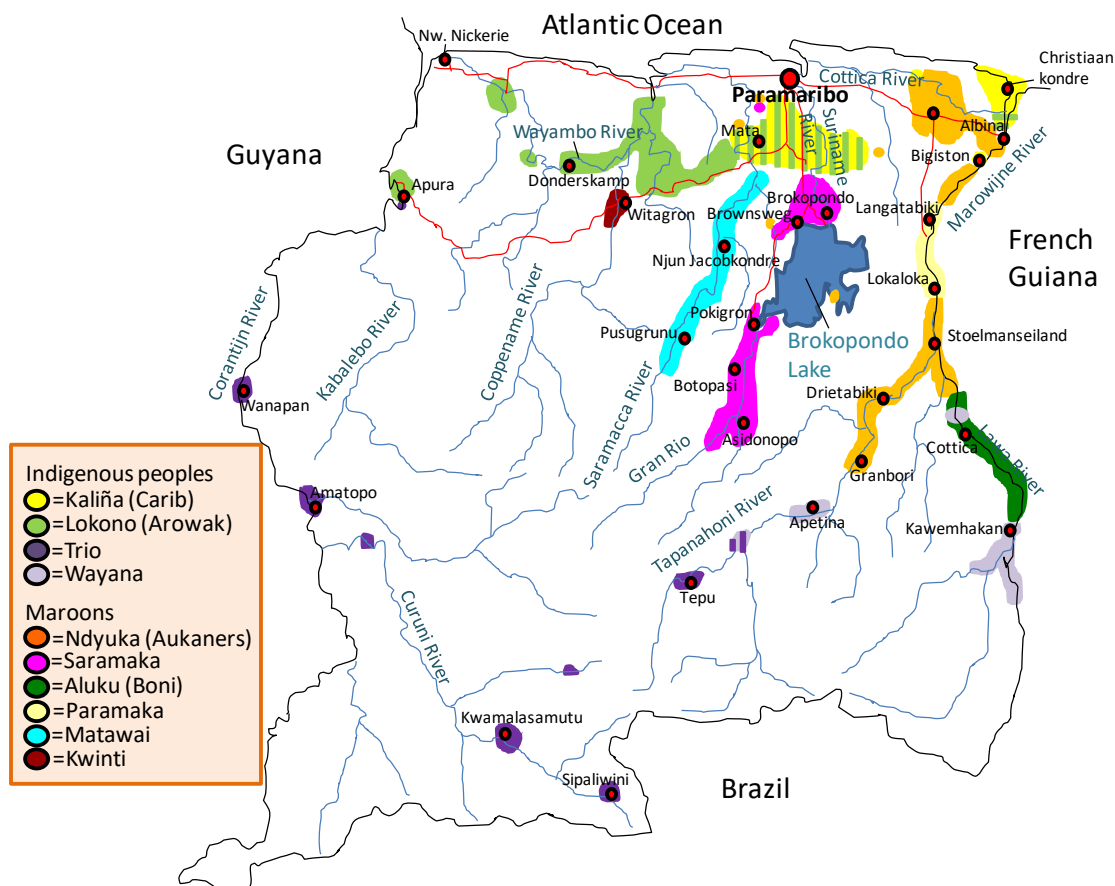


Figure 5.2: Map of various Indigenous and Maroon Groups in Suriname

Source: ITPPF Suriname SCSD Project, 2019

RINA prepared the following table indicating the point of contact for each of the groups identified.

Table 5.6: Indigenous and Maroon Organizations and representations

Type of Stakeholder	Name	Point of Contact
Indigenous community / Organization/ Representation	De Vereniging van Inheemse Dorpschoufden in Suriname (VIDS)	Ms. C. France, Director E-mail: infovids@vids.sr Phone: (597) 520130
Indigenous community / Organization/ Representation	Organization of Indigenous Communities of Suriname – OIS	Mr. Jan Bolman, Primary Contact. E-mail: info@oisur.org Phone: (597) 465618
Indigenous community / Organization/ Representation	TRIJANA. Indigenous and Maroon organizations Palumeu	Ms Fendi Makadepuung, Secretary. Phone: (597) 8841079
Indigenous community / Organization/ Representation	Traditional authority of Palumeu	Ms Fendi Makadepuung, Secretary. Phone: (597) 8841079
Indigenous community / Organization/ Representation	Traditional authority of Apetina	Ms Fendi Makadepuung, Secretary. Phone: (597) 8841079
Indigenous community / Organization/ Representation	KLIM: Organization of Kaliña and Lokono in Lower Marowijne For Alfonsdorp, Pierrekondre, Marijke dorp, Erowarte, Langamankondre, Christiaankondre (Galibi) and, Bigiston	Mr. J. Gunther, Village Head Phone: (597) 8642411
Maroon Community / Organization/ Representation	Association of Saamaka Traditional Authorities (VSG)	Mr. Stiefen Petrusi, Chairman E-mail: bureauvsg@hotmail.com Phone: +(597) 8888146
Maroon Community / Organization/ Representation	KAMPOS (Organization for Kwinti, Aluku, Matawai, Paamaka, Okanisi and Saamaka Maroons)	Via: Mr. Stiefen Petrusi, Chairman VSG Phone: +(597) 8888146
Maroon Community / Organization/ Representation	Cabinet of the Saamaka granman	Mr. I. Adjako, Village Head Phone: (597)8599795
Maroon Community / Organization/ Representation	Okanisi (Ndyuka) traditional Authorities	Mr. J. Lesina, Head Captain Drietabbetje Mobile: +597 7210082
Maroon Community / Organization/ Representation	Cabinet of the Okanisi (Ndyuka) granman	Mr. J. Lesina, Head Captain Drietabbetje Mobile: +(597) 7210082
Maroon Community / Organization/ Representation	Drietabbetje women's organization "Uma Fu Du".	Via: Mr. J. Lesina, Head Captain Drietabbetje Mobile: +597 7210082
Maroon Community / Organization/ Representation	Cabinet of the Matawai granman	Mr. D. Sebastiaan, Head Captain Phone: +(597) 8666218
Maroon Community / Organization/ Representation	Small scale Goldminers organizations: Makambo in New Koffiekamp	Mr. J. Plein and C. Plein (597) 8960936; 8100644

Maroon Community / Organization/ Representation	New Koffiekamp traditional authorities	Captain Schalwijk. (597) 8919003
Maroon Community / Organization/ Representation	Langatabbetje community, Pamaka tribe, District Sipaliwini	Ms. G. Loseng (SDOP) g.loseng@dna.sr (597) 8642411 Mr. Adam Ceder: (597)8732518
Maroon Community / Organization/ Representation	Adjoema Kondre, Ndjuka/Okanisa Maroon Village, District of Marowijne	Mr. Th. Palata, Village Head Phone: (597)8687247
Maroon Community / Organization/ Representation	Kawina community members; Ndjuka or Okanisi tribe/community of the Commewijne District (Java, Peninika, Gododrai,)	Mr. Michel Noordzee Phone: (597)8547588
Maroon Community / Organization/ Representation	New Jacobkondre and surroundings; Matawai tribe/community	Ms T. Elmond, Village Head Phone: (597)8576620
Maroon Community / Organization/ Representation	Companykreek (Ndjuka of Okanisi tribe/community in Brokopondo	Mr. A. Boobe, Village Head Phone: (597)8887241 Mr. Ijzer, Village Head Phone: (597)8796515
Maroon Community / Organization/ Representation	Balingsoela community (Samaka village), Brokopondo	Ms. C. Sinei, Village Assistant Phone: (597)8882806

Source: RINA; 2022

5.1.3.2 FPIC and Engagement

For all projects financed by the WB and that affect Indigenous Peoples, the Bank requires that free, prior and informed consultations (FPIC) be conducted with potentially affected Indigenous Peoples. Particularly if the project involves the commercial development of natural resources (such as minerals, hydrocarbon resources, forests, water, or hunting/fishing grounds) on lands or territories that Indigenous Peoples traditionally own, or customarily use or occupy, the borrower must ensure that the affected communities are informed in a culturally appropriate way using their local dialect and in a clear language to ensure of (a) their rights to such resources under statutory and customary law; (b) the scope and nature of the proposed commercial development and the parties interested or involved in such development; and (c) the potential effects of such development on Indigenous Peoples' livelihoods, environments, and use of such resources. A local translator will participate in this whole process.

The protocol for meaningful engagement, consultation and grievance redress in this Project thus requires that Indigenous Peoples and Maroons affected by the project will have the opportunity to provide their views and feedback in a culturally appropriate manner during Project implementation, as well as to ensure access to appropriate Project benefits.

When approaching the Indigenous Peoples and Maroons, all Project-related communication will be executed according to FPIC principles. Guidelines for the practical application of these principles in Suriname have been described by the Association of Indigenous Village Leaders (VIDS) and the Association of Saramaka Dignitaries (VSG) in the "Community Engagement Strategy for the Government" (2016, version 1.1).⁴

In line with this document, the ITPPF report proposes the following principles guide engagement with Indigenous and Maroon communities which will be followed:

- Respect;
- Capacity building and empowerment - the community should not merely participate as a recipient but should be empowered to independently execute and lead projects;
- Rights-based - any project or program must respect human rights, including the collective rights of Indigenous Peoples and Maroons;
- Full information provision, optimal and continuous communication and transparency;
- Effective participation, in all phases and at all levels;
- Mutual trust, meet commitments made;
- Cultural sensitivity - respect for each other's cultures and way of life, including respect for traditional authorities and local decision-making processes;
- Gender sensitivity - respect for the differentiated roles and responsibilities of women and men, in a way that is both equitable and culturally sensitive; and
- Taking different rights, needs and interests into account of different segments of Indigenous and Maroon communities, including elderly, youth, and so forth.

To start the engagement process, contact will be made with the traditional authorities of the different communities to request permission for a first visit to explain the Project. This first contact will be made by our local coordinator (Salomon Emanuels). In each community, a community contact person may be appointed to serve as the liaison between the company and the community. These persons may help schedule follow-up meetings.

Later, RINA's team will visit the communities according to the agenda and pre-agreed dates to present the Project, inception, and scoping reports. These meetings will be held in the communities, and community members will have an opportunity to ask questions and provide feedback, which will later be integrated into the final reports. Women's participation will be promoted during the meetings and separate consultations will be organized is necessary.

⁴ https://vids.sr/wp-content/uploads/2021/02/community-engagement-strat_compressed.pdf

To ensure continuous engagement with the IP and Tribal communities, communication will be ensured through cellular phone communication and WhatsApp. Although not everyone in the community uses WhatsApp, there is always someone who does. As such, in coordination with the Captain of the community, we will identify one person in the village that will be responsible for receiving information that can then be shared with the Captain and with the other members of the community. Considering the unlikely case that a community does not have anyone using a cell phone or WhatsApp, information will be hand-delivered by Mr. Salomon Emanuels to community members either in Paramaribo or within the community itself.

5.1.4 TASK 4. Establishment of SESA Goals and Objectives

During the engagement process, a key series of questions and corresponding desired outcomes related to sustainable mining development will be raised. Example of preliminary questions are detailed below:

- ✓ How can the development of mining sector affect artisanal and small-scale (ASM) and informal mining?
- ✓ Is there a desire to transition artisanal miners to other trades or perhaps hire them into the overall mining effort? Or perhaps hire them to conduct works that would help preserve the forests of the country and its biodiversity? What are the different options these groups have?
- ✓ How can the mining sector in Suriname be developed while preserving environmental and social integrity at the same time? Particular emphasis will be on answering this question as it relates to Indigenous Peoples, tribal communities, and other vulnerable populations.
- ✓ How can effective stakeholder participation and involvement be incorporated into the process of sustainable mining development as part of Suriname's long term strategic planning?
- ✓ What reforms need to be made to Suriname's legislative framework to ensure that mining development is carried out in accordance with applicable standards and industry best practices?
- ✓ How can mitigating changes be monitored and responded to in order to allow for immediate corrective feedback action?
- ✓ How can all parties involved in or affected by mining development in Suriname be engaged throughout the SESA process?
- ✓ How can coordination between stakeholders to collaborate in resource exploitation be improved?
- ✓ How can transparency in decision making in the mining sector be improved?
- ✓ How can equity in the access/use of resources and distribution of benefits be dealt with?
- ✓ How can existing and potential conflicts in the sector and with other sectors be addressed?

During the stakeholder engagement process, themes of concern, main risks and impacts, strategic options, key restrictions, and major interests will be defined through conversations including questions and answers between RINA and stakeholders. Moreover, consistency of the various sectors objectives and alternatives will be discussed in meetings, workshops, and site visits. The following Table shows the anticipated concern themes, risks, and strategic objectives. A finalized Table will be presented in the consultation.

Table 5.7: Preliminary Strategic Objectives for Main Risks and Impacts from Mining

Theme	Risks & Impacts from Mining (informal and formal)	Strategic Objectives
Water	• Water contamination by mercury and sediments in river ecosystems	• Protect quality of surface fresh water and groundwater

Theme	Risks & Impacts from Mining (informal and formal)	Strategic Objectives
	- Potable water shortages - Surface run-off - Water quality - Water supply	- Protect surface water and aquifer resources - Ensure sustainable water resources management - Protect and preserve marine water quality - Prevent conflicts (including transboundary) over water use, quality and quantity - Promote clean technologies for mining
Air	- Air quality pollution - GHG emissions	- Enhance and/or maintain air quality - Control GHG and pollutant emissions to acceptable standards
Geology and soils	Soil and subsoil contamination	- Protect quality of seabed and sediments - Preserve the hydrogeological environment - Maintain / recover / improve soil quality
Flora and fauna	- Loss of biodiversity - Physical damage to biotopes - Fragmentation of habitats - Removal of wildlife and forest cover	- Preserve terrestrial and river ecosystems and their biodiversity - Preserve critical habitats and protected or endangered species - Ensure that protected areas (e.g., parks) are not affected by mining development - Contribute to conservation of wildlife and wildlife habitats - Implement an effective monitoring program - Include the IP and maroon societies in preservation activities of flora and fauna
Socio-economic	- Water conflicts - Job creation and revenue sharing - Disturbance to family networks, community structures and cultural and ethnic identities - Increase in drug trade and prostitution - Malaria and HIV	- Preserve social structure integrity - Sustain and enhance agricultural and cattle economic activities - Enhance community development - Ensure benefits from mining development - Avoid resource conflicts - Minimize expectations over benefits of mining development
Land use and land tenure	- Allocation of land for mining activities - Economic and physical displacement	Avoid, mitigate and/or compensate land acquisition in accordance with best practices

Theme	Risks & Impacts from Mining (informal and formal)	Strategic Objectives
Indigenous Peoples (IP) Maroons, tribal communities	- Involuntary resettlement of IP - Livelihood impacts of IP	- Protect Indigenous People and Tribal People and other vulnerable groups - Recognize cultural diversity
Health and safety	- Potential for significant effects on human health - Malaria and other diseases spread (including sexually transmitted diseases) - Exposure to mercury during processing and its presence in food and water	- Prevent disruption, disturbance and nuisance to communities - Ensure public and community safety in areas of mining operations - Ensure a safe work environment for mining operators and employees
Institutional, legal and regulatory	- Conflicting roles and responsibilities for environmental protection and management - Poor inter-institutional coordination - Unknown training requirements and capabilities - Capacity for SESA implementation	- Implement effective, transparent inter-institutional co-ordination - Reduce and remove institutional barriers and duplication of effort - Improve institutional capacity and training - Establish clear, practical regulatory standards - Implement an efficient environmental assessment process - Implement monitoring, follow-up and compliance initiatives - Ensure that progressive rehabilitation and mine closure planning are part of Life-of-Mine Planning
Risk management	- Cumulative effects - Potential pollution of remote uncontaminated areas	- Ensure that risks associated with mining development are identified and managed. - Implement major hazards / Environmental risk management
Cultural resources	Loss or alteration of cultural, archaeological and historical sites	Preserve and valorize cultural heritage
Regional	Trans-boundary pollution	- Manage trans-boundary effects - Manage cumulative effects

Source: RINA, 2022

5.1.5 TASK 5. Conducting the validation workshops

Task 5 aims to ensure the participation of all stakeholders through consultative and validation workshop and meetings. The first workshop will be held upon the submission of the Inception Report draft version. The draft will be submitted to MNR and WB for review and comments. After a maximum of five working days, comments will be returned to RINA for their inclusion in the final Draft Inception Report. Upon completion of the Draft Inception Report, RINA will present the Draft Scoping Report to MNR and WB for review and comments. After a maximum of five working days, comments will be returned to RINA for finalization of the Draft Scoping Report.

The RINA team will then travel to Suriname to present the Inception and Scoping Reports to the identified stakeholders for consultation and validation. The presentation will include an explanation of the SESA methodology and its main benefits for Suriname. Invitations will be sent to stakeholders for their participation in the consultation and validation workshop. RINA will collect all comments, suggestions, and recommendations from stakeholders for inclusion in the reports.

Field visits to Indigenous and tribal communities will be conducted to better understand the realities of mining communities, and determine the main interests, concerns, and mitigation measures if required. These visits will further ensure the contributions of Indigenous and tribal communities to the SESA and their validation of the Reports. All collected information from the workshop and the field visits will help RINA define the strategic options to be examined, key restrictions, major interests, and how consistency of the various sector objectives and alternatives can be ensured. Details for the validation workshops are presented below. The agenda is included as an Appendix B of this report.

5.1.5.1 First Validation Workshop

The first workshop will be held between October 24 and November 5, 2022, following the submission of the Draft Inception and Scoping Reports.

The objectives of the first workshop will be to explain to the invited stakeholders what a SESA is, present the proposed methodology, present the main benefits of the SESA for Suriname, and to collect comments, concerns and recommendations from the stakeholders for the development of the SESA. Findings from the Scoping Report will be presented to complete data in the Scoping Report and verify the data and RINA's understanding of Suriname's mining sector. A list of stakeholders attending the workshop will be prepared.

During the first workshop, RINA will:

- ✓ Present the Inception and Scoping Reports;
- ✓ Collect and identify the main interests, concerns, and advice of each stakeholder group regarding the SESA work plan;
- ✓ Define strategic options to be examined, key restrictions, major interests, and how the consistency of the various sector objectives and alternatives can be ensured; and
- ✓ Disclose the process for stakeholder participation throughout the SESA, including Indigenous, tribal and Maroons peoples.

5.1.5.2 Second Validation Workshop

The objective of the second workshop will be to:

- ✓ Present the SESA's main findings and recommendations; and
- ✓ Record and address doubts, concerns, and recommendations to ensure its applicability in Suriname.

The second workshop will be held upon the submission of the SESA draft report which is expected to take place towards the end of the Project, between the months of August and October of 2023, with the exact dates to be confirmed.

5.1.5.3 Development of the Workshops

Both workshops will start with an initial expository and informative session and PowerPoint slides summarizing the main points of the Inception and Scoping Reports during the first workshop, and the Draft SESA and its results during the second workshop.

Stakeholders will be consulted in various events in large groups or in focus groups based on availability⁵. The participation of women during the workshops will be ensured. Anticipated events for consultation are as follows:

- ✓ Workshop with Government officials.
- ✓ Workshop with NGOs, CSOs, Associations, Foundations, Consultation with Indigenous people and Maroon groups and representatives per FPIC protocol.
- ✓ Workshop with NGOs, CSOs, Associations, Foundations.
- ✓ Workshops with mining companies mining organizations and private companies.

5.2 STAGE 2. DESCRIPTION OF THE REGULATORY AND INSTITUTIONAL FRAMEWORK OF THE GOS CAPACITY FOR ENVIRONMENTAL, SOCIAL AND OCCUPATION HEALTH AND SAFETY ASSESSMENT AND MANAGEMENT

5.2.1 TASK 6. Development of an Institutional, Legal and Regulatory Framework and GOS Capacity Assessment

The development of an institutional, legal and regulatory framework and the GOS capacity assessment for the mining sector in Suriname will comprise the following activities:

- ✓ Identify the institutions responsible for policy, planning, regulating, and enforcing legislation in the mining sectors.
- ✓ Compile and describe the environmental, social, safety and occupational health legal, regulatory, and institutional framework. A preliminary compilation of Suriname's legal framework is presented as Appendix C.
- ✓ Assess existing capacity of the MNR and NIMOS. To assess the capacity of MNR and NIMOS, RINA will perform the following activities:
 - Design an evaluation methodology for institutional capacity to fulfill mission driven mining sector responsibilities. Relying as a reference on publicly available good practices and methodologies from other entities, propose a capacity assessment tool to be applied to MNR and NIMOS.
 - Assessment tool calibration. The designed methodology and tool should be presented to the technical team of mining experts and other international and local consultants designated by RINA evaluation before being implemented in the field.
 - Technical assistance for the application of the tool. Orientation will be provided to the RINA consultant team⁶ that will apply the tool in the field and any questions that arise during implementation will be answered remotely.
 - Analysis of information and preparation of evaluation reports. Once the field collection stage is completed, RINA will analyze the information collected and prepare a report

⁵ As some stakeholder groups may not always feel comfortable sharing their opinions with government authorities, the consultation workshops are to be divided into different focal groups.

⁶ 1 Team Leader, 1 Project Coordinator, 1 Public Participation Specialist, 1 Mining Specialist, 1 Institutional Assessment Specialist, 1 Artisanal mining Specialist and 1 Social Specialist.

detailing results and conclusions that shows the main strengths and weaknesses identified, as well as recommendations to improve institutional capacity.

- ✓ Prepare a compilation of relevant national policies, laws, regulations, guidelines, and contracts, including environmental licensing requirements. This will include all relevant WB standards, guidelines and policies including all the operational policies as referenced earlier in this report.
- ✓ Assess the adequacy of the national framework, based on international experience, and the capacity of public agencies at national and local levels, governmental organizations at national and community levels to address the potential impacts of developments in mining, especially for licensing and enforcement of legal ESHS requirement.
- ✓ Assess mandates, capacity, incentives, and transparency in licensing/permitting, monitoring and enforcing environmental, social and occupational health and safety regulations in the above-mentioned sectors.
- ✓ Assess the adequacy of the existing grievance mechanisms to protect landowners and affected people, Indigenous and tribal communities; and, if applicable, we will provide recommendations to improve such mechanisms.
- ✓ Assess the adequacy of existing mechanisms for the social distribution of benefits to affected people; If applicable, provide recommendations to improve such mechanisms.
- ✓ Assess the adequacy of the EIA system in Suriname, and identify potential deficiencies in the environmental permitting processes, EIA review, EIA follow-up and enforcement in the above-mentioned sectors. Recommendations to improve the EIA system, EIAs, EIA follow-up and enforcement.
- ✓ Identify potential gaps regarding international good practices, not only limited to the WB policies and safeguards but also practices from ILO, such as ILO 169 and others.
- ✓ Identify the existing environmental, social, occupational health and safety requirements that are included in mining sector contracts with the GOS.
- ✓ Verify compliance with Suriname's ratified conventions including all eight ILO fundamental conventions (C029, C087, C098, C100, C105, C11, C138 and C182) and three of the four governance conventions (C081, C129 and C144).
- ✓ RINA will evaluate whether mining projects are equipped with appropriate tools to ensure compliance with these core labor rights, specifically the prohibitions on discrimination, child labor, forced labor, and limitations on freedom of association and collective bargaining. The tools should include labor rights policies, work regulations, worker contracts, worker codes of conduct, indirect worker policies, among others. For example, one potential labor compliance gap in Suriname mining project policies is related to the failure to cover indirect workers. A significant gap in Suriname's current legislation is the absence of national legislation on Indigenous and tribal rights and the failure to sign ILO Convention 169 - Indigenous and Tribal Peoples Convention, 1989. The absence of such legislation allows for tensions to emerge among Indigenous communities and mine operators without a framework to mitigate and address conflict.
- ✓ Finally, RINA will Review of the Mining Sector Contracts with GOS, and will:
 - Identify the existing environmental, social, occupational health and safety requirements that are included in mining sector contracts with the GOS, and assess if additional environmental, social, and occupational health and safety safeguards need to be incorporated in such contracts.
 - Provide the information necessary and specific recommendations on environmental, social, occupational health and safety minimum requirements that the GOS should include in the Contracts with private sector companies.
 - Review contracts between mining companies in other countries in the region that can be used as a comparison.

5.3 STAGE 3. IDENTIFICATION OF THE GENERAL, STRATEGIC AND SOCIAL CONSEQUENCES RESULTING FROM FORECASTED FUTURE DEVELOPMENT OF MINING

5.3.1 TASK 7. Analysis of Future Developments in Mining Environment and Social Consequences

During this stage RINA will conduct the following:

- ✓ Develop forecasts of future expansions in mining sectors, including mapping of potential development scenarios. These will forecast developments over a defined period of time and include mapping of potential and actual sites that are being explored for mining, or that may be explored. Maps will be prepared and presented in the scoping report showing overlap of Mining Exploration and Exploitation polygons
- ✓ Identify and evaluate the environmental and social consequences and risks, including climate change risks associated with existing developments and plans for investment in mining, and safety and occupational health risks and issues relevant to mining, considering the forecasts and scenarios developed. This will include mapping to identify overlapping areas of natural resources/forests/ other areas of high biodiversity value, with potential and actual extraction sites.
- ✓ Include a particular focus on Indigenous and tribal communities and on the possible impacts of potential future mining sector developments on social consequences and risks for these vulnerable groups.
- ✓ Evaluate cumulative / synergistic / secondary impacts as well as impacts of individual developments in mining should also be included. Coupled with this the consultants should estimate or model the potential impacts to the ecosystems and vulnerable populations such as indigenous and tribal communities from expansion in the mining sector over a defined period of time.
- ✓ Investigate, without exclusion, strategic environmental, social and occupational health and safety issues related to increased private sector operations in mining and:
 - Campsites (increased demand for land conversion, potential GHG emissions from land use conversion, increased demand for water and potable water, loss of biodiversity; transformation of natural ecosystems and consequent loss of environmental services associated with these ecosystems; increase in greenhouse gas emissions in the face of possible deforestation of forest ecosystems; increase in the consumption and contamination of water; social conflicts (IPs and Maroons) over access to local resources; community safety); associated access roads and infrastructures (site clearing, loss of biodiversity, fragmentation of habitats, resettlement and loss of livelihoods, community safety); processing facilities (discharge of contaminants; air and water pollution; impacts of accidental spills; community safety).
 - Labor influx and associated issues related to gender-based violence, security forces, community-level violent conflict, forced and child labor, and human trafficking. This should include identification of appropriate mitigation measures, including for example review or creation of labor and civil rights protection authorities or regulatory frameworks.
- ✓ Gather and process information from the informal ASM sector to understand how this sector may be impacted by a more formalized set of procedures and approaches; present an overview of small-scale mining ASM good practices initiatives in neighboring Latin American countries. Explain international certification seals such as Fairmined, Fairtrade, Green Gold and Craft that guide ASM formalization processes.

- ✓ Develop criteria to prioritize environmental, social, and occupational health and safety issues in the context mining in Suriname. To the extent possible, such criteria will allow for quantification to objectively compare and categorize competing issues.
- ✓ Consider resettlement practices, whether positive or unintentionally negative, as a result of mining activities.

5.4 STAGE 4. PROPOSED MITIGATION AND MONITORING MEASURES

5.4.1 TASK 8. Submit Mitigation Measures Including Indicators for Monitoring and Evaluation

During this stage RINA will:

- ✓ Identify potential mitigation measures and monitoring requirements for each potential strategic impact on mining. Mitigation measures for strategic impacts may include, among others: implementation of appropriate legislation, including legal environmental discharge standards for emissions into air, water and soil; measures specifically targeting engagement with indigenous and tribal communities, drawing on initial mitigation measures to enable effective benefit sharing with such communities identified in the ITPPF and Rapid Social Assessment; revised institutional organization, streamlining environmental responsibility, creation of new GOS agency/ies; strengthening of specific GOS agencies; review or creation of environmental and social licensing agencies or mechanisms; enforcement of international best practices and operational standards guidelines for each sector; creation of protected areas or exclusion zones (No Go zones, buffer zones); etc. Compensation and sustainable ideas involving the IP communities in mitigating the effects of mining through protection of forests and biodiversity with potential cash transfer will be addressed in the SESA process.
- ✓ Select specific indicators to allow for follow up on the effectiveness of proposed mitigation measures. Indicators may be individual/stand-alone or combination (complex index), such as established indices (representing poverty reduction, migratory bird species, water quality, etc.).
- ✓ Consider good practice international standards, like the WB Safeguards and ILO policies when developing the mitigation and monitoring measures, by proposing how these practices can enhance the existing Environmental and Social Framework for mining within the Surinamese context. This will draw on lessons learned and good practices, such as effective stakeholder engagements with indigenous and tribal communities by existing mining companies such as Newmont.
- ✓ Consider resettlement practices as part of the mitigation hierarchy resettlement good practices since this could help to enhance the Environmental and Social Framework.

5.5 STAGE 5. FORMULATION OF RECOMMENDATIONS AND A SESA ACTION PLAN

5.5.1 TASK 9. Development of the SESA Action Plan

Based on the analysis conducted, and the recommendations developed, an Action Plan will be developed by RINA to facilitate implementation. Priorities will be structured into immediate, short term (1-2 years) and medium term (3-5 years), including an Action Plan Cost Estimate.

The objectives of the SESA Action Plan will be:

- ✓ To present a series of recommendations arising from key findings of the SESA;
- ✓ To suggest a priority timeline for implementation of the recommendations;
- ✓ To identify the GOS Agency responsible for implementation;

- ✓ To provide some preliminary costs that will require confirmation and adjustment by applicable GOS Authorities.
- ✓ To formulate a policy matrix comprising of concrete policy, institutional and governance recommendations (short-term, mid-term and long-term), verifiable indicators as part of a monitoring program, action plans, capacity building plans and budgets for its implementation and monitoring. The policy matrix, implementation and monitoring plans will also be validated in this national workshop.
- ✓ Organize a consultation workshop in line with FPIC procedures with all stakeholders to validate the draft report and action plan.

The Action Plan will contain the preliminary related recommendations organized as follows:

- ✓ Finding: refers to a summary of the SESA key finding and forms the basis for the Recommended Actions.
- ✓ Priority: organized in three priorities: immediate (2023-2024), short-term (2024-2025) and medium to long term (2026-2028). These priorities will define the ideal time span within which to follow up on the Action Plan, and therefore the implementation of the recommendations.
- ✓ Justification: represents the justification for implementing the action.
- ✓ Recommended Action: a series of action(s) which should be undertaken to implement the recommendations.
- ✓ Anticipated Outcomes: describes the anticipated outcome if the action is fully implemented.
- ✓ Responsibility: assigns the responsibility for implementation of the action.
- ✓ Cost: presents a preliminary cost estimate which should be re-determined by the responsible Suriname authority.

This Action Plan will include a suggested prioritization of activities or reforms that could be directly financed or supported by the Suriname Competitiveness and Sector Diversification (SCSD) Project during its implementation. This would include potential support to MNR/NIMOS or other GOS agencies, as well as to representatives of social or environmental groups to strengthen their capacity to engage in future discussions and planning about mining sector development.

RINA will use the interim outputs prepared at the end of each phase to compile the draft SESA report and Action Plan. Findings, recommendations, and reports will be submitted for consultation to the various stakeholders. The consultation will include IP and Maroons as guided by the FPIC protocols. The draft SESA Report and Action Plan will be presented for final consultation and validation.

5.5.2 TASK 10. Realization of the National Consultative Workshop and Validation of SESA Action Plan and Report

The objective of the second workshop is to present the SESA's main findings and recommendations and record and address doubts, concerns, and recommendations. The second workshop will be held upon the submission of the Draft SESA Report and Action Plan.

A workshop will be organized to validate the Draft SESA Report and Action Plan. Invitations will be sent to all stakeholders for their participation in the workshop. Stakeholders will be consulted, reviewing the Draft SESA report and Action Plan, along with its findings and recommendations. All doubts, concerns and recommendations from stakeholders will be compiled and reported in a stakeholder consultation report. Field visits will be organized to ensure the equal participation of Indigenous and Tribal communities that could not attend the national workshop that is to be held in Paramaribo. All comments, observations and contributions by the stakeholders are to be considered for the preparation of the final draft SESA report.

5.6 STAGE 6. FINAL CONSULTATIONS, REVIEW AND APPROVAL

5.6.1 TASK 11. Preparation of the Final Report

The final SESA report and Action Plan will be prepared and will include the Stakeholder Consultation Report, the Policy Matrix, recommendations, implementation and monitoring plans with indicators, institutional arrangements, and budget. RINA will be available on request to advise on the environmental, social, occupational health and safety implications of any policy and/or regulatory measure proposed as part of the SESA.

5.7 CONCURRENT STAGE - CAPACITY BUILDING AND ADVISORY SERVICES

5.7.1 TASK 12. Capacity building and advisory services

The capacity-building program represents an important and critical tool towards the successful implementation of the SESA. As such, the capacity-building program will be implemented during the development of the various Stages and Tasks of the Project.

5.7.1.1 Objectives

The general objective of the sector administration capacity-building is to assist the GOS, staff from the MNR, the Minerals Institute, NIMOS and other relevant ministries including the Ministry of Finance, to build the required technical skills and managerial capabilities to regulate and monitor the mining sector in the country.

The specific objective is the transfer of knowledge on SESA to the Project Implementation Units, implementing ministries, associated agencies, and other GOS officials as appropriate. This includes on-the-job-training, presentations of international experience, and participatory workshops.

The capacity-building program focuses on the reinforcement of:

- ✓ Technical understanding of exploration and operation mining activities;
- ✓ A full grasp of environmental and social issues; and
- ✓ Capacity to effectively supervise, control and measure potential environmental and social effects.
- ✓ Cultural competency trainings for GOS staff

The participants will learn how to reach the capacity-building objectives for each Stage and Task of the Project to develop the SESA.

5.7.1.2 Overview

The purpose of the capacity-building program is to:

- ✓ Provide advisory services and knowledge transfer to GOS in:
 - Assessing potential environmental and social impacts, and occupational health and safety risks;
 - Understanding existing laws and regulations related to the mining sector;
 - Assisting in the implementation of potential policy, legislative, or regulatory changes under consideration.
- ✓ Monitor the progress of the implementation of the SESA recommendations.
- ✓ Identify the main difficulties and bottlenecks faced by the implementing officer and advise on how to resolve and manage them.

- ✓ Provide recommendations for technical assistance support that environmental and social advocacy and representative groups, particularly Indigenous and tribal communities, can use to help manage environmental and social impacts of future mining developments.
- ✓ Provide technical advice to GOS officials through meetings, video calls and emails, during the development of the various Stages and Tasks of the Project. Anticipated activities are as follows:

Table 5.8: Capacity Building Program

Preliminary subject	Presenter	Date	Participants
- Stakeholder engagement according to IFC - Public consultation and involvement	Social Expert	TBD	TBD
- Other countries experiences and samples - Capacity assessment - General mining technical training - Mercury - Mining resources	Anthony Sanford Nancy del Prado	TBD	
- Environmental and Social Impact Assessment - Database development and management - Environmental damage and risk assessment - Resettlement, land acquisition, grievance, and compensation	Anthony Sanford Jacqueline Castro	TBD	
- Monitoring, compliance, regulations, audits, etc. - Environmental and social monitoring methods and practices	Julie Evans	TBD	
Monitoring and follow up.	Julia Evans	TBD	

Source: RINA, 2022

6 PROPOSED TOC FOR THE SESA REPORT

Below is the proposed table of contents for the SESA report:

- ✓ Executive Summary
 - Project Context
 - SESA Objectives
 - SESA Methodology
 - Key Findings
 - Institutional, Legal and Regulatory Framework
 - Mining Sector Development Context
 - SESA Action Plan
 - Immediate Action
 - Short Term Action
 - Medium to Longer Term Action
- ✓ Project Background
 - Background and context
 - SESA ToR
 - SESA Objectives
 - Vision for Sustainable Management of Suriname Mining Sector
 - Project Team
 - Public Consultation
 - SESA Methodology
 - Monitoring and Follow-Up
 - Reporting
 - Organization of the SESA Report
- ✓ Mining Potential Trend Scenarios
- ✓ Critical Environmental and Social Risks
- ✓ Mitigation and monitoring measures
- ✓ Key SESA Findings
 - Institutional Capacity
 - Policy, Legal and Regulatory Framework
 - Mining Development Context
 - Concluding Remarks
- ✓ SESA Action Plan

- ✓ Action Plan
- ✓ Capacity Building Plan
- ✓ SESA Implementation
- ✓ List of Tables and Figures
 - ✓ Appendices
 - ✓ References



7 TIMELINE

The following chart describes the timeline for the accomplishment of the assignment.

Tasks	Activities	2022					2023											
		Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	
STAGE 1. SCOPING																		
TASK 1. PROJECT KICK-OFF MEETING (KoM)	Define communication channels																	
	Review project schedule																	
	Confirm collaborative and consultative methodology																	
	Finalize contracting procedures with sub contractors																	
TASK 2. DESKTOP REVIEW	Set up the context for the SESA																	
	Assemble preliminary information relevant to the scope of the SESA																	
	Forecast future mining development scenarios to be developed subsequently																	
	Identify the environmental and social risks and vulnerabilities to be considered and assessed																	
	Identify the environmental and social characteristics of the areas																	
TASK 3. STAKEHOLDERS MAPPING	Review existing policies, programs, plans, studies, and public information material																	
	Conduct a stakeholder analysis to identify key stakeholders																	
	Identify and map stakeholders in the mining sector that should be consulted																	
TASK 4. ESTABLISHMENT OF SESA GOALS AND OBJECTIVES	Analyze stakeholder's interests, concerns and incentives.																	
	Prepare consultation questionnaire																	
	Confirm goals and objectives of the SESA																	
	Define themes, risk, impacts, interest																	
	Develop a work plan for consultations that will ensure stakeholders' interests, concerns and advice																	
TASK 5. REALIZATION OF THE CONSULTATIVE AND VALIDATION WORKSHOP AND MEETINGS WITH THE STAKEHOLDERS	Secure meetings with Stakeholders																	
	Organize invitations for stakeholders to the workshop																	
	Coordinate the consultation and validation WORKSHOP and meetings																	
	Coordinate field visits for validation in remote indigenous and tribal communities																	
	RINA Team in Suriname for field visits and Validation workshop (See detailed program)																	
	Explain what is a SESA, its methodology, its main benefits for Suriname, and collect the stakeholders' recommendations																	
	Define the process for stakeholder participation through the SESA, in particular with representatives of indigenous and tribal groups																	
	Ensure FPIC guidelines for consultation of indigenous and tribal groups throughout the SESA																	
	Collect and identify stakeholders' main interests, concerns and advice																	
	Define the strategic options to be examined, the key restrictions, major interests, and how consistency of the various sectors objectives and alternatives can be ensured.																	
	Conduct in person capacity building for identified GOS officers																	
	Validate the scoping with stakeholders																	
Finalize the Scoping and Inception Reports																		
Presentation of Draft Inception report, PPT, methodology, work plan and stakeholders list																		
Presentation of Draft scoping report																		
Deliverable 1.	Submission and acceptance of the Inception Report																	
	Presentation on the Inception report, methodology and approach																	
	Stakeholders list																	
Submission and acceptance of the Scoping Report																		



Tasks	Activities	2022					2023										
		Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
STAGE 2. Description of the regulatory and institutional framework																	
TASK 6. DEVELOPMENT OF AN INSTITUTIONAL, LEGAL AND REGULATORY FRAMEWORK AND GOS CAPACITY ASSESSMENT	Identify the institutions responsible for policy, planning, regulating, and enforcing legislation in the mining sectors.																
	Compile and describe the environmental, social, safety and occupational health legal, regulatory, and institutional framework																
	Prepare a compilation of relevant policies, laws, regulations, guidelines, and contracts, and environmental licensing including relevant WBG standards, guidelines and policies including all the operational policies.																
	Assess the adequacy of the framework, to address the potential impacts of developments in																
	Assess the adequacy of the existing grievance mechanisms																
	Assess the adequacy of existing mechanisms for the social distribution of benefits to affected																
	Assess the adequacy of the EIA system in Suriname																
	Identify potential deficiencies in the environmental permitting processes , EIA review, and EIA follow-up and enforcement																
	Identify potential gaps regarding the international good practice																
	Identify the existing environmental, social, occupational health and safety requirements that are included in mining sector contracts with the GOS																
	Verify compliance with Suriname’s ratified conventions																
	Evaluate whether mining projects have broad tools that enable them to ensure compliance with core labor rights																
	Review Mining Sector Contracts with GOS																
	Assess existing capacity of the relevant regulatory institutions																
	Assess mandates, capacity, incentives, and transparency in licensing/permitting, monitoring and enforcing environmental, social and occupational health and safety regulations																
	Prepare the first Interim Report																
Draft report Regulatory and Institutional framework																	
Draft report Assessment of the Government of Suriname capacity																	
Deliverable 2.	Submission and acceptance of the Regulatory and Institutional framework																
	Assessment of the Government of Suriname capacity																



Tasks	Activities	2022					2023										
		Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
STAGE 3. Identification of the general, strategic environmental and social consequences resulting from forecasted future development of mining.																	
TASK 7: ANALYSIS OF FUTURE DEVELOPMENT OF MINING AND SOCIAL CONSEQUENCES	Develop forecasts of future expansions in mining sectors including mapping of potential and actual sites that are being explored for mining or that could be explored.																
	Identify and evaluate the environmental and social consequences and risks																
	Include a particular focus on indigenous and tribal communities and of the potential impacts of potential future mining sector developments on these vulnerable groups.																
	Evaluate cumulative / synergistic / secondary impacts as well as impacts of individual developments in mining.																
	Estimate or model the potential impacts to the ecosystems and vulnerable populations such as indigenous and tribal communities from expansion in the mining sector																
	Identify best practices and lessons learnt in neighboring countries																
	Analyze initiatives from institutions and NGOS i.e. Green Gold initiative.																
	Collect and process information on artisanal mining and small-scale mining (ASM) in Suriname																
	Explain how the informal sector may be impacted by a more formalized set of procedures and approaches																
	Develop criteria to prioritize environmental, social, and occupational health and safety issues																
	Quantify and objectively compare and categorize competing Environmental, social H&S																
	Review resettlement practices as a result from mining activities.																
	Prepare the Second interim report																
Draft report Identification of the general, strategic environmental and social consequences																	
Deliverable 3.	Submission of the Identification of the general, strategic environmental and social consequences																



Tasks	Activities	2022					2023										
		Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Stage 4: Proposed mitigation and monitoring measures																	
TASK 8 MITIGATION MEASURES INCLUDING INDICATORS FOR MONITORING AND EVALUATION	Identify potential mitigation measures and monitoring requirements for each potential																
	Review institutional organization, streamlining environmental responsibility, creation of new GOS agency/ies; strengthening of specific GOS Agencies																
	Review or create environmental and social licensing agencies or mechanisms																
	Provide compliance mechanisms for enforcement of international best practices and																
	Propose implementation of international best practices and operational standards guidelines																
	Develop specific indicators to allow for the follow-up of the effectiveness of proposed mitigation measures.																
	Develop mitigation and monitoring measures that can enhance the existing Environmental and Social Framework for mining considering good practice international standards like the World Bank Safeguards and ILO policies																
	Consider resettlement Practices as part of the mitigation hierarchy resettlement good																
	Prepare the third interim report																
Draft report Proposed mitigation and monitoring measures																	
Deliverable 4.	Proposed Mitigation and monitoring measures report.																



Tasks	Activities	2022					2023										
		Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Stage 5: Formulation of recommendation and a SESA Action Plan																	
TASK 9: DEVELOPMENT OF THE SESA ACTION PLAN	Develop an Action Plan																
	Formulate and present recommendations																
	Suggest a priority timeline for implementation of the recommendations																
	Identify the GOS Agency responsible for implementation																
	Provide some preliminary costs that will require confirmation and adjustment by applicable GOS Authorities.																
	Formulate a policy matrix																
	Organize a consultation workshop in line with FPIC procedures with all stakeholders to validate the draft report and action plan.																
TASK 10. REALIZATION OF THE NATIONAL CONSULTATIVE WORKSHOP	Organize invitations for stakeholders to the workshop																
	Coordinate the national workshop with all stakeholders																
	Consult stakeholders on the draft Report																
	Present SESA's main findings and recommendations;																
	Record and address doubts, concerns, and recommendations																
	Coordinate field visits for validation in remote indigenous and tribal communities where																
	Prepare Draft SESA report and the Action Plan																
Draft report SESA Action Plan																	
Deliverable 5.	Submission and acceptance of the of the Draft SESA Report and Action Plan.																



Tasks	Activities	2022					2023										
		Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Stage 6: Final consultations, review and approval																	
TASK 11 PREPARATION OF THE FINAL REPORT	Present SESA main findings and recommendations																
	Prepare stakeholder consultation report																
	Review information gathered from consultation																
	Prepare the final version of the SESA report, policy matrix, recommendations and implementation and monitoring plans with indicators, institutional arrangements and budget																
Deliverable 6.	Submission and acceptance of Final SESA Report and SESA Action Plan																

Tasks	Activities	2022					2023										
		Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
CONCURRENT STAGE: Capacity Building and Advisory Services																	
TASK 12. CAPACITY BUILDING AND ADVISORY SERVICES	Identify capacity building needs of GOS based on the results of the GOS assessment																
	Develop a capacity building plan for advisory services and knowledge transfer for GOS																
	Capacitate GOS with hands-on activities following the development of the SESA																
Draft Capacity Building Program																	
	Complete Capacity building program to be included in the SESA report																

8 REFERENCES

LEGAL FRAMEWORK

FAO; Country Programming Framework 2016 2019; 2015; [FAO Country Programming Framework 2016 2019.pdf](#)

Government of Suriname; The Constitution of the Republic of Suriname; 1987; [Suriname Constitution Republic of Suriname English.pdf](#)

FAO; Summary Suriname Environmental framework act 2020 EN; 2020; [FAO Summary Suriname Environmental framework act 2020 EN.pdf](#)

IFC (World Bank) ; E&S Performance Standard 1 Assessment and Management of Environmental and Social Risks and Impacts; 2012; [IFC E&S Performance Standard 1 Assessment and Management of Environmental and Social Risks and Impacts.pdf](#)

IFC (World Bank); E&S Performance Standard 2 Labor and Working Conditions; 2012; [IFC E&S Performance Standard 2 Labor and Working Conditions.pdf](#)

IFC (World Bank); E&S Performance Standard 3 Resource Efficiency and Pollution Prevention; 2012; [IFC E&S Performance Standard 3 Resource Efficiency and Pollution Prevention.pdf](#)

IFC (World Bank); E&S Performance Standard 4 Community Health, Safety, and Security; 2012; [IFC E&S Performance Standard 4 Community Health, Safety, and Security.pdf](#)

IFC (World Bank); E&S Performance Standard 5 Land Acquisition and Involuntary Resettlement; 2012; [IFC E&S Performance Standard 5 Land Acquisition.pdf](#)

IFC (World Bank); E&S Performance Standard 6 Biodiversity Conservation and Sustainable Management of Living Natural Resources; 2012; [IFC E&S Performance Standard 6 Biodiversity.pdf](#)

IFC (World Bank); E&S Performance Standard 7 Indigenous Peoples; 2012; [IFC E&S Performance Standard 7 Indigenous Peoples.pdf](#)

IFC (World Bank); E&S Performance Standard 8 Cultural Heritage; 2012; [IFC E&S Performance Standard 8 Cultural Heritage.pdf](#)

IFC (World Bank); EHS Environmental, Health, and Safety General Guidelines; April 2007; [IIFC Environmental, Health, and Safety guidelines.pdf](#)

IFC (World Bank); EHS Environmental, Health and Safety Guidelines for Mining; December 2007; [IFC Environmental, Health and Safety Guidelines for Mining.pdf](#)

IFC PERFORMANCE STANDARDS ON ENVIRONMENTAL AND SOCIAL SUSTAINABILITY

ILO; Indigenous and Tribal Peoples Convention No.69; 1991; [ILO Convention 169 Indigenous and Tribal Peoples.pdf](#)

ILO; Handbook for ILO Indigenous and Tribal Convention; 2013; [ILO Handbook for ILO Indigenous and Tribal Convention.pdf](#)

OHSAS 18001; BS OHSAS 18001 has been withdrawn – upgrade to ISO 45001 now; 2021; [OHSAS 18001 REQUIREMENTS replaced.pdf](#)

World Bank ; Operational Policy OP 4.01 - Environmental Assessment ; January 1999; [World Bank Operational Policy OP 4.01 - Environmental Assessment.pdf](#)

World Bank; Operational Policy OP 4.01 - Annex A; January 1999; [World Bank Operational Policy OP 4.01 - Annex A.pdf](#)

World Bank; Operational Policy OP 4.01 - Annex B; January 1999; [World Bank Operational Policy OP 4.01 - Annex B.pdf](#)

World Bank; Operational Policy OP 4.01 - Annex C; January 1999; [World Bank Operational Policy OP 4.01 - Annex C.pdf](#)

World Bank; Operational Policy OP 4.10 Indigenous Peoples; July 2005; [World Bank Operational Policy OP 4.10 Indigenous Peoples.pdf](#)

World Bank; Operational Policy OP 4.10 Indigenous Peoples Annex A ; July 2005; [World Bank Operational Policy OP 4.10 Indigenous Peoples Annex A.pdf](#)

World Bank; Operational Policy OP 4.10 Indigenous Peoples Annex B; July 2005; [World Bank Operational Policy OP 4.10 Indigenous Peoples Annex B.pdf](#)

World Bank; World Bank Operational Policy OP 4.10 Indigenous Peoples Annex C; July 2005; [World Bank Operational Policy OP 4.10 Indigenous Peoples Annex C.pdf](#)

World Bank; Operational Policy OP 4.11 Physical Cultural Resources; July 2006; [World Bank Operational Policy OP 4.11 Physical Cultural Resources.pdf](#)

World Bank; Operational Policy OP 4.12 Involuntary Resettlement; December 2001; [World Bank Operational Policy OP 4.12 Involuntary Resettlement.pdf](#)

World Bank; Operational Policy OP 4.12 Involuntary Resettlement Annex A ; December 2001; [World Bank Operational Policy OP 4.12 Involuntary Resettlement Annex A.pdf](#)

INSTRUMENTS PREPARED FOR SCSD

World Bank; Suriname Competitiveness and Sector Diversification Project - SCSD Project summary; June 13 2019; [World Bank SCSD Project summary.pdf](#)

World Bank; World Bank SCSD Project A Assessment Document PAD; June-2019; [World Bank SCSD Project A Assessment Document PAD.pdf](#)

World Bank; World Bank SCSD Rapid-Social-Assessment; June2019; [World Bank SCSD Rapid-Social-Assessment.pdf](#)

World Bank; World Bank SCSD Resettlement Policy Framework (RPF); June2019; [World Bank Country Partnership Strategy.pdf](#)

World Bank; SCSD- ENVIRONMENTAL AND SOCIAL MANAGEMENT FRAMEWORK ESMF-for-disclosure-; June-2019; [World Bank SCSD-ESMF-for-disclosure-June-2019.pdf](#)

World Bank; SCSD TOR Strategic Environmental and Social Assessment (SESA); June 2019; [World Bank Terms-of-Reference-for-Strategic-Environmental-and-Social-Assessment.pdf](#)

World Bank; SCSD Indigenous and Tribal Peoples Planning Framework ITPPF; February 8, 2019; [World Bank SCSD - ITPPF - for disclosure9.docx](#)

World Bank; Country Partnership Strategy; 2014; [World Bank Country Partnership Strategy.pdf](#)

SURINAME AND SURINAME MINING DOCUMENTS AND REPORTS

Asesoramiento Ambiental Estratégico (AAE); Report of the Strategic Environmental and Social Assessment (SESA) accompanying the development of the National REDD+ Strategy of the Republic of Suriname; 2017; [AAE sesa report REDD+.pdf](#)

CHALLENGES TO SUSTAINABLE SMALL-SCALE MINE DEVELOPMENT IN SURINAME; Marieke Heemskerk¹ and Rachael van der Kooye²; n/d; [Heemskerk Challenges to sustainable small scale mine development in Suriname.pdf](#)

Del Prado, Nancy; Support Sound Land Use Planning in Suriname; A review of the Legal and Institutional Framework for LUP; 2015; [Del Prado Nancy Support Sound Land Use Planning in Suriname 2015.pdf](#)

EITI (Extractive Initiative Industries Transparency Initiative) International Secretariat; Validation of Suriname: Report on initial data collection and stakeholder consultation; 2021; [EITI Validation of Suriname Report on initial data collection and stakeholder consultation.pdf](#)

EUROPEAN COMMISSION; Multi-annual Indicative Programme 2021-2027 Republic of Suriname; 2021; [EC Multi-annual Indicative Programme 2021-2027 Suriname.pdf](#)

GEOSTRATUM WATER MANAGEMENT; EIA Merian Mine NIMOS Reply Kupari Maraba; 2021; [GEOSTRATUM WATER MANAGEMENT EIA Merian Mine NIMOS Reply Kupari Maraba.pdf](#)

Government of Suriname; Land Degradation Neutrality High-Level Note; 2020; [Suriname Land Degradation Neutrality High-Level Note 2020.pdf](#)

Government of Suriname; Minimata Initial Assessment Report 2020; 2020; [Suriname Minimata Initial Assessment 2020.pdf](#)

Government of Suriname; Gmt Suriname Multi-year Development Plan 2022-2026 DUTCH; ; [Gmt Suriname Multi-year Development Plan 2022-2026 DUTCH.pdf](#)

Government of Suriname; Suriname REDD+ Status summary; n/d; [Suriname REDD+ Status summary.pdf](#)

Government of Suriname; Suriname Biodiversity Action Plan 2012 2016; 2013; [Suriname Biodiversity Action Plan 2012 2016.pdf](#)

Government of Suriname; Suriname Environmental Framework Act 2020 DUTCH; 2020; [Suriname Environmental Framework Act 2020 DUTCH.pdf](#)

Government of Suriname; Suriname Stabilization and recovery plan_2016_2018 DUTCH; 2016; [Suriname Stabilization and recovery plan 2016 2018 DUTCH.pdf](#)

Government of Suriname; Suriname_national_development_plan_2012-2016 DUTCH; 2012; [Suriname national development plan 2012-2016 DUTCH.pdf](#)

Government of Suriname; SurinamePolicy-Development-Plan2017-2021-PartI; 2017; [SurinamePolicy-Development-Plan2017-2021-PartI.pdf](#)

Heemskerk, Marieke University of Florida; Gender and Gold Mining: The Case of the Maroons of Suriname; July 2000; [Gender and Gold Mining The Case of the Maroons of .pdf](#)

IDB; Indigenous Peoples and Mining in West Suriname SHIFTING GROUNDS; 2008; [IDB Indigenous Peoples and Mining in West Suriname SHIFTING GROUNDS.pdf](#)

IDB; State of the Climate Report: Suriname; 2021;

IDB - Kambel; Policy Note On Indigenous Peoples and Maroons in Suriname; 2006; [IDB Indigenous-Peoples-and-Maroons-in-Suriname.pdf](#)

IFC; Environmental, Health and Safety Guidelines for Mining; 2007; [IFC EHS mining guidelines.pdf](#)

IICA ; Annual Report 2016; 2016; [IICA Annual report 2016.pdf](#)

IICA ; Significant achievements 2017; 2017; [IICA significant achievements 2017.pdf](#)

IISD IGF Intergovernmental forum on Mining Minerals metals and sustainable development; IGF MINING POLICY FRAMEWORK ASSESSMENT; May 2017; [IGF Suriname-mining-policy-framework-assessment-en.pdf](#)

ILO; SURINAME Decent Work Country Programme 2019 to 2021; 2019; [ILO Suriname Decent Work Country Programme 2019 2021.pdf](#)

Marieke Heemskerk¹ and Rachael van der Kooye; CHALLENGES TO SUSTAINABLE SMALL-SCALE MINE DEVELOPMENT IN SURINAME; n/d; [CHALLENGES TO SUSTAINABLE SMALL-SCALE MINE DEVELOPMENT IN SURINAME.pdf](#)

Ministry of Health of Suriname; National Health Sector Plan 2011 2018; 2011; [Ministry of Health National Health Sector Plan 2011 2018.pdf](#)

Ministry of Public Transport; Republic of Suriname Saramacca Canal System Rehabilitation Project Resettlement Policy Framework (RPF); 2018; [Ministry of Public Works Canal Saramarca resettlement policy.pdf](#)

Netherlands Journal of Geosciences ; Gold-deposits-of-suriname-geological-context-production-and-economic-significance; 2016; [Netherlands Journal of Geosciences gold-deposits-of-suriname-geological-context-production-and-economic-significance.pdf](#)

NIMOS; NIMOS NAP Project – Mercury use limitation; 2020; [NIMOS NAP Project – Mercury use limitation.pdf](#)

Suriname Extractive Industries Transparency Initiative; Suriname EITI (SEITI) Work plan 2016 2018; 2017; [Suriname EITI \(SEITI\) Work plan 2016 2018.pdf](#)

UNDP; Environment Statistics; 2016; [UNDP Suriname Environment Statistics 2016.pdf](#)

UNDP; 9th Environment Statistics publication 2015-2019; 2020; [UNDP 9th Environment Statistics publication 2015-2019.pdf](#)

UNDP; ENVIRONMENT STATISTICS; 2014; [UNDP Environment Statistics-2014 Suriname Environment Statistics 2014.pdf.pdf](#)

UNDP; Midterm Review Improving Environmental Management in the Mining Sector of Suriname, with Emphasis on Artisanal and Small-Scale challenges (ASGM); 2021; [UNDP Final Midterm Review Improving Env mining sector Suriname.pdf](#)

UNDP GEF; UNDP GEF The Sixth National Report to the UN Convention on Biological Diversity; 2019; [UNDP GEF The Sixth National Report to the UN Convention on Biological Diversity 2019.pdf](#)

UNEP; UNEP 6th National Report for Convention on Biological Diversity 2020; 2020; [UNEP 6th National Report for Convention on Biological Diversity 2020.pdf](#)

UNEP; UNEP MERCURY RELEASE INVENTORY IN SURINAME 2016; 2016; [UNEP MERCURY RELEASE INVENTORY IN SURINAME 2016.pdf](#)

UNEP; MERCURY RELEASE INVENTORY, WASTE STORAGE AND DISPOSAL IN THE REPUBLIC OF SURINAME; 2016;

UNIDO; Artisanal Gold Mining Activities in Suriname; December 1997; [UNIDO Artisanal Gold Mining Activities in Suriname.pdf](#)

USA Government; USA Investment Climate Statement Suriname 2021; 2021; [USA Investment Climate Statement Suriname 2021.pdf](#)

World Bank; SCSD-ESMF-for-disclosure-June-2019 WORKING DOCUMENT; 2019; [World Bank SCSD-ESMF-for-disclosure-June-2019 WORKING DOCUMENT.pdf](#)

World Bank; Suriname-Sector-Competitiveness-Analysis-Report 2017; 2017; [World Bank Suriname-Sector-Competitiveness-Analysis-Report 2017.pdf](#)

World Trade Organization; TRADE POLICY REVIEW Suriname; 7 August 2019; [WTO Trade Policy Review of Suriname 2019.pdf](#)

WWF CHRISTOPHER HEALY and MARIEKE HEEMSKERK for WWF; SITUATION ANALYSIS OF THE SMALL-SCALE GOLD MINING IN SURINAME Reforming the sub-sector to promote sound management; June 2005; [WWF SITUATION ANALYSIS OF THE SMALL-SCALE GOLD MINING IN SURINAME.pdf](#)

GENERAL MINING AND OTHER REFERENCE DOCUMENTS

GEF; GEF Nigeria_ASGM_NAP_2021_EN; 2021; [GEF Nigeria ASGM NAP 2021 EN.pdf](#)

IDB; IDB GROUP COUNTRY STRATEGY WITH SURINAME 2021 - 2025; September 2021; [IDB GROUP COUNTRY STRATEGY WITH SURINAME\(2021-2025\).pdf](#)

Integrated Environments (2006) Ltd. D 'Appolonia S.p.A.; Strategic Environmental and Social Assessment of Oil and Gas Development in Mauritania; June 2011; [sesa Mauritania d'appolonia.pdf](#)

International Institute for Environment and Development; Participation as policy: time to formalize artisanal and small-scale mining in Colombia; February 2015; [IIED Participation as policy time to formalize artisanal and small-scale mining in Colombia.pdf](#)

Ministry of Electricity and Renewable Energy Egypt; Strategic Environmental and Social Assessment of Renewable Energy Projects in the East Nile Region Scoping Report–Rev. 03–Final Report; October 2016; [Strategic Environmental and Social Assessment of Renewable Energy Projects in the East Nile Region.pdf](#)

UNDP IFG; SGP Suriname Country Programme Strategy OP6; June 2016; https://rinagroup.sharepoint.com/:b:/s/82779/ESX_f-FDd_ZEoSqOTnQjx_4BWDIBG6EFK4XdSja-K6OR5g?e=0FBgKJ

USAID; MINING AND THE GREEN ENERGY TRANSITION REVIEW OF INTERNATIONAL DEVELOPMENT CHALLENGES AND OPPORTUNITIES; November 2021; [USAID Green-Energy-Minerals-Report FINAL.pdf](#)

World Bank; The WORLD BANK Environmental and Social Framework; 2017; [The WORLD BANK Environmental and Social Framework.pdf](#)

World Bank ; 2020 State of the Artisanal and Small-Scale Mining Sector; 2020; [World BANK 2020 State of the Artisanal and Small-Scale Mining Sector.pdf](#)

World Bank; Oil, Gas, and Mining A SOURCEBOOK FOR UNDERSTANDING THE EXTRACTIVE INDUSTRIES; 2017; [World Bank Oil-Gas-and-Mining-A-Sourcebook-for-Understanding-the-Extractive-Industries.pdf](#)

World Bank Loayza and Albarracin ; Mining Sector Strategic Environmental and Social Assessment (SESA); March 2010; [World Bank Mining Sector Strategic Environmental and Social Assessment \(SESA\).pdf](#)

Appendix A

RINA Health and Safety Plan

Doc. No. P0032241-IR Rev. 4 – December 5,
2022



COVID Protocol for Fieldwork of RINA Teams

The last information by the Suriname Ministry of Health: <https://gov.sr/covid-19-update-suriname/> indicates that since March 2022, the Ministry of Health has started phasing out Covid-19 restrictions on the basis of epidemiological developments. The Ministry of Health strongly recommends wearing a face mask (surgical mouth-nose mask) in the presence of a large crowd and in poorly ventilated areas. The use of sanitizer is still highly recommended. RINA will follow the CDC guidelines and those of the government of Suriname. For visits RINA will use a face masks and utilize alcohol for disinfection.

Based on the best practices and recommendations of the Suriname Ministry of Health, World Health Organization, and US Center for Disease Control RINA has prepared the following protocol to prevent transmission of the virus that causes COVID-19. RINA's team will comply with the following precautionary measures:

1. All RINA's employees and consultants have to be vaccinated with vaccines approved by World Health Organization.
2. The RINA team will bring masks and self-test kits and will be required isolation in the event of a positive test
3. Wear a surgical mouth-nose mask at all times, especially when meeting with stakeholders. Face coverings, if used, will be disinfected with 5% bleach, washed, rinsed, and air-dried before being used again.
4. Wash hands frequently, especially before drinking water and after touching surfaces that may contain the virus: car handles, door handles, and the like. The field team will carry water for hand washing and a container to collect and disinfect liquid waste with bleach after hand washing, before disposing of it. Where it is not feasible to carry water, hand sanitizer and disinfectant wipes can be used.
5. The field team will be advised to have a good breakfast before leaving home and not to consume solid food while working in the field to avoid removing mask and reduce bathroom stops. If deemed appropriate, the afternoon activity could take place at a restaurant on the way back where the field team can have a late working lunch.
6. Maximum occupancy in cars is 4 (four) persons unless a van is provided. If no security risks are identified, the windows of the cars should be opened. In case driving with the windows open is not feasible, the occupants should wear a mask.
7. Alert the supervisor when a team member has signs or symptoms that suggest COVID-19. Based on test availability and test response times, the supervisor will need to discuss with the team member (it is not an option to continue working, although the Suriname Ministry of Health informs that people who show mild complaints and are able to work can work with a surgical mouth-nose mask) what he or she can do, in accordance with local labor laws; if a reliable rapid test is available, take a day of sick leave until a negative COVID-19 test result is reported.
8. If a field team member has a reliable positive COVID-19 rapid test, this person will need to be quarantined. All field team members will be tested for COVID-19 in the event that a field team member has a reliable positive COVID-19 test result
9. In the event of an ongoing outbreak, i.e., several confirmed cases of COVID-19 in this community or among field team contacts, the supervisor will immediately contact management to discuss the possibility of delaying fieldwork, the same applies to local closures and regulations that do not allow the field team to visit the community.
10. Every person to be interviewed should be asked the following question and if any answer is positive, they will not be interviewed:

Have you recently had any of the following symptoms of COVID-19?

✓ Cough

- ✓ Fever or chills
- ✓ Shortness of breath or difficulty breathing
- ✓ Muscle or body aches
- ✓ Sore throat
- ✓ Loss of taste or smell
- ✓ Diarrhea
- ✓ Headache
- ✓ Fatigue
- ✓ Nausea or vomiting
- ✓ Congestion or runny nose

11. When performing the interviews with stakeholders and in meetings, maintain a distance of approximately two meters from the others and avoid physical contact like greetings with a handshake.

Appendix B

Agenda for validation workshop and site visits

Doc. No. P0032241-IR Rev. 4 – December 5,
2022



**RINA TEAM IN SURINAME - AGENDA FOR VALIDATION WORKSHOP AND
SITE VISITS
10/24/2022 – 11/05/2022**

Week of Oct 17, 2022	
Invitations to be sent to stakeholders MNR to send invite to Stakeholders as proponent of the Project. RINA to be copied to follow up for attendance. We will provide the list with emails and phone numbers.	
Monday October 24, 2022 – Departure to Paramaribo	
Time according to individual flight	RINA team to leave from individual locations (see RINA team flight details).
1:00- 2:30 pm	RINA team meets in Miami
6:35 pm (Departs from Miami)	RINA team leaves from Miami to Paramaribo Arrives Paramaribo 25 Oct 12:22 AM AA 2867
Tuesday October 25, 2022	
12:22 am	RINA team arrives in Paramaribo Driver to pick-up RINA Team at the airport
Approx. 2:00 am	RINA team checks into hotel
12:00 pm - 3:00 pm	RINA team meeting with local consultants (Salomon, Nancy) Review of schedule, meetings, logistics Location: Hotel
3:30 pm - 5:30 pm	RINA team meeting with MNR. - Introduction of RINA team - Final logistics for site visits and workshop. - Presentation of PPT to be shown to stakeholders. (It is anticipated that the Ministry will receive the Inception report by Oct 5 and Scoping report by Oct 14). Location. MNR office.
Wednesday October 26, 2022	
9:00 am - 12:00 am	Meeting with Governmental stakeholders (per stakeholder list) to present Draft Inception and Draft Scoping report Location: TBD
12:30 pm - 1:30 pm	Lunch
1:30 pm - 1:50 pm	Travel by car to VIDS to present Draft Inception and Scoping report
2:00 pm - 3:30 pm	Meeting District Paramaribo: VIDS – Vereniging Inheemse Dorpen in Suriname

3:30 pm - 3:50 pm	Travel by car to Mulokot (Wayana of Kawemhakan)
4:00 pm - 5:30 pm	Meeting District Paramaribo: de Mulokot Foundation (Wayana Amerindians) – impact gold mining.
Thursday October 27, 2022	
9:00 am - 10:00 am	Travel by car to New Koffiekamp (District Brokopondo)
10:00 am - 12:00 pm	Meet with Makamboa Foundation – Maroons ASM gold mining (106 km) and Traditional Authority New Koffiekamp
12:30 pm - 1:30 pm	Lunch in the field
2:00 pm - 3:00 pm	Travel back to Paramaribo
3:30 pm – 6:00 pm	Meet with Cabinet Matawai Traditional Authority Meet with Cabinet of Okanisi granman – Tribal people
Friday October 28, 2022	
9:00 am - 10:30 am	Travel by plane to Tapanahoni area (District Sipaliwini)
10:30 am - 12:30 am	Site visit and meeting Wayana Traditional Authority and local organizations: Lawa- Kawemhakan/Anapaike (Wayana Indigenous people) – gold mining
12:30 pm - 1:30 pm	Lunch
1:30 pm - 4:00 pm	Travel by plane Return to Paramaribo (organize Passover mining sites to view by plane)
4:00 pm – 5:00 pm	Debrief to client Location: Hotel or MNR
Saturday October 29, 2022	
	RINA Team – organize inputs from stakeholders, available for stakeholders
Sunday October 30, 2022	
	Internal meetings
Monday October 31, 2022	
8:00 am - 10:00 am	Travel by car to KLIM
10:30 am - 12:00 pm	Site visit and meeting with KLIM (District Paramarib): KLIM (Kali'na & Lokono Inheemsen beneden Marowijne – impact bauxite mining (same day as Adjoema kondre).
12:00 pm - 1:00 pm	Lunch
1:00 pm - 3:00 pm	Travel to Adjoema kondre – Aukanisa Maroon village District Marowijne
3:30 pm – 5:00 pm	Site visit and meeting with Adjoema kondre (Aukanisi) – former bauxite mining.

Tuesday November 01, 2022	
8:00 am - 12:00 pm	Travel by car to Pamaka community (District Sipaliwini – Pamaka resort)
12:00 pm - 12:30 pm	Lunch
12:30 pm - 2:30 pm	Site visit and meeting with Pamaka: Traditional Authority and Sustainable Development Foundation Pamaka – Pamaka Maroon Community.
2:30 pm - 6:00 pm	Travel to Paramaribo
Wednesday November 02, 2022	
9:00 am - 12:00 pm	Workshop in Paramaribo for donor agencies, development aid agencies, UN and international organizations
9:00 am - 10:30 pm	Presentation of the Inception and Scoping reports for validation by RINA
10:30 pm - 10:50 am	Refreshments
10:50 am - 11:50 pm	Open discussion with stakeholders for inputs and comments to be incorporated into the Inception and scoping reports
11:50 pm – 12:00 pm	Closing of workshop
12:00 pm – 2:00 pm	Lunch time
2:00 pm - 5:00 pm	Workshop in Paramaribo for Government organizations
2:00 pm - 3:30 pm	Presentation of the Inception and Scoping reports
3:30 pm - 3:50 pm	Refreshments
3:50 pm - 4h50	Open discussion with stakeholders for inputs and comments to be incorporated into the Inception and scoping reports
4:50 pm - 5h00	Closing of workshop
Thursday November 03, 2022	
9:00 am - 12:00 pm	Workshop in Paramaribo Organizations/Representations, NGOs, CSOs, and associations
9:00 am - 10:30 pm	Presentation of the Inception and Scoping reports for validation by RINA
10:30 pm - 10:50 am	Refreshments
10:50 am - 11:50 pm	Open discussion with stakeholders for inputs and comments to be incorporated into the Inception and scoping reports
11:50 pm - 12:00 pm	Closing of workshop
12:00 pm - 2:00 pm	Lunch time
2:00 pm - 5:00 pm	Workshop in Paramaribo for Mining companies and private organizations
2:00 pm - 3:30 pm	Presentation of the Inception and Scoping reports for validation by RINA
3:30 pm - 3:50 pm	Refreshments
3:50 pm - 4h50 pm	Open discussion with stakeholders for inputs and comments to be incorporated into the Inception and scoping reports
4:50 pm - 5h00 pm	Closing of workshop
Friday November 04, 2022	
9:00 am - 12:00 pm	Capacity building at MNR
10:30 pm - 10:50 am	Refreshments
10:50 am - 11:50 am	Continuation of capacity building course
11:50 am - 12:00 pm	Closing of capacity building workshop
12:00 pm - 2:30 pm	Lunch time
3:00 pm - 5:00 pm	Wrap-up meeting RINA international team with local consultants
10:00 pm	Departure RINA team to airport

Saturday November 05, 2022	
1:55 am	Departure RINA Team from Paramaribo

Appendix C

Legal, regulatory and institutional framework Suriname – Mining sector

Doc. No. P0032241-IR Rev. 4 – December 5,
2022



This following preliminary legal, regulatory and institutional framework will be further developed in the scoping report.

4 LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK IN SURINAME'S MINING SECTOR

4.1 INTRODUCTION

This section provides a description of the relevant regulations set by the Republic of Suriname for social planning and environmental protection relevant and applicable to the mining sector. Therefore, the legislative, regulatory and institutional requirements guiding the mining sector development project, are discussed in more detail below.

Suriname is signatory to a number of international conventions that address environmental and social issues. These international conventions are the key drivers behind for the development of legislation in most countries, including Suriname.

The Republic of Suriname have different ministries and institutions over which fall the responsibility for the management of the environment and natural resources, and social themes.

4.2 LEGAL, REGULATORY AND INSTITUTIONAL FRAMEWORK

The legal and regulatory framework of Suriname is composed of the national regulatory framework composed from the national constitution, laws, decrees, policies, strategies and plans. The regulatory framework is also supported by international agreements, treaties, conventions, etc. The following sections describe in detail the legal and regulatory framework of Suriname.

4.2.1 NATIONAL CONSTITUTION AND LAWS

Suriname is governed in accordance with its National Constitution⁷ established in 1987 and last amended in 1992. The Suriname Constitution sets specific social objectives related to environmental aspects, which include:

- Article 6 (a) – “the identification of the potential for development of its own natural environment and the augmentation of the capacity to increasingly expand that potential”; and
- Article 6 (g) – “creating and improving the necessary conditions for the protection of nature and for the preservation of the ecological balance”.

Suriname's legislation at the national level is exercised through Laws or Acts of Parliament (Wet, also called Verordening and Landsverordening prior to 1975), Decrees (Decreet)⁸, Government Decree (Staatsbesluit), Presidential Decree (Resolutie), Presidential Orders (Presidentieel Besluit) or Ministerial Orders (Ministeriële Beschikking) targeting various sectors including industry, tourism, nature conservation, etc. (ERM, 2016)

In 2020, an Environmental Framework Act⁹ was established (No. 97 of 2020) for the protection and sustainable management of the environment in Suriname and the implementation and carrying into effect of obligations deriving from the membership of Suriname to international agreements, notably the UN Framework Convention on Climate Change, the UN Convention on Biological Diversity, the Paris Agreement and the Stockholm and Rotterdam (Conventions on POPs and PIC respectively). Another important matter is the investigation, prosecution and trial of offences defined in this Act as environmental offences. (UNEP, 2022)

The Act is composed of 77 articles divided into 11 Chapters: General provisions (I); The National Environment Authority (II); Duty of Care (III); Environmental Strategy (IV); Activities and Environmental Consequences (V); Control of Pollution – Environmental Pollution and Standards (VI); Waste and

⁷ S.B. 1987 no.116, as lastly amended by S.B. 1992 no.38

⁸ Decrees date from the period of 1980 – 1986 and have the same status as past or present Laws

⁹ <https://leap.unep.org/countries/sr/national-legislation/environmental-framework-act-no-97-2020>

Hazardous Substances (VII); Legal Protection Mechanisms (VIII); Environmental offences and Sanctions (IX); Provisions on Enforcement and further provisions (X); Transitional and Final Provisions (XI). (UNEP, 2022)

The Act establishes the National Environment Authority (NMA) as an Independent administrative body. It shall be the successor to the Environmental Coordination Office of the Presidential cabinet and the National Institute for Environment and Development in Suriname (NIMOS) shall be placed under it. The NMA shall prepare the National Environment Strategy and be the main focal point for environmental protection in Suriname. It shall also ensure that the FPIC principle is applied in decision-making processes affecting the habitat of the native and tribal peoples. Key chapters include (UNEP, 2022):

- Chapter III defines a duty of care for the environment. This duty means that everyone knows or can reasonably suspect that his or her actions or omissions shall have adverse consequences for the environment, must omit such action insofar as this can be reasonably required.
- Chapter IV requires the NMA to elaborate the environmental strategy as incorporated in the National Development Plan for Strategy into a national environmental policy plan. The Act sets out the priorities of such policy. One priority is the inventory, protection and regulation of endemic flora and fauna and traditional and medicinal knowledge. The national plan shall be elaborated in environmental programs, which shall also be established at district level. The NMA shall publish every four year an environmental report. The Act establishes an Environment Fund. Activities for which an environmental impact assessment is required shall be defined by a State Decree. This Act sets out the procedures for such environmental impact assessment.
- Chapter V, the NMA shall be informed about government activities that may affect the environment and shall carry out an environmental impact analyses for such activities or policies or a strategic impact analysis for economic sectors such as forestry, water and agriculture.
- Chapter VI is about environmental pollution and rehabilitation of the environment. The NMA shall define polluting activities and be responsible for environmental authorizations.
- Chapter VII indicates that the NMA shall establish standards and procedures for waste treatment, reuse and recycling as well as the processing of waste. The NMA establishes rules for handling with hazardous substances. It is also charged with monitoring activities in this area prevention of environmental pollution.
- Chapters VIII and IX are about environmental offences. Penal law shall be the ultimate resort for such offences. Article 47 defines environmental offences and for this purpose indicates various other laws including, amongst others the Agriculture Act, the Forestry Act and the Water Resources Act.

Other relevant environmental legislation in Suriname includes:

- The Hindrance Act (Hinderwet, 1930, 1944, and 1972) defines the permit requirements to control noise and air pollution for industrial development projects. The permits are issued and enforced by local District Commissioners (Buursink, 2005; SRK, 2007); however, the Act's effectiveness has been negatively impacted by outdated and inadequate regulations concerning inter alia pollution standards and waste management, and a lack of sufficient resources to conduct monitoring inspections (ERM, 2011).
- The Nature Conservation Act (Natuurbeschermingswet, 1954) defines the procedures to establish and manage conservation areas and protect wildlife. According to the Act, the only allowable activities in the protected areas are activities of scientific, educational or cultural importance. Conservation areas can be established only by Presidential decree and managed by the Nature Conservation Division of the Forest Service (LBB). The Act does not provide for the protection of sensitive areas outside the established conservation areas. (ERM, 2011).
- The Forest Management Act (Wet Bosbeheer 1992) replaced the Timber Act of 1947 as the governing regulation for forest exploitation and conservation. The Act classifies forests in Suriname in three ways: permanent, temporary (maintained for the time being), or one-time exploitation forest (land intended for other future use where tree cover will be completely cleared). Resource exploitation in public forests (domain land) is allowable with a permit or forest concession granted by the Ministry of Spatial Planning, Land and Forestry Management

(RGB), which issued regulations for logging, deforestation, and processing-related activities. The Forest management Act also allows the Surinamese government to establish conservation forests. (ERM, 2011).

A summary of other major laws, acts, decrees, and orders concerning the mining or extractive industries sector in Suriname are described in Table 4-1.



Table 4-1: Regulatory Framework for National Environmental Management in Suriname

Title	Objective(s)	Implementing Agency	Comments	Relevance
GENERAL				
Kruispunt, Regeringsverklaring 2011-2015	Overall national sustainable development strategy	Government Statement		Describes the development strategy of the country into which all industry and projects must fit. The developmental impacts of the project will be compared to the requirements of this strategy and efforts will be made to incorporate regional strategy into any developmental projects that Newmont may invest in.
MINERAL RESOURCES				
Mining Decree 1986 SB. 1986 no. 28	1. Governs exploration and exploitation of mineral resources. 2. Article 2 stipulates that all raw materials in and above the ground, including the territorial sea, are property of the State.	MNR	1. Articles 2, 4, 16, 43, 45 are applicable to environmental protection. 2. Contains requirements for consideration of affected communities of Indigenous Peoples. 3. Several implementation regulations have been issued under this decree.	Provides legal expectations and requirements to exploit minerals nationally including consideration of environmental and social impacts.
Government Decree on Mining Installations S.B. 1989 no.38	Provides provisions for offshore mining installations,	MNR	1. Formulated according to: UNCLOS, SOLAS, MARPOL conventions (see Section 2.2.2 on International Agreements). 2. Chapter 3 addresses environmental protection.	Provides legal expectations and requirements to exploit minerals.
Petroleum Law 1990 S.B. 1991 no. 7	Provides provisions for the exploration and exploitation of hydrocarbons.	MNR	Article 6e deals with the management of adverse impacts on the environment.	Provides specific requirements for the management of environmental impacts
Drilling Law G.B. 1952 no. 93	Provides provisions for drilling in Suriname	Head of MNR		Provides specific requirements for drilling programs.
Brokopondo Agreement 1958	Agreement between the Government of Suriname and Suralco LLD concerning the development of hydropower potential.	MNR	No provisions with regard to environmental protection.	Provides context to pre-existing mineral agreements in Suriname.
Gross Rosebel Agreement S.B. 1994no. 22	Agreement between the Government of Suriname, Grassalco NV and Golden Star Resources Limited for the exploration and exploitation of minerals.	MNR	1. Environmental Impact Statement is required. 2. All activities to comply with Californian legislation.	Provides context to pre-existing mineral agreements in Suriname.
LAND/LAND USE				
Planning Law 1973 GB. 1973 no. 89	Provisions for national and regional planning e.g., land-use policy issues.	Minister of Planning, Planning Coordination Commission and Planning Council.	Contains the mechanism to establish Special Management Areas, to be developed as MUMAs.	Provides planning and land-use requirements.



Title	Objective(s)	Implementing Agency	Comments	Relevance
Law on the Issuance of State-Owned Lands 1937 G.B. 1937 no. 53	Provides general regulations on the issuance of State-owned lands,	The President	Under this Law, STINASU received a long-term lease for the Brownsberg Nature Park.	Provides regulations and requirements to acquire state land.
Construction Law 1956 G.B.1 1956 no.30	Provides requirements for the construction of buildings.	Ministry of Public Works		Provides regulations and requirements to construct buildings.
City- Construction Law G.B. 1972 no.96	Provisions for urban development.	Ministry of Public Works		Provides regulations and requirements to affect development. .
Law on Ecological Circumstances in Residential Areas S.B. 1980 no.68	To improve the ecological circumstances in residential areas	District Commissioners		Provides regulations for ecological circumstances in residential areas.
Police Criminal Law GB. 1915 no. 77 as amended.		Ministry of Justice and Police	Article 39a penalizes the disposal of waste in public places.	Provides regulations and requirements for waste disposal.
Civil Code G.B. 1860 no. 4			Article 625 deals with ownership/proprietary rights as well as expropriation for the general good prior to compensation.	Provides regulations and requirements for expropriation.
AIR QUALITY AND NOISE				
Hindrance Law G.B. 1930 no. 64 as amended	Controls industrial pollution(noise and air).	District Commissionersissue permits inconsultation with Ministry of Health	Permits are required for industrial developmentprojects.	Provides regulations and requirements controlindustrial pollution.
WATER/MARITIME				
Water Supply Law G.B. 1938 no. 33	Contains prohibitions with respect to water wells, etc. that serve as water supply sources.	MNR, Ministry of Public Health	According to this Law the President is responsible for its implementation, but in practice the ministries assume the role.	Provides regulations and requirements for water pollution.
Water Board Law G.B. 1932 no. 32 as amended	To establish water boards in charge of maintenance of waterways and waterworks within designated areas.	Execution by the water boards, on behalf of the MRD	The manner in which water boards execute their tasks is of importance to the protection, improvement of the living conditions and the maintenance of the quality of the natural environment.	Provides regulations and requirements maintenance of waterways.
Harbors Decree 1981 S.B. 1981 no. 86	Provisions for harbor activities.	Maritime Authority Suriname and District Commissioners, assisted by the Prosecutors office, the Police and the Ministry of Trade and Industry	Prohibits the discharge of waste, oil, and oil-contaminated water and condemned goods into public waterways and harbors.	Provides regulations and requirements for the discharge of water and oil.
Maritime Safety Law SB. 2004 No. 90	Provisions for safety of ships and harbors.	Maritime Authority Suriname, Ministry of Trade		Provides regulations and requirements for ship movements.

Title	Objective(s)	Implementing Agency	Comments	Relevance
		and Industry, Maritime Safety Council		
Decree Multi- Purpose Corantijn Project — Management S.B. 1984 no. 14	To supervise the water management system and to develop, stimulate and manage agricultural activities in the northwest of Suriname.	Multi-Purpose Corantijn Project Management on behalf of LVV		Provides regulations and requirements to control water management for agriculture.
Police Criminal Law GB. 1915 no.77 as amended		Ministry of Justice and Police	In terms of Article 51 the polluting of a water source or water well is liable to a fine.	Provides regulations and requirements for water pollution.
Penal Code G.B. 1911 no.1 as amended		Ministry of Justice and Police	In terms of Articles 224 and 225, contamination of water resources is penalized.	Provides regulations and requirements for water pollution.
Government Decree on Mining Installations SB. 1989 no. 38	Provisions for offshore mining installations,	MNR	It is prohibited to drain or throw overboard substances in concentrations that are hazardous to the marine environment. Protection of the marine environment should be taken into consideration during dismantling.	Provides regulations and requirements for releases to marine environments.
Government Decree on Pesticides G.B. 1974 no. 89 as amended	To implement article 13 of the Pesticides Law	LVV, ATM, Ministry of Public Health	Article 13 Part 2 forbids the removal or destruction of empty containers or remainders of undiluted pesticides in such a manner that water procurement areas or surface waters are polluted.	Provides regulations and requirements for pesticide disposal.
NATURAL ECOSYSTEMS (VEGETATION, FISH, AND WILDLIFE)				
Forest Management Law S.B. 1992 no.80 (replaced the Timber Law of 1947)	Provides a framework for forest management, exploitation, and related sector activities (e.g., primary processing and export) to guarantee sustainable utilization of forest resources. Provides for establishment of conservation forests.	MNR, SBB	1. Permits are required for the exploitation of public forests. 2. Currently 13 implementing resolutions have been issued under the Law 3. The Law also contains a requirement to respect the traditional rights of tribal communities.	Provides regulations and requirements for forest management.
Nature Conservation Law 1954 G.B. 1954 no. 26 as amended	Wildlife protection, establishment and management of protected areas. Prohibits any activities that may impact on protected areas, except for activities of scientific, educational or cultural importance.	LBB manages nature reserves with the Nature, Protection Commission in an advisory role.	1. Forms the basis for the establishment of nature reserves. 2. Several state-owned lands have been designated nature reserves by Government Decree. 3. No provision for protection of sensitive areas outside of established conservation areas.	Provides regulations and requirements for wildlife protection.
Government Decree 1998 on Nature Protection S.B. 1998 no.65	Establishment of the Central Suriname Nature Reserve.	MNR, NCD		Provides regulations and requirements for Nature Reserves.
Balata Law G.B 1914 no 51	Provides provisions for research on the availability of 'bolletrie' trees (rubber trees) and	The President and Administrator of Finance	No current balata (rubber) exploitation activities in Suriname.	Provides regulations and requirements for protection of rubber trees.

Title	Objective(s)	Implementing Agency	Comments	Relevance
	the exploitation of balata on state land as well as private land.	(implementation), with State Council in an advisory role.		
Plant Protection Law G.B. 1965 no. 102 as amended	- Provides for control of diseases and pests affecting plants. - Prohibits import of plants, compost, soil, etc. without permission. - Provides for declaration of protected areas for the protection of plants.	LVV		Provides regulations and requirements for plant protection.
Police Criminal Law GB. 1942 no. 152 as Amended.			Article 44 requires a permit in order to hunt or to capture wildlife in certain areas of the country's domain, which affords these areas protection.	Provides regulations and requirements for wildlife protection.
Law on Territorial Sea and the Contiguous Economic Zone 1978 SB. 1978 no.26	Determines the economic zone (200 nautical miles) and the extension of the territorial sea to 12 nautical miles in relation with the exploration and exploitation, the maintenance and management of natural resources, living and non-living.	MNR, LVV	The provisions of this Law and requirements thereto can by Government Decree be declared applicable for the protection of the marine environment.	Provides regulations and requirements for protection of the marine environments.
Sea Fisheries Decree SB. 1980 no. 144 (repeals the Fishing Vessels Decree of 1971)	- Provides for registration, licensing, seaworthiness certification and other requirements for fishing vessels - Provides for issuing of regulations for the protection of fish stocks.	LVV	- Permits required for fishing in the territorial waters and the economic zone. - Provides for creation of Advisory Council for Sea Fisheries.	Provides regulations and requirements for releases to marine environments.
Government Decree on Fish Stock Protection GB. 1961 no.144 as amended	Protection of fish stocks in inland waters.	LVV		Provides regulations and requirements for protection of fish stocks.
Game Act 1954 G.B. 1954 no. 25 as amended	Protection of fauna and game management.	MNR	The Economic Offences Law is also applicable.	Provides regulations and requirements for protection of fauna and game.
Hunting Act 1954	- Prohibits hunting of certain protected animals. - Regulates hunting and fishing.	LBB	- Permits/licenses are required to hunt certain species. - Implies that Indigenous Peoples need permission to hunt and fish on State land.	Provides regulations and requirements for protection of fauna and game.
Hunting Decree 2002	Provisions for various animal species, including hunting seasons and numbers allowed,		Protection clause for Indigenous Peoples in the south of Suriname is included.	
CULTURAL HERITAGE				
Law on Historical Monuments GB. 1963 no. 23	Provisions for the preservation of historical monuments, art and architecture in Suriname.	Ministry of Education & Community Development		Provides regulations and requirements for preservation of cultural heritage.

Title	Objective(s)	Implementing Agency	Comments	Relevance
Law of 7 February 1952 G.B. 1952 no. 14	Controls the export of objects that have historical, cultural and scientific value.	Ministry of Education & Community Development	A permit is required to export objects of historical, cultural and scientific value.	Provides regulations and requirements for protection of cultural heritage.
OCCUPATIONAL HEALTH & SAFETY/PUBLIC HEALTH				
Occupational Safety Law G.B. 1947 no. 142 as amended	To advance safety and hygiene in enterprises so that the chance of accidents and occupational diseases can be reduced to a minimum.	Ministry of Labor	Nine regulations have been issued for the implementation of this Law.	Provides regulations and requirements for health and safety in the workplace.
Labor Inspection Law S.B. 1983 no. 42	Outlines the tasks and responsibilities of the Labor Inspector.	Ministry of Labor	In cases where the safety of persons is in danger, the Inspector has the authority to close the enterprise in question.	Provides regulations and requirements for protection of cultural heritage.
Pesticide Law G.B. 1972 no. 151 as amended	Regulates the trade and use of agrochemicals.	LVV and the Ministry of Public Health	It is prohibited to sell, have in store or use agrochemicals which were not imported according to this Law.	Provides regulations and requirements for the use of agrochemicals.
Mosquito Control Law	To combat mosquitoes and other insects judged to be destructive to the health of human beings and animals.			Provides regulations and requirements for the control of mosquitos.
Water Supply Law G.B. 1938 no. 33	Establishes prohibitions with respect to water wells, etc. that serve as water supply sources.	MNR, Ministry of Public Health	According to this Law the President is responsible for its implementation, but in practice the ministries assume the role.	Provides regulations and requirements for protection of water wells.
Penal Code G.B. 1911 no 1			Article 226 and 227 deals with harmful products. Article 393 pertains to adulterated foodstuffs and medicines.	Provides regulations and requirements for protection of human consumption of foodstuffs.
Movement of Goods Law SB. 2003 no. 74	Provides general rules for international trade	Ministry of Trade & Industry		Provides regulations and requirements for international trade.
Government Decree Resolution Negative List 2003 SB. No. 74	Regulates the international traffic of goods.	Ministry of Trade & Industry	Import and export of chemical waste, pesticides, animals, mercury, radioactive materials, etc. are allowed only after the approval of the Government concerning.	Provides regulations and requirements for shipping of goods.

Key: MNR (GMD) – Geologische Mijnbouwkundige Dienst (Geological Mining Service); MNR – Ministerie van Natuurlijke Hulpbronnen (Ministry of Natural Resources); MRD – Ministry of Regional Development; LVV – Ministerie van Landbouw, Veeteelt en Visserij (Ministry of Agriculture, Animal Husbandry and Fisheries); LBB – Lands Bosbeheer (Forestry Service)

Source: (ERM, 2011) modified by RINA, 2022



Table 4-2: Environmental Management Responsibilities of Key Institutions in Suriname

Environmental Management Aspects	Institutions							
	ATM	Sectoral Ministries	NIMOS	STINASU	SCF	District Commissioners	NCD (LBB)	SBB
Nature conservation and management			Facilitation of National Biodiversity Strategy	Browns- berg Nature Park	CSNR and Sipaliwini Nature Reserve		Creation of new protected areas, management, and control	Management of forests
ESIA processes		Approval of development projects and enforcement	Guidance and review of reports. Provides advice regarding environmental planning, but the final decision is still the responsibility of permitting agency.					
Industrial pollution			Technical advice; Enforcement			Permit approval	Enforcement	
Monitoring		Implementation	Supervision; Enforcement				Implementation	Implementation
Environmental Planning (Management Plans)		Participation in design and plan implementation	Coordination of design and plan implementation				Participation in design and plan implementation	
Environmental Regulations	Approval	Enforcement	Drafting and stakeholder consultation					
Forest Development							Permits approval and control of forest management plans	Monitoring of logging activities

Key: ATM – Ministry of Labor, Technological Development and Environment; NIMOS – National Institute for Environment and Development in Suriname; CSNR – Central Suriname Nature Reserve; STINASU – Foundation for Nature Conservation in Suriname; SCF – Suriname Conservation Foundation; NCD – Nature Conservation Division; LBB – Suriname Forest Service; SBB – Foundation for Forest Management and Production Control

Source: (ERM, 2011)

4.2.2 INTERNATIONAL AGREEMENTS

Suriname is signatory to a number of international agreements and conventions relating to environmental management, OHS and labor rights and Indigenous Peoples (refer to Table X). The international conventions are not always translated into national legislation. Some of the key agreements are listed in the following Table 4-3.

Table 4-3: International Agreements Relevant to Environmental and Social Issues in Suriname

Agreement / Convention	Notes / Comments	Status	Relevance
CLIMATE CHANGE/AIR QUALITY			
Vienna Convention for the Protection of the Ozone Layer, 1985	Protection of the ozone layer, came into force in 1988,	Suriname acceded in 1997.	Sets international standards for protection of the ozone layer; emissions from project potential to harm ozone layer
Montreal Protocol on Substances that Deplete the Ozone Layer, 1988	Protection of the ozone layer. came into force in 1988,	Suriname acceded in 1997 but subsequent amendments not yet ratified.	As above
United Nations Framework Convention on Climate Change (UNFCCC), 1994	Control of greenhouse gas emissions.	Ratified by Suriname in 1997.	Sets international guidelines on restrictions of GHG emissions in order to prevent climate change; Project will emit greenhouse gases from power generation through heavy fuel combustion
Kyoto Protocol (Framework Convention on Climate Change), 1997	The Kyoto Protocol to the United Nations Framework Convention on Climate Change came into force in 2005.	signed by Suriname on 22 April 2016	It applies the principle of common but differentiated responsibilities and sets binding targets for reducing greenhouse gas emissions for industrialized countries, recognizing them as those primarily responsible for the high levels of emissions currently present in the atmosphere.
Paris Agreement (Framework Convention on Climate Change), 2015	The 21 st Conference of Parties to the United Nations Framework Convention on Climate Change (COP 21), held in Paris in 2015, concluded with the adoption of the Decision and the Paris Agreement.	Ratified by Suriname on 13 Feb 2019	Its goal is to keep the global rise in temperature well below 2 °C. It aims to achieve this by increasing adaptability to the adverse effects of climate change and promoting climate resilience and low carbon development. In order to achieve the agreed targets, the Paris Agreement establishes a reinforced transparency framework aimed at fostering mutual trust and promoting the effective implementation of the Agreement, increasing clarity and facilitating the monitoring of progress.
BIODIVERSITY/PROTECTED AREAS			
Convention on Nature Protection and Wildlife Preservation in the Western Hemisphere (Western Hemisphere Convention), 1942	Provisions to establish a set of protected areas; national parks to provide recreational and educational facilities; strict wilderness areas to be maintained inviolate; co- operation in the field of research between governments; species listed in annex to enjoy special protection and controls to be imposed on trade in protected fauna and flora and any parts thereof.	Signed by Suriname in 1985. Suriname has three Western Hemisphere Shorebird Reserves: - Coppenamernonding - Bigi Pan - Wia	Sets international requirements for the protection and preservation of nature/wildlife; project is in a high-biodiversity area and has potential to impact local wildlife and natural environment.
Convention on Wetlands of International Importance especially as Waterfowl Habitat (Ramsar Convention), 1971	Suriname acceded to this convention in 1985. Suriname has multiple Multi Use Management areas (MUMA) and protected areas	Came into force for Suriname in 1985. The Coppenamernonding Nature Reserve is a Ramsar site.	Sets international requirements for the protection of wetlands; project has potential to impact local wetland area.

Agreement / Convention	Notes / Comments	Status	Relevance
	hosting wetlands, including the following: • Bigi Pan MUMA • Noord Coronie MUMA • Noord Saramacca MUMA • Noord Commewijne/ Marowijne MUMA • Wanekreek Nature Reserve • Galibi Nature Reserve		
The Cartagena Protocol on Biosafety to the Convention on Biological Diversity	The Cartagena Protocol, in force since September 2003, was adopted as a complementary agreement to the Convention on Biological Diversity.	Accession since 27 March 2008	The Protocol seeks to protect biological diversity by managing the movements of Live Modified Organisms resulting from the application of modern technology (LMOs) between countries. It establishes a procedure for prior informed agreement to ensure countries have the necessary information to make decisions about the importing of LMOs into their territory.
Convention on the International Trade of Endangered Species of Wild Fauna and Flora (CITES), 1973	In force since 1975, to ensure that international trade in specimens of wild animals and plants does not threaten their survival and it accords varying degrees of protection to more than 33,000 species of animals and plants.	Ratified by Suriname in 1980. New game regulations to comply with CITES requirements were passed in 2002.	Sets international restrictions/bans on trade of certain wild animals/plants. Project takes place in high biodiversity area
United Nations Convention on Biological Diversity, 1992	Promotes development of national strategies for the conservation and sustainable use of biological diversity. Often seen as the key document regarding sustainable development.	Ratified by Suriname in 1996. A National Biodiversity Strategy (2006) has been compiled as framework for a National Biodiversity Action Plan.	Sets guidelines for protection and promotion of biological diversity. Project takes place in high biodiversity area.
United Nations Convention to Combat Desertification, 1994	In force since 1996, to combat desertification and mitigate the effects of drought through national action programs that incorporate long-term strategies supported by international cooperation and partnership arrangements.	Ratified by Suriname in 2000. As part the implementation of this convention, a national report has been compiled. Currently discussions are underway on how to implement the convention in a more effective and efficient manner, The development of national action plan is prerequisite.	Sets guidelines to combat desertification. Project has potential to impact local water resources and quality and land use
FOREST MANAGEMENT			
International Tropical Timber Agreement, 1994 (Tropical Timber 94)	To ensure that (by the year 2000) exports of tropical timber originate from sustainably managed sources and to establish a fund to assist tropical timber producers in obtaining the resources necessary to reach this objective.	Suriname acceded in 1998.	Sets international targets for sustainable export of tropical timber; Project has potential to impact local forested area
American Convention on Human Rights, 1969	International human rights instrument to establish a system of personal liberty and social justice based on respect for the essential rights of man, within the framework of democratic institutions. The bodies responsible for overseeing compliance are the Inter- American Commission on Human Rights and	Ratified by Suriname in 1987	Sets framework for protection of human rights; project has impact to affect lives of those living in local community

Agreement / Convention	Notes / Comments	Status	Relevance
	the Inter-American Court of Human Rights, both of which are organs of the Organization of American States.		
CULTURAL HERITAGE			
UNESCO Convention Concerning the Protection of the World Cultural and Natural Heritage, 1972 (World Heritage Convention)	Promotes cooperation among nations to protect heritage from around the world that is of such outstanding universal value that its conservation is important for current and future generations.	Ratified by Suriname in 1997. Suriname has two World Heritage Sites: - Central Suriname Nature Reserve; and - Historic Inner City of Paramaribo	Sets guidelines for the conservation of significant heritage/cultural sites.
INDIGENOUS PEOPLE			
UN formulation of Declaration on Rights of Indigenous People, 1994. UN approval of Declaration on Rights of Indigenous People, 2007.	Promotes rights of Indigenous people	Signed by Suriname	Sets standards for protection of Indigenous people. Project has potential to impact tribal groups.
REGIONAL DEVELOPMENT/COOPERATION			
Amazon Cooperation Treaty, 1978 (amended in 1998)	Aimed at the promotion of sustainable development of the Amazon Basin. Member states are Bolivia, Brazil, Colombia, Ecuador, Guyana, Peru, Suriname and Venezuela.	Suriname is a founding member,	Sets guidelines to promote sustainable development of the Amazon basin; project taking place in the Amazon
LABOR/HEALTH/SAFETY			
Constitution of the International Labor Organization	Promotes opportunities for women and men to obtain decent and productive work, in conditions of freedom, equity, security and human dignity.	Suriname member since 1976.	Sets international labor standards; project will employ large workforce
Constitution of the Pan American Health Organization	To improve health and living standards of the countries of the Americas.	Suriname member since 1976.	Sets guidelines to improve health and living standards; project has potential to impact local health/living standards.
POLLUTION PREVENTION AND CONTROL			
Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal	The agreement, in force since May 1992, seeks to protect the health of people and the environment from the harmful effects of hazardous wastes.	Suriname acceded on 20 Sept 2011	The provisions of the Convention revolve around the reduction of hazardous waste generation and the promotion of the environmentally sound management of hazardous wastes, the restriction of transboundary movements of hazardous wastes, and the application of a regulatory system for permissible movements of hazardous waste.
Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade	In force since February 2004. It aims to promote shared responsibility and cooperation among Parties in addressing the international trade of certain hazardous chemicals, in order to protect human health and the environment from potential harm.	Suriname acceded on 20 May 2000	The agreement establishes a prior informed consent (PIC) procedure for the import of hazardous chemicals.

Agreement / Convention	Notes / Comments	Status	Relevance
Stockholm Convention on Persistent Organic Pollutants	Came into force in 2004. It aims to protect human health and the environment from Persistent Organic Pollutants (POPs).	Ratified on 20 Sept 2011	The Convention requires Parties to take steps to eliminate or reduce the production, use, import, export and emissions into the environment of POPs and includes provisions on access to information, public awareness, training and participation in the development of implementation plans.
Minamata Convention on Mercury	The Minamata Convention, in force since August 2017, was adopted at the 2013 Plenipotentiary Conference in Kumamoto, Japan	Suriname Acceded on 02 August 2018	The aim of this global treaty is to protect human health and the environment from anthropogenic and other releases of mercury and mercury compounds. It includes provisions on public information, environmental education, promotion of participation and capacity building.

Source: SRK, 2007 modified and updated by RINA, 2022

Suriname has ratified eight (out of ten) fundamental conventions, three (out of four) governance conventions and 22 (out of 176) principal technical conventions. These and other relevant OIT conventions are presented in Table 4-4.

Table 4-4 Relevant OIT Conventions

Type	Category	#	Convention (C) - Protocol (P)	Date	Status
Fundamental	Freedom Association	1	C087 - Freedom of Association and Protection of the Right to Organise Convention, 1948 (No. 87)	15-Jun-76	In Force
		2	C098 - Right to Organise and Collective Bargaining Convention, 1949 (No. 98)	05-Jun-96	In Force
	Force labor	3	C029 - Forced Labour Convention, 1930 (No. 29) P029 - Protocol of 2014 to the Forced Labour Convention, 1930 ratified on 03 Jun 2019 (In Force)	15-Jun-76	In Force
		4	C105 - Abolition of Forced Labour Convention, 1957 (No. 105)	15-Jun-76	In Force
	Discrimination	5	C100 - Equal Remuneration Convention, 1951 (No. 100)	04 Jan 2017	In Force
		6	C111 - Discrimination (Employment and Occupation) Convention, 1958 (No. 111)	04 Jan 2017	In Force
	Child labor	7	C138 - Minimum Age Convention, 1973 (No. 138) Minimum age specified: 16 years	15 Jan 2018	In Force
		8	C182 - Worst Forms of Child Labour Convention, 1999 (No. 182)	12 Apr 2006	In Force
	OHS	9	C155 - Occupational Safety and Health Convention, 1981 (No. 155)	/	not ratified
		10	C187 - Promotional Framework for Occupational Safety and Health Convention, 2006 (No. 187)	/	not ratified
Principal Governance	Labor	1	C081 - Labour Inspection Convention, 1947 (No. 81)	15-Jun-76	In Force
	Labor	2	C122 - Employment Policy Convention, 1964 (No. 122)	15-Jun-76	In Force
	Labor	3	C144 - Tripartite Consultation (International Labour Standards) Convention, 1976 (No. 144)	16-Nov-79	In Force
Principal	Labor	1	C014 - Weekly Rest (Industry) Convention, 1921 (No. 14)	15-Jun-76	In Force

Technical	OHS	2	C017 - Workmen's Compensation (Accidents) Convention, 1925 (No. 17)	15-Jun-76	In Force
	OHS	3	C019 - Equality of Treatment (Accident Compensation) Convention, 1925 (No. 19)	15-Jun-76	In Force
	OHS	4	C042 - Workmen's Compensation (Occupational Diseases) Convention (Revised), 1934 (No. 42)	15-Jun-76	In Force
	Labor	5	C095 - Protection of Wages Convention, 1949 (No. 95)	15-Jun-76	In Force
	Labor	6	C118 - Equality of Treatment (Social Security) Convention, 1962 (No. 118) Has accepted Branch (g)	15-Jun-76	In Force
	Freedom Association	7	C135 - Workers' Representatives Convention, 1971 (No. 135)	15-Jun-76	In Force
	Freedom Association	8	C154 - Collective Bargaining Convention, 1981 (No. 154)	05-Jun-96	In Force
	Labor	9	C097 - Migration for Employment Convention (Revised), 1949 (No. 97)	/	not ratified
	OHS	10	C102 - Social Security (Minimum Standards) Convention, 1952 (No. 102)	/	not ratified
	Labor	11	C143 - Migrant Workers (Supplementary Provisions) Convention, 1975 (No. 143)	/	not ratified
	OHS	12	C148 - Working Environment (Air Pollution, Noise and Vibration) Convention, 1977 (No. 148)	/	not ratified
	OHS	13	C152 - Occupational Safety and Health (Dock Work) Convention, 1979 (No. 152)	/	not ratified
	OHS	14	C161 - Occupational Health Services Convention, 1985 (No. 161)	/	not ratified
	OHS	15	C162 - Asbestos Convention, 1986 (No. 162)	/	not ratified
	Labor	16	C168 - Employment Promotion and Protection against Unemployment Convention, 1988 (No. 168)	/	not ratified
	Indigenous People	17	C169 - Indigenous and Tribal Peoples Convention, 1989 (No. 169)	/	not ratified
	OHS	18	C170 - Chemicals Convention, 1990 (No. 170)	/	not ratified
	OHS	19	C176 - Safety and Health in Mines Convention, 1995 (No. 176)	/	not ratified
	Labor	20	C183 - Maternity Protection Convention, 2000 (No. 183)	/	not ratified
	OHS	21	P089 - Protocol of 1990 to the Night Work (Women) Convention (Revised), 1948	/	not ratified
	OHS	22	P155 - Protocol of 2002 to the Occupational Safety and Health Convention, 1981	/	not ratified

4.2.3 INSTITUTIONAL FRAMEWORK

Suriname's mining sector is governed by the Mining Decree (1986) and regulated by the Ministry of Natural Resources, the Ministry of Finance, the Ministry of Regional Development and the Ministry of Trade, Industry and Tourism. The legal and fiscal framework for the oil, gas and mining sector is also supported by other legislation, including the Corporate Income Tax Act and Dividend Tax Act. (EITI, 2022). Applications for mining rights are processed by the Geological Mining Service (GMD) and are reviewed and approved by the Ministry of Natural Resources in accordance with the Mining Decree. Suriname does not have a public mining license register, however some information about issued licenses is included in EITI Reports. (GLiM, 2022). EITI Suriname is administered by the Suriname Multi-Stakeholder Group (MSG), which is hosted by the Ministry of Natural Resources. The MSG is chaired by the Minister of Natural Resources. (EITI, 2022)

On the other hand, the environmental approval process is administered under the new Environmental Framework Act 2022, the National Environment Authority (in Dutch: Nationale Milieu Autoriteit, NMA) and the National Institute for Environment and Development in Suriname (in Dutch: National Instituut voor Milieu en Ontwikkeling in Suriname, NIMOS) together form the National Environment Structure in Suriname. Through this structure the provisions included in the Environmental Framework Act will be implemented and applied in an efficient and effective manner. (NIMOS, 2022)

4.2.3.1 [Ministry of Natural Resources - MNR](#)

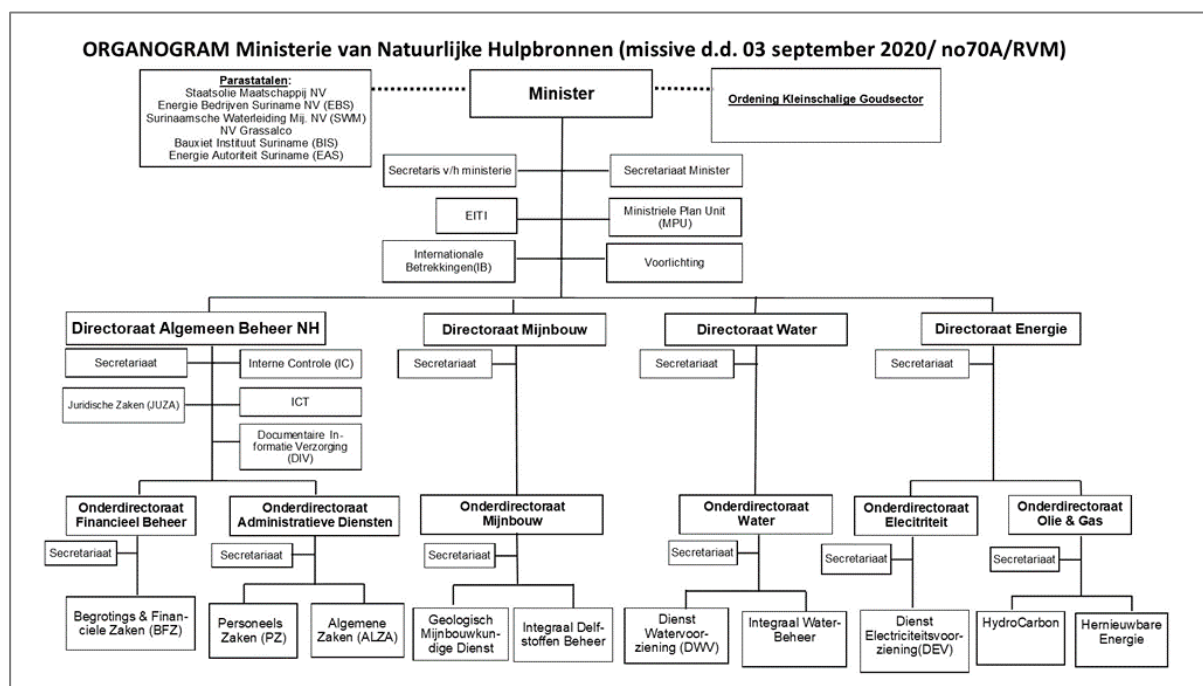
The aim of the Ministry of Natural Resources (MNR) is to ensure an integral sustainable and efficient management and development of the production of the natural resources present in Suriname, taking into account the environment, safety and rights of local communities. The organizational structure of the MNR has recently changed. The Ministry now consists of four (4) directorates, as further detailed in the organizational structure, namely (MINR, 2022a):

1. Directorate General Administration NH
2. Directorate of Water
3. Directorate of Energy
4. Mining Directorate

The organization has been expanded to include directorate and departmental tasks. The determining organizational objectives for this are:

5. Planning, monitoring and control of the spending of the budget and other resources.
6. The optimization of the operational execution of tasks.
7. Application of integrated management in the various policy areas.
8. Application of advancing technology within the various policy areas.
9. Continuous quality improvement and quality maintenance of the execution capacity.
10. Entering into regional and international partnerships.
11. Sustainable professional and personal development of staff.

Based on the objectives described above, the organization chart of the MNR is presented in Figure 4-1.



Source: (MINR, 2022a)

Figure 4-1: MNR organizational structure

4.2.3.1.1 Extractive Industries Transparency Initiative (EITI) Suriname

EITI Suriname is administered by the Suriname Multi-Stakeholder Group (MSG), which is hosted by the Ministry of Natural Resources. The MSG is chaired by the Minister of Natural Resources. (EITI, 2022)

Suriname was found to have made *meaningful progress* in implementing the 2016 EITI Standard in March 2021, following its first Validation. The Validation identified 19 corrective actions to be addressed by the country's next Validation, expected to commence in April 2023. (EITI, 2022)

4.2.3.2 National Council for the Environment – NMR

National Council for the Environment (Nationale Milieuraad – NMR) was established in 1997 with a mandate to advise the government of Suriname on the development and implementation of national environmental policies. The NMR consists of a chairperson and 5-10 members representing government, private sector, Amerindian and Maroon communities, labor unions, consumer rights, and other Non-Governmental Organizations (NGOs). Executive support for the NMR is provided by the Nationale Instituut voor Milieu en Ontwikkeling in Suriname (NIMOS – the National Institute for Environment and Development in Suriname).

4.2.3.3 National Environment Authority– NMA

Under the new Environmental Framework Act 2022, the National Environment Authority (in Dutch: Nationale Milieu Autoriteit, NMA) will be mandated to formulate the Environmental Policy and coordinate the effective implementation. The State of the Environment Report will be one of the key documents to support informed policy and decision making. This Report will be developed by NIMOS. (NIMOS, 2022)

4.2.3.4 National Institute for Environment and Development in Suriname (NIMOS)

NIMOS was established on March 15, 1998, as the executive body of the National Environmental Council (NMR). NIMOS is an autonomous government foundation, supervised by a Foundation Board

and consists of the following seven board members. The board is appointed for two years by the President of the Republic of Suriname. The day-to-day management and overall coordination of NIMOS is undertaken by the General Director. Currently NIMOS has 27 staff members, of which 20 are employed on a permanent basis and the remaining on a project basis. (NIMOS, 2022)

NIMOS consists of several offices, each tasked with specific responsibilities, namely (NIMOS, 2022):

1. Administration
2. Environmental & Social Assessment
3. Environmental Monitoring & Enforcement
4. Environmental Legal Services
5. Environmental Education & Public Outreach
6. Environmental Planning & Information Management
7. Environmental Funding & Investments (not operational).
8. Environmental Research (not operational).

The mission of NIMOS is to “*initiate the development of a national legal and institutional framework for environmental policy and management in the interest of sustainable development*” and driven by the following goals (NIMOS, 2022) :

1. achieving National Environmental Legislation in the broadest sense of the word.
2. Preparing and implementing Environmental Protection legislation.
3. Coordination and monitoring of Compliance.

Under the new Environmental Framework Act, NIMOS will be in charge to develop the State of the Environment Report that will be used to inform policy and decision making of the NMA (NIMOS, 2022). In addition, the new Act will require the development of Implementation Regulations to facilitate its implementation. A total of 21 implementation regulations (i.e., Decrees) have been identified, however, the top six or most important ones were selected to be drafted first and currently under development, including (NIMOS, 2022):

1. EIA Activities,
2. EIA Procedure,
3. Environmental Permit,
4. Hazardous Substances,
5. Environmental Fund,
6. Rules and Procedures for the Environmental Controllers.

NIMOS has also developed Generic Environmental Assessment Guidelines as well as specific Guidelines. These Guidelines carefully explain the procedures as how they should be carried out if the Environmental Framework Law and the Environmental Regulations were in place (NIMOS, 2022).

The EIA process described by NIMOS is divided in five phases, namely (NIMOS, 2022):

1. Screening phase: This is the phase in the EIA process, in which NIMOS decides whether an Environmental Impact Assessment is needed or not. If yes, NIMOS will indicate the nature and extent of the analysis. NIMOS acknowledges three categories of EIA:
 - a. Category A: an EIA is mandatory
 - b. Category B, either:
 - i. An EIA is required (Category B, path 3);
 - ii. Another environmental document is needed (e.g., Environmental Management Plan) (Category B, path 2);
 - iii. No EIA is needed, but some environmental information is required before a decision can be taken (Category B, path 1).
 - c. Category C: No EIA is required, but the project proponent will have to keep with the minimal guidelines.
2. Scoping phase: In this phase the Terms of Reference (ToR), sometimes also called Scoping report, for the EIA-study has to be prepared.

3. Assessment phase: In the assessment phase the EIA-study is to be carried out by a qualified consultant or consultancy firm by order of the project proponent.
4. Review phase: During the review process the submitted Environmental Impact Statement (EIS) is reviewed by NIMOS
5. Decision-making phase: After the EIS is reviewed, NIMOS should send the environmental advice regarding approval or denial of the project to the permitting agency.

4.2.3.5 Minerals Institute

The government of Suriname wants to promote a regulated and responsible expansion of the mining industry, including improved control over informal goldmining. However, weak institutional capacity in the public sector and the absence of an adequate legal and regulatory framework have inhibited prospects for growth. Moreover, weak corporate governance in the small-scale mining activity and the lack of social and environmental standards have diminished the sustainability and positive social impacts of the sector. The current mining policy aims to create a Minerals Institute – through the merger of the Geological Mining Service (GMD), the Bauxite Institute of Suriname (BIS) and the Suriname Gold Sector Regulation Commission (OGS) - and to enact a new mining law. Since, these currently are incapable to meet legal deadlines for licensing, develop regulations and procedures for their activities, enforce the regulatory framework for artisanal and small-scale mining. (MINR, 2019)

The government of Suriname by State Decree plans to create the "Delfstoffen Instituut Suriname" (DIS) or Minerals Institute Suriname for the promotion, coordination, regulation, monitoring and inspection of the mining industry aimed at an optimal contribution to Suriname's sustainable development. Its creation is a response to the expected growth of sector activities in the country, the changing focus (from bauxite to gold and other minerals), the increasing complexity of financial, social and environmental challenges, and the increased demand for governance and accountability in managing the sector. (MINR, 2019)

In March 2022, the Minister for Natural Resources officially launched the mineral reform for the restructuring process of the mining sector. The Suriname Competitiveness and Sector Diversification Project (SCDS-Project) started in December 2021 and should ultimately form a good basis for setting up the Mineral Institute of Suriname. The Minister indicated that a legal and institutional framework will be needed to monitor mining areas and restructuring the small-scale gold sector. The Ministry of Natural Resources is holding workshops with stakeholders and a number of online discussions with various organizations. (MINR, 2022b)

The Minerals Institute Suriname will be given the functions of Land Registry, Geological Survey and Inspectorate in order to regulate the mining sector in an efficient and transparent manner. This on the basis of forms of cooperation and transparent work processes in which technical and capacity issues are analyzed in a participatory manner, the digitization of the geological survey and capacity enhancement. (MINR, 2022b).

5 REFERENCES

- EITI. (2022, Sept 06). *Extractive Industries Transparency Initiative (EITI)*. Retrieved Sept 06, 2022, from Suriname: <https://eiti.org/countries/suriname>
- ERM. (2011). *SCOPING REPORT AND TERMS OF REFERENCE: Environmental and Social Impact Assessment Study of Surgold's Merian Project in Suriname*. Environmental Resources Management (ERM). Retrieved September 29, 2022, from https://s24.q4cdn.com/382246808/files/doc_downloads/operations_projects/south_america/documents/5-Merian_TOR-Final-Merian-ESIA-TOR_27Sept11.pdf
- ERM. (2016). *DRAFT-Environmental Impact Assessment Addendum*. Retrieved September 29, 2022, from <https://paperzz.com/doc/9382693/draft-environmental-impact-assessment-addendum>
- GLiM. (2022). *University of Hamburg, Global Lithological Map*.
- MNR. (2019). *Note on Minerals Institute Suriname*. Ministry of Natural Resources (MONR) of Suriname, Republic of Suriname. Retrieved Sept 05, 2022, from <https://documents1.worldbank.org/curated/zh/715551576654339922/Note-on-Minerals-Institute-Suriname.docx>
- MNR. (2022a). *Ministry of Natural Resources*. Retrieved September 29, 2022, from About us: <https://gov.sr/ministeries/ministerie-van-natuurlijke-hulpbronnen/over-ons/>
- MNR. (2022b, March 09). *Ministry of Natural Resources*. (Republiek Suriname) Retrieved Sept 06, 2022, from Officiële doorstart Suriname competitive and sector diversification project; duurzame ontwikkeling mijnbouwsector, naar delfstoffen instituut: <https://gov.sr/officiële-doorstart-suriname-competitive-and-sector-diversification-project-duurzame-ontwikkeling-mijnbouwsector-naar-delfstoffen-instituut/>
- NIMOS. (2022). *National Institute for Environment and Development in Suriname (NIMOS)*. Retrieved September 29, 2022, from About Us: <http://nimos.org/en/about-us/>
- UNEP. (2022). *Environmental Framework Act (No. 97 of 2020)*. FAO, FAOLEX. Retrieved September 29, 2022, from <https://leap.unep.org/countries/sr/national-legislation/environmental-framework-act-no-97-2020>

Appendix B

Macro-Economic Development Outlook.

Doc. No. P0032241-SR Rev. 2 – January, 2023



1 ECONOMIC IMPORTANCE OF THE MINING SECTOR

In economic terms, Suriname's economy has traditionally been dependent on extractive industries, which play a dominant role in driving growth, exports, employment, and government revenues. Gold, oil, and bauxite resources have historically accounted directly for around 30% of GDP and as much as 90% of exports. About 10% of government revenues are directly linked to gold production in the form of corporate income taxes, royalties and dividends. More than 30 tons of gold are exported annually, accounting for about 60% of the country's exports. The services sector accounts for more than 50% of the economy and is dominated by construction, retail, trade, and transport activities that are closely linked to income earned from extractives. The Government of Suriname redistributes extractive industries revenues through significant public sector employment, accounting directly for about 40% of total formal employment. (MINR, 2019) However, other extractive industries and sectors for Suriname include Oil and Gas and agricultural sector. In 2018, the agriculture sector (composed by agriculture, livestock, and fisheries) contributed 8.1% to Suriname's GDP and it has been designated by the government as priority area (ITA, 2022) In terms of the forestry sector, Suriname's abundant rainforest contains many valuable hardwood tree species. A sizeable portion of production remains in the local market as the industry has been unable to compete in quantity, quality, or price with large logging companies in the region. First Timber N.V. and Houtexploitatie en Verwerkings Onderneming Soekhoe & Zn N.V. are the largest timber-exporting companies in Suriname. In 2019, the total export value of timber was US\$ 52.2 million, and Suriname exported hardwoods to Europe, Asia, and the Caribbean region, with Asia (95 percent) representing the largest market (ITA, 2022).

1.1 GENERAL OVERVIEW OF CURRENT ECONOMIC AND BUSINESS ENVIRONMENT

A brief snapshot the current macroeconomic development is provided in the sections below.

1.1.1 Imports and Exports

Details of Imports and Exports are provided in the following two charts in the following figure

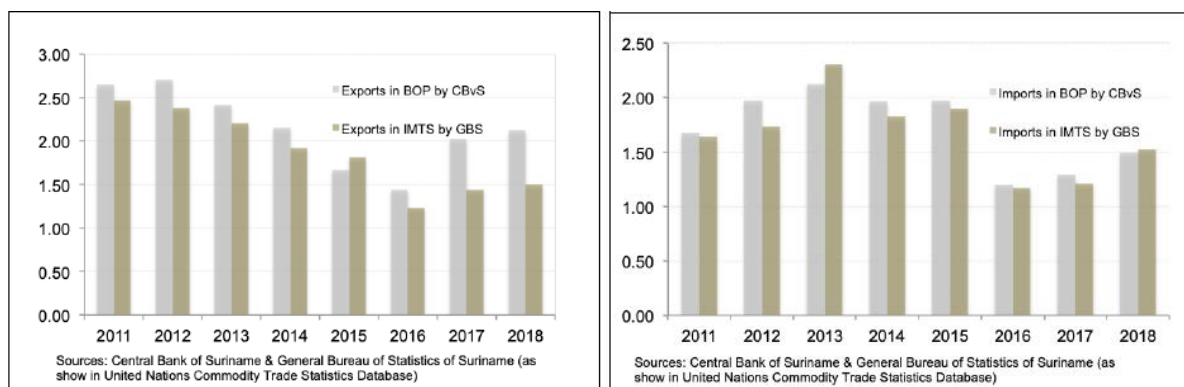


Figure 1.1: Good Exports (left) and Imports (right) values compiled in the balance payment by CBvS and the IMTS by GBS.

It is likely however that, in particular, the level of exports will increase again in the near future, due to the substantial amounts of oil and gas that have been discovered in offshore waters, as well as potential development in the gold and bauxite industries. In parallel, a further increase in imports is expected to service these mining sector developments

1.1.2 Gross Domestic Product - GDP

In 2021, Suriname had a GDP per capita of US\$5,166, below its peak of US\$9,350 in 2014 and well below the Latin America Country (LAC) regional average of US\$15,092. Suriname had strong economic growth between 2001 and 2014, averaging 4.4%, mainly due to favorable commodity prices. From 2015, Suriname's economy contracted with Government revenue from mining fell sharply (public sector revenue and gold representing more than 80 percent of total exports), foreign reserves were drained, and GDP growth contracted. By 2020, a severe fiscal and balance of payments crisis began to unfold in Suriname, which continued into 2021. Domestic vulnerabilities were exacerbated by the COVID-19 pandemic, leading to a sharp GDP contraction (15.9 percent in 2020) and increasing unemployment and poverty. In 2021, Suriname had a GDP per capita of US\$5,166, below its peak of US\$9,350 in 2014 and well below the LAC regional average of US\$15,092 (WBG, 2022a). The annual GDP growth (%) for Suriname is presented in Figure 1.2: (WBG, 2021). Suriname ranked 87th of 180 countries and territories on Transparency International's Corruption Perceptions Index (CPI) in 2021, putting it on par with Colombia, Guyana, Ethiopia, Kosovo, Morocco, Tanzania and Vietnam (Transparency International, 2021).

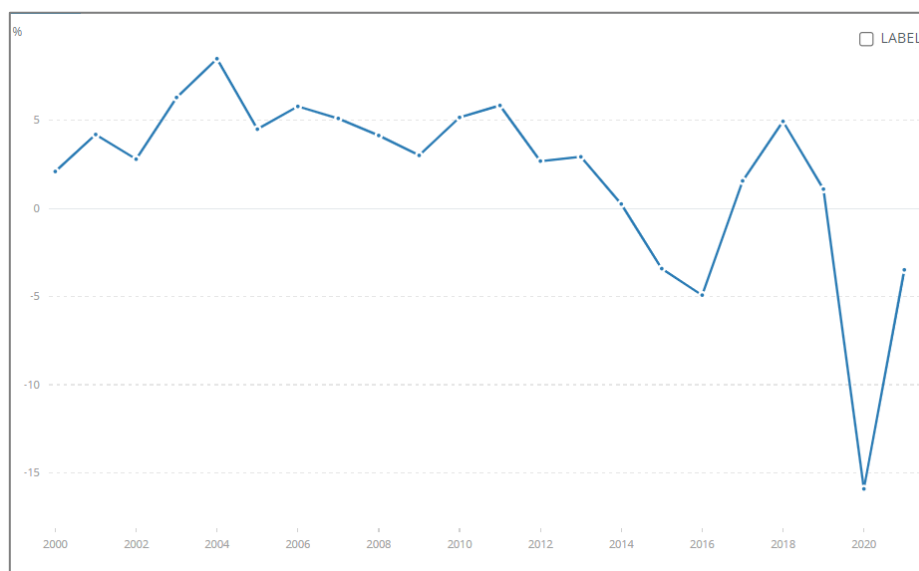


Figure 1.2: GDP Growth (annual %) – Suriname (2000-2021)

Source: (WBG, 2021)

1.1.3 Short-term development

Suriname's main short-term challenge relates to improving its macroeconomic performance, particularly as it relates to fiscal imbalances and increasing debt levels. Thus, addressing energy and water service provision challenges, along with strengthening the Public-Private Partnerships (PPP) framework, can help the country realize important benefits in the short to medium term in regard to both fiscal performance and closing the country's infrastructure gap in an efficient and cost-effective manner. Pursuant to these goals, the government's ongoing fiscal strengthening program calls for the creation of a PPP Unit, along with improvements to public investment system and strengthening public financial management.

1.1.4 Medium Term Development

The medium term reported that Suriname faces limited fiscal space over the medium-term to increase public investment. Although gross fixed capital formation steadily increased from 38 percent of GDP

(BNP) in 2010 to 71 percent of GDP in 2014, it has declined to 57 percent of GDP in 2017, as reported by the World Bank¹. Details of Medium-Term growth Plan are provided in the Figure below:

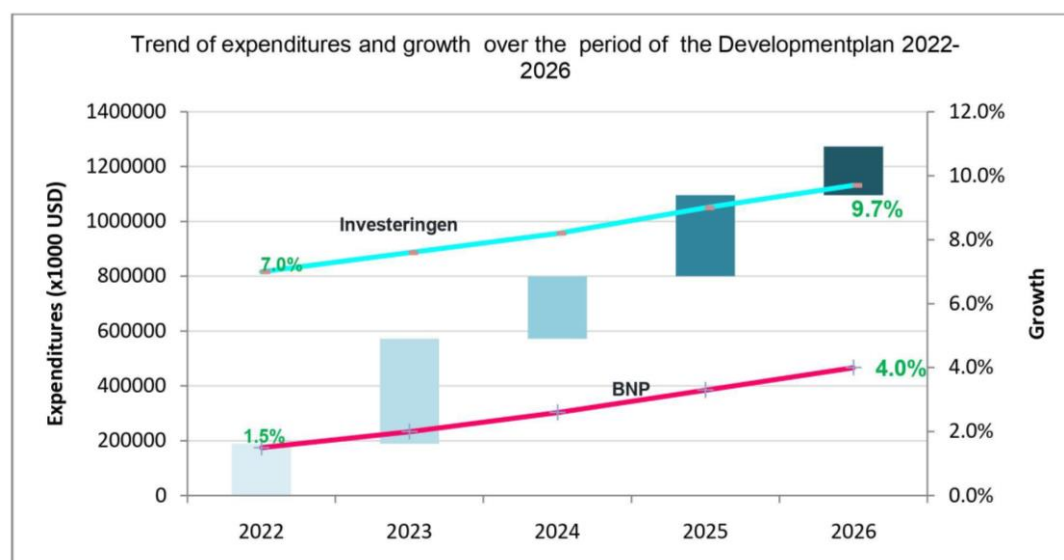


Figure 1.3: Trend of expenditure and growth over the Development Plan period 2022-2026

Source: (GOS, 2022)

In this context, a public private partnerships (PPP) framework could help the country foster long-term infrastructure investments and close the infrastructure gap in energy, telecommunication, transportation and water and sanitation. However, Suriname does not have a PPP framework, and the World Bank (2014) found that there are no laws, policies, detailed guidelines, dedicated units or dedicated project preparation funding in place and that the percentage of staff with PPP experience was classified as low. Although the country lacks a framework, it does have some PPP experience, especially in the energy sector, through a 189MW hydroelectric plant, built by Alcoa Aluminium in 1958, that provided up to 60 percent of Suriname's electricity under a power purchase agreement (World Bank, 2014).

However, it is not clear whether current medium and long-term development plans take into account the promising developments in the oil and gas industries (the exploration / development potential of the Stabroek block in neighboring Guyana alone will lead to a rapid increase GDP there. Suriname has the better deep-water location (because of deeper natural port depths) and the exploration potential in its own blocks is also looking increasingly positive.

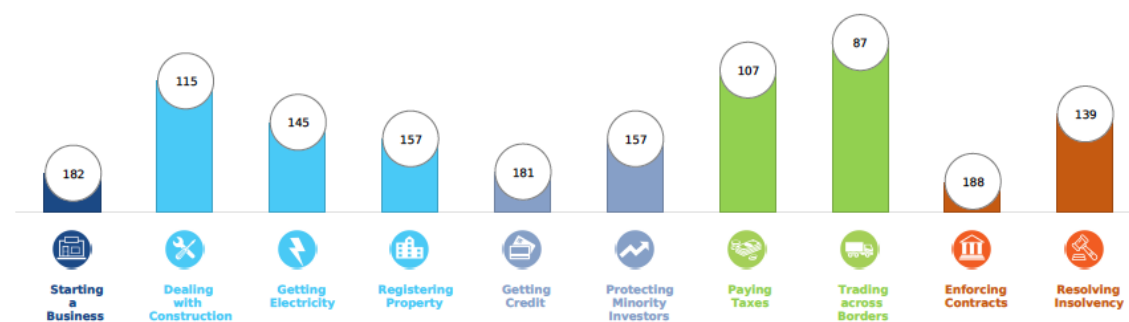
1.1.5 Ease of Doing Business/ Export Environment

1.1.5.1 Ranking on Doing Business Topics; Suriname

An overview of the WBs statistics on the Ease of Doing Business in Suriname is provided in the following Figure. The figure provides a score across ten separate Ease of Doing Business parameters, providing a visual summary of Suriname's scores.

¹ IMF Country Report No. 22/90

Rankings on Doing Business topics - Suriname



Topic Scores



Starting a Business (rank)	182	Getting Credit (rank)	181	Trading across Borders (rank)	87
Score of starting a business (0-100)	61.6	Score of getting credit (0-100)	10.0	Score of trading across borders (0-100)	75.0
Procedures (number)	8	Strength of legal rights index (0-12)	2	Time to export	
Time (days)	66	Depth of credit information index (0-8)	0	Documentary compliance (hours)	12
Cost (number)	93.5	Credit registry coverage (% of adults)	0.0	Border compliance (hours)	84
Paid-in min. capital (% of income per capita)	0.0	Credit bureau coverage (% of adults)	0.0	Cost to export	
				Documentary compliance (USD)	40
Dealing with Construction Permits (rank)	115	Protecting Minority Investors (rank)	157	Border compliance (USD)	468
Score of dealing with construction permits (0-100)	66.3	Score of protecting minority investors (0-100)	28.0	Time to export	
Procedures (number)	10	Extent of disclosure index (0-10)	1.0	Documentary compliance (hours)	24
Time (days)	223	Extent of director liability index (0-10)	0.0	Border compliance (hours)	48
Cost (% of warehouse value)	0.3	Ease of shareholder suits index (0-10)	6.0	Cost to export	
Building quality control index (0-15)	6.5	Extent of shareholder rights index (0-6)	4.0	Documentary compliance (USD)	40
		Extent of ownership and control index (0-7)	2.0	Border compliance (USD)	658
		Extent of corporate transparency index (0-7)	1.0		
Getting Electricity (rank)	145	Paying Taxes (rank)	107	Enforcing Contracts (rank)	188
Score of getting electricity (0-100)	57.7	Score of paying taxes (0-100)	69.4	Score of enforcing contracts (0-100)	25.9
Procedures (number)	4	Payments (number per year)	30	Time (days)	1,715
Time (days)	113	Time (hours per year)	199	Cost (% of claim value)	37.1
Cost (% of income per capita)	895.8	Total tax and contribution rate (% of profit)	27.9	Quality of judicial processes index (0-18)	3.5
Reliability of supply and transparency of tariff index (0-8)	0	Postfiling index (0-100)	48.4		
				Resolving Insolvency (rank)	139
Registering Property (rank)	157			Score of resolving insolvency (0-100)	33.8
Score of registering property (0-100)	46.8			Recovery rate (cents on the dollar)	7.7
Procedures (number)	6			Time (years)	5.0
Time (days)	46			Cost (% of estate)	30.0
Cost (% of property value)	14.2			Outcome (0 as piecemeal sale and 1 as going concern)	0
Quality of the land administration index (0-30)	13.5			Strength of insolvency framework index (0-16)	9.5

Figure 1.4: WBGs statistics on the Ease of Doing Business in Suriname

Source: (WBG, 2022)

1.1.5.2 Trading Across Borders: Suriname

An overview of the WBGs statistics on the Ease of Trading Across Borders in Suriname is provided in the Figure below.

Doing Business 2020

Suriname

Trading across Borders - Suriname

Indicator	Suriname	Latin America & Caribbean	OECD high income	Best Regulatory Performance
Time to export: Border compliance (hours)	84	55.3	12.7	1 (19 Economies)
Cost to export: Border compliance (USD)	468	516.3	136.8	0 (19 Economies)
Time to export: Documentary compliance (hours)	12	35.7	2.3	1 (26 Economies)
Cost to export: Documentary compliance (USD)	40	100.3	33.4	0 (20 Economies)
Time to import: Border compliance (hours)	48	55.6	8.5	1 (25 Economies)
Cost to import: Border compliance (USD)	658	628.4	98.1	0 (28 Economies)
Time to import: Documentary compliance (hours)	24	43.2	3.4	1 (30 Economies)
Cost to import: Documentary compliance (USD)	40	107.3	23.5	0 (30 Economies)

Figure - Trading across Borders in Suriname - Score

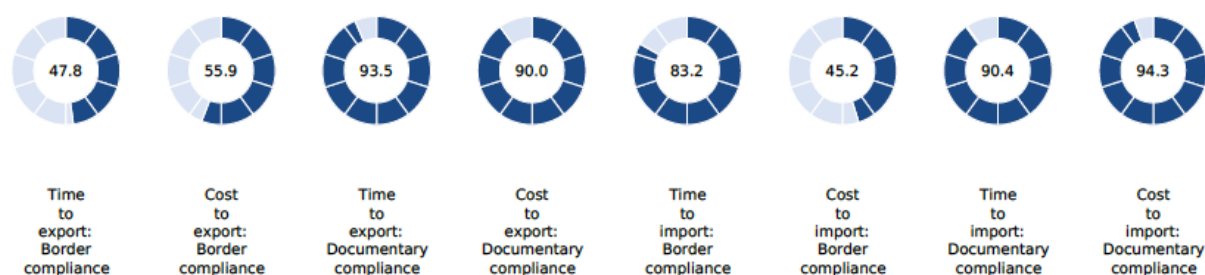
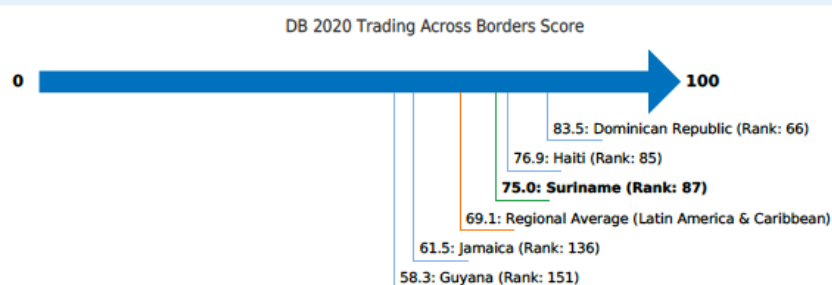


Figure - Trading across Borders in Suriname and comparator economies - Ranking and Score



Note: The ranking of economies on the ease of trading across borders is determined by sorting their scores for trading across borders. These scores are the simple average of the scores for the time and cost for documentary compliance and border compliance to export and import.

Figure 1.5: Key indicators of Ease of Trading Across borders in Surinam

Source: (WBG, 2022)

Although some Ease of Doing Business (EODB) indicators are relatively low for instance ease of getting credit (10), Suriname score on Cross border trade (75) is good, particularly valuable in view of the potential increase in mine sector development (Oil and gas).

1.2 LINKAGES WITH DONOR AND GOS DEVELOPMENT PLANS

1.2.1 Country's Sector Strategy

The 2006-2011 Development Plan (DP) recognized the importance of physical infrastructure as a means to facilitate transport and logistics, stimulate development within the coastal areas in the N.E. of the country and facilitate cross-border links with French Guiana. The 2011-2015 DP continues to emphasize the east-west corridor and the Government Declaration on infrastructure for the period 2010 to 2015 given to Parliament on October 1, 2010 states that the connections between Paramaribo and Commewijne will be improved. A later plan is indicated in the section below.

Furthermore, the GOS had included the MACRP as one of the two anchor projects of Suriname in the Initiative for Regional Infrastructure Integration in South America (IIRSA), within which regional transportation linkages between the neighbouring countries of Venezuela, Guyana, French Guyana and Brazil are to be established.

1.2.1.1 Enabling Sectors: The role of Physical Infrastructure in the 2017-2022 Plan²

The physical infrastructure for national and international transactions and transport in the form of roads, waterways, ports, airports, railways and associated facilities, mostly requires long-term investments, which usually are of service for decades to the national and international community, in the course of which in addition to the transport of people, production, export, transit trade, tourism and trade can be done in an efficient manner. The development of the infrastructure is therefore embedded in long-term development plans³, which eventually must increase and stabilize the capacity of the economy in a sustainable manner. However, substantiating the implementation of basic infrastructural works in Suriname must be linked to the development of the national production, integration of Suriname in the region and the rest of the world. This is the case, due to the very small size of the population and the restricted local implementation capacity:

1. Dredging the Suriname River with the development of Paranam as industrial centre with all facilities to handle big Panamax ships and oil tankers to forward in transit, products to the South American continent.
2. Dredging the Nickerie River with the development of New-Nickerie as industrial centre, facilities handling big Panamax ships and all tankers.
3. The bridge over the Marowijne River with the development of Albina as industrial and tourist centre. The bridge over the Corantijn River with the development of Apoera as industrial port. Construction of a railway between Bakhuis and Apoera or Paranam for the export of oil, gas, alumina, and aluminium, timber and agrarian products.
4. Construction of a centre for the gas and oil industry along the Suriname River near Paranam after more certainty has been obtained about the gas and oil reserves in the territorial waters.

The decision about the implementation of the aforementioned large-scale projects will be made based on consensus⁴.

² 2017-2021 POLICY DEVELOPMENT PLAN National Planning Bureau

³ We have referenced the detailed plan for 2017-2021 for this Report. If further detailed information becomes available during the study, we will reference it.

⁴ The economy declined substantially over the Covid period hence implementation has been slowed.

Harmonic development of the national regions strongly depends on a suitable road network which opens up both residential and production areas. Construction and improvement of the physical Infrastructure is therefore approached as an important instrument to increase the competitiveness and development of the local economic activities.

Improvement of the North – South and the East – West access roads will facilitate more efficient transport and open up future development poles. At the same time, during the coming planning period, Government will work on the physical integration of Suriname in the South American region by connecting to the neighbouring countries.

The biggest part of the economic activities of the three Guyanas happens to take place in the coastal area where this connection is available. Participation of Suriname in the CARICOM, Initiative for the Integration of the Regional Infrastructure of South America (IIRSA), FTAA and MERCOSUR Offers the opportunity to enter into a gigantic market of millions of people. The construction of permanent shore connections with adjacent neighbouring countries will enhance the integration process of Suriname. This will create new opportunities for trade and economic collaboration.

It is expected that the SESA study will contribute a refinement of some of these infrastructure plans, based on the green infrastructure development principles and environmental and social protection

1.2.1.2 Existing Submodal Development Plan: Roads⁵

The implementation of a National Road Plan in conjunction with the opening of residential areas and communities across the country, will create the conditions for access and above all affordable land transport for every Surinamese. There is a structure and a road plan, based on the long-term national strategic development plan (Vision 2035), in which the planning is included for more efficient passenger travel and trucking transport in our country, also as part of the regional development and population planning.

1.2.1.3 Existing Submodal Development Plan: Ports⁶

The increase in the quality of the inland shipping with better facilitation will stimulate the domestic water transport. The legal and technical facilities will be adapted to ensure the safety and quality of water transport. The following outcomes will be further elaborated in the policy in results and action programs of the relevant ministries and institutions:

- The shipping movements and destinations have increased both nationally and internationally.
- Better infrastructure facilities, modern equipment and adequate qualified expertise increase the safety and the capacity of the sea traffic to provide services to a larger number of shipping movements and destinations.
- The Suriname river is more navigable and thus the capacity of the ports of Paramaribo and Paranam has increased.

1.2.1.4 Existing Submodal Development plans: Airports⁷

In order to improve flight safety and accessibility of local communities in the interior, the infrastructure of the airstrips in the interior will be improved. Within the process of integration and liberalization, geographically Suriname has a very strategic location in the northern part of South America, since the Zanderij International Airport has the longest runway in the north of South America. The following

⁵ 2017-2021 POLICY DEVELOPMENT PLAN National Planning Bureau

⁶ 2017-2021 POLICY DEVELOPMENT PLAN National Planning Bureau

⁷ 2017-2021 POLICY DEVELOPMENT PLAN National Planning Bureau

outcomes will be further elaborated in the policy in results and action programs of the relevant ministries and institutes:

- The flight movements and destinations have increased both nationally and internationally.
- Better infrastructural facilities, modern equipment and sufficiently qualified experts increase the safety and capacity of the airports to render services to a bigger number of flight movements and destinations.

1.2.1.5 Existing Submodal Development plans:⁸ Production, Transport and Storage of Hazardous Substances

Partly on the basis of existing laws and regulations, the existing institutions will conduct an inventory from the company plant that could be classified as "hazardous installations" because an accident or incident may cause a disaster. These hazardous substances include chemicals, toxins and gases, explosives and radioactive materials. It will also set up a mechanism on the basis of this inventory, which can monitor:

- the emission of harmful substances in the production process.
- the transport and the storage of hazardous materials.

1.3 NATURAL RESOURCES AND MINERAL POTENTIAL

The Surinamese economy is dominated by natural resources: oil, gold and bauxite (IGF, 2017). In terms of mineral potential, Suriname's mineral and oil potential remains largely underexploited. Suriname is located on the Guiana Shield geological formation and has a very high potential for a range of minerals, including gold, diamonds, and other metallic minerals in addition to kaolin, rare earth elements and dimension stones (MINR, 2019).

Suriname is located on the Guiana Shield in the north- eastern sector of South America. Precambrian rocks make up about 80% of the country, forming a crystalline basement, and the remaining 20% consists of Cretaceous to Recent sediments that were deposited along the northern fringed of the Guiana Shield (Coastal Plain) as depicted in Figure 1.6 (Monsels, 2016). Large-scale gold-mining started in 2004, through IAMGOLD's subsidiary Rosebel Gold Mines, currently expanding its capacity. In the fourth quarter of 2016, Surgold, a subsidiary of Newmont Mining, opened the Merian mine. In addition to these two large-scale industrial mines, the small-scale gold sector makes a major contribution to gold production and exports. A gold refinery - the Kaloti Suriname Mint House (KSMH) – was commissioned in February 2015, with an annual theoretical processing capacity of 60 tons of gold. (MINR, 2019).

⁸ 2017-2021 POLICY DEVELOPMENT PLAN National Planning Bureau

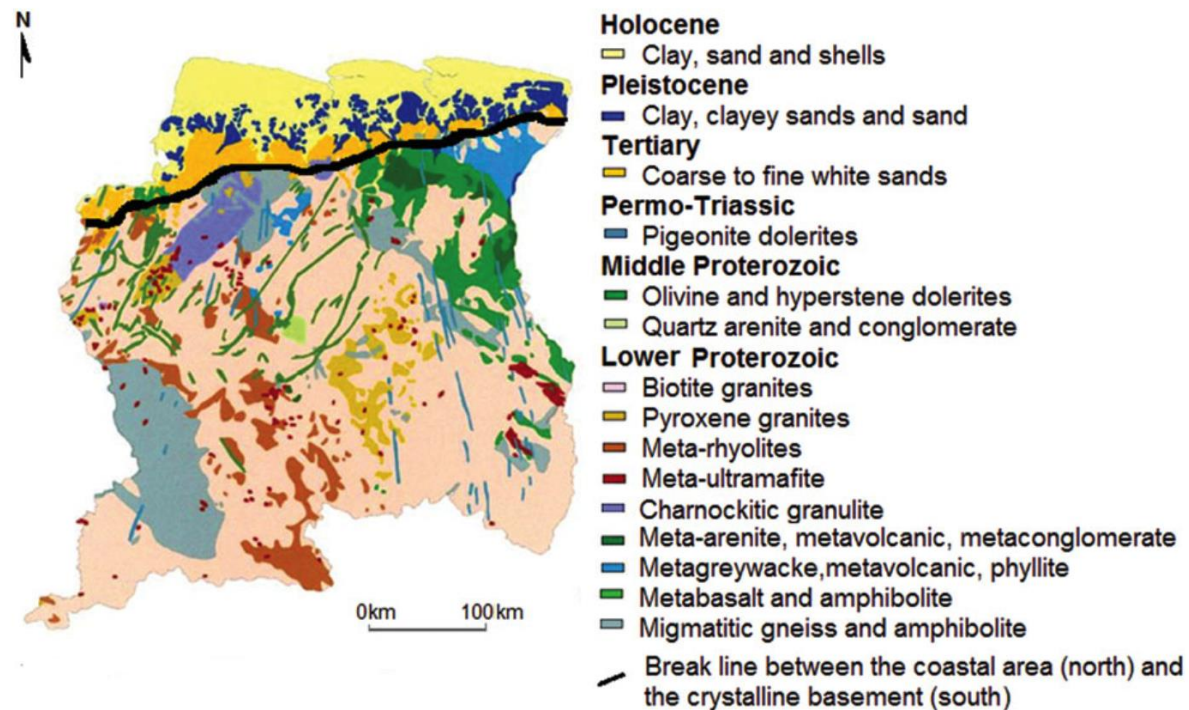


Figure 1.6: Geological map of Suriname with Boundary Line between the Sedimentary Coastal Area to the North and the Crystalline Basement to the South

Source: Modified after www.staatsolie.com and Kroonenberg et al., 2015, in Monsels, 2015

1.3.1 Mining Industry

Suriname's economy is highly concentrated in the extractive industries (gold, oil, and bauxite), which play a dominant role in driving growth, employment, and government revenues while also exposing economic performance to commodity price fluctuations. Historically, extractive industries have accounted directly for around 30 percent of GDP and as much as 90 percent of exports (IBRD, 2019).

The production clusters in the mining and extractive industry mainly consist of gold, oil, bauxite and kaolin. The mining sector substantially contributes to the economy in terms of exports, infrastructural development, creation of employment and economic growth. However, the government revenues are marginal compared to the export proceeds. The gold sector is subdivided in the large-, medium- and small-scale gold sector. The largescale gold sector develops the economically extractable large gold ore deposits with foreign capital, in the course of which the government participates as a shareholder. The medium- and small-scale gold sectors, in which in addition to Surinamese also a large number of foreigners participate, exploit gold ore deposits with local capital. (GOS, 2017)

The GOS for the 2017 to 2021 planning period, included the following actions related to the mining sector (GOS, 2017):

- The government is in favor of a regulated and responsible expansion of the mining industry, including an improved regulation of the informal gold-mining activities. The mining policy is aimed at the preparation and adoption of legislation for the minerals institute and the mining act. The preparations of the minerals Institute will be substantiated in this planning period. The goal is to merge the Geological Mining Department (GMD), the BIS and the Suriname Gold Sector Regulation Commission (OGS – Dutch abbreviation).
- Increasing the transparency and reporting about the income and expenses from the mining sector will also be worked on. In this context, Suriname will join the Extractive Industries Transparency Initiative (EITI). The EITI standards will be laid down in the new act.
- In this planning period, preparations will be made for the restart of the alumina and aluminum industry. In this respect, the exploitation of the bauxite deposits in West Suriname is concerned.
- For the coming period, the small-scale gold-mining will be formalized in order to guarantee the revenues for the State in the form of royalties and taxes. Efforts will be aimed at eliminating the use of mercury to the benefit of the environment and the health of the citizens.

1.3.1.1 Gold Mining

Gold has been a major economic asset for Suriname for more than a century. The long history of gold mining, concentrated in large parts of a greenstone belt in the northeast of the country, began with small-scale artisanal extraction activities and has more recently seen the development of major open-pit operations. Despite the range of mining activities, Suriname's gold deposits and occurrences are under-explored from a scientific point of view. Primary gold mineralisations in the greenstone belt occur in multiple forms, and although their origin is commonly related to the Palaeoproterozoic Trans-Amazonian orogeny, the controls of ore formation in specific cases often remain obscure (Kioe-A-Sen et.al., 2016).

From the distribution of gold concessions on old maps, Brinck (1955) (in Kioe-A-Sen et.al., 2016) distinguished three areas of gold occurrences: (1) an E–W-trending area from the Marowijne River to the Saramacca River (Nassau, Rosebel, Mindrinetti-area), (2) a NW– SE-trending area with the Sara Creek and Boven-Grankreek prospects and (3) the Drietabbetje creek and Oelemari area. From recent exploration and mining activities, gold enrichments appear to follow two major zones: one with an E– W orientation (including Rosebel, Nassau and Maripaston) and the other with a NW–SE tendency (including Sara Creek, Lely Mountains and Benzdorp). These areas in the eastern green- stone belt host the majority of large-, medium- and small-scale operations, but gold also occurs in the Apoera area in west Suriname (Kioe-A-Sen et.al., 2016).

Since the beginning of the 1990s a new gold rush has developed, involving intensive, often illegal artisanal mining of lateritic and alluvial deposits in many major creeks in the greenstone belt, as well as dredging in the Marowijne River and the Afobaka storage lake (Kioe-A-Sen et.al., 2016).

Gold production in Suriname reached its lowest point in the middle of the 20th century and a new stage of growth driven by the rising gold price beginning in the 1970s was disrupted by the Interior War (1986–1992). Since the end of the war, jointly with the rise of the price of gold, the now safer gold fields attract immigrants, foreign prospecting companies and urban Surinamese to the interior of the country (UNEP, Mercury Release Inventory in Suriname 2016).

There is a long history of gold production in Suriname, which in recent years has culminated in the development of major mining operations. Although geological research into the origin and setting of gold mineralisation in the country has made considerable progress, many areas remain virtually untouched in terms of modern scientific exploration. The apparent diversity of primary and secondary deposits in the greenstone belt calls for an expansion of research efforts, preferably in a regional context, to better understand the local complexity of gold-producing processes and the geological controls of ore formation in the Guiana Shield in general. (Kioe-A-Sen et.al., 2016). In accordance with the NDP (GOS, 2022), the gold mining sector represents the following to Suriname:

- Contribution to GDP (gross domestic product) is 21.5%
- Contribution to the state treasury in 2020 is USD 125 million
- Number of people employed in the (formal) gold sector in 2020 was 2,829
- The average gold production from a large mining was about 24,000 kg in 2020 and from small and medium-sized mining companies approximately 15,000 kg.

One of the potential impacts associated with the mining sector development in Suriname is associated with deforestation, the associated Land Use Land Cover (LULC) for post-deforestation in mining is reported to represent 67% of the post-deforestation from 2000 to 2019 (see Table 1-1), followed by infrastructure (20%) and Agriculture (5%). The trend of gold mining activity shows a relatively constant progression of land deforested between 2001 and 2014, however, the level of deforestation was higher during the period 2008-2014 (around 26,000 ha) compared to 2001-2008 (approximately 19,000 ha). This seems to follow the annual gold production, which was relatively stable between 2004 and 2013 with a slight increase since 2008 (Rahm, et al., 2014, in UNEP, Mercury Release Inventory in Suriname 2016).

Table 1-1: Total Post-deforestation area, Land Use Land Cover (LULC) (classes in ha), 2000-2015, 2000-2017 and 2000-2019

Land Use and Land Cover (LULC) classes	2000-2015	2000-2017		2000-2019	
	Area (ha)	Area (ha)		Area (ha)	Area (%)
Secondary vegetation	1,205	1,642		1,642	1%
Urban Areas	3,424	2,931		3,807	3%
Infrastructure	12,964	18,843		24,400	20%
Mining	62,102	72,261		83,739	67%
Agriculture	2,213	4,956		5,932	5%
Pasture*	455	1,114		1,495	1%
Others	281	769		1,124	1%

Land Use and Land Cover (LULC) classes	2000-2015	2000-2017		2000-2019	
	Area (ha)	Area (ha)		Area (ha)	Area (%)
Burned areas	2,502	2,302		2,922	2%
Total Area	85,146	104,818		125,061	100%

Source: (GBS, June 2022) – Tab: 18.1 - Land Use – Modified by RINA, 2022

The spatial distribution of post-deforestation per LULC categories is presented in the Figure below. It is noted that the mining post-deforestation LULC (dark yellow areas) is spread over the central-eastern section of Suriname (around Brokopondok Reservoir⁹) and extending south along the border with Guyana.

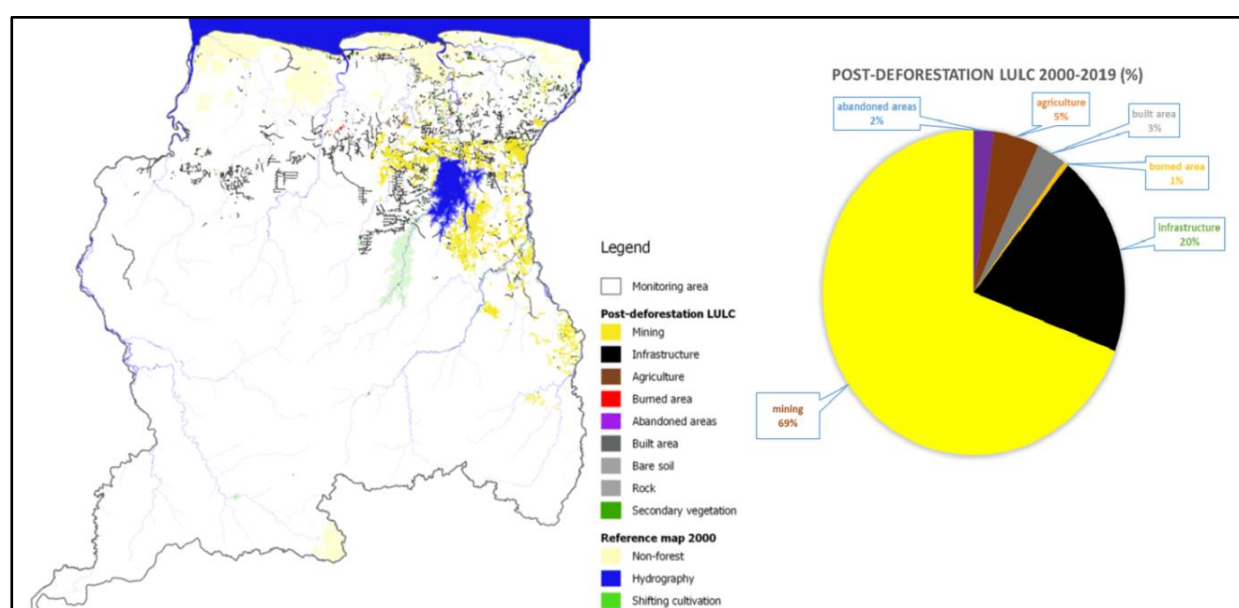


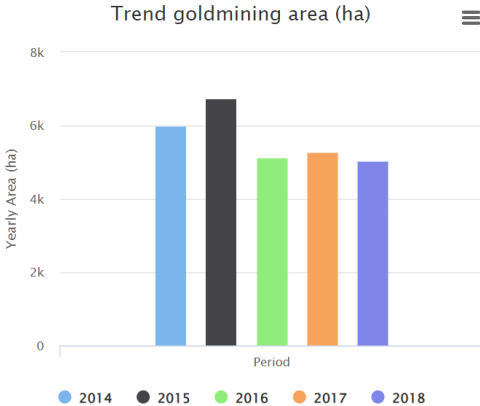
Figure 1.7: Post-Deforestation LULC 2000 – 2019

Source: (GBS, June 2022) – Tab: 18.1 - Land Use.

In terms of gold mining, since it can be considered the most representative mining activity in Suriname at this moment, the national statistics indicate a constant annual trend of land clearing between 5,139 ha in 2015 and 5,047 ha in 2018. The total land area utilized for gold mining in Suriname extends over an approximately 75,519.25 ha (see Table below).

⁹ officially known as Professor Doctor Ingenieur W. J. van Blommestein Meer.

Table 1-2: Trend of goldmining areas in Suriname (2014-2018)

#	Year	Area (ha)	Yearly Area (ha)	Yearly Goldmining Area Trend (2014 -2018)
1	2001	5,113.89	0	
2	2008	12,152.34	0	
3	2014	36,041.22	6,006.87	
4	2015	6,747.21	6,747.21	
5	2016	5,139.93	5,139.93	
6	2017	5,277.53	5,277.53	
7	2018	5,047.13	5,047.13	
Total area		75,519.25		

Source: (KOPI, 2022) Goldmining Info10.

As stated by the NDP (GOS, 2022), in addition to deforestation other environmental impacts associated with gold mining includes:

- Forest degradation causing the biodiversity of ecosystems to decline
- Severe (mercury) pollution that poisons/ pollutes the land and aquatic life in the rivers and creeks.
- Turbidity of the water with (health) effects on the aquatic environment and on the people who depend on that water
- Disruption of the natural water management system, including by damming creeks, causing flooding
- Construction of hinterland infrastructure making it more accessible to local residents but also
- further expanding other illegal activities such as logging and hunting endangered species.

In terms of gold production, during 2019 the gold exported by big mining companies (Newmont Suriname and Rosebel Gold Mines) and local small exporting companies was 39,372,016 grams, with an export value of US\$ 1,721,555,932 (see Table 1-3). The share of the local gold exporters in 2019 was 15,114,213 grams (GBS, March 2020). According to Suriname's Planning Office, small scale goldmining produced 14,800 kg of gold in 2020 (ITA, 2022)

Table 1-3: Gold exports (in grams and US\$), 2015-2019

Year	Export volume (in grams)	Export value (in US\$)
2015	25,525,871	916,552,075
2016	26,135,611	995,754,011
2017	40,085,103	1,579,165,972

¹⁰ <https://kopi.sbb.sr/index.php?r=goldmining%2Findex>

2018	41,054,199	1,625,150,645
2019	39,372,016	1,721,555,932

Source: (GBS, March 2020)

Suriname acceded to the Minamata Convention on 02 August 2018. In its process towards accession, the Government of Suriname started in 2013 with the legal and institutional analysis of the then current situation towards the use of mercury and the role of the various actors. Mercury is a persistent heavy metal, hazardous to human health and the environment, whose release from mining activities is contaminating lands, rivers and oceans on a global scale and harming human health (UN, 2022). For Suriname one of the important obligations under this Convention is to formulate a National Action Plan (NAP) for the Artisanal and Small-Scale Gold Mining (ASGM) (GOS, 2020). The government of Suriname has yet to act on the project's findings or implement the Minamata Convention (ITA, 2022). While the Minamata Convention addresses anthropogenic mercury pollution by regulating trade in mercury, phasing out mercury-related products and processes and reducing industrial mercury emissions, it still allows the trade and use of mercury in small-scale gold mining (UN, 2022).

1.3.1.1.1 Industrial Gold Mining

In Suriname, there are two industrialized mining companies namely the multinational IAMGOLD, which operates at the concession of Rosebel Gold Mines N.V., and Newmont, the majority shareholder of Newmont Suriname (UNEP, Mercury Release Inventory in Suriname 2016). In addition, a range of smaller foreign and domestic companies (e.g., Golden Star, Canarc, Wylap, NanaResources, Suparna Gold, Hunter Bay Minerals) have been active in exploring and exploiting gold mineralisations in many parts of inland Suriname since the 1990s. (Kioe-A-Sen et.al., 2016).

1.3.1.1.1.1 Grassalco state-owned mining company

Grassalco is a 50 year old state-owned company with mining rights to produce crushed stone, gold, granite and sand (Grassalco, 2022). Grassalco began recovering gold from reprocessed ore in 2014 and started exporting gravel to the Caribbean region (ITA, 2022). Grassalco owns the mining rights to explore and exploit the Maripaston concession, a gold exploration site located 90 km southwest of Paramaribo, in the district of Para. Grassalco also explores other sites for gold such Goliath Mountain. The Goliath concession is located 65 to 90 km south-east of Paramaribo and covers approximately 244 km² in the Bigi Poika area in Para in the central part of North Suriname (Grassalco, 2022).

Grassalco has begun recovering gold from reprocessed ore since 2014 and has started exporting gravel to the Caribbean region (ITA, 2022). In 2019, Grassalco exported 102,031 metric tons of gravel (GBS, March 2020).

1.3.1.1.1.2 Merian Mine - Newmont

The Colorado-based U.S. multinational, Newmont, began operations in eastern Suriname in October 2016 (ITA, 2022). Newmont Suriname LLC is a fully-owned subsidiary of Newmont Goldcorp Corporation, and operates the Merian mine on behalf of Suriname Gold Project CV, a Suriname limited partnership. Newmont Suriname, LLC ("Newmont Suriname") is the managing partner, owning a 75% interest in the limited partnership, and Staatsolie Maatschappij Suriname N.V., Suriname's State-owned oil company, the limited partner, owns the remaining 25 % interest (Newmont, 2019)

The Merian gold mine is located approximately 60 kilometers south of Moengo and approximately 150 kilometers southeast of Paramaribo, the capital of Suriname. Commercial production at Merian was achieved in October 2016. During 2019 the Merian mine produced 393,000 attributable ounces of gold (Newmont, 2020).

Over the next 10 years (2018-2028) and assuming today's gold prices, the GOS expects to receive about o US\$1.3 billion from the operations of the Merian mine. This representing back in 2018, an additional US\$4.2 billion in gold over the 10 years period. The Merian contains gold reserves of 400,000

5.2 million ounces and is expected to produce an average of ounces of gold annually over a mine life of at least Continued exploration is showing promising results (GOS, 2018).

1.3.1.1.1.3 Rosebel Gold Mines - IAMGOLD

Rosebel Gold Mines, a subsidiary of the Canadian company, IAMGOLD, started gold production in 2004 (ITA, 2022). IAMGOLD holds 95% of the share capital of Rosebel, while the Republic of Suriname owns the remaining 5%. The Rosebel gold mine is located in the mineral rich Brokopondo district in northeastern Suriname, approximately 85 kilometres south of the capital city of Paramaribo. The mining concession covers 170 square kilometres, with the Suriname River to the east, the Saramacca River to the west and the Brokopondo reserve to the south. Mining at Rosebel has been carried out by conventional open pit methods (using excavators and trucks). A total of eight open pits to date: Royal Hill, Pay Caro, East Pay Caro, Koolhaven, Rosebel, Roma, J-Zone and Mayo have been developed to date (IAMGOLD, 2022).

In 2016, Rosebel acquired additional property from the Government of Suriname in the Saramacca area, the Pikin Saramacca project, in which the Surinamese Government participates for 30% and Rosebel for 70% (RG, 2022). In 2017, IAMGOLD announced that the higher concentrations of gold reserves at Saramacca will extend the life of the mine until at least 2038 (ITA, 2022). In October 2019, the first ore from Pikin Saramacca was successfully delivered to the Rosebel mill, using an alternate road. The new mine was scheduled to begin full production in 2020, with an operation time of 15 years (RG, 2022). IAMGOLD holds 95% of the share capital of Rosebel, while the Republic of Suriname owns the remaining 5%. In 2020, Rosebel produced 220,544 troy ounces of gold (ITA, 2022).

1.3.1.1.1.4 Gold Refining – Kaloti Suriname Mint House

Dubai-based Kaloti Precious Metals (Kaloti) opened a gold refinery in Suriname in February 2014. The refinery is part of a joint venture between the Kaloti, the Government of Suriname and local gold traders (ITA, 2022). The Kaloti Suriname Mint House (gold refinery) opened to promote and enhance the production of gold within Suriname and the wider Latin American region. It provides a number of key services and solutions including melting, assaying, evaluation, vaulting, bullion trading, coins, medallions and investment bars distribution, global logistics, and liability coverage and insurance for the international transportation of precious metals. The gold will be sourced from Suriname and neighboring countries like Guyana and French Guiana (GTR, 2015).

1.3.1.1.2 Artisanal and Small-Scale Gold Mining

The international term artisanal and small-scale gold mining (ASGM) refers to a wide range of activities, ranging from the mere use of pans to sift through the soil, to the deployment of modern machinery such as tractors, hydraulic engines, and sluice boxes. Both forms are prevalent in Suriname, but the latter makes up the majority of small-scale gold mining production in the country. ASGM is often illegal, but there are a few Surinamese companies like Sarafina NV, Bishumbar, and Grassalco NV that are working with a valid exploitation license facilitating small-scale mining operations. Although Maroons have been mining gold for over 100 years, the modern small-scale mining techniques were introduced to Suriname by garimpeiros, Brazilian gold miners. It is estimated that up to 60% of miners in Suriname are garimpeiros; Maroons and indigenous (33%), urban Surinamese (5%) and other foreigners (2%) make up the remainder (Marieke Heemskerk, personal communication) (ACT, 2015).

Traditionally ASGM has always been a source of income for parts of the community. The negative impact of mercury on health and environment in Suriname cannot be overlooked and needs to be addressed. (GOS, 2020).

The actual harm caused by small-scale gold mining to human rights and the environment of countless individuals, communities, peoples and ecosystems cannot be justified by the economic gains of the mining industry or the poverty of miners. Everyone has the right to an adequate standard of living, but no one has the right to harm the health or physical integrity of others to earn a living. The harm caused by mercury is amplified by its persistence, leading to the proliferation of contaminated sites and

accumulated mercury pollution in the food chain that will negatively affect the rights of future generations. (UN, 2022).

Millions of men, women and children around the world are engaged in ASGM. For most, it provides a subsistence livelihood based on hard labour in difficult conditions with many hazards. They use hand tools, panning, sluices, floating dredges and rudimentary mining equipment to extract tiny fragments of gold from low-grade ore. Those supported by small investors may have excavators. Their activities are destroying vast areas of jungle, forests and riverbanks, leaving wastelands of tailings and mining pits (UN, 2022).

The contribution of mercury releases and emissions of mercury from the small-scale gold mining sector has continued to grow, with grave consequences for millions of miners, vulnerable women and children, indigenous peoples, ecosystems and aquatic life. There is a growing flow of mercury into the rainforests of the Amazon basin (UN, 2022).

Small-scale informal gold mining in the interior has increased rapidly since the 1990s. Small-scale gold miners generally use hydraulic equipment and mercury in their mining with severe consequences to the environment. According to Suriname's Planning Office, small scale goldmining produced 14,800 kg of gold in 2020 (ITA, 2022).

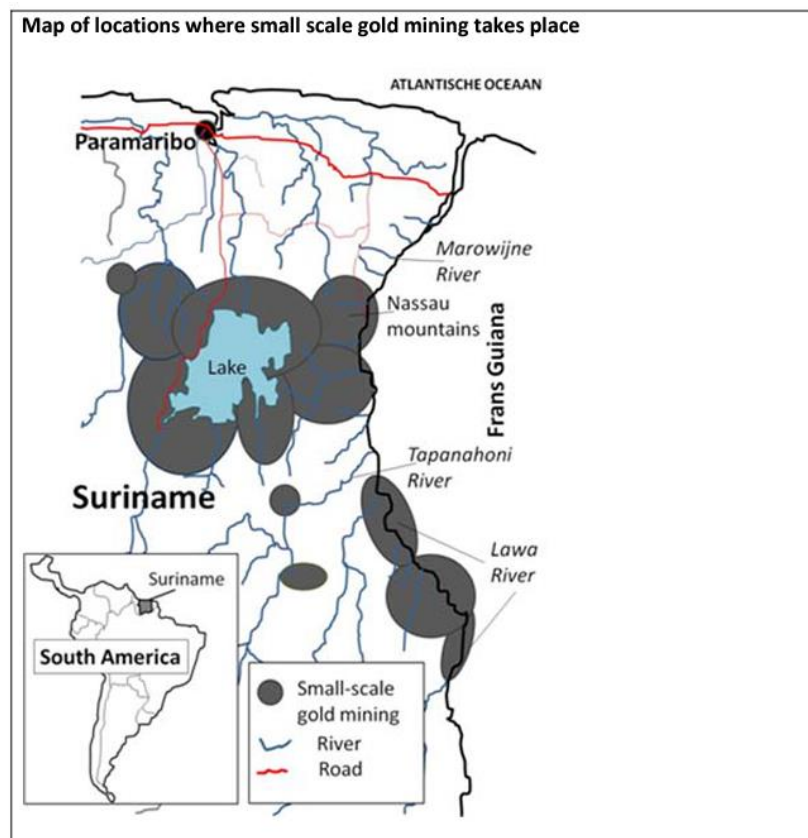


Figure 1.8: Small Scale Gold Mining Areas

Source: (GOMIAN, 2011)

Virtually all Suriname (not migrant) miners are Maroons; tribal peoples of African descent. Maroons traditionally live in the Suriname interior where the country's gold reserves are located. Throughout history, Maroons have mined for gold when they needed cash to supplement subsistence resources.

Today SMGM has become the primary source of subsistence for large share of Maroon households (GOMIAN, 2011).

One of the most significant environmental aspects of ASGM are the occupational health risks, environmental contamination of rivers and food sources (fish) and its consequences in public health from use of mercury. In 2011, GOMIAN (2011) estimated that for 1kg of gold produced, small-scale gold miners may use up to 1k of mercury, with most of the mercury lost into the environment since there is little mercury recovery methods (few miners work with retorts. A practical guide fo the use of retorts and exposure reduction to mercury in ASGM was developed by the Artisanal Gold Council (AGC, 2014) that is applicable to this risk in Suriname.

Since 2017, the government of Suriname has been working in the developed a National Action Plan (NAP) for the ASGM in Suriname project (NAP project) with the support and finance of the UNDP and GEF. The aim of this project was to formulate a NAP to enable the government to determine national requirements, needs, goals and schemes to limit the use of mercury (Hg) in the small-scale gold mining sector (ASGM) and processing or stop, as reported by GEF (2016).and NIMOS (2022a).

1.3.1.2 Bauxite Mining

Bauxite, the raw material of aluminum, has been one of the economically vital natural resources for Suriname. Mining operations started about a century ago, and subsequent development of a refinery industry and hydro-electric power made Suriname one of the foremost bauxite and alumina producers worldwide for a prolonged period of time. Alumina dominated the export revenues of Suriname until a decade ago.

The Surinaamsche Bauxiet Maatschappij (SBM) was established in 1916 after the discovery of the Moengo bauxite hills in 1915 by the Pittsburg Reduction Company, later renamed the Aluminum Company of America (Alcoa). In 1939, the N.V. Billiton Maatschappij obtained a concession for bauxite exploration in the Para district, which led to the discovery of the Onverdacht bauxite deposit. Suriname was one of the leading bauxite-producing countries in the world during the Second World War (1939–1945), which generated a boost in bauxite exploration and research on the local deposits. In 1941, the SBM opened the Paranam bauxite processing plant, named after the Para and Suriname Rivers bordering the mining concession areas. In 1958, SBM signed the Brokopondo Agreement with the Surinamese Government to create a fully integrated aluminum industry in the country and became the Suriname Aluminum Company (Suralco). The most important objectives were the construction of a dam in the Suriname River and a hydro-electric power facility at Afobakka, and the establishment of an aluminum smelter and an alumina refinery at Paranam. The smelter operated from 1964 until 1999 and was dismantled one year later and the refinery closed in November 2015. In 1993, Suralco and Billiton signed a joint venture about mining and refining activities, but Billiton halted its exploration and mining activities in Suriname in 2009, while Suralco also later reduced its activities. Suralco finally mining the remnants of some bauxite deposits in the Moengo area and the Lelydorp 1 deposit before closing in 2015 (Monsels, 2016).

Bauxite deposits of economic interest occur in four districts (see Figure 1.9). The lateritic bauxite deposits are spread across the northern part of the country and developed on various parent rocks during Late Cretaceous–Early Tertiary times. Bauxites in the coastal lowlands formed on Cenozoic sedimentary deposits, whereas plateau bauxites originated on various crystalline rocks in inland regions of the Precambrian Guiana Shield. The more accessible bauxite deposits in the coastal lowlands are almost mined out, the areas of economic interest being (Monsels, 2016):

1. the Paranam–Onverdacht–Lelydorp district (Coastal), which includes the Lelydorp 1, Kankantrie and Para bauxite deposits
2. the Moengo–Ricanau–Jones district (Coastal) with the Coermotibo deepseated bauxite deposit.
3. The plateau bauxites have been extensively explored but have not been brought into production to date. These areas of potential future exploration interest being:
4. the Bakhuis district (Plateau) with the Bakhuis bauxite deposit

5. the Nassau district (Plateau), which encompasses the Nassau, Brownsberg, Lely bauxite and Wintiwaai laterite deposits.

For economic and environmental reasons, the future of the bauxite industry in Suriname is considered to be uncertain (Monsels, 2016) and will be determined by the quality and size of the reserves, geological conditions, available production technology, targeted locations in relation to transport facilities, world market price and political factors. Existing Surinamese bauxite deposits have not been mined yet for different reasons. The Coermotibo deposit has not been economically attractive due to the high sulfur content in the bauxite and the 40 m thick overburden. The heterogeneous properties of the bauxite within the Bakhuis and Nassau bauxite deposits will make it necessary to drill a very close-spaced grid for mine planning. The remote location of the Nassau Mountains is also an economic and strategic problem. Political and economic factors have hampered mining of the Bakhuis deposit. A final and most sensitive obstacle is the environmental impact, as the Nassau Mountains have a unique flora and fauna, including critically endangered endemic species (Ouboter et al., 2007; Bánki et al., 2008; Alonso & Mol, 2007, in (Monsels, 2016).

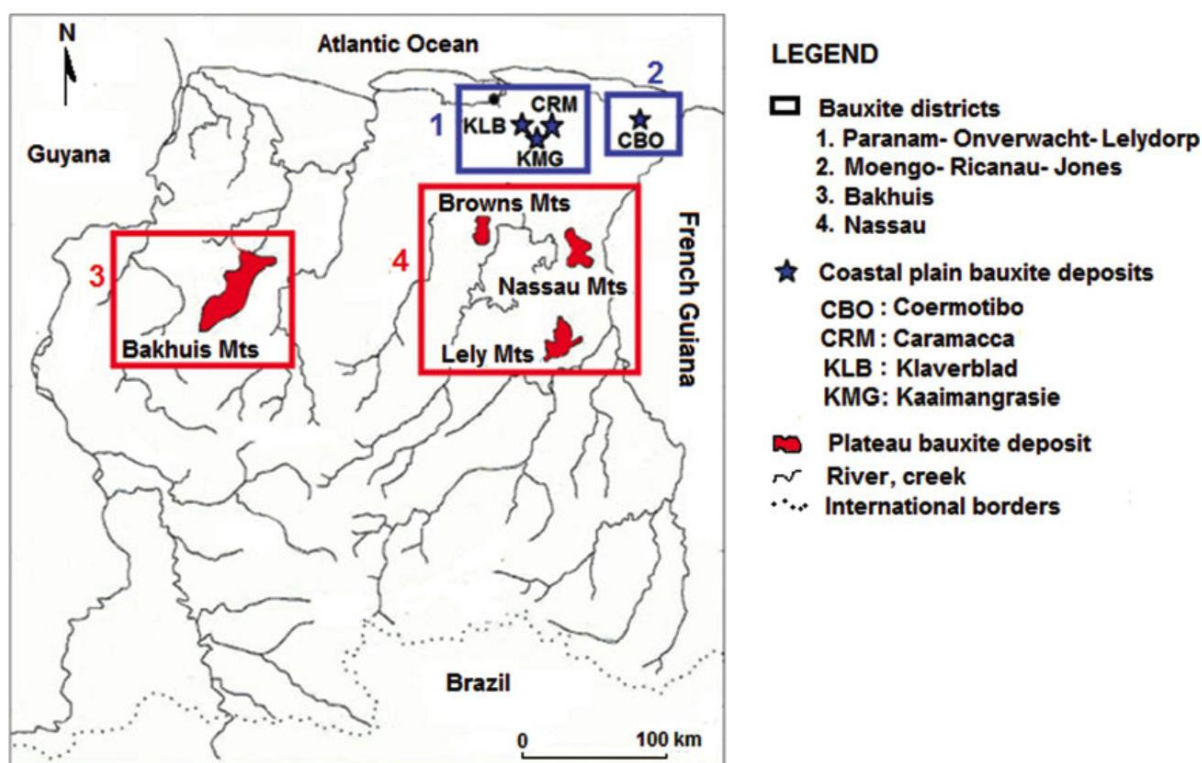


Figure 1.9: Districts of Coastal- and Plateau-type Deposits of Bauxite in Suriname

Source: (Monsels, 2016)

1.3.1.3 Rare Earth Elements

The mineral monazite—one of the main sources of rare earth elements—occurs in several locations in Suriname. Also, indications of rare earth elements such as neodymium, cerium, scandium, and yttrium in Bakhuis have been found (SITA, 2022). No significant exploration is known to have been carried out for Rare Earth Elements in Suriname.

1.3.1.4 Other resources with potential

Suriname's Investment and Trade Agency - SITA reports that other resources with potential include diamonds, crushed stone, sand kyanite, phosphate, and manganese (SITA, 2022), but no significant deposits of these minerals have been identified.

Except for diamonds, the other minerals and resources used for construction that are mentioned are required in deposits with large reserves and, if present in Suriname, are unlikely to be discovered and brought to market in the short term and will require significant exploration effort to identify them.

1.3.2 Oil and Gas Industry

Oil was first discovered in Suriname in 1928 on the coastal plain, close to the western border, but systematic exploration both onshore and offshore only started in the 1960s, when various international oil companies became interested. Oil became the primary industry by the mid-1990s, overtaking bauxite. Total oil production in the last 35 years reached 100 million barrels of crude oil which has boosted the total value of exports (SITA, 2022).

State-owned entity Staatsolie is the predominant oil company in Suriname. Until 2017, they have been mainly active in onshore oil production. For offshore exploration, Staatsolie is considered, on behalf of the Government of Suriname, the contracting party for International Oil Companies (IOCs) and the oil sector is regulated and supervised by them. The onshore and offshore areas granted to Staatsolie are shown in the following Figure (EITI, 2017).

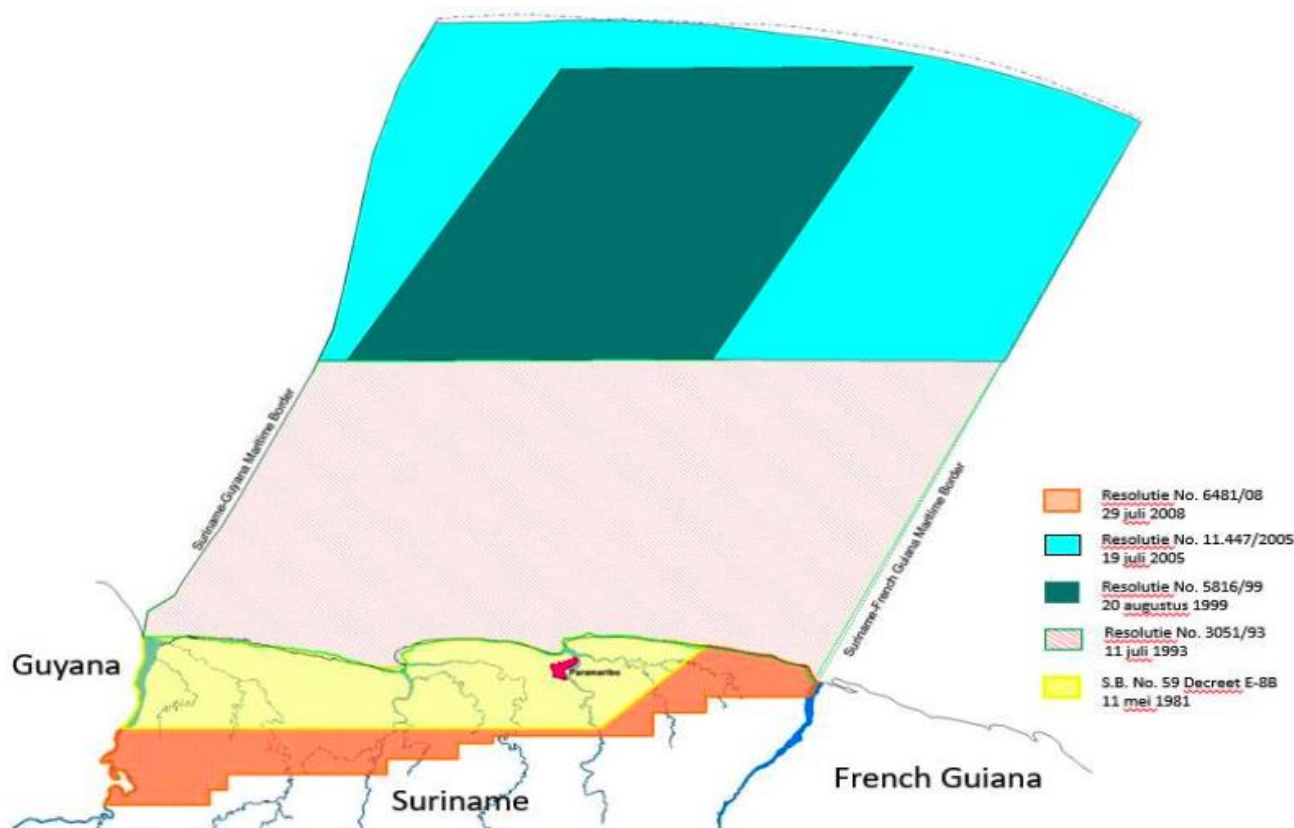


Figure 1.10: Overview of Areas Granted to Staatsolie

Source: EITI Report of the Republic of Suriname, 2017

Through petroleum agreements, Staatsolie transfers its rights to explore, develop and produces hydrocarbons in a contract area (block) to a contractor, being an International Oil Company (IOC). The situation in 2017 is given in the Figure below.

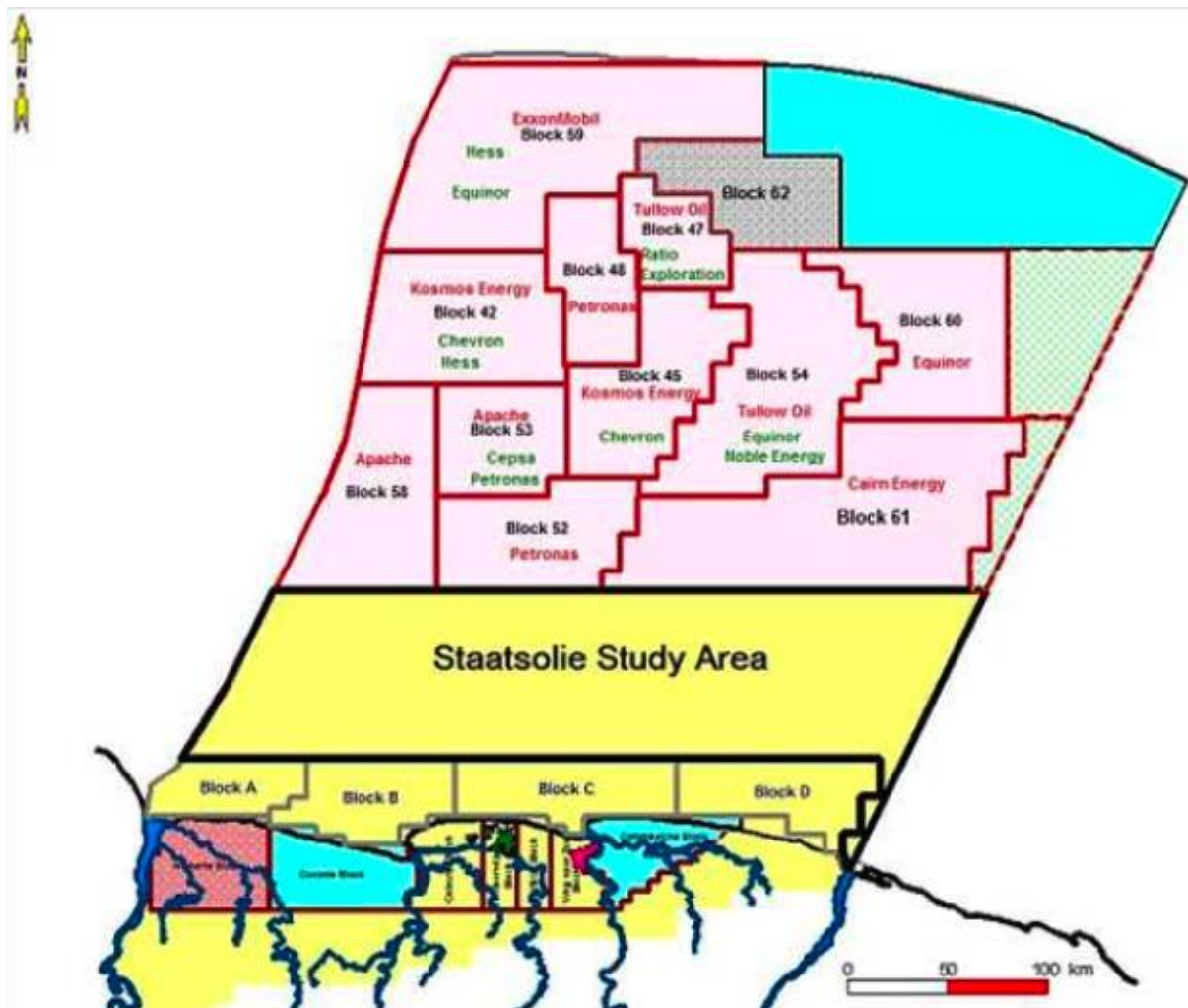


Figure 1.11: Overview of the Identified Onshore, Nearshore and Offshore Exploration and Exploitation Blocks at Year End 2017

Source: (EITI, 2017)

Suriname is rapidly growing as a key hotspot in the Caribbean region, as recent discoveries in offshore Blocks 52 and 58 have detected new wells with excellent oil properties (an API gravity ranging from 27° to 37°). Moreover, even if three reserves are still under evaluation, the prospections performed until 2021 show reserves accounting for more than 1.300 million barrels.

Suriname produces 16,440 barrels of oil per day (bopd) and 692 thousand cubic feet of gas per day (Mcfgd) from two mature onshore fields, but with five discoveries over the past two years, the country is projected to begin offshore production no later than 2025 (Redden, J., 2022). Suriname also has potentially large oil reserves on-and offshore, which numerous studies have shown, and the discovery of large reserves in Guyana strengthen the suspicion that there are also large oil reserves in our part of the Guyana Shield.

Active onshore and offshore exploration has been taken up to compensate the under-exploration of the Suriname-Guyana Basin. The United States Geological Survey (USGS) ranked in 2000 the Guyana Suriname Basin as the 2nd in terms of prospectivity among the world's unexplored basins and estimated its potential in:

- 15.2 billion barrels of crude oil
- 42.058,37 billion cubic feet of gas, and
- 2,329.62 million barrels of natural gas liquids.

Suriname exploration has primarily focused on Block 58 where two fields (green) have been discovered. For now, Block 58 is shaping up as Suriname's version of Stabroek, where operator TotalEnergies SE is in a 50/50 partnership with Houston's APA Corp. (Apache). The partners have hit five exploration discoveries since January 2020 (Redden, J., 2022).

Production costs, at about US\$ 30 to US\$ 40 dollars per barrel are considered to be low and large production of oil allows the country to have high levels of primary energy self-sufficiency. The result is a very low electricity price at around 7 cents per kWh.

There is a significant spin-off industry around oil with types of activity with potential being exploration companies, prime contractors, subcontractors, service providers, consulting, and investment firms and there is supportive skills development with the existence of training and management programs and initiatives such as the Master's Program in Petroleum Geology in the Anton de Kom Universiteit van Suriname and Hydrology and Engineering in the Polytechnic College of Suriname (SITA, 2022).

1.4 CURRENT DEVELOPMENT IN THE INFRASTRUCTURE SECTOR

A brief overview of current networks is provided below, by sub-sector (mode). The main transport sub-sectors include Ports, Roads (Highways, Bridges and Local Roads). Possible secondary sectors might include Domestic/National Airports, depending on the mine locations, and the volumes and values of products exported.

1.4.1 Road transport

The main interregional roads are the east-west corridor which links Paramaribo with Nieuw Nickerie, at the border with Guyana, and Albina, at the border with French Guiana. The major elements of the road network are this east-west corridor paralleling the coast, and the north –south corridor between Paramaribo and Brokopondo, the major hydro-electric power station as indicated in the figure below. The network is linked to the sub region by the ferries across the Corantijn River to Guyana in the west, and in the east to French Guyana across the Marowijne River at Albina.

The road network in Suriname is comprised of 4,500 km of roads, of which 1,300 km are primary paved roads. The urban areas of Paramaribo, Albina and New Nickerie, contain 40% of all paved roads, mostly located within the capital Paramaribo. The primary roads are mainly interregional roads in the coastal area and forest roads often laterite roads. The vehicle fleet has grown steadily from 148,000 to over 190,000 registered vehicles from 2005 to 2010 contributing to congestion in urban areas and along with deficiencies in roadway design and maintenance, to a high level of traffic fatalities

The road network is affected by a high motorization rate (303 vehicles/1,000 population in 2015 compared to an average of 204/1,000 population for Latin American and the Caribbean (LAC), relatively high population density in the coastal area, and capacity and maintenance challenges, which all contribute to congestion in urban areas. Congestion is also related to an inefficient public transportation system and parking arrangements for private and public buses in the city. Limited availability of designated lanes for motorcyclists and bicyclists and a lack of pedestrian walkways, risky practices such as speeding, and limited capacity for law enforcement are the key factors that contribute to safety issues and a high fatality rate (19.1 road fatalities/100,000 population in 2013 compared to an average of 16.7 for LAC, as reported by the World Health Organization), with most victims being motorcyclists and auto drivers.

The road infrastructure in Suriname is characterised by a gradually deteriorating network with a few rehabilitation programmes under way but generally insufficient maintenance activity on primary and secondary roads. The tropical climatic conditions, medium to poor quality soils and subgrades, the

unavailability of sufficient first quality road building material and the poor state of maintenance, contribute to the low quality of the roads, reduced fatigue life of roads and bridges and their rapid deterioration. The overloading of trucks and poor drainage contributes to a shortened life span.

In 1997 the construction of the bridges across the Coppename and the Suriname river was initiated with the view to open up the production areas in the east and west completely. The road system is being destroyed because there are no rules and regulations preventing the transportation of heavy loads on the roads. Transportation of heavy loads on water, which is environmentally friendly and cheaper transport, should be stimulated and a policy and strategy should be developed.

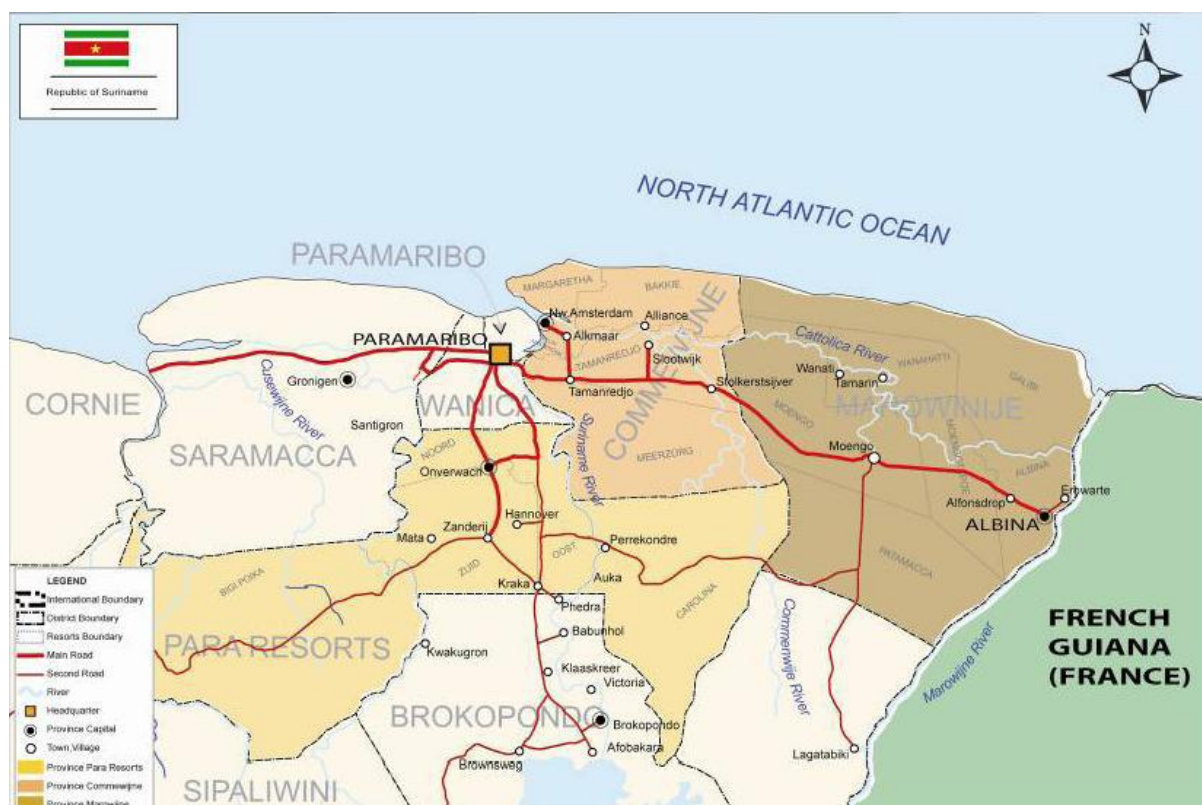


Figure 1.12: Main Road Network in Suriname

Source: EU Country Strategy, 2022

The rehabilitation and possible new construction of roads for mining program support purposes will be investigated later in the SESA study.

1.4.2 Water transport sector

The water transport sector includes the following transport systems:

- Sea transport and ports,
- Coastal navigation,
- Interior waterways and
- ferry services.

Sea transport is the main mode of transport of goods (import and export). The Harbours of Paramaribo, Paranam/Smalkalden, Coermotibo/Moengo, Albina and Nw. Nickerie are the main ports for import and export of goods and produce and serving ocean-going vessels. Of these, only Paramaribo has significant installations for loading and unloading of ocean freighters. The harbour of Paramaribo is however not yet optimally equipped to handle containers.

The Paramaribo Harbour also known as the Jules Sedney Harbour, is managed by “N.V. Havenbeheer” (NVHB), a parastatal. The Harbour is situated along the Suriname River, is a tidal port and the principal wharf for on and off-loading goods for the domestic economy. The concrete pier is (1815 feet) 600 meter long and the approximate depth alongside at Low Water At Spring Tide (LWS) is 20 feet (6.10 metres). The tidal range averages from 8.85 feet (2.70 m) at springs to 5.00 feet (2.00 m) at neaps.

Some of the operations are private sector owned such as the ESSO bunkering facilities and the Suralco bauxite terminal. Due to the shallowness of the waterway, vessels cannot be loaded to their full capacity. Dredging of the river is needed to allow bigger vessels to enter the port so port usage can be increased. indicates the current dredging project. The objective of the dredging project is to achieve a minimum depth of 4.5 m LWS from the entrance to the Paramaribo Harbour (see following Figure 1.13) also known as Sedney Harbour, by the end of 2021

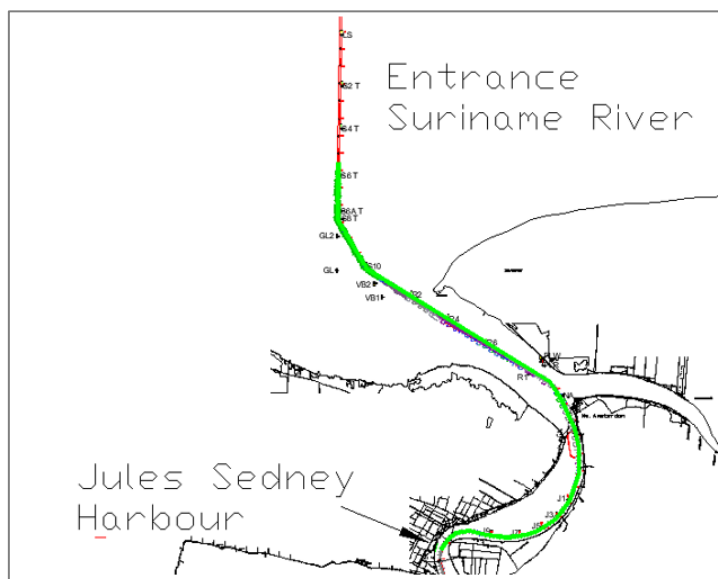


Figure 1.13: Phase I Suriname River Dredging Project

Source: (MAS, 2020)

There are 4 small ports in the district of Nickerie in the far west of Suriname. One is a government wharf also managed by NVHB and mostly used to load rice and as an intermediate stop for goods going to Guyana. The others are a private wharf; an oil jetty; and a private dock of SML for rice loading, which is currently not in use.

A small private port owned and operated by Suriname Port Services located at La Vigilantia, one of 3 business managed by Integra Marine & Freight Services, mostly used by IAMGOLD, a Canadian multinational active in gold mining, and a number of construction companies active in the vicinity of this port. This port is located in the Paranam Harbour area (see Figure below).



Figure 1.14: Identified Main Harbours/Ports in Suriname

Source: RINA, 2022

The Moengo Port is located in district Marowijne, alongside the Cottica river. For more than 30 years, Moengo was the center of mining activities performed by then owner, Suralco. The Moengo Port has played an important role in this part of Suriname's history as it was known as Suriname's liveliest Port for longer than 10 years, exporting one of the country's valuable commodities as Bauxite. As the years went by, Suralco decided to move their operations to the Coermotibo river; leaving the Port non-operational and being purchase by Traymore Moengo Port (TMP, 2022). In October 2015, Coermotibo bauxite mine operations ceased operation. The port operation supported a mining operation covering an area of 32,800 hectares and adjacent to and within International Union for Conservation of Nature protected area; threatened species (International Union for Conservation of Nature listed) (Alcoa, 2019).

Noted that the other modes of water transport have lost importance in economic terms, they are still an important mode of transportation of people and goods (shipping of sand, crushed stones and timber is also done by inland water transport. In the past small vessels have been used for the bauxite transport from East Suriname to Paranam).

The largest port in 2020 is Paramaribo, from where most exports and imports also take place. It concerns the imported and exported freight since the amount of freight exported in 2018 was; 1,200,000 tonnes in 2019; 1,089,000 tons in 2020; 975,000 tons in 202. A decrease every year is observed.

Table 1-4 Exported Freight via International Navigation by Harbour Place (x 1,000 tons),

Harbour	Paramaribo	Nickerie	Para	Other	Total
2017	784	30	60	586	1460
2018	752	47	140	233	1172
2019	681	37	98	273	1089
2020	477	46	272	180	975

Source; 2017-2021 Policy Development Plan Suriname

Table 1-5 Imported Freight via International Navigation by Harbour Place (x 1,000 tons),

Harbour	Paramaribo	Nickerie	Para	Other	Total
2017	830	16	58	55	959
2018	793	15	1057	122	1987
2019	939	10	3871	92	4912
2020	904	22	11	204	1141

Source; 2017-2021 Policy Development Plan Suriname

On the other hand, in 2017 the amount of cargo imported was 959,000 and in 2020 1,141,000 tons. There is an increase of 28% over the period. (2017-2020). There was little change in the number of vessels docked annually in Surinamese ports between 2017 and 2020 about 884/ 885 in each year.

Details of port imports by commodity will be sought¹¹. Small-scale gold miners generally use hydraulic equipment and mercury in their mining with severe consequences to the environment. In particular details of imports of these commodities will be sought.

There appear to be considerable prospects for Oil and Gas, when the wells are in production, they will require both imports of materiel inwards and the transport of oil and gas export outwards.

In maritime transportation, challenges relate to port capacity constraints and infrastructure quality. Moreover, few direct connections and a small number of service providers result in low connectivity. According to the United Nations Conference on Trade and Development (UNCTAD), in 2018 Suriname had a liner ship connectivity index of 6.01 (on a 100-point scale), one of the lowest in LAC. Moreover, the entrance to the Suriname River channel prevents large ships from entering the Jules Sedney Port, which in part has a draft of only 6.5 meters. Although the country's internal waterways are already being used to move people and materials, their use could be formalized and better structured to form a comprehensive network for accessibility.

It is intended that the rehabilitation and possible new construction of ports for mining support purposes will be investigated later in the SESA study.

1.4.3 Air transport sector

The air transport sector has an international, regional and domestic dimension. Suriname has one major airport, the Johan Adolf Pengel International Airport at Zanderij, about 45 km south of Paramaribo. The route Paramaribo – Amsterdam is the main source of income for the Air transport sector. Air transport is an important mode of transport of goods and people to the region and to effectively establish regional integration, one of the main objectives of the Caribbean Community (CARICOM or CC) of which Suriname is a Member State. The connections by air are however not the most appropriate and a

¹¹ Not available at this point.

(integrated) transport/transportation policy and strategy should be developed to reach the objectives of one single market, regional integration and unity.

The domestic flights have their base at Zorg en Hoop airport and their destination is mainly the interior where 50 airstrips are present of which 18 are used intensively.

Suriname Airways (SLM) is the national carrier, which frequents the Caribbean and the United States, but its main route is the Mid-Atlantic to The Netherlands. Apart from passengers, the main cargo concerns fruit and vegetables from Suriname to cater for almost 342,000 expatriates. Smaller airlines that mostly offer their services domestically and in the region are Blue Wings, Gonini and Gum Air. Even though they are not immediately part of the main supply chain, they play an important part in supplying companies in the extractive sector of investment goods and merchandise, mostly from abroad.

Next to Surinam Airways some small private operators dominate this market as mentioned. The domestic flights are an important mode of transportation for the gold sector (for staff and materiel). The "Vliegveld Zorg en Hoop" (VZH), an airstrip in the south-east of the capital, used to transport passengers and commercial goods into the interior, mostly for entrepreneurs active in gold mining.

Air service is challenged by issues related to terminal and passenger services, plane services, and ground facilities. Challenges also extend to unreliability and infrequent flights and logistical issues for both international and domestic flights. It is intended that the rehabilitation and possible new construction of domestic interior airports for mining support purposes will be investigated later in the SESA study.

1.4.4 Railways

Suriname has two major railways, none of which are currently operational. The first railway (Suriname Landspooweg SL) was constructed in the beginning of the 20th century and connected Paramaribo with the gold fields in the District of Sipaliwini. The Lawa railway has not been used from the time the Blommenstein reservoir was created, around 1965. There are still some remains of the railway visible in the District of Para but a great deal of the railway has disappeared in the vast Blommenstein hydraulic lake. In 1995 the Ministry of Transport, Communication and Tourism rehabilitated part of the railway and some locomotives to provide tourist and citizens a nostalgic trip along the old railway route. The project did not take off.

The track was lightly-laid, with sand used as ballast. At Kabel, the line was interrupted by the Suriname River, so a cableway was built to connect both ends. This opened in 1911 and was replaced by a ferry in 1923. The 86 km section between Onverwacht and Brownsweg remained in use mainly to serve the small village of Kwakoeegron, a gateway to the gold mining camps up the Saramacca River, and the Dutch military camp at Brownsweg. In May 1986, twice weekly trains still operated to Kwakoeegron to the following schedule: Mo/Fr Onverwacht 08.00 – Kwakoeegron 11.00/14.00 – Onverwacht 17.30. Shortly thereafter, the railway was closed (by 1988). The other railway, which connects Apoera to the Bakhuisgebergte¹², was constructed in the period of 1975-1980 to exploit the (substantial bauxite reserves in the Bakhuisgebergte. The railway has never been used due to the military conflict and was subsequently neglected. It appears that some of the original infrastructure remains unutilised. It is intended that the potential resurrection of the railways for mining support purposes will be investigated later in the SESA study.

1.4.5 Energy

The National Power System (NPS) of Suriname consists of seven isolated power networks served by state-owned N.V. Energie Bedrijven Suriname (EBS¹³) (BID, 2020). Suriname's energy mix comes from Hydro (Afobakadam) and thermal generation (diesel generated power). The MINR is responsible for the administration, regulation, monitoring and development of the energy policy in Suriname. The energy

¹² <https://www.youtube.com/watch?v=y5khSp69rVc>

¹³ N.V. EnergieBedrijven Suriname - EBS

generators of the country are the EBS, the State Oil Company of Suriname (Staatsolie¹⁴) and Suralco LLC (Suralco) (GBS, March 2020).

EBS is responsible for the design, construction, operation and maintenance of the country's electric transmission and distribution network (GBS, March 2020). EBS provides and sells electricity to 9 districts in the coastal area and also distributes propane gas through its state-owned subsidiary OGANE.

The electrical energy needs of the country are covered by four generating plants, two of which are managed by the EBS (i.e., thermal power plants with a generating capacity of 133 MW) and the other two generating plants (Suralco and Staatsolie) are managed through Power Purchase Agreements (PPAs). In 2019, the country had an installed capacity of 220.2MW and available capacity of 167.7MW. (GBS, March 2020).

EBS operates six additional power systems that use Heavy Fuel Oil (HFO) and diesel.

- The largest is in Nickerie (also referred as ENIC), which serves around 10,489 customers with electricity supplied by a 20.6 MW thermal power plant at Clarapolder running on HFO and premium diesel.
- The remaining five rural power networks served by EBS are Albina, Apoera, Coronie, Moengo and Wagenigen, with an approximate installed capacity of 23 MW of diesel power plants, serving roughly 5,857 customers in the coastal zone.

In isolated and remote communities of the Hinterland, where the NPS does not reach customers, the Electricity Supply Service (DEV¹⁵) or Department of Rural Energy, part of the MINR, is responsible for rural electrification. Approximately 130 villages (close by the Marowijne river, origin of the Suriname and Saramacca river and the Cottica river) are being intermittently served with small diesel generators. The electricity service provision is designed for an average time of six hours per day, from 5:00 pm to 11:00 pm, but this is seldom the case due to irregular provision of diesel or, very often, because the units are out of commission (GBS, March 2020) and (BID, 2020). A 500 kWp solar power generation and battery storage system was installed in 2018 and integrated with the diesel plant for the Atjoni and Pokigron villages.

The results of unreliable electricity, or lack thereof, on rural productivity have been widely documented with impacts on income, employment, school enrollment and firm productivity¹⁶ (BID, 2020).

1.4.5.1 [Energy generated for the Bauxite sector](#)

The energy that is generated by the Afobaka Hydro Power Plant, the Paranam (thermal) Power Plant and the Moengo plant was in the first instance intended to facilitate the bauxite sector and in the second instance for the government (via EBS). The power plants of Paranam and Moengo stopped operation in 2015 (see table X).

1.4.5.2 [Energy generated for the Gold mining sector](#)

The Rosebel Gold Mines Solar Energy Plant has an installed capacity of 5 megawatts and was commissioned in 2014. This Plant is the first large scale solar energy plant in Suriname. Rosebel uses the generated energy for its company activities in Brokopondo.

The power supply for the Merian goldmine is provided by a 62.3 MW on-site HFO power plant.

¹⁴ Staatsolie Maatschappij Suriname - Staatsolie

¹⁵ Dienst Elektriciteitsvoorziening - DEV

¹⁶ Jimenez, Raul (2017) "Development effects of Rural Electrification". IDB Policy Brief No. IDB-PB-261.

2 REFERENCES

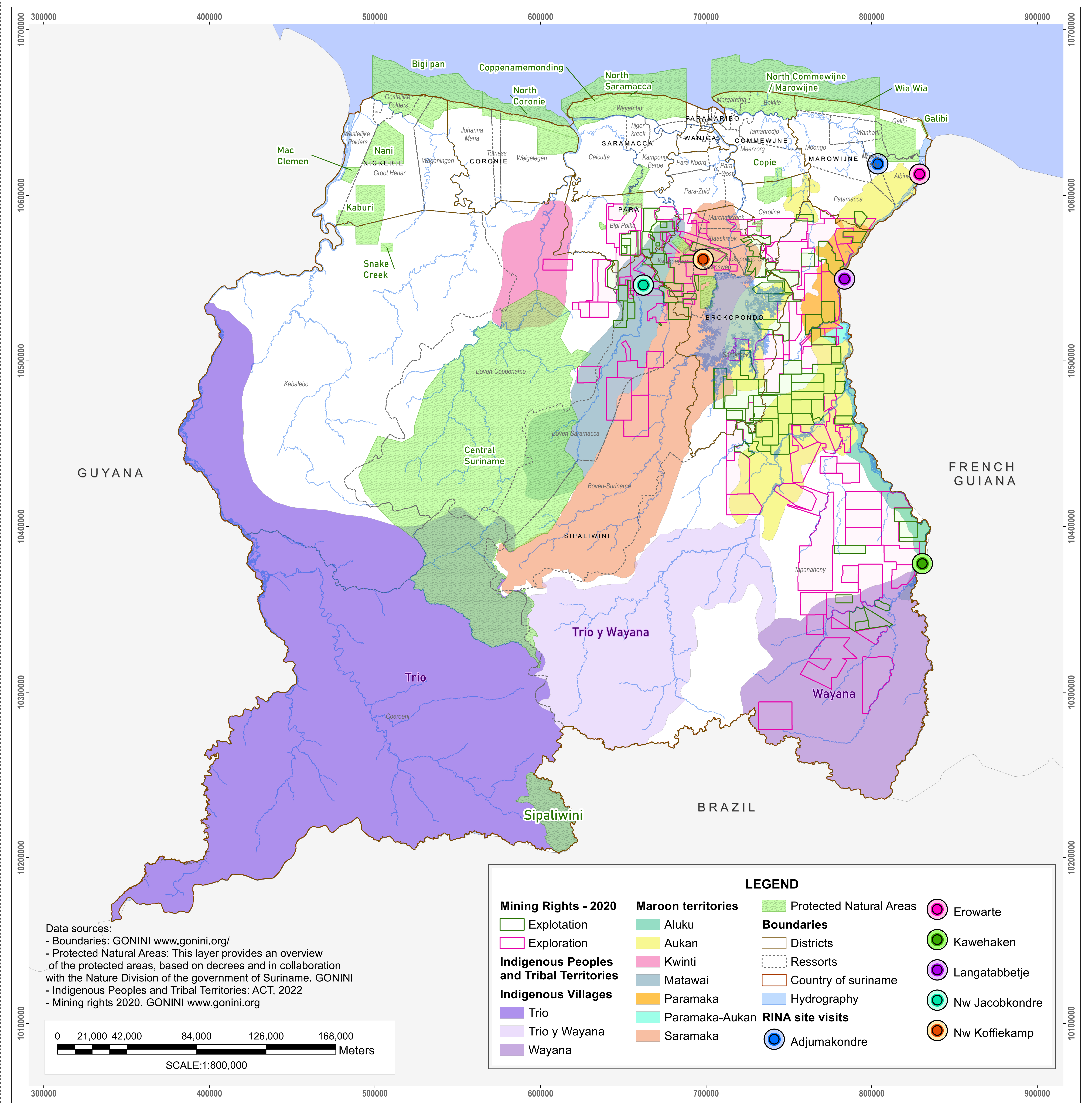
- EITI. (2019a). *EITI Suriname - History of Hydrocarbon Industry*. Tratto da EITI Suriname: <https://eitipuriname.org/en/about-suriname/history-of-hydrocarbon-industry/>
- GOS. (2022). *Surinam Multi-Year Development Plan 2022 - 2026*. Paramaribo, Suriname: Republic of Suriname. Tratto da <https://observatorioplanificacion.cepal.org/en/plans/multi-year-development-plan-2022-2026-suriname>
- IGF. (2017). *IGF MINING POLICY FRAMEWORK ASSESSMENT Suriname*. Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development (IGF). Winnipeg, Manitoba Canada: The International Institute for Sustainable Development. Tratto da <https://www.iisd.org/system/files/publications/suriname-mining-policy-framework-assessment-en.pdf>
- Redden, J. (2022, July). *Guyana-Suriname Regional Report: The overly generous PSA may be history*. Tratto da World Oil Online Magazine: <https://www.worldoil.com/magazine/2022/july-2022/features/guyana-suriname-regional-report-the-overly-generous-psa-may-be-history/>
- Transparency International. (2021). *Corruption Perception Index (CPI) 2021: Surinam*. Tratto il giorno October 2022 da Transparency International - Corruption Perception Index: <https://www.transparency.org/en/cpi/2021/index/sur>
- WBG. (2021). *GDP growth (annual %) - Suriname*. Tratto il giorno October 05, 2022 da World Bank Group (WBG): <https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG?end=2021&locations=SR&start=1961&type=shaded&view=chart>
- WBG. (2022). *EASE OF DOING BUSINESS IN SURINAME*. World Bank Group (WBG). Tratto da <https://archive.doingbusiness.org/en/data/exploreconomies/suriname>
- WBG. (2022a). *Overview*. (World Bank Group (WBG)) Tratto il giorno October 13, 2022 da The World Bank In Suriname: <https://www.worldbank.org/en/country/suriname/overview#1>

Appendix C

General display map

Doc. No. P0032241-SR Rev. 2 – January, 2023





Appendix D

Stakeholders' inputs for the Scoping Report

Doc. No. P0032241-SR Rev. 2 – January, 2023



COMMENTS, SUGGESTIONS AND INPUTS RECEIVED FROM STAKEHOLDERS ON THE SCOPING REPORT

NAME OF STAKEHOLDER ORGANIZATION/INSTITUTION	NAME OF PERSON THAT GAVE THE INPUTS	METHOD FOR RECOLLECTION OF INPUT (Email, WhatsApp, In person)	COMMENTS / SUGGESTIONS / INPUTS	ACTION TAKEN (incorporated in report, discarded because not applicable...)
Bauxite Institute	Raymondo Emanuels	Email	✓ Page 41-par.4.2.2. no. 6: Mining law amendment. Why not in 2023 when DIS is pre-starting phase? And before the new election in 2025. ✓ Page 58 : The Minerals Institute Surinameand Inspectorate within the mining sector in order to regulate it in an efficient and transparent manner. Other land register is done by ROM. ✓ Page 117- par.6.1.1.: This scenarioOil and Gas sector in Suriname or by the significant gas finds in ✓ Page 123 – point 3 Constraints: New contracts would.....taken on by the Suriname government. The GOV of Suriname doesn't have the financial capacity and it will be better in a joint venture with a multinational for a new contract for bauxite exploration and exploitations. ✓ Page 123- point Opportunities: Suriname has to enter in Joint venture with a multinational to restart the bauxite Industry. Otherwise, this part of the case Scenario is not realistic.	The issues addressed in the email are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study.

			✓ Page 126: what is the abbreviation of LSM and SSM	
Newmont mining	Bernadette Cederboom	Email	✓ p.23 Add Table 1.3.1 as Appendix. ✓ p.23. Explicitly clarify in the notes below the table what a and b mean. ✓ p.25 Please confirm with NIMOS and the Environmental Coordination Office of the Presidential cabinet if this is still the case. ✓ p.29 What about the Newmont Mineral Agreement? ✓ p.67 Incorrect map used for Suriname ✓ p. 74 Why are not all data available for the Commewijne river? ✓ p.96 Ensure correct map of Suriname is used. ✓ p.180 Las t Box Delet Goldcorp ✓ p.255 Delete Goldcorp ✓ p.255 – 256 *last line and first line (Rephrase sentence.	The issues addressed in the email are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study. Maps will be changed and updated in the SESA report.
NIMOS	<i>Quan Tjon-Akon</i>	Email	✓ p. 15 World Bank Operational Policies	The issues addressed in the email are included in the

			Comment. It seems that the World Bank has no procedures on SESA? So, will this SESA basically be conducted like some sort of ESIA? ✓ p. 16 Additional to, but not within the scope of the ESMF, a Social Assessment will be conducted. Comment. But, will it be conducted within the scope of the SESA? Cause logically it should be part of the SESA? ✓ p. 17 It should be noted that in August 2016, the WBG adopted a new set of environment and social policies called the Environmental and Social Framework (ESF). As of October 1, 2018, the ESF applies to all new WBG investment project financing. These policies will be required by borrowing governments seeking financing by the WBG. Comment. Is the mentioned ESF also relevant for this SESA? If so, in which manner? ✓ p. 25 It shall be the successor to the Environmental Coordination Office of the Presidential cabinet and the National Institute for Environment and Development in Suriname (NIMOS) shall be placed under it. Comment. Regarding NIMOS the formulation is not correct. NIMOS will not be placed under the NMA, but will transition into the NMA. ✓ p. 26 Draft Regulations Comment. Where NIMOS is mentioned (points 1, 2 and 3) you should change it into NMA. The draft regulations mentioned	scoping report either directly or indirectly and will be further investigated and developed during the SESA study. Maps will be changed and updated in the SESA report.
--	--	--	---	---

			under point 1 and 2 have been combined in 1 draft regulation. ✓ p. 52 On the other hand, the environmental approval process is administered under the new Environmental Framework Act 2022, the National Environment Authority (in Dutch: Nationale Milieu Autoriteit, NMA) and the National Institute for Environment and Development in Suriname (in Dutch: National Instituut Comment. The National Environmental Authority is not yet operational. Also, NIMOS will transition into the National Environmental Authority. voor Milieu en Ontwikkeling in Suriname, NIMOS) together form the National Environment Structure in Suriname ✓ p. 96 Figure 5-21 Comment. This figure contains a map with incorrect Surinamese borders. Please revise this map. ✓ p. 97 The great diversity of the population of Suriname is composed by indigenous tribes³³ (also Amerindians) and tribes (descendants of African slaves) Maroon diaspora, the European diaspora (Dutch descendants), Javanese diaspora (descendants of Indonesia), Hindostani diaspora (descendants of India) and the greatest contribution of all these ethnic groups is the preservation of cultural values, norms and traditions. Comment. Although you included creoles in the next Alinea, you did not include them here. As they are a significant	
--	--	--	--	--

			percentage of the population they should also be mentioned here.	
Suriname Environmental and Mining Foundation - SEMiF	Vishal Premchand	Email	No specific comments on the document but he sent 10 PDF documents that he says can be useful in the development of the SESA, list as follows: (1) Concept-Rapport-Versterking-NGOs-en-Locale-Organisaties-in-het-Binnenland-SSDI (2) Concept-Rapport-Versterking-Traditioneel-Gezag-SSDI (3) District-Strategisch-Ontwikkelingsplan-2015-2025_Brokopondo-DLGP (4) de-bloeiende-toekomst-van-brokopondo (5) Final-Version-Nat-Jaarplan-2019_V3-22-nov-2018 (6) OP-2017-2021-Ontwikkelingsprioriteiten-van-Suriname-1 (7) Onderzoek-Regionaal-Plan-Brokopondo-2015-2040 (8) 20130312-Cuaderno-Small-Scale-Gold-Mining-in-the-Amazon (9) Deelstudie (10) 7aba8-cultuur-binnenland	References will be taken into consideration for the development of the SESA
Workshop participants from donor organizations, development aid, financial assistance, international organizations agencies	General participants	Direct conversation	✓ Clearly identify the agency responsible for the implementation of the SESA ✓ Clarify the international mining standards to be used in Suriname. ✓ Maps should show the area that is being contaminated by mining activities. ✓ Clearly identify the impacts of mining on the daily activities of the people and on the environment	The issues addressed during the validation workshop are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study. Standardization of agreements with the mining companies will be addressed in the SESA report.

			✓ The lack of clear land rights is of utmost preoccupation for IP and Tribal communities. ✓ It is necessary to clarify that a lot of ASM are formalized by the government in Suriname. ✓ There is a need for more transparency in the sharing of information between public institutions and to the public. Information should be readily available online. ✓ Gonini should offer o\updated information. ✓ There should be a study on the social impacts of large mining companies in IP and tribal communities, as to verify the use of alcohol, drugs and increased sexually transmitted diseases, among other things. ✓ The report should emphasize on the benefits of mining as a source of income for the government through the payment of income taxes. However clear laws and regulations need to be set into place for this purpose, and monitoring and enforcement need to be strengthened. ✓ Agreements with mining companies should be standardized. ✓	
Workshop participants from NGO / CSO / Associations / foundations / traditional authorities	General participants	Direct conversation	✓ The SESA should clearly explain the implications of the Minamata convention. ✓ Continues consultation will be ensured with all visited and other communities, mainly those involved and directly impacted by mining. ✓ The SESA will recommend that communities be invited to participate in	The issues addressed during the validation workshop are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study.

			the environment and social monitoring of mining activities. ✓ A biodiversity study should be conducted as to verify the impacts of mining on the fauna and flora of Suriname. ✓ The SESA will review other projects involving mercury, the Minamata convention, infrastructure projects and agriculture development, among others. MNR will provide this information to RINA. ✓ The SESA should clarify the legal and illegal mining concessions. ✓ Communities affected by mining, particularly the IP and tribal communities, should have access to a grievance mechanism to file their complaints and get them resolved. ✓ Rulings from the three cases *the Moiwasa case; (ii) the Saramaka vs. Suriname case from 2005; and (iii) the Kalina Lokono case from 2015 (should be taken into consideration in the institutional and legal framework section of the SESA. ✓ The SESA process should be available online on the MNR website. MNR is to provide a platform for those who wish to participate and give their opinion on whether the SESA is working in practice or not during its implementation. ✓	MNR will post the reports and information on its website.
Workshop participants from government organizations, mining and private companies	General participants	Direct conversation	✓ The SESA report will be translated into Dutch to facilitate its understanding to the stakeholders of Suriname.	The issues addressed during the validation workshop are included in the scoping report either directly or indirectly and will be further investigated

workshop participants			✓ Depending on the size of the document, a minimum of two weeks is required to review and comment. ✓ Monitoring and training will be required to implement the SESA. Government authorities must ensure a budget for this purpose. ✓ It is important that Suriname continues to comply with the Minamata convention. ✓ Small mining companies must be included in the SESA report. ✓ It would be necessary to review the existing and required Bauxite infrastructure to verify the feasibility of reopening the Bauxite mines. ✓ Land tenure is a main issue since without this, communities cannot receive any compensations for mining on their land. ✓ The SESA should clearly explain the different types and levels of mining in Suriname. ✓ Mining is an important income generating activity in Suriname. Should mining be eliminated, what would be the economic consequences. ✓ It is necessary that the SESA clarifies the terminology used in Suriname regarding the indigenous populations, and the maroon or tribal communities. ✓ Mine closure plans should be clearly defines and reinforced by law. ✓ The environmental agreements and conventions in Suriname will be confirmed in the SESA report.	and developed during the SESA study. Budget required for the implementation of the SESA will be addressed in the SESA report. Appropriate and updated terminology will be used in the SESA report. A glossary of terms will be included in necessary.
-------------------------------------	--	--	---	---

			✓ A clear explanation of the tax payment for large medio, small and artisanal mining activities needs to be included in the SESA. ✓	
Njuka (Aucaners/Aukanisi) Maroon tribe	Kapt Schalwijk +5978919003	Visit to New Koffiekamp community Direct conversation	✓ - Obtain land tenure rights. ✓ - That lamgold carries out community engagement activities. That it fulfills its promises and that there is transparency. ✓ - That lamgold provides training so that people from the community can produce and sell products to the company's camp. ✓ - That the government has better control over the migration of artisanal mining workers. The community does not want foreign artisanal miners working within their territory. Comment. The community members want to be independent miners. Most of them are not interested in working directly for the large mining companies, but they are not against them either.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study. Small-scale mining, independent mining and large mining companies will be well explained in the SESA report.
Njuka (Aucaners/Aukanisi) Maroon tribe	- Palata Theodorus, village head +597 8687247 - Lino Mathews - Gazon Lando	Visit to Adjoema Kondre community Direct conversation	✓ They want to have tenure of their ancestral lands. ✓ They are interested in mining in the area as long as the companies generate work for the community members and fulfill their promises. ✓ They want to develop their agriculture.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study.
Njuka (Aucaners/Aukanisi) Maroon tribe	Msiedjan Meredith. Foundation for Forestry + control prod. +597 7466170.	Meeting with Kawina community authorities and community members from Jaffa, Peninica, and Gododrai Direct conversation	✓ They want to have tenure of their ancestral lands. ✓ That Newmont provide job opportunities for the community members and pay royalties to the community.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further

	meredithlelibi@gmail.com - Nyman Climpie. +597 8921234 - Hersih Yapare +597 715978 - Lanaki Lynda lyndalanaka72@gmail.com - Michael Noordzee maikelnoordzee@gmail.com		✓ That there be good faith in negotiations and that agreements be fulfilled. ✓ That road and river roads be improved. ✓ Receive training in the proper use of mercury.	investigated and developed during the SESA study.
Matawai Maroon tribe	- Elmont Marieffe, +597 7150262 - Achmet Abeneti, +597 8826079 - Endro Black, +597 8423349 - Gerrit Benker	Visit to New Jacobkondre community Direct conversation	✓ They want to have tenure of their ancestral lands. ✓ They want to have technology/machinery to be able to develop small-scale mining. ✓ Interest in receiving training in the proper use of mercury. ✓ That lamgold fulfills its promises, such as improving the access road to the community and improving the bridge.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study.
Paramaka Maroon tribe	- Josef Foster (Grandman +5978406308) - Josef van Gerder - Adam Ceder - Daniel Soena - Michel Pinas - Henri Kamil - Johannes Foster - Johannes Josis - Adje Amouthtem	Visit to Langatabbetje community Direct conversation	✓ The people want to develop their artisanal mining. They indicate that they have 3 needs to be able to better develop in gold mining: (i) obtain tenure of their land, (ii) that Newmont allows them to work within the Marian concession, (iii) access financing to acquire technology that will allow them to work in mining in a much more efficient way. ✓ - Receive training in the proper use of mercury.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study.
Saamaka Maroon tribe	Rudolf Pansa. Village head. +597 8791843 – Contact person for Whatsapp	Visit to Balingsoela community Direct conversation	✓ The young men of the community are actively working in ASGM within their territory. There are concerns within the	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further

	communications: Carmen Sinei. Wife of Rudolf Pansa. +597 8882806.		community around the high use of mercury by ASGM. ✓ A small number of community members are hired by large-scale mining companies (IAMGOLD and Newmont). The community is interested in large mining companies to hire more people from the community. ✓ The community is concerned with the future of gold mining and interested to develop other income generating activities through cooperatives. ✓ The community is interested to receive technical assistance for agricultural development in their territory, including training, capacity building and equipment. Including training on IT and technology for youngsters and elder. Comment. The community is interested on receiving support for the development of sporting infrastructure and playgrounds for young and kids.	investigated and developed during the SESA study.
Saamaka Maroon tribe	• Boobe Arthur. Village head, captain +5978887241 boobearthur@gmail.com • R. Leyman. Miner +597 8766340 • Freddy Basied. Ex miner. +597 8560577 • Zeegenaar Ocenó. Miner. +597 8725846 • Urvi Boobe. Miner. +597 8921882		✓ The community is concerned of land rights and concession granted within their territory without being consulted, and benefits not flowing to the community. ✓ Agriculture after mining it is not an alternative since the land is not fertile for agriculture. ✓ The community is interested to develop other income generation activities through training and/or capacity development, particularly for agriculture and other income generation activities for the village population.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study.

	• Ardilles Landveld. Miner. +597 8631710 • Regilio Asadang. Miner. +597 8514883		✓ In addition, the miners of the community are interested on receiving training on mercury use and alternative mining techniques. ✓ The community is interested in receiving support for the development of key infrastructure in their village. ✓	
Wayana Indigenous peoples	• Jupta Itoewaki. Chairperson +597-817-7535/+597-434-790 juptaitoewaki@mulokot.com • Andre Verhoogt	Visit to the representative office Mulokot Foundation Direct conversation	✓ Obtain Wayana's land tenure rights. ✓ Mining brings social benefits to IPs, It is proposed that 10% of mining profits be destined to improve education, health, decontamination of river water, and employment generation. ✓ That the FPIC process be transparent and effective, and not just something that is done eventually and without commitment. ✓ Implement a quality and culturally appropriate educational system in indigenous communities, which is the basis for creating leadership. Comment Concern about the high levels of crime in Wayana. Every day people are killed in shootings. ✓ Concern about the contamination of the river by the use of mercury by Chinese and Brazilian artisanal miners. This affects the health of the indigenous population, who fish and use the river water for human consumption.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study. Benefits for IPs will be addressed within the legal framework and through recommendations.
Kalina and Lokono Indigenous peoples	1. Jonh Gunther. President of KLIM (Kali'na, Lokono in Marowijne)	Visit to Kali'na & Lokono Inheemsen beneden Marowijne (KLIM), Erowarte community	✓ The number of fish in the river has decreased significantly; fishing used to be abundant.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further

	jona.gunther@hotmail.com Biswane Lowis. Secretary of KLIM. lowisbiswane@gmail.com	Direct conversation with KLIM and community representatives	✓ The river water has a different color. The community perceives that it is contaminated because of artisanal mining. ✓ The trees planted in the area rehabilitated by Alcoa do not reach the expected height for their species. The community indicates that this is because the topsoil layer left by Alcoa is thin. They also indicate that Alcoa did not reforest with native species but used other species. Comment. During the time that Alcoa exploited bauxite, there was an inflation in the community of approximately 60%. ✓ The community is interested in developing tourism (one of the villages that is part of the community is a turtle nesting site) and fishing (they need to make ice to preserve the fish until it is sold).	investigated and developed during the SESA study.
Ministry of Natural Resources (MNR)	Clyde Griffith Component Coordinator SCSD C1 clydegriffith1977@gmail.com +5978532104	Direct conversation	✓ Requested that building materials mining be included in the SESA. ✓ Perception that the mining sector is under regulated and growing.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study.
Ministry of Regional Development and Sports	Dhr. Drs. Mavrick Boejojoe MBA Director Regional Development	Visit at office of the Ministry of Regional Development and Sports Direct conversation	✓ Land rights issues ✓ Support entrepreneurship ✓ Education ✓ Health (STDs, prostitution, contamination of environment) ✓ Water and sanitation ✓ recommendations on preparing a baseline on the mining industry.	The issues addressed during the stakeholder engagement meeting are included in the scoping report either directly or indirectly and will be further investigated and developed during the SESA study.

			✓ All communities should have a buffer zone of some 5 km or more where no mining activities or rights should exist. This is not respected. ✓ It's important to respect the local protocol when visiting communities.	
WB and MNR	Aradhna Mathur	Email	p.4 Missing Appendix C	Appendix C was included in the Draft report presented in October 2022. It will be added to the Final Scoping Report.
			✓ P. 19 As the ESF does not directly apply to this Project, as a standard for good international practice, it can be referred to.	Corrected. These policies will be required by borrowing governments seeking financing by the WBG and will be referred to in the SESA report as a benchmark for good international practice.
			✓ P. 27 In 2020, the Environmental Framework Act was adopted, but the by legislations that would allow for implementation have not yet been approved. The National Environmental Authority has not yet been installed. In addition, a Bill to amend the EFA has now been prepared. When is it likely that the Act will come into force and all its provisions enacted?	Amended. . Although the Environmental Framework Act was adopted in 2020, the legislation that would allow for its implementation has yet to be approved. As such, the National Environmental Authority still needs to be installed. In addition, a Bill to amend the EFA has now been prepared. The date for the Act to enter into force along with its provisions has yet to be confirmed but was expected toward the end of 2022 or early 2023.
			✓ P.27 End of 2022 or 2023?	Included
			✓ P.48 When was this signed? And have initiatives/activities started under this?	Addressed and answered in the Scoping Report.

			✓ P.59 Has this happened?	Addressed and answered in the Scoping Report.
WB and MNR	Isaac Alberto Mera Chehin	Email	P.113 Only the Wayana people are affected by the gold mining? Is this only small-scale gold mining or also illegal mining? Please add a paragraph explaining that these issues will be addressed in the SESA and RPF for the project. Like it has been done in other previous sections.	Addressed and amended in the Scoping report.
WB and MNR	Aradhna Mathur	Email	✓ P.119 What does extent of mining in the legend of the map mean?	Addressed and answered in the Scoping Report.
			P. 122 A map overlapping this information with other data such as protected areas, Forests, IP lands, rivers, archaeological, etc would be very useful going forward for planning purposes. If mining is developed, associated infrastructure such as roads will be needed. Having a map that overlays key features can help plan the development.	This will be considered in the SESA report
			✓ P.123 Please include	Appendix B was included in the Draft report presented in October 2022. It will be added to the Final Scoping Report.
			✓ P.126 Is the Oil and Gas sector being actively developed?	Addressed and amended in the Scoping report.
			✓ P.126 These are significant investments that will require planning, coordination, stakeholder engagement, costs etc. I don't think its correct to state "No real constraints".	Addressed and amended in the Scoping report.
			✓ P.129 Globally mercury use is being phased out of AGSM. Cheap technologies are available.	Addressed and amended in the Scoping report.

			✓ P.135 Another impact from mines is the development of Associated Infrastructure and induced impacts. Building roads, railways, opening up areas, etc. can have significant impacts. This needs to be captured.	Addressed and amended in the Scoping report.
			✓ P.136 What about Waste management?	Addressed and amended in the Scoping report.
			✓ P.148 Missing	Appendix C was included in the Draft report presented in October 2022. It will be added to the Final Scoping Report.
MNR observations received for the SESA report	Clyde Griffith	Email	✓ There needs to be more information on the mitigation measures to address issues such as the two largest-scale mining companies having been involved in litigations over land rights and land use issues with local inhabitants (see section 5.3), which highlights a significant challenge for the people and the government over land use and rights. ✓ The SESA should clarify the legal and illegal mining concessions. There is discussion about small scale and large scale mining and their impacts on health and land right, but there is no mention of illegal mining, specially illegal gold mining that requires the use of mercury. ✓ In the SESA there needs to be a chart where the World Bank OPs are described with a column that establishes the GAP between World Bank requirements and the legal framework of Suriname, as well as a column that clearly states the mitigation measures for the identified impacts. ✓ A map overlaying mining sites, concessions with key information such as	Most observations and recommendations received for the SESA report have already been addressed in the Scoping report and will be further developed in the SESA report.

			protected areas, forests, IPs, rivers should be included. ✓ The SESA must develop a clear explanation of types of mining and their impacts in Suriname. ✓ The SESA should have an explanation of the impacts on labor when the mining sector starts to be more regulated. Changes in this sector can be troublesome, in countries like Peru it has led to strikes that took the lives of police officers and people from the affected communities. ✓ Land tenure must be a central focus of the consultations for the SESA. However, both documents should state that World Bank policies will apply in case of discrepancies with the national law. Especially when it comes to indigenous land rights. ✓ FPIC must be explained in both documents, the scoping report and the SESA.	
--	--	--	---	--

